

- B. oxygen
- D. temperature

C. rainfall

D. salinity

33. UME/97/29

The groups of instruments that can be used in the measurement of ecological factors in a habitat is

- A. secchi disc, microscope and pooter
- B. potentiometer, hygrometer and chemical balance
- C. anemometer, photometer and hygrometer
- D. telescope, quadrat and wind vane

34. UME/2000/30/TYPE: A

A pooter is used in ecological studies to collect

- A. flying insects
- B. small insects
- C. worms
- D. rodents

35. UME/2003/22/TYPE: N

A maximum and minimum thermometer is suitable for measuring the

- A. temperature of the soil
- B. temperature of water at different depth
- C. highest and lowest body temperatures of an animal
- D. highest and lowest temperatures of the day

36. PCE/93/36

The estimation of the population of Tilapia in a pond is best done by

- A. the quadrant method
- B. counting
- C. capture and recapture
- D. transect method

MISCELLANEOUS QUESTIONS (15)

37. All members of a species inhabiting a given area make up the

- A. community
- B. population
- C. biomass
- D. ecosystem (PCE/91/37)

38. In an ecological study, the biotic factors are.

- A. rainfall, light and temperature
- B. general atmospheric conditions
- C. the effects of living things on the ecosystem
- D. the structures of the earth's surface (PCE/91/36)

39. The functional position of an organism in the ecosystem is referred to as its

- A. habitat
- B. niche
- C. community
- D. micro-habitat (PCE/94/34)

40. The habitat of barnacles and limpets is

- A. freshwater
- B. arboreal
- C. marine
- D. terrestrial (PCE/94/35)

41. The ecological factors common to both terrestrial and aquatic habitats are

- A. tides, current, salinity and rainfall
- B. salinity, rainfall, temperature and light
- C. pH, salinity, rainfall and humidity
- D. rainfall, temperature, light and wind (PCE/2001/32/TYPE: 4)

42. The most important abiotic factor in the desert biome is

- A. temperature
- B. light
- C. rainfall
- D. wind (PCE/2002/15/TYPE: 5)

UME 2008/32

remain ecological problem facing intertidal organisms is

- B. floatation
- D. humidity

UME 2009/32

freshwater, marshes and swamps, the most important

factor that organisms have to adapt to is

- B. high salinity
- D. low pH

UME 2010/37/TYPE: D

The activity of an organism which affects the survival of

another organism in the same habitat constitutes

- A. an abiotic factor
- B. a biotic factor
- C. physiographic factor
- D. an edaphic factor

METHODS OF MEASURING ABIOTIC/BIOTIC FACTORS

UME/84/42

An anemometer is an instrument for measuring

- A. relative humidity
- B. altitude
- C. wind speed
- D. turbidity
- E. salinity

UME/84/33

The transect method can be used in ecology to show the

- A. number of plants and animals in a habitat
- B. population of a particular plant species
- C. distribution of organisms along a line
- D. heights of trees in a section of a forest
- E. number of young plants across a forest

UME/88/38

The turbidity of a pond can be measured using the

- A. anemometer
- B. secchi disc
- C. theodolite
- D. hydrometer

UME/93/40

Which of the following instruments is NOT used in measuring abiotic factors in any habitat?

- A. microscope
- B. thermometer
- C. hygrometer
- D. wind vane

UME/95/31

The depth of illumination in a water body can be measured with a

- A. photometer
- B. secchi disc
- C. hydrometer
- D. anemometer

UME/97/33

The speed of wind can be measured with an instrument called

- A. hygrometer
- B. secchi disc
- C. anemometer
- D. wind vane

UME/2001/36/TYPE: T

The hygrometer is used for measuring

- A. relative humidity
- B. specific gravity

2019-08-14 13:49

CHAPTER 15

1. E	2. B	3. C	4. B	5. C
6. C	7. B	8. A	9. C	10. B
11. B	12. A	13. C	14. A	15. A
16. D	17. A	18. A	19. C	20. B
21. B	22. C	23. B	24. D	25. B
26. C	27. D	28. B	29. A	30. B
31. C	32. A	33. C	34. B	35. D
36. C	37. B	38. C	39. B	40.
41. D	42. A			

Understand the following after reading this chapter:

Food chains, food webs and trophic levels
Energy flow in the ecosystem
Nutrient cycling in nature

SYMBIOTIC INTERACTIONS OF PLANTS AND ANIMALS

Symbiosis: Symbiosis is a close association between two organisms in which both of them benefit from each other. Examples are: Algae and fungi in Lichen, Algae and plant roots in mycorrhiza, Protozoa in the intestine of termite, Nitrogen fixing bacteria in the root nodules of leguminous plants.

Commensalism: Commensalism is an association between two organisms living together in which only one (commensal) benefits from the association while the other neither benefited nor is harmed. Examples are: Remora fish and shark, Oyster and crab, Man and intestinal bacteria, Epiphytes growing on plants

Parasitism: Parasitism is a close association between two organisms in which one, known as the parasite, lives in or on and feeds at the expense of the other organism which is known as the host. Examples include: Man and the tapeworm, Mistletoe and flowering plant.

Predation: Predation is a type of association between two organisms in which the predator kills the other, called the prey and directly feeds on it. Examples are: Cat feeding on rats, Hawk and chicken.

FOOD CHAINS, FOOD WEBS AND TROPHIC LEVELS

Food chains is a linear feeding relationship in which an organism feeds on the one before it in the sequence e.g. Grass → Grasshoppers → Lizards → Snakes

Food web is a complex feeding relationship consisting of interconnected food chains in a community.

Trophic level refers to any step in the transfer of energy and matter in a community. This describes a specific level in a food chain. The term *trophic* refers to **nutrition**.

There are four important levels in most food chains:

- **Producers:** Organisms which convert some of the energy from the sun into stored chemical energy (usually plants).
- **Primary consumers:** Organisms that obtain energy by consuming producers. They are herbivores.
- **Secondary consumers:** Organisms which obtain energy by consuming primary consumers. They are carnivores.

- **Decomposers:** These organisms form the end point of every food chain. They are bacteria and fungi that obtain their energy by breaking down dead organisms from the other trophic levels. Each description of a trophic level will describe an organism's role in the ecosystem. Organisms may occupy more than one trophic level, (e.g. when acting as omnivores).

ENERGY FLOW IN THE ECOSYSTEM

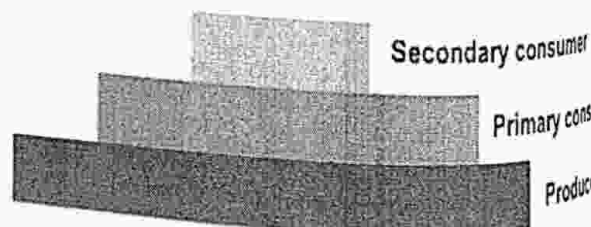
The sun is the ultimate source of energy on earth. Producers i.e. green plants produce their food through photosynthesis. The consumers i.e. animals and non-green plants depend on already made food by the producers. Decomposers i.e. microbes feed on dead plants and animals and in the process help to recycle nutrients.

Pyramids in ecology

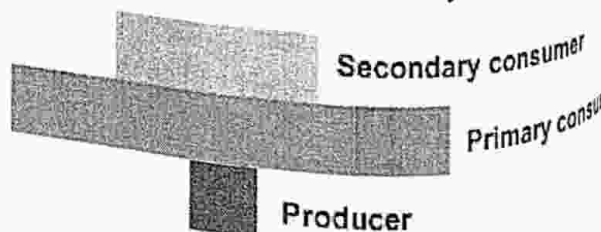
Ecological pyramids are used as a tool to illustrate the feeding relationships of the organisms, which together make up a community.

Pyramid of numbers

This is the simplest way of illustrating the feeding relationships within a community. The commonest form shows that the numbers of organisms occupying each trophic level decreases from producers to secondary consumers and beyond.



Two problems with this form of pyramid are that the numbers involved may be huge (in the hundreds of thousands) and some pyramids may be inverted.

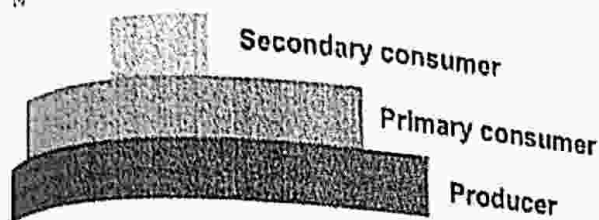


Pyramid of biomass

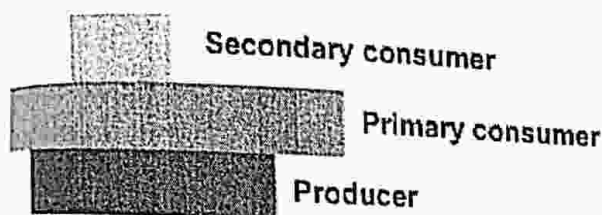
This indicates the feeding relationship between organisms occupying different trophic levels with reference to biomass.

Biomass can be measured as either wet mass or dry mass. Measuring the dry mass is more accurate as it does not include the variable water content of organisms. The commonest form of the pyramid of biomass shows...

the total biomass of organisms occupying each trophic level decreases from producers to secondary consumers beyond.

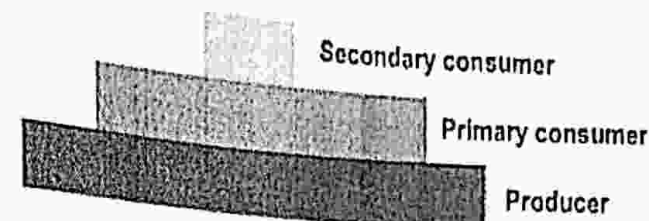


It is still the problem that a pyramid of biomass can be misleading and also it does not take account of changes over time. The sampling must all be carried out at one moment and therefore indicates the standing crop and not productivity.



Pyramid of energy

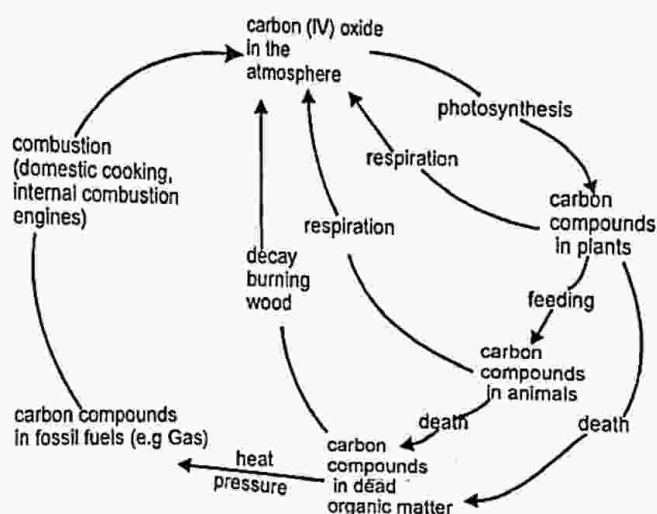
This is the most accurate representation of the feeding relationship between the organisms at different trophic levels. It takes into account the energy gains and losses over a period of time.



NUTRIENT CYCLING IN NATURE

CARBON CYCLE

Plants get their carbon from carbon dioxide in the atmosphere while animals obtain carbon through feeding on plants. Carbon dioxide is added to the atmosphere by: Respiration, combustion, decay; and volcanic eruption and removed by photosynthesis



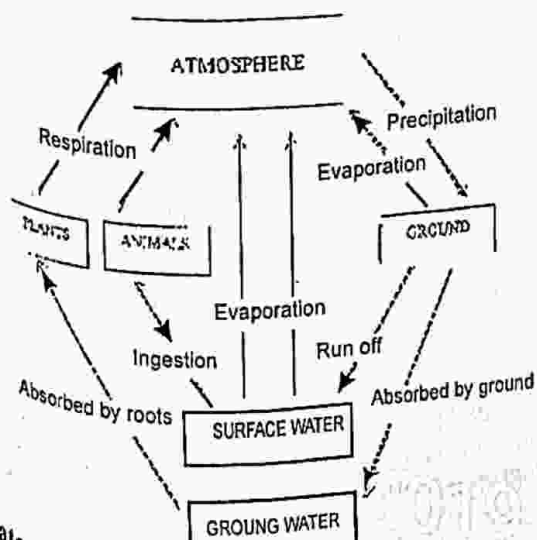
Carbon cycle

HYDROLOGICAL CYCLE

The hydrological cycle or water cycle is the continuous movement of water between the land, the sea and the air

- Precipitation** - All water released from clouds such as rain, snow, hail, sleet & snow
- Surface Runoff** - Water flowing across the surface of the land, whether in a channel or over the land
- Interception** - When trees or man made objects get in the way of rain reaching the ground surface
- Infiltration** - When water soaks into the soil

- Evaporation** - When water is heated by the sun and rises into the sky as water vapour
- Transpiration** - When moisture from plants and leaves is lost to the atmosphere
- Condensation** - When water vapour is cooled and turns into water droplets to form clouds
- Water Table** - The level of saturated ground in the soil - it rises and falls depending on the amount of rain



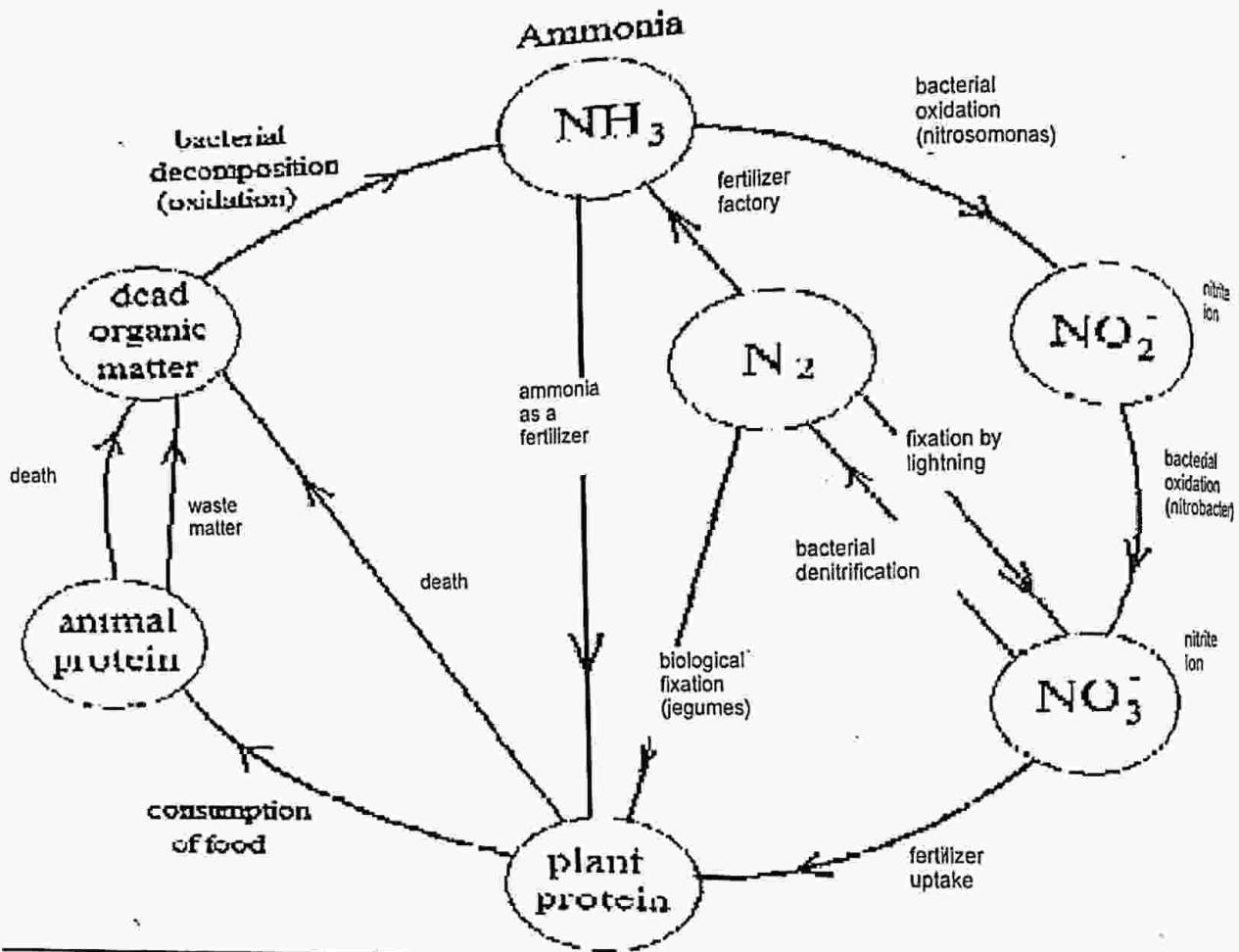
NITROGEN CYCLE

Nitrogen cycle involves a complex biochemical transformation which adds and removes nitrogen and its compound from the soil.

Nitrification (process that add nitrogen to the soil)

- Nitrifying bacteria**- Nitrosomonas and nitrobacter convert ammonia and nitrites to nitrates respectively.
- Nitrogen fixation** by;
 - Lightening or electrical discharge in the atmosphere
 - Symbiotic nitrogen fixation by some bacteria e.g. Rhizobium
 - Non-symbiotic nitrogen fixation by some free-living bacteria e.g. Clostridium and Azotobacter
 - Application of nitrogen containing fertilizers e.g. NPK

Denitrification (process that remove nitrogen from soil): uptake by plants, erosion, leaching, burning denitrifying bacteria.



Nitrogen cycle

Revision Questions

16.1. INTERACTIONS OF PLANTS AND ANIMALS

1. UME/79/19

In which of these associations is much harm done to one of the partners?

- A. Symbiosis
- B. Epiphytism
- C. Commensalisms
- D. Parasitism
- E. Mutualism

2. UME/80/39

An organism X lives entirely on the waste products in another organism Y. In this association, X is a

- A. Symbiont
- B. Commensal
- C. Saprophyte
- D. Parasite
- E. Epiphyte

3. UME/81/28

Which of the following statements is NOT true of symbiosis?

- A. Symbionts must be living
- B. It is an association of give and take
- C. The association may involve two plants
- D. Association between two similar species
- E. symbionts derive mutual benefit

4. UME/87/46

Epiphytes growing on the branches of trees provide an example of the relationship known as

- A. parasitism
- B. commensalisms
- C. saprophytism
- D. holophytism

5. UME/88/41

Which of the following relationship involves only one organism?

- A. saprophytism
- B. commensalism
- C. parasitism
- D. symbiosis

6. UME/89/41

Lichen is an example of

- A. a saprophytic organism

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Asymbiotic association
Canepiphytic plant
Acarnivorous plant

7. UME/91/50
Nitrogen fixing micro-organisms in leguminous plants
symbiotically in the
A. root nodules
B. tap roots
C. branch roots
D. root hairs

8. UME/2003/32/TYPE: A
Mycorrhiza is an association between fungi and
A. roots of higher plants
B. filamentous algae
C. bacteria
D. protozoans

9. UME 2007/4
Which of the following is an example of parasitism?
A. fungi growing on a dead tree branch
B. a squirrel living in an abandoned nest of a bird
C. mistletoe growing in an orange tree
D. cattle egrets taking ticks from the body of cattle

10. UME 2008/30
Which of the following associations is an example of mutualism?
A. *Hydra vividis* and *zoochlorellae*
B. Human and lice
C. Shark and remora fish
D. Bread and *Rhizopus stolonifer*

11. UME 2009/30
The association between bacteria residing in the caecum and the ruminant is
A. parasitism
B. predator
C. saprophytism
D. mutualism

12. UME 2010/29/TYPE: D
Mycorrhiza promote plant growth by
A. protecting it from infection
B. helping it to utilize atmospheric nitrogen
C. serving as a growth regulator
D. absorbing inorganic ions from the soil

16.2. FOOD CHAINS, FOOD WEBS AND TROPHIC LEVELS

13. UME 2010/28/TYPE: D
A food chain always begins with a
A. decomposer
B. producer
C. primary consumer
D. consumer

14. UME/79/43
Grasses → Grasshoppers → Lizards → Snakes → Hawks
In a food chain, the organism which are the least

A. grasses
C. lizards
E. hawks

B. grasshoppers
D. snake

15. UME/79/44
Which of these is NOT true? Grasses in the above food chain
A. trap all the sun energy
B. trap a small percent of the sun energy
C. are primary producers
D. are eaten by primary consumers

16. UME/81/14
Choose the sequence which represents the correct order of organism in a food chain
A. Grass → snake → toad → grasshopper → hawk
B. Grass → grasshopper → toad → snake → hawk
C. Grass → grasshopper → snake → toad → hawk
D. Grass → snake → grasshopper → toad → hawk
E. Grass → toad → snake → grasshopper → hawk

17. UME/86/46
In a community, bacteria and fungi are referred to as
A. producers
B. decomposers
C. scavengers
D. consumers

18. UME/90/42
Which organism is an omnivores?
A. Praying mantis
B. Hawk
C. Mouse
D. Grasshopper

19. UME/90/43
Which of the organisms will have the lowest population in an ecosystem?
A. Hawk
B. Cowpea
C. Praying mantis
D. Mouse

20. UME/91/42.
Organisms in an ecosystem are usually grouped according to their trophic level as
A. carnivorous and epiphytes
B. consumers and parasites
C. producers and consumers
D. producers and saprophytes

21. UME/93/38
At which trophic level would the highest accumulation of a non-biodegradable substance occur?
A. primary productions
B. tertiary consumers
C. primary consumers
D. secondary consumers

22. UME/94/32
Which of the following set is made up of decomposers?
A. rhizopus, earthworm and protozoa
B. mushroom, rhizopus and bacteria
C. bacteria, earthworm and nematodes
D. earthworm, sedges and platyhelminthes.

23. UME/2000/36/TYPE: M
In a food chain, each succeeding level in a forward direction, represent?
A. an increase in the number of individuals

- B. a decrease in the number of individuals
 C. an increase in the biomass of individuals
 D. a gain in the total energy being transferred

16.3. ENERGY FLOW IN THE ECOSYSTEM

24. UME/2001/32/TYPE: T

The highest percentages of energy in an ecosystem occurs at the level of the

- A. secondary consumers B. decomposers
 C. producers D. primary consumers

25. UME/82/36

A pyramid of numbers can be defined as

- A. the number of plants and animals in an ecosystem
 B. an arrangement of organisms according to their habitats
 C. the numerical relationship of a food chain
 D. the number of plants and animals in a population
 E. the total number of species and genera in a community.

26. PCE/2002/21/TYPE: 5

The unidirectional flow of energy in an ecosystem is known as

- A. feeding relationship B. energy pyramid
 C. food chain D. food web

27. UME/89/40

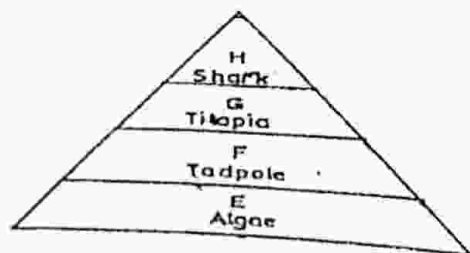
In an ecosystem, the LEAST efficient transfer link is from

- A. producers to primary consumers
 B. sun to producers
 C. primary consumers to secondary consumers
 D. secondary consumers to decomposers

28. UME/92/34

In a typical predator food chain involving secondary and tertiary consumers, the organisms become progressively

- A. smaller and more numerous along the food chain
 B. equal in number and size along the food chain
 C. larger and fewer along the food chain
 D. parasitized along the food as consumers get bigger



29. UME/92/35

Which level of the pyramid has the least total stored energy?

- A. E B. F C. G D. H

30. UME/92/36

Which organisms in the pyramid functions as a tertiary consumer?

- A. Algae B. Shark
 C. Tadpole D. Tilapia

31. UME/98/32

In a food chain involving a primary producer, a primary consumer, a secondary consumer, a tertiary consumer, the flow of tropic energy is in the form that the

- A. primary consumer has more energy than the primary producers
 B. secondary consumer takes up all the energy contained in the primary consumers
 C. energy is shared equally between the three groups of organisms
 D. secondary consumers gets only a small portion of the energy contained in the primary producer.

32. UME/98/29

Which of the following is the direct consequence of transferring energy from one trophic level to another?

- A. an increase in biomass
 B. a decrease in the efficiency of energy conversion
 C. an increase in the total numbers of resulting individuals
 D. a decrease in the resulting biomass

16.4. NUTRIENT CYCLING IN NATURE

16.4.1. CARBON CYCLE

33. UME/95/33

In spite of the removal of carbondioxide from the atmosphere, its amount remains more or less constant because

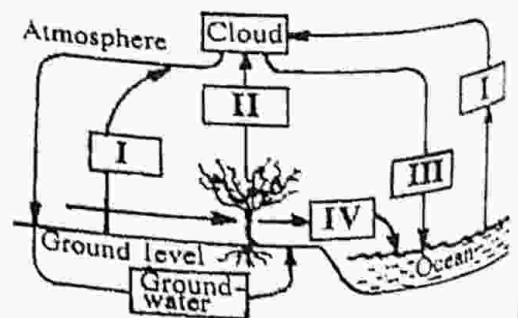
- A. it is produced by green plants during photosynthesis
 B. it is produced during respiration by animals
 C. it is absorbed in ocean water
 D. green plants release it during the day.

34. PCE/91/15

The carbon dioxide content of the atmosphere is kept constant by the processes of

- A. decay, photosynthesis and transpiration
 B. respiration and decay
 C. photosynthesis and combustion
 D. decay, respiration and photosynthesis

16.4.2. HYDROLOGICAL CYCLE



35. UME/2005/34/TYPE: D

Evaporation and transpiration are respectively represented by the components labelled

- A. I and II
 B. II and III
 C. III and IV
 D. IV and I

36. UME/2005/35/TYPE: D

The main reservoir of water in the cycle is the

- A. cloud
- B. groundwater
- C. plant
- D. ocean.

37. PCE/2003/29/TYPE: N

Water cycle is maintained in nature mainly by

- A. evaporation and precipitation
- B. precipitation and condensation
- C. transpiration and evaporation
- D. transpiration and precipitation

38. PCE/95/40

Which of the following is readily available to plants in the soil?

- A. gravitational water
- B. hygroscopic water
- C. capillary water
- D. run-off water

16.4.3. NITROGEN CYCLE

39. UME/80/44

During thunderstorms, the energy of lightning discharge cause

- A. oxygen and nitrogen to combine
- B. more carbon dioxide to be formed
- C. nitrites to be converted to nitrates
- D. nitrates to be converted to nitrogen
- E. protect the microorganism in soil.

40. UME/2000/29/TYPE: M

Atmospheric nitrogen is converted to soil nitrogen for plant use by

- A. nitrification and combustion
- B. putrefaction and lightning
- C. lightning and nitrification
- D. combustion and putrefaction

41. UME/81/32

Nitrification means

- A. conversion of nitrates to nitrogen
- B. fixing nitrogen into plant
- C. conversion of nitrates to nitrites
- D. changing of ammonia to nitrites, then nitrates
- E. nitrogen cycle.

42. UME/80/43

Leguminous plants, mucosa, are usually planted in cultivated farmlands because they

- A. enrich the soil with phosphates
- B. provides animals with food
- C. enrich the soil with organic nitrogen
- D. protect the soil from being overheated
- E. protect the micro-organisms in the soil.

43. UME/85/49

Atmospheric nitrogen is directly replenished in nature through

- A. activities of denitrifying bacteria
- B. the breakdown of ammonium salts in the soil
- C. the activities of nitrifying bacteria
- D. the activities of nitrogen-fixing bacteria in root
- E. activities of nitrogen-fixing bacteria in root

44. UME/87/49

Nitrifying bacteria are important because they

- A. release nitrogen to the atmosphere
- B. convert atmospheric nitrogen to ammonia
- C. combine ammonia with nitrogen.
- D. oxidize ammonium salts to nitrates

45. UME/88/50

Dead plants and animal are decomposed by bacteria and fungi into

- A. nitrates
- B. nitrites
- C. amino acids
- D. ammonia

46. UME/94/15

The nitrifying bacteria, nitrosomonas, convert ammonia to

- A. nitrites
- B. nitric acid
- C. nitrates
- D. nitrous oxide.

47. UME/98/30

The condition that encourages denitrification is

- A. low soil oxygen
- B. high soil nitrogen
- C. absence of soil bacteria
- D. lightning and thunderstorm

48. UME/84/48

Denitrifying bacteria nature liberates gaseous nitrogen directly from

- A. ammonium salt
- B. soil nitrates
- C. thunderstorms
- D. soil nitrites
- E. plant and animal proteins.

49. One of the ways in which the nitrogen of the atmosphere is converted into nitrates for plants is by the action of:

- A. Putrefying bacteria
- B. Lightning
- C. Denitrifying bacteria
- D. Decay
- E. None of the above

50. The most important bacteria that convert nitrogen into nitrates are:

- A. Nitrobacter
- B. Nitrosomonas
- C. Azotobacter
- D. Denitrifiers
- E. Bacterium radicicola

51. The symbiont that lives in the root nodules of leguminous plants is:

- A. Azotobacter
- B. Bacteria
- C. Nitrosomonas
- D. Nitrogen
- E. None of the above

52. While the leguminous plant gains nitrates from the bacteria in its root nodules, it supplies the bacteria with:

- A. Protein
- B. Starch
- C. Carbohydrates
- D. Air
- E. Fats

53. The nitrates absorbed from the soil by plants are built up into:

- A. Plant nitrites
- B. Plant nitrates
- C. Animal proteins
- D. Ammonium compounds
- E. Plant proteins

CHAPTER 16

1. D	2. C	3. D	4. B	5. A
6. B	7. A	8. A	9. C	10. A
11. D	12. D	13. B	14. E	15. A

54. The first product of putrefaction of plants and animal protein is:

- A. Nitrites
- B. Nitrates
- C. Ammonium compounds
- D. Nitrogen of atmosphere
- E. Animal nitrate

MISCELLANEOUS QUESTIONS (16)

55. Two organisms of different species, living in close association but not dependent on each other are referred to as.

- A. Parasites
- B. commensals
- C. Symbionts
- D. autotrophs (UME/93/39)

56. An association between the root nodules of a leguminous plant and rhizobium sp is known as

- A. commensalisms
- B. mycorrhiza
- C. parasitism
- D. symbiosis (UME/2001/1/TYPE:T)

57. Bacteria which reduce nitrates in the soil into gaseous nitrogen are called

- A. putrefying bacteria
- B. nitrifying bacteria
- C. saprophytic bacteria
- D. denitrifying bacteria (PCE/2005/39/TYPE:P)

58. The presence of bacteria in the caecum of many herbivores is an example of

- A. parasitism
- B. commensalism
- C. saprophytic
- D. symbiosis (PCE/95/32)

59. Two organisms of different species living in close association but not totally dependent on each other are referred to as

- A. parasites
- B. commensals
- C. symbionts
- D. autotrophs (PCE/2001/31/TYPE:A)

60. A typical food chain, at which trophic level would the highest amount of energy occur?

- A. producer
- B. primary consumer
- C. secondary consumer
- D. decomposer (PCE/98/33/TYPE: C)

61. A correct food chain in a aquatic habitat is illustrated as

- A. fish → worm → tadpole
- B. phytoplankton → tadpole → fish
- C. zooplankton → spirogyra → tadpole
- D. algae → fish → tadpole (PCE/98/32/TYPE: C)

62. The release of inorganic salts from the decomposition of dead and animals in the soil is called

- A. mineralization
- B. volatilization
- C. composting
- D. humification (PCE/99/28/TYPE: E)

63. Which of the following organisms contribute to the process of denitrification?

- A. Nitrosomonas
- B. Nitrobacter
- C. Rhizobium
- D. Pseudomonas (PCE/99/30/TYPE: E)

64. The progressive loss of energy at each level in a food chain leads to

- A. a decrease in biomass at each successive level
- B. an increase in the number of organisms at successive level
- C. an increase in the total weight of living matter at successive level
- D. an increase in biomass at each successive level (UME/2004/42/TYPE 4)

16. B	17. B	18. C	19. A	20. C
21. B	22. B	23. B	24. C	25. C
26. B	27. D	28. C	29. D	30. B
31. D	32. D	33. B	34. C	35. A
36. B	37. A	38. C	39. A	40. C
41. D	42. C	43. A	44. D	45. D
46. A	47. A	48. B	49. B	50. A
51. E	52. C	53. E	54. C	55. D
56. D	57. D	58. D	59. D	60. A
61. C	62. A	63. D	64. A	

CHAPTER 17

stand the following after reading this chapter:

Aquatic
Terrestrial

is any environment in which an organism lives. It is an area where physical and chemical factors required by organism are met. There are two types of habitats:

Aquatic habitat: the habitat related to water.

Terrestrial habitat: the habitat that relate to land

AQUATIC HABITAT

Aquatic habitat comprises marine, estuarine and fresh

Marine Habitat: It is either ocean or high sea water

Characteristics of Marine Habitat

- 1. It has a high concentration of salt.
- 2. Water pressure increases with depth up to 1000 atmosphere.
- 3. Surface current is driven by wind.
- 4. Tides occur due to alternative rise and fall in water level.
- 5. The high sea is relatively alkaline with a pH range of 8.0 to 8.4 at the surface.
- 6. Light can only penetrate up to about 200m or more.

They are (A) Continental shelf (B) Abyssal region

(A) Oceanic region

- 1. Continental shelf comprises of:
 - Splash Zone:** Is just above the high tide mark. It is wetted by the spray of water from breaking waves.
 - Intertidal Zone:** Water stays in this area temporarily, especially when there is high tide, and gets exposed to air when the tide goes down.
 - Littoral Zone:** It is found between high tide and low tide zones. It is about 200m deep.
 - Subtidal Zone:** Extends beyond 200m.
- 2. **Abyssal region:** This zone is more than 2000 metres to the ocean floor.
- 3. **Oceanic region:** It is beyond the edge of continental shelf over the deep waters.

Estuarine Habitat: It is a transition between a river and the sea.

Characteristics of Estuarine Habitat

- 1. The salinity is always fluctuating.
- 2. Low species diversity: species are fewer in terms of diversity.
- 3. Turbidity increases especially during rainy season.
- 4. Very low oxygen content.
- 5. Very high productivity because of high nutrient level.

Fresh Water Habitat: It is known to have low salt content and can either be running (i.e. lotic) or stagnant (lentic). Example are rivers and ponds respectively.

Characteristics of Fresh Water Habitat

1. The salt content is very low.
2. Stratification of temperature is rare especially in running water.
3. Oxygen at the surface of the water is more than at the bottom.
4. Current can affect the distribution of gases, salts and small organisms e.g. in rivers or stream.
5. The waters are shallow.
6. Light penetrate more in running water than in stagnant water.
7. Low density of water.
8. The water bodies are always small.

TERRESTRIAL HABITAT

Soil and rain influence the nature of terrestrial habitat. They are divided into the following:

1. Marsh
2. Rain Forest
3. Grass Land
4. Arid Land

Marsh

1. It is a low land soil that is usually flooded.
 2. It stands between aquatic and terrestrial habitats. Examples can be found in dried ponds or lakes as well. Examples of plants are sedges, water drop-wart, marsh pennywort, balsam, hornwort (ceratophum), water lilies and duck weed.
- Examples of animals are fishes, salamanders, frogs, toads, lizards, snakes, birds and mammals.

Rain Forest

Plants:

1. Scattered very tall trees of about 40m or above.
2. Continuous layer of trees of about 30m.
3. Possession of draw-out tips by trees for easy dripping.
4. The search for sun make the trees to be tall.
5. Possession of thin bark by trees so as to fasten gas exchange as well as transpiration
6. Plants that have the ability to climb. They do so in search of light e.g. Wild yam.
7. Possession of buttress root for strong anchorage e.g. silk cotton.
8. The broadness of the leaves enhances the rate of transpiration.
9. Clasp plants are used for support.

Animals:

1. Animals are protected by camouflage.
2. Some that are adapted to climbing do so with grasping pads tails to hold on to tree branches e.g. monkeys
3. Others have grasping scales e.g. snakes
4. The birds make use of their feathers

Grassland

They are divided into these groups namely:

1. Temperate grassland
2. Savanna or Tropical grassland (southern guinea, northern guinea and sahel savanna as found in Nigeria)
3. Meadow grassland

Characteristics

1. Presence of intensive sunshine and high temperature
2. Rainfall is unevenly distributed
3. Humidity is always low
4. Deciduous plants are prominent
5. Trees which are fire resistant are common since there is constant bush burning
6. The habitat harbors a whole range of herbivores such as squirrel, grass cutters, antelopes and rodents.
7. Some carnivores which prey on herbivores are common e.g. Leopards and Lions.

Arid Land

They are divided into:

1. Hot arid lands (temperature as high as 77°C) e.g. hot deserts and semi desert.
2. Cold arid (temperature never up to 0°C).

Characteristics

1. Rainfall is always very low (below 25.4cm per annum)
2. There is high evaporative stress
3. Ecosystem is unstable
4. Extremely hot in the day and extremely cold at night.
5. Soil may be rocky or sandy
6. Extremely windy all year round.
7. Mobile sand dunes are common.

Revision Questions

17.1. AQUATIC

1. UME/82/50

The absence of stomata shows that a leaf may be

- A. from a floating plant B. from a submerged plant
C. variegated D. from a terrestrial plant
E. from a parasitic green plant

2. UME/88/43

Which of these groups of animals is likely to be found in fresh water?

- A. blood worm, pondskater and scorpion
B. bloodworm, pondskater and dragon fly larva
C. pondskater, scorpion and dragonfly larva
D. pond skater, bloodworm and ant – lion

3. UME/90/44

The salinity of a brackish environment

- A. increases immediately after rain
B. increases at the end of the rainy season
C. decrease with increase in micro-organism
D. increase during the dry season

4. UME/93/41

Plants adapted for life in salty marsh are called

- A. hydrophytes B. xerophytes
C. halophytes D. epiphytes

5. UME/98/31

A fresh water plant such as water lily can solve the problem of buoyance by the possession of

- A. aerenchymatous tissues
B. dissected leaves
C. thin cell walls of the epidermis
D. water-repelling epidermis

6. UME/2001/45/TYPER: M

The stem of a typical aquatic plant usually has many

- A. air cavities B. intercellular space
C. water cavities D. water conducting cells

7. PCE/93/40

The roots commonly found in red mangroves are the

- A. buttress roots B. breathing roots
C. clasping roots D. stilt roots

8. PCE/96/31

Production of the viviparous seedlings is an adaptive feature of

- A. mangrove forest plants
B. rain forest plants
C. savanna plants
D. desert plants

9. PCE/97/32

Pneumatophores are adaptations for

- A. support
B. breathing
C. feeding
D. rainforest

10. PCE/97/33

Species diversity is highest in the

- A. savanna
B. mangrove swamp
C. desert
D. rainforest

11. UME 2008/31

In a typical freshwater habitat, the edge of the stream or pond constitutes the

- A. tidal zone
B. intertidal zone
C. littoral zone
D. euphotic zone

12. UME 2010/32/TYPER: D

Which of the following organisms is mainly found in the marine habitat?

- A. Tilapia B. Dog fish
C. Tortoise D. Achatina

17.2. TERRESTRIAL

13. UME/90/41

- In a savanna ecosystem, the abiotic factors include
- A. legumes, temperature and sandy soil
 - B. water, temperature and soil
 - C. minerals, oxygen and reptiles
 - D. water, soil and grasses

14. PCE/96/34

Which of the following plants is most adapted to the grassland ecosystem?

- A. mahogany
- B. pine
- C. date-palm
- D. baobab

15. PCE/2000/34/TYPE: A

The most important adaptation required by terrestrial organisms is the ability to

- A. search for food
- B. escape from predators
- C. conserve water
- D. procure space

16. PCE/2003/26/TYPE: N

The most limiting factor in a terrestrial habitat is

- A. high solar radiator
- B. high wind action
- C. the availability of water
- D. fluctuating temperatures

17. UME 2009/29

The most important ecological factor in a terrestrial environment is

- A. rainfall
- B. humidity
- C. wind
- D. soil

18. PCE/98/31/TYPE: C

The greatest hazard faced by terrestrial organisms is

- A. predation
- B. dehydration
- C. temperature changes
- D. food scarcity

MISCELLANEOUS QUESTIONS (17)

19. One adaptation shown by terrestrial plants for combating drought is the

- A. poor development of conducting tissues
- B. possession of shorter roots which are less branched
- C. the reduction in the number of stomata on the leaves
- D. reduction in the intake of water from soil by roots (PCE 91/38)

20. In a forest habitat, emergent are best described as the

- A. layer of shrubs
- B. ground layer of shade-loving plants
- C. layer of tall trees towering above others at intervals
- D. layer of small trees than 10 metres tall (PCE/95/38)

21. A major difference between the marine and fresh water habitats is the

- A. abundance of fishes
- B. colour of the water
- C. abundance of phytoplankton
- D. concentration of salts (PCE/98/36/TYPE: C)

22. The most widespread arboreal mammal is the

- A. monkey
- B. squirrel
- C. pangolin
- D. bat (PCE/99/37/TYPE: E)

CHAPTER 17

1. B	2. B	3. D	4. C	5. A
6. A	7. D	8. A	9. B	10. D
11. C	12. B	13. B	14. C	15. C
16. A	17. A	18. B	19. C	20. C
21. D	22. A			

CHAPTER 18

Understand the following after reading this chapter:

Mangrove swamps
Tropical rain forest
Guinea savanna (Southern and Northern)
Sahel (Sudan) savanna
Desert

LOCAL (NIGERIA) BIOMES

In Nigeria there are three distinct biomes viz: Mangrove swamps or estuarine, Tropical rainforest and Savanna.

MANGROVE SWAMPS

In Nigeria, they are found in the delta regions of Lagos, Edo, Rivers, Cross Rivers and Delta states. The coastal areas of these states are muddy and water logged. The climate is hot and wet throughout the year. The areas support forests of small, evergreen broad-leaved trees growing in the shallow brackish water or muddy soil.

Characteristics of Mangrove Vegetation

1. Plants of the mangrove swamp possess specialised breathing roots called pneumatophores (in White mangrove) which grow into the air above the water surface. On it are tiny openings called lenticels which serve for gaseous exchange.
2. Many of the plants that float on the water surface have air-spaces in their leaves, stems and roots e.g. water lettuce and water lettuce.
3. Many mangrove trees have prop roots for support e.g. Red mangrove (*Rhizophora*).

Animals in Mangrove Swamp

The animals include worms (e.g. earthworms), crabs, frogs, snakes, insects, many fish including some species of Tilapia and the mud-skipper which can move rapidly over wet mud. Birds include herons, palm nut vultures while mammals include bats and monkeys.

TROPICAL RAINFORESTS

In Nigeria, the tropical rainforests cover parts of Rivers, Cross Rivers, Akwalbom, Imo, Abia, Edo, Delta, Ogun, Ondo and Oyo states.

Characteristics

1. The forest is dense and contains tall trees which branch at the top to form a canopy. This cuts off light so that the forest interior has low light intensity, high humidity, still air and damp floor.
2. Most of the trees are broad-leaved and evergreen i.e. the leaves are green throughout the year, and they are gradually dropped and new ones grown to replace them.
3. The trees have tall, un-branched stems which are supported by large buttress roots at the base.
4. They have thin barks and they flower and form fruits

in the wet season.

5. They are rich in epiphytes (plants that depend on others for support) and woody climbers known as lianas.
6. The vegetation on the forest floor is sparse because not enough light reaches it. Only small plants such as liverworts, mosses and ferns are found on the forest floor.

Notable rainforest trees in Nigeria include Obeche (*Triplochintonscleroxylon*), Iroko (*Chlorophora excelsa*), Mahogany timber trees (*Khaya ivorensis*); Silk-cotton tree (*Ceibapetandra*); Teak (*Naucleadiderichii*); Afara (*Terminalia species*) etc.

N.B:

The trees are arranged in layers called storeys. There are five storeys or layers of vegetation, each consisting of plants with different light requirements.

1. **Upper storey:** This is the topmost layer with the tallest trees also called emergent. The trees may be 60 metres or more in height. They have an open canopy i.e. their canopies do not touch and form a continuous column.

2. **Middle storey:** This is made up of trees with dense and closed canopy i.e. their canopies touch and form a rather continuous column. The trees may be between 20-30 metres in height.

3. **Lower storey:** This layer is made up of trees of about 10-20 metres in height. The trees are scattered and do not form any canopy.

4. **Shrub layer:** This is made up of small plants below 10 metres in height (about 5 metres).

5. **Ground flora:** Only very little light reaches the ground flora as most of it is filtered off by the leaves of the trees and shrubs. Therefore only the smallest plants are found on the forest floor.

Animals of the Tropical Rainforest

1. There are 'arboreal' mammals including monkeys, squirrels, pottos and tree-hyrax. These mammals have long limbs and tail that enable them to climb and jump from tree to tree.
2. There are large forest birds including fruit-pigeons, hornbills, parrots and turacos.
3. Bats are the only flying mammals. They feed on fruits and insects. Their skin is modified to form wings which connect to fore and hind limbs.
4. There are insects including plant-bugs, beetles and the caterpillars of moths and butterflies.
5. Big mammals which move about in herds to protect

themselves from predators include apes and elephants. Many invertebrates which live in the litter or upper soil layer to prevent desiccation include earthworms and snails.

Savanna

Most parts of Nigeria outside the rainforest belt is covered by tropical Savanna vegetation. This is divided into three zones:

Guinea Savanna (South and North)

This borders the rainforest belt. In Nigeria, it is found in parts of Kaduna, Benue, Kogi and Kwara states. Because this zone is moist, the grasses grow tall during the rainy season. The trees occur quite close together especially on the fringes of the rainforest. There is the habit of burning the grasses during the dry season, hence Savanna trees are fire-resistant.

Sudan Savanna

This occurs to the North of the Guinea Savanna. It is therefore drier than the Guinea savannah and has shorter grasses, fewer and more scattered trees. It covers Kano and parts of Sokoto, Borno, Bauchi and Niger states.

Sahel Savanna

This borders the Sahara desert. The grasses here are generally very short and in clumps or tussocks. The trees and shrubs are few and sparsely distributed. Bush fire is a major problem here. During the dry season, when the grasses are burnt, the underground parts survive and grow again during the rainy season. Sahel occurs around the lake Chad Basin e.g. Borno, Yobe.

Animals of the Savanna Region

The Savanna supports an abundance of herbivores (plant eaters) and carnivores (flesh eaters). Examples of herbivores are insects (termites and grasshoppers), birds, rodents, antelopes, goats, cows, camels, elephant etc. Some of these graze the grass and browse on shrubs and trees. The carnivores include leopards, lions, tigers, cheetah etc. The scavengers are mostly vultures and hyenas. These eat up or clean up the remnants of animals killed by the carnivores.

DESERT

A desert is an area with less than 25.5cm rainfall a year or no rain fall at all for many years. Because of this, trees are scarce or completely absent i.e. the vegetation is very poor and consequently, the animal population is also poor.

There are two types of deserts; Hot and dry desert: the temperature is very high (e.g. 800C or 1780F) e.g. Sahara desert. Cold and frozen desert: the temperature here is usually below 00C during the winter and the top soil is frozen all year round e.g. Gobi desert in China.

There is no desert in Nigeria but the Sahel Savanna which is the narrow, most northern border to the Sahara desert, outside the northern Nigerian border to the West Africa. The Sahara desert is the only desert in the West Africa.

Tropical rainforest: This has tall trees and climbers which form canopies. It is characterized by high rain fall and high temperature. It is usually stratified as follows:

A or uppermost layer, B or middle layer, C or lower layer, Shrub layer and Floor layer.

Common plants are: iroko, oil palm, rubber, obeche, wall nut etc. Common animals are snails, squirrel, snakes, monkeys etc.

Rain forest is found in Southern Nigeria i.e. Oyo, Lagos, Ogun, Edo, Ondo, Delta, Rivers, Enugu, Abia, Anambra, Imo, Cross Rivers and Bayelsa states.

Savanna: The savanna has more grasses than trees and shrubs. It is characterized by low rainfall, high temperature and a long dry season. It is further subdivided into:

- Southern Guinea savanna:** Niger, Benue, Kwara States.
- Northern Guinea savanna:** Kano, Sokoto States.
- Sahel savanna** found in Borno, Yobe state

Swamp or Estuarine: It is solid if saturated and lacks oxygen. Common plants include mangrove, Raphia and coconut. It is found in the Niger Delta, Calabar, Lagos

Revision Questions

.PCE\99\36\TYPE: E

In which of the following local biomes is Makurdi, Kaduna and Enugu located?

- Sudan savanna
- Sahel savanna
- Guinea savanna
- Rainforest

2. UME\98\32

The sequence of the biomes in Nigeria from Port Harcourt to Damaturu is

- estuarine → rainforest → Guinea savanna → sahel savanna
- rainforest → savanna → rainforest → sahel savanna
- estuarine → Guinea savanna → rainforest → sahel savanna
- rainforest → estuarine → Guinea savanna → desert

3. UME\2003\36\TYPE: A

The correct sequence of biomes from northern to southern Nigeria is

- estuarine → tropical rainforest → guinea savanna → sahel savanna
- sahel savanna → sudan savanna → guinea savanna → tropical rainforest → estuarine
- sahel savanna → tropical rainforest → estuarine → guinea savanna
- guinea savanna → sudan savanna → tropical rainforest → sahel savanna → estuarine.

18.1. TROPICAL RAINFOREST

4. UME\95\34.

In a tropical rainforest, non-epiphytic ferns and fern allies occurs as

- mid-story species
- upper-story species
- shade-loving species
- emergent species.

5. PCE/98/35/TYPE: C

Epiphytes and climbers are mostly found in the

- A. sahel savanna
- B. sudan savanna
- C. rainforest
- D. mangrove swamp

18.2. GUINEA SAVANNA (SOUTHERN AND NORTHERN)

6. UME/88/44

One of the characteristics of plants in the savanna is the

- A. possession of thin, smooth barks
- B. possession of large tap roots
- C. production of seedlings on mother plant.
- D. possession of thick flaky barks.

7. UME/97/37

The southern Guinea savanna differs from the Northern Guinea savanna in that it has

- A. lower rainfall and shorter grasses
- B. less grasses and scattered trees
- C. more rainfall and taller grasses
- D. less arboreal and burrowing animal

8. PCE/2004/38/TYPE 9

Grasses are better adapted than trees in the savanna because of their possession of

- A. corms
- B. bulbs
- C. rhizomes
- D. tubers

9. UME/2005/37/TYPE: D

In Nigeria, the Guinea savanna belt borders the

- A. mangrove swamps and the sahel savanna
- B. rainforests and the sudan savanna
- C. deserts and the sudan savanna
- D. rainforests and the deserts

Use the list of biome below to answer questions 10 and 11

- 1. Deserts
- 2. Rainforest
- 3. Southern Guinea Savanna
- 4. Northern Guinea Savanna

10. UME/94/35

A biome with a low annual rainfall, few scattered trees within dense layer of grasses are found in Kano and Katsina states is

- A. 1
- B. 2
- C. 3
- D. 4

11. UME/94/36

A biome where there are many cacti and the small mammals undergo aestivation for long period is

- A. 1
- B. 2
- C. 3
- D. 4

12. UME 2006/42

Grasses recover quickly from bush fires in the savanna because of their

- A. perennating organs
- B. rapid growth rate
- C. fibrous roots
- D. succulent stems

18.3. SAHEL(SUDAN) SAVANNA

13. UME/2002/21/TYPE: Y

In a population study using the transect method a student is likely to record the highest number of species in

- A. a tropical rainforest
- B. a guinea savanna
- C. a sahel savanna
- D. an estuarine swamp

14. PCE/92/34

Sokoto state is located in the

- A. rainforest
- B. southern guinea savanna
- C. sahel savanna
- D. desert

15. UME 2006/50

The main purpose of establishing shelter belts in the Sahel region is to

- A. check desert encroachment
- B. provide wood fuel
- C. beautify the region
- D. break the harmattan wind

18.4. DESERT

16. UME/97/36

Adaptive feature of plant to desert condition include

- A. thick barks, succulent stems and sunken stomata
- B. thin barks, succulent stems and sunken stomata
- C. thick barks, air floats on stems and sunken stomata
- D. air spaces in tissues, adventitious root and thick barks

Ecological Zone	Temperature(°C)	Rainfall(mm)
I	45	300
II	32	2000
III	30	2200
IV	15	800

17. UME/2004/44/TYPE: 4

Which of the zones is likely to be a desert?

- A. III
- B. IV
- C. I
- D. II

18. UME/2004/45/TYPE: 4

High relative humidity will be expected in zones

- A. II and III
- B. II and IV
- C. I and IV
- D. III

19. UME/2004/45/TYPE: 4

A state in Nigeria that is most susceptible to desert encroachment is

- A. Kaduna
- B. Katsina
- C. Kwara
- D. Taraba

MISCELLANEOUS QUESTIONS (18)

20. The least adaptive feature for arboreal life is the

- A. possession of four limbs
- B. possession of claws
- C. development of long tail
- D. counter shading of coat colour

(UME/99/46/TYPE: D)

21. In which of the following zones of the marine habitat is the intensity of light at its minimum?

- A. littoral
- B. intertidal
- C. benthic
- D. splash

(PCE/96/32)

2019-08-14 15:00

22. The guinea savanna is mainly characterized by

- A. grass and thorny shrubs
- B. tall grasses and sparse trees
- C. tall grasses and shrubs
- D. short and sparse grasses (PCE/2003/31/TYPE:N)

23. In which of the following local biomes is Makurdi, Kaduna and Enugu located?

- A. Sudan savanna
- B. Sahel savanna
- C. Guinea savanna
- D. rain forest (PCE/99/36/TYPE: E)

24. The typical features of the grassland of the Obudu and Jos Plateaux in Nigeria are

- A. heavy rainfall, rocky highlands and high temperature
- B. low temperature, thick forests and low rainfall
- C. rocky highlands and low temperature
- D. thick forests and grassland (PCE/06/47/TYPE: A)

25. The cactus plant is structurally adapted to survive by the possession of

- A. swollen leaves that store food and water
- B. swollen roots that store food and water
- C. thick corky bark that reduces loss of water
- D. swollen stems that store food and water.

(PCE/2002/26/TYPE: 5)

26. The mambilla plateau is a unique Nigerian biome located in

- A. Plateau state
- B. Borno state
- C. Taraba state
- D. Benue state.

(UME/2005/36/TYPE:D)

27. The mangrove swamp in Nigeria is restricted to the

- A. Sahel savanna
- B. Sudan savanna
- C. Guinea savanna
- D. Tropical rain forest (UME 2007/8)

28. UME 2007/10

In which of the following Nigerian States can Montana vegetation be found?

- A. Taraba
- B. Enugu
- C. Bauchi
- D. Plateau

29. Which of the following biomass could be characterized by very low rainfall, cold night, hot days and fast blooming plants?

- A. Northern Guinea Savanna
- B. Southern Guinea Savanna
- C. Tropical desert
- D. Montana forest (UME 2009/33)

30. Low annual rainfall, sparse vegetation, high diurnal temperatures and cold nights are characteristic features of the

- A. desert
- B. montana forest
- C. guinea savanna
- D. rain forest (UME 2010/23/T 22 E 5)

2019-08-14 13:51

21. D

CHAPTER 18

2. A	3. B	4. C	5. C
7. C	8. C	9. B	10. D
12. A	13. A	14. C	15. A
17. C	18. A	19. B	20. D
22. B	23. C	24. A	25. D
27. D	28. A	29. C	30. A

21. A 27
26. C 2

CHAPTER

1. E
6. B
11. E
16. B
21. E
26.

CHAPTER 19

Understand the following after reading this chapter:

Patterns of growth in population size
Factors affecting population sizes
Ecological succession

CHARACTERISTICS OF A POPULATION

Population density or size: This is the number of individuals occupying a given habitat.

Natality or birth rate: This is the number of new individuals produced per unit time.

Mortality or death rate: This is the number of individuals dying per unit time.

Growth rate: This is the increase in number of organisms in a community.

Dominance: This refers to the species which is dominant over others in terms of cover or density or both.

Age distribution: This is the proportion of individuals in different age range within a population.

Dispersability: This is the ability of members of a population to disperse or spread.

ECOLOGICAL SUCCESSION

This is a long term, gradual or progressive series of changes occurring in the structure, composition, variety or diversity and number of species in an area till a stable or climax community is established.

Pioneer species to climax communities

Pioneer species: These are the first species to occupy a new habitat, starting new communities. They have rapid reproductive strategies, enabling them to quickly occupy an uninhabited area. Many have an asexual stage to their reproduction.

Seres: These are the various stages that follow on from the pioneer species.

Climax community: This is the stable community that is reached, beyond which, no further succession occurs.

TYPES OF SUCCESSION

Primary succession

This occurs when the starting point is a bare ecosystem, (e.g., following a volcanic eruption or a landslide). The pioneer species are usually lichen, then by moss or algae. They are able to penetrate the bare surface, trap organic material and begin to form humus.

Over several generations soil begins to form. The soil can be used by a more diverse range of plants with deeper root systems. Gradually larger and larger plants occupy the ecosystem along with a diversity of animals.

Finally a climax community is reached and the species present do not change unless the environment changes in some way.

An example of primary succession forming woodland:

1. Bare rock is colonised by lichen and then moss.
2. Small plants, ferns and grasses take over.
3. Larger plants with deeper roots appear.
4. Bushes and shrubs replace non-woody plants.
5. Fast growing trees form a dense, low woodland.
6. Larger, slow growing oak trees create the woodland.

Secondary succession

This occurs when the starting point is bare, existing (e.g., following a fire, flood or human intervention). The type of succession proceeds in the same way as primary succession except that the pioneer species tend to be grasses and fast growing plants.

An example of secondary succession forming an woodland:

1. Bare soil is colonised by grasses and pioneer plants.
2. Grasses begin to predominate with time.
3. Shrubs replace the grasses.
4. Fast growing trees appear.
5. Slow growing oaks create the climax community.

FACTORS AFFECTING A POPULATION

These are divided into:

Density dependent factors e.g. predation, competition, diseases etc.

Density independent factors e.g. flood, bush fire, earthquake, erosion etc.

PROBLEMS AFFECTING THE POPULATION

1. Overcrowding
2. Food shortage

Causes of Overcrowding

1. Lack of space
2. Increase in birth rate
3. Absence of predators
4. Immigration

Effects of Overcrowding

1. Decrease in natality
2. Stress and death
3. Spread of disease
4. Shortage of food
5. Competition

Causes of Food shortage

1. Drought
2. Pests
3. Diseases
4. Over population

Effects of Food shortage

1. Competition
2. Emigration
3. Decline in natality

FAMILY PLANNING AND BIRTH CONTROL

Family planning is a device by which couples determine the number of children they want and when they want them. Birth control is used to prevent a woman from becoming pregnant, for as long as she wishes.

Control methods include:
 Withdrawal method
 Coitus interruptus method or safe period
 Condom (sheath) in males
 Diaphragm in females
 Spermicidal cream or tablet
 Intra-uterine device (IUD)
 Contraceptive pill
 Injection
 Sterilization

Revision Questions

1.1. PATTERN OF GROWTH IN POPULATION SIZE

1. UME/92/33

A population is defined as a collection of similar organisms that

- A. behave in the same way
 B. interbreed freely
 C. are found in the same habitat
 D. eat the same food

2. PCE/2004/34/TYPE: 4

A combination of density-dependent factors that can limit the size of a population is

- A. predation, behavioural adaptations and competition
 B. parasitism, physiological adaptations and competition
 C. predation, parasitism and competition
 D. competition, behavioural adaptations and physiological adaptations.

3. UME/2000/30/TYPE: M

- I. High birth rate and high immigration rate
 II. Low mortality rate and low emigration rate
 III. Low birth rate and high immigration rate
 IV. High mortality rate and high emigration rate. Which combination above can cause rapid overcrowding in climax biotic communities and human settlements?

- A. II and III
 B. I and III
 C. I and IV
 D. I and II

4. UME/2002/14/TYPE: Y

Plants tend to prevent overcrowding by means of efficient

- A. water uptake
 B. seed germination
 C. pollination
 D. seed dispersal

5. UME/2003/33/TYPE: A

A density-dependent factor that regulates the population size of organisms is

- A. sudden flood
 B. disease
 C. fire outbreak
 D. drought

6. PCE/93/36

The estimation of the population of Tilapia in a pond is best done by

- A. the quadrant method
 B. counting
 C. capture and recapture method
 D. transect method

7. PCE/99/31/TYPE: E

The combination of factors most likely to cause over crowding is

- A. competition, immigration and high death rate
 B. immigration, natality and high survival rate
 C. natality, competition and migration
 D. high survival rates, scarce resources and migration

8. PCE/99/32/TYPE: E

The factor that affects distribution of organisms in an aquatic habitat is

- A. humidity
 B. rainfall
 C. wind
 D. turbidity

9. UME 2006/49

A population that doubles in size at constant intervals is an indication of

- A. sigmoid growth
 B. exponential growth
 C. population explosion
 D. rapid growth

10. UME 2010/36/TYPE: D

The average number of individuals of a species per unit area of the habitat is the

- A. population frequency
 B. population size
 C. population distribution
 D. population density

11. PCE/07/32

A population is said to have reached equilibrium if it

- A. no longer faces competition
 B. has attained its carrying capacity
 C. is declining with time
 D. has exceeded its carrying capacity

19.2. FACTORS AFFECTING POPULATION SIZES

12. UME/2004/47/TYPE: 4

The scarcity of food causes a sudden decrease in populations in size by

- A. minimizing the rate of competition
 B. raising the mortality rate
 C. bringing about immigration
 D. decreasing the reproductive rate

13. PCE/2004/32/TYPE: 9

The study of changes in population size over time is known as

- A. population census
 B. population dynamics
 C. population control
 D. population estimate

19.3. ECOLOGICAL SUCCESSION

14. UME/93/42

Which group of plants would be the first colonizers in an ecological succession changing rocks to soil?

- A. Mosses
 B. Ferns
 C. Lichen
 D. Grasses

15. UME/95/35

Colonization of a bare rock surface is termed

- A. evolution
 B. speciation
 C. primary succession
 D. secondary succession

16. UME/98/43

The biological association that contributes directly to succession in a community is

- A. competition
- B. predation
- C. parasitism
- D. commensalism

17. UME/97/38

After the heavy rainfall and the formation of a large pond, the most likely sequence of changes in the vegetation of the food is

- A. Euglena → Waterlily → Spirogyra
- B. Waterlily → Spirogyra → Grass → Euglena
- C. Spirogyra → Euglena → Grass → Waterlily
- D. Euglena → Spirogyra → Waterlily → grass

18. UME 2006/43

Climax communities in a biotic succession are usually characterized by

- A. a constant change in the appearance of the communities
- B. different species that are constantly changing
- C. a stable composition of plant and animal species
- D. rapid change in the plant and animal species

MISCELLANEOUS QUESTIONS (19)

19. Secondary succession is much faster than primary succession because

- A. pioneer coloniers are more in number
- B. soil is already present
- C. secondary succession requires less nutrients
- D. species competition is increased (UME/2005/39/TYPE:D)

20. In ecological succession, older communities are characterized by greater productivity as a result of increasing

- A. Stability
 - B. biomass
 - C. population
 - D. species turnover
- (PCE/2005/35/TYPE:P)

21. Which of the following is true of a climax community?

- A. species diversity increases
- B. species diversity decreases
- C. the community is almost in a state of equilibrium
- D. the community keeps changing (PCE/92/46)

22. Succession that takes place on an abandoned farmland is known as

- A. primary succession
 - B. climax
 - C. animal succession
 - D. secondary succession
- (PCE/93/41)

23. One effective means of reducing intra-specific competition in a population is to

- A. decrease the resources competed for
 - B. increase the resources competed for
 - C. increase the number of competitors
 - D. increase the number of species in the community
- (PCE/95/35)

24. In a normal succession, species diversity

- A. decreases
- B. increases
- C. is constant
- D. is zero (PCE/96/35)

25. Species diversity is highest in the

- A. savanna
- B. mangrove swam
- C. desert
- D. rainforest (PCE)

26. In a bare rock environment, which of the likely to be the pioneer occupant?

- A. lichens
- B. ferns
- C. spermatophytes
- D. bryophytes (PCI)

27. The correct formula for determining the organisms in a habitat is

- A. $\frac{\text{Number of organisms}}{\text{Area of habitat}}$
- B. $\frac{\text{Area of habitat}}{\text{Number of organisms}}$
- C. $\frac{\text{Number of organisms}}{\text{Mass of organisms}}$
- D. $\frac{\text{Density}}{\text{Number of habitats}}$ PCE/99/35/TYPE:E

28. In nature, organisms prevent overcrowdingt

- A. Increase in the rate of reproduction
- B. Territorial behaviour
- C. Immigra
- D. Avoiding predators (PCE/01/35/TYPE:A)

CHAPTER 19

2. C

3. D

4. D

5. B

7. B

8. D

9. B

10. D

12. B

13. B

14. C

15. C

17. C

18. C

19. B

20. A

22. D

23. B

24. B

25. D

27. B

28. B

CHAPTER 20

SOIL

Understand the following after reading this chapter:

Components and characteristics of different types of soil
Loss of soil fertility
Renewal and maintenance of soil fertility

Soil gives plant support, in addition to harbouring animals, supply of mineral salts, and production of vegetation.

Types of soil are as follows:

Sandy Soil: Their particles are coarse, loosely bound and have poor nutrient and water-holding capacity. They have very low capillarity and are well drained.

Loamy Soil: They are mixtures of clay and sand particles, with a higher proportion of organic matter. They are more moist, rich in terms of plant nutrients; and drain lesser than sandy soil.

Clay Soil: They are made up of smallest and finest particles. They have water-retaining capacity. They reduce and form lumps when dry. They are damp and sticky to touch when wet.

Laterites: They are earthy, granular or concretionary mass, chiefly of iron and aluminium oxides and hydroxides, occurring as a layer or as scattered nodules in tropical soils. They are impervious to water and contain little nutrients.

To determine the particle sizes in a soil, shake up about 100g of soil with same water in 500ml measuring cylinder and allow settling. The particles settle down in the following order: gravel, sand, silt and clay; clay particles are also suspended in the water. The humus floats on the surface.

SOIL FERTILITY

A fertile soil supports the growth of plants adequately. It is usually a loamy soil with a good humus content, good drainage and aeration and adequate nutrient content. Soil fertility can be increased by adding fertilizer. Soil can lose its fertility through loss of organic matter (humus), soil erosion and leaching.

EXPERIMENTS ON SOIL

Experiment: 1 To determine the amount of air in the soil.

Materials required. A plastic container, two measuring cylinders of 500cm³ each, water, glass rod.

Method. The internal volume of the plastic container is first determined. This is done by pouring water into the

container until it is filled to the brim. The water is poured from the container into one of the cylinders. Fresh soil is then packed into the container. The soil is made tight to the top of the container. Next, the soil is levelled off at the top. The volume of the soil and its component air, is taken to be x . The soil is scraped into the second measuring cylinder which contains 200ml of water. The soil and water are then stirred continuously with a glass rod to drive away any gas in the soil. The volume of the soil and the water in the cylinder is then noted. This volume is known as y . The amount of air in the soil is calculated on a percentage basis as shown below:

$$\begin{aligned}\text{Volume of soil + Air} &= x \\ \text{Volume of soil - Air} &= y - 200 \\ \therefore \text{Volume of air} &= x - (y - 200) \\ \therefore \text{Percentage of soil air} &= \frac{x - (y - 200)}{x} \times 100/1 \\ &= \frac{x - y + 200}{x} \times 100\end{aligned}$$

Experiment: 2 To determine the amount of water in the soil.

Materials required. A small dish, small amount of soil, an oven.

Method. A small dish is weighed and the weight is recorded as a gram. The dish and some amount of soil in it again is weighed and this weight is also noted and it is taken as b gram. Next, the dish with the soil is placed in an oven whose temperature is 100°C, to dry the soil. After some hours, say 12 hours, the dish is removed from the oven. It is then allowed to cool down in a desiccator. Next, the dish and the dried soil are weighed and recorded. Drying, cooling as before and weighing are continued until a more or less constant weight is obtained. This weight is taken as c gram.

Observations. There is a loss in weight of the oven dried dish and soil.

Conclusion. The loss in weight is due to water given off by evaporation. This shows that water is present in the soil.

Calculation. The amount of water in the soil can be calculated on a percentage basis as follows:

$$\begin{aligned}\text{Weight of dish} &= a \text{ gm} \\ \text{Weight of dish + soil} &= b \text{ gm} \\ \text{Weight of soil only} &= (b - a) \text{ gm} \\ \text{Weight of dish + soil from oven} &= c \text{ gm} \\ \therefore \text{Loss in weight} = \text{Weight of water} &= (b - c) \text{ gm} \\ \therefore \text{Percentage of water in the soil} &= \frac{b - c}{b - a} \times 100/1\end{aligned}$$

Experiment: 3 To determine the amount of humus in the soil.

Materials required. A lump of soil, a platinum crucible.

Method. A platinum crucible weighing b gram is filled with soil. The soil is then dried at 100°C as in experiment 2 until all the water in the soil is evaporated. The weight is taken to be a gram. Heat is then applied to the platinum and

the soil is stirred with a wire. Smoke comes out of the soil but heating is continued until the smoke stops and the soil turns red. The soil is then allowed to cool in a desiccator. The red soil with the crucible is then weighed to be c gram.

Observations. There is a loss in weight of the soil which received very strong heat until it turned red.

Conclusion. The loss in weight is due to humus (organic matter) burnt during heating. This shows that humus is present in the soil.

Calculation. The amount of humus in the soil can be calculated on a percentage basis as follows:

Weight of crucible = b gm
 Weight of crucible + dry soil containing humus = a gm
 Weight of dry soil with humus = $(a - b)$ gm
 Weight of crucible + dry soil without humus = c gm
 $\therefore$ Weight of humus lost = $(a - b) - (c - b) = a - c$
 $\therefore$ Percentage of humus lost = $\frac{a - c}{a - b} \times 100/1$

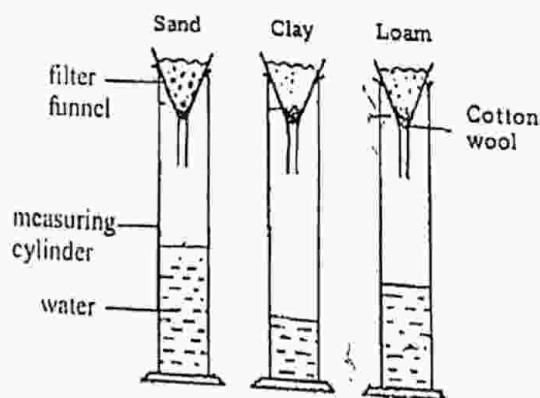
Experiment: 4 To compare the porosity of different samples of soil (sandy, clay, and loamy).

Materials required. Three measuring cylinders, sandy soil, clay soil, loamy soil, 3 filter funnels, water, labels.

Method. Three measuring cylinders of size 200cc are taken. They are labelled A, B, and C. a filter funnel is placed in the mouth of each. Next, soil is put in the funnel of each cylinder; sandy soil in A, clay soil in B and loam soil in C. A known volume of water, say 70cc is then poured onto the soil in each funnel at the same time. The time it takes the water to pass through in the different samples is recorded.

Observations. The water drains through the soil at different rates, being fastest in sandy soil, followed by loam and least in clay soil.

Conclusion. There is difference in porosity in different samples of soil.



Experiment :5 To compare the capillarity of different samples of soil.

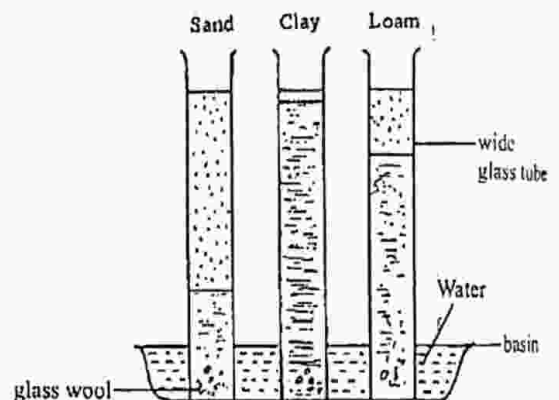
Materials required. Three long glass tubes, cotton wool, labels, sandy soil, loamy soil, clay soil, a trough of water, and maize grains.

Method. Three long glass tubes are collected and at one end of each, some cotton wool is placed. The tubes are

labelled A, B, and C. the tubes are filled with soil (about up) as follows: A to contain sandy soil. B to contain loam soil. C to contain clay soil. The tubes are stood in the trough of water as shown in Fig. 40.12. Next, two or three maize grains are sowed in each. The tubes are then left for some days.

Observations. The grain in A germinates first, after a few days. This is followed by grains in B. Those in C germinate last.

Conclusion. The observations made as above are because water first got to the grains in A before B and this shows that capillarity varies with different samples.



Revision Questions

0.1. COMPONENTS AND CHARACTERISTICS OF DIFFERENT TYPES OF SOIL

1. UME/92/38

Soil with the finest particles is called

- A. silt
- B. clay
- C. sand
- D. gravel

2. UME/79/26

Which of these is NOT a type of soil?

- A. sand
- B. granite
- C. loam
- D. clay
- E. sandy loam

3. UME/85/48

The origin of mineral particles in the soil is

- A. humus
- B. water
- C. micro-organisms
- D. weathered rock
- E. organic matter

4. UME/97/39.

In a soil, the relative amounts of the different soil particles can best be determined by the process of

- A. filtration
- B. centrifugation
- C. precipitations
- D. sedimentation

5. UME/2004/43/TYPE: 4

The soil type that is difficult to plough in a wet season is one that is

- A. sandy
- B. loamy
- C. salty
- D. clayey

6. UME/80/41.

If three 30cm lengths of glass tubing are tightly packed with clay, sand and loamy soil respectively and then stood in a beaker of water for one week. The level of water will be

- A. lowest in the tubes with clay
- B. the same in all the tubes
- C. lowest in the tube with loamy soil
- D. highest in the tube with sandy soil
- E. lowest in the tube in the with sandy soil

7. UME/80/45.

A few grams of dried soil were first heated until red and hot and then further heated until no more smoke was released. This experiment was to determine the

- A. amount of water in soil
- B. percentage of water in soil
- C. presence of humus in soil
- D. resistance of laterites to heat
- E. release of smoke from the soil

8. UME/84/47.

If a handful of soil is shaken with water and left to settle, the soil particles will settle from light to heavy particles as follows

- A. humus, clay, silt, sand, stones
- B. humus, silt, clay, sand, stones
- C. humus, silt clay, stones, sand
- D. humus, sand, silt, clay, stones
- E. clay, humus, silt, sand, stones,

9. UME/85/47

Water retention is highest in soils which are rich in

- A. sand, poor in humus and devoid of clay
- B. clay and sand, but poor in humus
- C. clay and humus but poor in sand
- D. clay, poor in humus and devoid of sand
- E. sand and humus, but poor in clay

10. UME/86/48

A large percentage of tropical soils tend to be acidic because they

- A. contain large quantities of potash
- B. contain large quantities of lime
- C. lose a high proportion of their organic matter to running water
- D. lose lime and potash from the top soil through rain action

11. UME/94/39

The water-retention capacity of a soil indicates its

- A. Fertility
- B. capillarity
- C. aeration level
- D. loss of organic matter due to exposure to direct sunlight
- E. washing out of chalk and limestone from upper layers of soil by heavy rains

12. UME/88/48

The mineral nutrient that easily gets leached out of the soil is

- A. phosphorus
- B. calcium
- C. magnesium
- D. nitrate

13. UME/89/47

Most irrigated lands often become unproductive in later years because

- A. loss of fertility
- B. increase in salinity
- C. soil erosion
- D. loss of water

14. UME/2004/50/TYPE: 4

A farm practice that results in the loss of soil fertility is

- A. continuous cropping
- B. mixed farming
- C. bush fallowing
- D. shifting cultivation

15. PCE/2001/36/TYPE: 4

Leaching of soil nutrients is mainly due to water

- A. flocculation
- B. percolation
- C. erosion
- D. capillarity

16. UME 2007/2

The increasing order of the particular size in the following soil type is

- A. clay → sand → silt → gravel
- B. silt → sand → clay → gravel
- C. clay → silt → sand → gravel
- D. silt → clay → sand → gravel

17. UME/2007/6

A crucible of 5gm weighed 10gm after filling with fresh soil. It is then heated in an oven at 100°C for 1 hour. After scaling in desiccators, the weight was 8gm. The percentage of water in the soil is

- A. 80%
- B. 60%
- C. 20%
- D. 40%

18. UME/2008/37

In an experiment to determine the percentage of air in a soil sample, the following readings were recorded:

- Vol. of water in a measuring cylinder = 500cm³
- Volume of soil added to water = 350cm³
- Volume of water and soil after stirring = 800cm³
- A. 6.25%
- B. 10.36%
- C. 14.28%
- D. 43.28%

19. UME/2009/36

Mass of a crucible = 10g

Mass of a crucible & soil before heating = 29g

Mass of a crucible & soil after heating = 18g

From the information above, determine the percentage of water in the given soil sample?

- A. 20%
- B. 25%
- C. 40%
- D. 50%

20. The soil of any locality is normally richer in

- A. Humus
- B. Rocks
- C. Carbonates
- D. Ammonia
- E. Bacteria

21. In an experiment to determine the proportion of air in sandy and clay soil, it is discovered that this former has much more air in the same volume of soil as the latter. This shows that:

- A. Clay particles being smaller are impervious to air
- B. Sand is better for the growth of crops than clay
- C. Sand can also retain much more water than clay
- D. Sand particles being coarser pack less tightly than the

clay, leaving much room for air
E. None of the above

22. Sand, though usually well drained and well aerated, is regarded as being "warm". This is due to:
A. its poor porosity
B. its good capillarity
C. Its poor imbibitory power
D. its low specific heat
E. none of the above

23. Crops grow on clay soils are usually poor and ripen late. This is due to the fact that:
A. clay soils are badly aerated and water-logged
B. clay soils are too warm
C. clay soils have low capillarity power
D. clay soils allow water to run through easily
E. Clay soils have fewer nutrients

24. The temperature of the soil is partly affected by:
A. The amount of water that evaporates from the soil
B. The amount of humus in the soil
C. The amount of carbon compounds in the soil
D. The texture of the soil
E. None of the above

20.2. RENEWAL AND MAINTENANCE OF SOIL FERTILITY

25. UME/84/46

Erosion can be reduced along a slope by
A. ridging across slope
B. ridging up slope
C. ridging down slope
D. crop rotation
E. bush fallowing system

26. UME/86/49

The following are methods of soil conservation EXCEPT
A. contour terracing
B. strip cropping
C. contour ploughing
D. mixed grazing

27. UME/87/50

The process by which lime is added to soils is known as
A. sedimentation
B. flocculation
C. leaching
D. manuring

28. UME/90/48

An acidic soil can be improved upon by
A. sedimentation
B. leaching
C. flocculation
D. watering

29. UME/98/33

Soil micro-organisms are beneficial because of their involvement in
A. photosynthesis
B. translocation
C. cycling of nutrients
D. respiration using soil air

30. UME/2002/29/TYPE:7

The addition of lime to clayey soil serves to
A. aid water retention
B. close up the texture
C. prevent water-logging
D. improve capillary action

31. UME/2003/35/TYPE:A

Soil fertility can best be conserved and renewed by the activities of
A. microbes
B. earthworms
C. man
D. rodents

32. PCE/03/21/TYPE: N

Excessive application of inorganic fertilizers on farmland leads to an
A. increase in soil fertility
B. alteration in soil
C. improvement in soil texture
D. increase in number of soil organisms

33. PCE/07/33

The organisms that help to increase soil fertility include
A. termite, butterfly, larvae, earthworms and bacteria
B. bacteria, termites, domestic fowls and fungi
C. earthworms, termites, bacteria and fungi
D. earthworms, spiders, bacteria and fungi

34. UME/2010/38/TYPE: D

The loss of soil through erosion can be reduced by
A. crop rotation
B. manuring
C. irrigation
D. watering

MISCELLANEOUS QUESTIONS (20)

35. In a community bacteria and fungi are referred to as:

A. producers
B. decomposers
C. scavengers
D. consumers (UME/86/4)

36. The importance of practicing crop rotation agriculture is to

A. maintain soil fertility
B. improve the maturational value of crops
C. control soil erosion
D. ensure the growth of crops
(UME/2005/41/Type: D)

37. Which of the following types of soil possesses the water retaining capacity

A. clayey soil
B. sandy soil
C. humus soil
D. loamy soil (PCE/93)

38. A soil sample maybe described as humus soil if it

A. it is black in appearance
B. has a greater proportion of sand than clay
C. allows water to run through it slowly
D. contains inorganic and organic materials
(PCE/94/41)

39. Which of the following is readily available to plants in the soil?

A. gravitational water
B. hygroscopic water
C. capillary water
D. run-off water (PCE/95)

40. The group that accurately describes the composition of the soil is

A. mineral matter, humus, water, air and organisms
B. silica, humus, magnesium and water
C. air, water organisms and carbondioxide
D. organisms, water, oxygen, humus and minerals
(PCE/97/35)

41. The best method of preventing leaching is

A. the planting of cover crop
B. mulching
C. terracing
D. strip-cropping (PCE/96/37)

2019-08-14

42. A soil with large particle size, little humus and coarse texture is a

A. clayey soil

B. loamy soil

C. sandy soil

D. humus soil

(PCE/2002/16/TYPE:5)

43. The most important function of soil water is to

A. facilitate absorption of nutrients

B. dissolve inorganic material

C. enhance plant growth

D. give a cooling effect (PCE/2000/37/TYPE:A)

44. Planting of leguminous cover crops increases soil fertility by

A. checking the growth of parasites

B. increasing soil nitrogen content

C. increasing soil colloids

D. adding humus to the soil (PCE/2001/37/TYPE:4)

45. In an experiment, the following results were obtained:

I. Original weight of soil = xg

II. weight of oven dried soil = yg

III. weight humus in soil = zg

The percentage of water in the soil is therefore given by the expression

A. $(x-y/z) \times (100/1)$

B. $(x-y/x) \times (100/1)$

C. $(z/x-y) \times (100/1)$

D. $(x/x-y) \times (100/1)$

(PCE/96/36)

46. A dry soil (x grams) is heated in a known weight of crucible until red hot without smoke. If the final weight of the soil is y grams, the percentage weight of humus in the soil is

A. $100(y-x)$ grams

B. $100(y-x)/y$ grams

C. $100(x-y)/x$ grams

D. $100(x-y)$ grams

PCE/06/39/TYPE: A

47. An adverse effects of excessive application of inorganic fertilizer on the soil is an increase in its

A. Microflora

B. aeration

C. Alkalinity

D. acidity PCE/06/48/TYPE: A

48. In an experiment to determine the percentage of water in a soil sample, the following data were obtained:

Mass of crucible = X, g

Mass of crucible and soil = X, g

Mass of crucible and soil after heating = X, g

What is the amount of soil water?

A. $(x, -x), g$

B. $(x, -x), g$

C. $(x, -x), g$

D. $(x, -x), g$ PCE/07/34

21. D

AFTER 20

- | | | | |
|-------|-------|-------|-------|
| 2. B | 3. D | 4. D | 5. D |
| 7. C | 8. A | 9. D | 10. D |
| 12. D | 13. A | 14. A | 15. B |
| 17. D | 18. C | 19. D | 20. A |
| 22. D | 23. E | 24. D | 25. A |
| 27. B | 28. C | 29. C | 30. D |
| 32. B | 33. C | 34. A | 35. B |
| 37. B | 38. D | 39. C | 40. A |
| 42. C | 43. C | 44. B | 45. B |
| 47. D | 48. B | | |

- | | |
|-------|-----|
| 51. B | 52. |
| 56. C | 57. |
| 61. D | 62. |
| 66. A | 67. |
| 71. C | 72. |
| 76. C | |

CHAPTER

- 1. A
- 6. C
- 11. D
- 16. A
- 21. A
- 26. A

CHAPTER 21

HUMANS AND THEIR ENVIRONMENT

Understand the following after reading this chapter:

Diseases
Pollution and its control
Conservation of natural resources

DISEASES

Disease simply means an illness affecting humans, animals or plants often caused by infections.

Diseases caused by micro-organism:

1a. Animal diseases caused by viruses: poliomyelitis, infective hepatitis, measles, common cold, influenza, small pox, chicken pox, yellow fever, rabies, Acquired Immune Deficiency Syndrome (AIDS)

1b. Plant diseases caused by viruses: rosette disease, cassava mosaic, maize streak, yam mosaic, cowpea mosaic.

2a. Animal diseases caused by bacteria: tuberculosis, leprosy, tetanus, typhoid, dysentery, cholera, pneumonia, anthrax, diphtheria, gonorrhea, syphilis.

2b. Plant diseases caused by bacteria: tomato rot, onion rot, banana wilt.

3a. Animal diseases caused by fungi: ringworm (tinea), aspergilosis, athletic foot, thrush.

3b. Plant diseases caused by fungi: maize smut, maize rust, rice blight, leafspot, cocoa black pod

4a. Animal diseases caused by protozoa: malaria, trypanosomiasis, trichomoniasis, coccidiosis.

CHOLERA

Cholera is caused by a bacteria called *Vibrio cholerae*. It is found in water contaminated with human faeces; no animal reservoirs. It is an important cause of severe infection in developing countries where portable water and sewage systems are poor.

It causes vomiting and frequent watery stools (diarrhoea) and may lead to death.

SCHISTOSOMIASIS (BILHARZIA)

Schistosomiasis is caused by Schistosomes, a fluke worm. Schistosome include: *Schistosomamansoni*, *S. japonicum*, *S. haematobium*.

Schistosomes have separate male and female worms. Infective cercariae penetrate the skin, enter the circulation and develop into male and female worms in the intestine (*S. mansoni*, *S. japonicum*) or bladder (*S. haematobium*). The female worm produces eggs which penetrate the intestinal or bladder wall and are passed in faeces or urine. Ova reach fresh water, hatch into larvae and infect snails.

ONCHOCERIASIS (RIVER BLINDNESS)

Onchocerciasis is caused by a roundworm, *Onchocerca volvulus*, through the bite of the blackfly, *Simulium damnosum*, which breeds in fast flowing streams.

AFRICAN TRYPANOSOMIASIS (SLEEPING SICKNESS)

African trypanosomiasis is caused by *Trypanosomabruceigambiense* and *T. bruceirhodesiense* and is transmitted by tsetse fly. The infective stage (trypomastigote) is present in salivary glands of the tsetse fly. Trypomastigotes enter the host via the insect bite and reach the lymphatic system, blood and CNS.

MALARIA

Malaria is caused by protozoan parasite, *Plasmodium* spp. through the bite of female *Anopheles* mosquito bites human and infectious plasmodia (sporozoites) are introduced into the blood stream.

POLLUTION AND ITS CONTROL

Pollution is the discharge of harmful substances into the environment in amount that can cause damage to man, animals and crops. There are three types of pollution: air, water and land.

Causes of Air Pollution: It is caused by discharge of gases and particles from factories, domestic fires, vehicle exhaust pipes, dust, storms, fog, noise and radioactive elements.

Effects of Air Pollution: Bronchitis, silicosis, eye and nose irritation, green house effect (excess CO₂ in atmosphere).

Control of Air Pollution:

1. Industries should be sited away from residential areas.
2. Tall factory chimneys should be built.
3. Banning of old cars.
4. The use of lead-free fuels.

Causes of Water Pollution: Discharge of untreated sewage, chemical wastes, fertilizers, heat etc.

Effects of Water Pollution: Contamination of drinking water, death of aquatic life, out-break of epidemic diseases like cholera and diarrhea.

Control of Water pollution:

1. Proper treatment of sewage before discharge.
2. Biological control should replace chemical control to avoid pollution.
3. Minimizing oil-spillage.

Causes of Land Pollution: Indiscriminate dumping of refuse, indiscriminate dumping of dilapidated vehicles and oil-spillage.

Effects of Land Pollution: Poor crop yield, breeding of pest.

Control of Land Pollution:

1. Recycling of wastes.

2. Refuse should be incinerated.
3. Public enlightenment campaign.

5. Avoidance of clean clearing which may expose the soil to erosion.

CONSERVATION OF NATURAL RESOURCES

Conservation of natural resources can be defined as the process of making full use of natural resources in order to ensure their continuous availability and to keep the quality or original nature of the environment safe from decaying.

Natural resources can be renewable and non renewable.

Renewable natural resources: These are natural resources that are recoverable. Examples are plants and animals.

Non renewable natural resources: these are resources which when exhausted cannot be replaced. Examples are mainly minerals, coal, oil, gas.

The Need for Conservation

1. To maintain and to promote the conservation of natural resources.
2. To prevent the destruction of natural environment
3. To prevent the destruction of natural ecosystem; this will allow the organisms in the ecosystem to survive
4. To promote the recycling of some scarce mineral resources eg water
5. To prevent naturally beautiful sceneries for their authentic value. Natural resources that need to be conserved are: wildlife, fresh water, forest, mineral resources and soil.

Method of Conserving of Wildlife

1. Establishment of game or forest reserves
2. Establishment of zoological gardens
3. Control of hunting to prevent extinction of some animal species
4. Prohibiting the killing of animals in game reserves.
5. Prohibition of bush burning as this may lead to migration or displacement of wildlife
6. Prohibition of deforestation and encouragement of a forestation or reforestation
7. Creation of awareness on the value of wildlife
8. Prevention of pollution to prevent the destruction of aquatic life.

Methods of Conserving Fresh Water Supply.

1. Use fresh water economically and prevent unnecessary wastage.
2. Reduce water pollution.
3. Prevent deforestation.
4. Prevent draining of marches and swamp so as to hold rain water and release it slowly to under ground water supplies, rivers and lakes.
5. Reduce unproductive irrigation schemes which causes Salinization of the land.

Methods of Conserving Soil

1. Prevention of over grazing which may cause soil erosion.
2. Prevention of bush burning which may expose the soil to erosion.
3. Prevention of indiscriminate falling of trees.
4. Prevention of pollution of land so as not to destroy useful soil organism.

Methods of conserving Mineral resources

Mineral resources, unlike other resources are non renewable resources because once they are exhausted, they cannot be replaced, hence the need to conserve. The method of conserving mineral resources are:

1. There should be legislation against indiscriminate mining of mineral resources.
2. There should be proper pricing of mineral resources and their product to ensure maximum value for the mineral products.
3. Effective and efficient extraction method of mining should be adopted to prevent wastage.

Revision Questions

1.1. DISEASES

1. UME/78/42

A farmer X working in a swamp did not eat any food nor drink any water. Which of these diseases can he not contract?

- | | |
|----------------------|--------------|
| A. Cholera | B. Bilharzia |
| C. River blindness | D. Malaria |
| E. Sleeping sickness | |

2. UME/80/47

Below are some groups of diseases. Which group is caused by bacteria?

- | | |
|-------------------------------|------------------------|
| A. tuberculosis, small pox | B. gonorrhea, measles |
| C. tuberculosis, polio | |
| D. sleeping sickness, measles | E. syphilis, gonorrhea |

3. UME/84/45

Which of the following lists of diseases, their causes a transmission is CORRECT?

- cholera, virus, severe diarrhea, infected water
- malaria, protozoan, high fever, contact with infected person
- syphilis, virus, venereal disease, sexual intercourse
- smallpox, virus, skin with blister, close contact with infected person
- sleeping sickness, bacteria, tiredness, headache, dozing, tsetsefly bite.

4. UME/85/43

The primary and secondary hosts respective bilharzia are

- | | |
|-------------------|------------------|
| A. fish and man | B. man and dog |
| C. snail and man | D. man and snail |
| E. fish and snail | |

5. UME/86/44

Which of these diseases CANNOT be prevented by immunization?

- | | |
|------------------|-------------------|
| A. Poliomyelitis | B. Tuberculosis |
| C. Cholera | D. Onchocerciasis |

6. UME/86/47
The swollen shoot disease of coconut tree is caused by a
A. virus B. fungus
C. bacterium D. protozoan
7. UME/88/45
Which of the following diseases can be contracted in areas with fast flowing rivers?
A. schistosomiasis B. elephantiasis
C. syphilis D. onchocerciasis
8. UME/90/46
Which set of diseases is spread mainly by insect vectors?
A. Cholera, tenia and gonorrhea
B. Cholera, malaria and tuberculosis
C. Poliomyelitis, tuberculosis and syphilis
D. Malaria, cholera and river blindness
9. UME/95/36
Which of the following is a measure for the control of bilharzia?
A. cutting low bushes around homes
B. application of molluscides in water bodies
C. screening windows and doors with mosquito net
D. application of herbicides in water bodies
10. UME/97/41
One of the most effective ways of controlling guinea worm is by
A. treating B. public enlightenment campaigns
C. accelerating rural development
D. provision of portable drinking water
11. UME/98/34
Which of the following groups of diseases are associated with water?
(i) Onchocerciasis (ii) Schistosomiasis
(iii) Dracunculiasis (iv) Elephantiasis (v) Taeniasis
A. i, ii and iii B. ii, iv and v
C. ii, iii and iv D. i, ii and v.
12. UME/99/41/TYPE: D
Vaccinations is carried out in order to
A. check the production of poison
B. increase the activity of white blood cells
C. increase the number of red blood cells
D. stimulate the production of antibodies
13. UME/99/42/TYPE: D
The construction of dams may lead to an increase in the prevalence of
A. typhoid fever, measles and yellow fever
B. tuberculosis, leprosy and trypanosomiasis
C. guinea worm, malaria and tuberculosis
D. malaria, bilharziasis and onchocerciasis
14. UME/2006/41
The causative agent of typhoid fever is
A. *Salmonella* B. *Entamoeba*
C. *Escherichia* D. *Shigella*
15. UME/2007/3
The causative agent of bird flu is a
A. protozoan B. virus
C. bacterium D. fungus
16. UME/2008/38
A boy who is found of swimming in a pond finds him-
passing urine with traces of blood. He is likely to be
contracted
A. schistosomiasis B. onchocerciasis
C. poliomyelitis D. salmonellosis
17. PCE/93/43
The vector of river blindness is
A. *Musca domestica* B. *Simulium damnosum*
C. *Anopheles gambiae* D. *Culex pipiens*
18. PCE/94/42
Which of the following diseases is caused by a virus?
A. Poliomyelitis B. Cholera
C. Biharzia D. Malaria
19. PCE/95/42
The goal of immunization is to
A. cure diseases
B. increase protection against certain diseases
C. enable a person to travel abroad
D. eradicate deadly diseases
20. PCE/97/38
The urinary bladder can become damaged when an
individual is infected by
A. *Schistosoma mansoni*
B. *Schistosoma japonicum*
C. *Schistosoma intercalatum*
D. *Schistosoma haematobium*
21. PCE/98/39
The infective stage of malaria parasite is
A. trophozoites B. schizonts
C. gametocytes D. sporozoites
22. PCE/99/33/TYPE: E
An example of an obligate parasite is
A. pediculi B. boophilus
C. mallophagus D. Taenia
23. PCE/02/18/TYPE: 9
Diseases that are caused by faecal pollution of water
include
A. dysentery, cholera and typhoid fever
B. dysentery, malaria and poliomyelitis
C. river-blindness, malaria and cholera
D. tuberculosis, typhoid fever and cholera
24. PCE/07/35
The spread of tuberculosis is favoured by
A. malnutrition B. contaminated water
C. air pollution D. overcrowding

ME/2009/37
 Schistosomiasis
 Mucocercariasis
 of the diseases listed above are associated with

II. Schistosomiasis
 IV. Meningitis

- B. II, III and IV
 D. II and IV

and II only
 and III

ME/2010/39/TYPE: D
 vector for yellow fever is
 Anopheles mosquito
 black fly

- B. tsetse fly
 D. aedes mosquito

ME/2010/40/TYPE: D
 protozoan plasmodium falciparum is transmitted by
 female aedes mosquitoes
 female culex mosquitoes
 female anopheles mosquitoes

- C. female blackfly

POLLUTION AND ITS CONTROL

1.2. UME/87/48.

one of the function of UNICEF is to

- A. prevent and control major diseases
 B. prevent disease outbreak by administering vaccines
 C. improve the health and nutrition of children and nursing mothers
 D. monitor environmental pollution

29. UME/88/46

Which of the following causes pollution?

- A. consumption of canned drinks
 B. the addition of fertilizers to farmland
 C. respiration of living organisms
 D. burning of refuse.

30. UME/89/43.

Which of the ways of controlling bilharzia can result in pollution?

- A. clearing waterweed on which the snails feed
 B. treating infected people with drugs
 C. preventing contamination of the water by infected urine and faeces
 D. applying chemicals to kill the snail

31. UME/90/47

Which of the following constitutes pollution?

- A. droppings from birds
 B. loud disco music
 C. a pack of cigarettes
 D. refuse in an incinerator

32. UME/93/45

Carbon monoxide poisons tissues by

- A. constricting the blood vessels
 B. killing the cells
 C. combining with haemoglobin
 D. rupturing the blood vessels

33. UME/94/40

People who suck petrol with their mouth run the risk of increasing in their blood the concentration of

- B. lead

34. UME/95/38

Environmental pollution which can work through the media of water, soil and air include

- A. carbon monoxide
 B. noise
 C. sulphur (iv) oxide
 D. smoke

35. UME/97/40

Which is the most important pollutant of the marine environment in Nigeria?

- A. insecticides
 B. sewage
 C. oil
 D. inorganic fertilizers

36. UME/2000/41/TYPE: M

The pollution that contribute to the depletion of the ozone layer in the atmosphere are

- A. radioactive materials
 B. oxides of sulphur
 C. oxides of carbon
 D. chlorofluorocarbons

37. UME/2002/20/TYPE: Y

The excessive use of agro-chemicals could lead to the pollution of

- A. the lithosphere
 B. the atmosphere
 C. Freshwater
 D. space

38. PCE/92/40

Which of the following may result in air pollution?

- A. incomplete combustion of fuel
 B. old vehicle
 C. sewage
 D. noise

39. PCE/94/43

Which of the following pollutions is carcinogenic?

- A. Detergents
 B. smoke
 C. carbon monoxide
 D. radioactive materials

40. PCE/97/36

An smoke from car exhausts is a poisonous air pollutant because it combines readily with

- A. haemoglobin
 B. antibodies
 C. leucocytes
 D. blood plasma

41. PCE/06/46/TYPE: A

A major effect of deforestation is

- A. a reduction in greenhouse effect
 B. an increase in greenhouse effect
 C. an increase in the ozone layer of the atmosphere
 D. a depletion of atmospheric carbon (iv) oxide

42. PCE/07/28

Excess carbon (iv) oxide in the atmosphere could lead to

- A. global warming
 B. high rainfall
 C. algal bloom
 D. condensation

43. UME/2008/39

The easiest way to establish the level of pollution in a local stream is to measure the level of

- A. Oxygen
 B. carbon (iv) oxide
 C. ammonia
 D. alkalinity

44. UME/2009/38

The major cause of global warming is the

- A. burning of fossil fuel
- B. construction of dams
- C. use of electricity
- D. exploration of space

21.3. CONSERVATION OF NATURAL RESOURCES

45. UME/2001/33/TYPE:T

The greatest influence on a stable ecosystem in nature is exerted by

- A. man
- B. pollution
- C. animal
- D. rainfall

46. UME/2003/34/TYPE:A

The most effective method of dealing with non-biodegradable pollution is by

- A. burying
- B. dumping
- C. incineration
- D. recycling.

47. PCE/95/39

Conservation of wildlife in Nigeria is carried out by

- A. cattle grazing
- B. annual bush burning
- C. changing mining techniques
- D. establishment of forest reserves

48. PCE/2000/40/TYPE:A

Conservation of wildlife in Nigeria is carried out by the

- A. World Wildlife Fund for Nature
- B. Nigeria Game Reserve Authority
- C. Federal Environmental Protection Agency
- D. Forestry Research Institute of Nigeria

49. PCE/98/38/TYPE:C

The most important resources in conservation is

- A. water
- B. wildlife
- C. soil
- D. forest

50. PCE/03/27/TYPE:N

Mineral resources need to be conserved because they are

- A. not renewable
- B. very expensive
- C. not abundant
- D. aesthetic in nature

51. PCE/07/36

Conservation of wildlife can be enhanced by

- A. intensified poultry farming
- B. legislating against poaching
- C. house-to-house-campaign
- D. planting various species of plants

52. UME/2008/29

The major consequence of bush burning in an ecosystem is

- A. the loss of water absorbing ability of the soil
- B. the loss of biological diversity
- C. a decrease in animal population
- D. an increase in soil fertility

MISCELLANEOUS QUESTIONS (21)

53. The greatest impact on a natural ecosystem is made by

- A. bacteria
- B. man

C. viruses
(PCE/2005/38/TYPE:P)

D. green plants

54. The blackfly is a vector of

- A. malaria
 - B. trypanosomiasis
 - C. onchocerciasis
 - D. yellow-fever
- (PCE/2005/11/TYPE:O)

55. One disease NOT caused directly by a bacteria is

- A. malaria
 - B. tuberculosis
 - C. pneumonia
 - D. tetanus
- E. cholera (UME/83/27)

56. Which of the following disease could be exclusively associated with a river basin?

- A. malaria
 - B. syphilis
 - C. onchocerciasis
 - D. cholera
- E. poliomyelitis (UME/83/33)

57. The recycling method of solid waste disposal is unsuitable for

- A. organic matter
 - B. glass
 - C. Plastic
 - D. metal scraps
- (UME/2005/42/TYPE:D)

58. Sources of air pollutants are

- A. industrial chimneys, burning fossil oils and river dams
 - B. sulphur dioxide, acid rain and pesticides
 - C. sulphur mines, vehicle exhausts and aerosols
 - D. sewage, smoke and old vehicles
- (UME/91/48)

59. A renewal natural resources is

- A. wildlife
 - B. gold
 - C. petroleum
 - D. coal
- (PCE/2004/40/TYPE:9)

60. A non-renewable alternative source of energy is

- A. a wind generators
 - B. solar panels
 - C. nuclear energy
 - D. hydroelectric power
- (UME/2005/43/TYPE:D)

61. Acid rain is due mainly to the release into the atmosphere of the oxides of

- A. carbon and phosphorus
 - B. sulphur and nitrogen
 - C. carbon and sulphur
 - D. phosphorus and nitrogen
- (PCE/2004/41/TYPE:9)

62. A causative agent of severe acute respiratory syndrome (SARS) is a

- A. protozoan
 - B. bacterium
 - C. fungus
 - D. virus
- (PCE/2005/32/TYPE:P)

63. In the human body, blood fluke inhabits the

- A. rectum
 - B. liver
 - C. stomach
 - D. bladder
- (PCE/96/39)

64. The clearing away of plants from water banks and draining of water collections are control measures targeted towards the eradication of

- A. Malaria
 - B. schistosomiasis
 - C. Onchocerciasis
 - D. draconiasis
- (PCE/96/40)

65. The stage of a parasite harboured by a definitive host is the

- A. egg
 - B. cyst
 - C. adult
 - D. larva
- PCE/97/37

66. The introduction of snails that feed on mosquito larvae in stagnant water is a

- A. cultural control
 - B. chemical control
 - C. biological control
 - D. mechanical control
- (PCE/01/38/TYPE: 4)

67. Which of these diseases can be controlled by immunization?

- A. bilharzias
 - B. polio
 - C. AIDS
 - D. diarrhoea
- (PCE/01/39/TYPE:4)

68. The best way to eradicate poliomyelitis is to

- A. isolate the affected individuals
 - B. inoculate children from zero to five years
 - C. keep a clean environment
 - D. treat patients at the emergence of symptoms
- (PCE/06/50)

69. Which of the following types of immunity do babies obtain from their mothers?

- A. Artificial passive immunity
 - B. Artificial active immunity
 - C. Natural active immunity
 - D. Natural passive immunity
- PCE/98/37

31. B	32. C	33. B	34. C	35. C
36. D	37. C	38. A	39. D	40. A
41. B	42. A	43. A	44. A	45. A
46. D	47. D	48. C	49. B	50. A
51. B	52. B	53. B	54. C	55. A
56. C	57. A	58. C	59. A	60. C
61. B	62. D	63. D	64. A	65. C
66. C	67. B	68. B	69. D	

CHAPTER 22

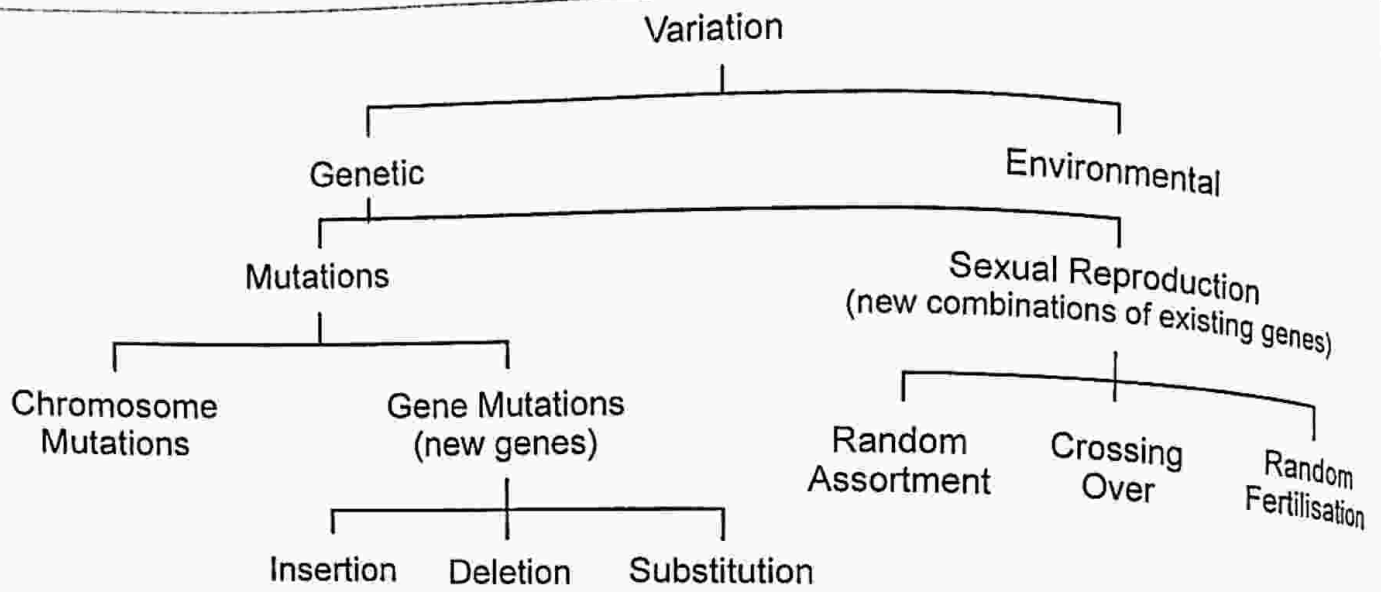
VARIATION IN POPULATION

Understand the following after reading this chapter:

Meaning of variation
Types of variation
Causes of variation
Application of variation

CAUSES OF VARIATION

There are many causes of variation as this chart shows



VARIATION

Variation means the differences in characteristics (phenotype) within a species.

Variation consists of differences between species as well as differences within the same species. Each individual is influenced by the environment, so this is another source of variation.

$$\text{genotype} + \text{environment} = \text{phenotype}$$

TYPES OF VARIATION

For each characteristic, the population may show either *discontinuous* or *continuous* variation.

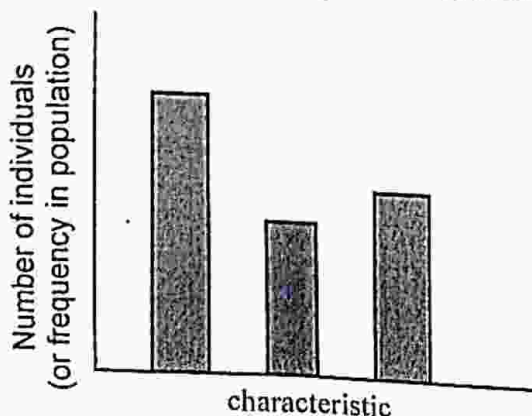
Discontinuous or physiological variation: limited or no intermediate forms. You either have the characteristic or you don't. Blood groups are a good example: you are either one blood group or another - you can't be in between. Such

data is called discrete (or categorical) data.

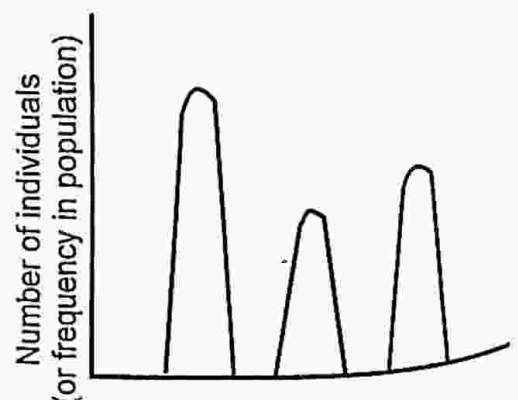
The characteristics of discontinuous variation:

- have distinct categories into which individuals can be placed
- tend to be qualitative, with no overlap between categories
- are controlled by one gene, or a small number of genes
- are largely unaffected by the environment

Discontinuous characteristics are rare in humans and other animals, but are more common in plants. Some examples are human blood group, ability to roll tongue, ability to detect phenylthiocarbamide (PTC), heart rate, muscle efficiency, intelligence, growth rate, rate of photosynthesis, etc.

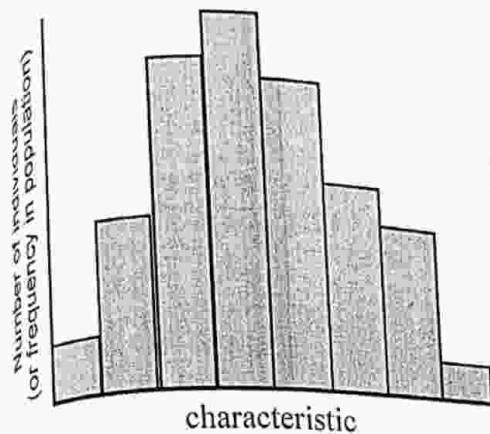


effect of environment
→

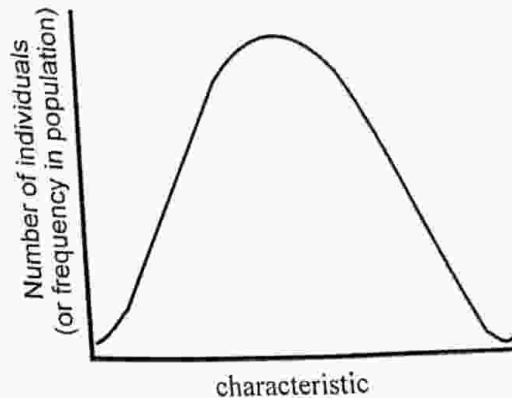


- continuous or morphological variation: prevalence of intermediates between the two extremes
- characteristics of continuous variation:
 - have no distinct categories into which individuals can be placed
 - tend to be quantitative, with overlaps between categories
 - are controlled by a large number of genes (polygenic)
 - are significantly affected by the environment

Continuous characteristics are very common in humans and other animals. Some examples are height, weight, shoe size, hand span, hair colour, milk yield in cow. Milk yield in cows, for example, is determined not only by their genetic make-up but is also significantly affected by environmental factors such as pasture quality and diet, weather, and the comfort of their surroundings. When plotted as a histogram, these data show a typical bell-shaped normal distribution curve, with the mean (= average), mode (= biggest value) and median (= central value) all being the same.



effect of environment



APPLICATION OF VARIATION

- Crime detection through analysis of chromosomes and genes.
- Crime detection through the use of finger prints
- Personal identification.
- In the determination of the paternity of a disputed child.
- In the determination of compatibility of the blood groups of prospective donors and recipients before transfusion.

ABO BLOOD GROUPS IN HUMANS AND BLOOD TRANSFUSION

There are four blood groups based on the presence or absence of antigens in the red blood cells. The blood plasma also contains antibodies which are sensitive to the antigens. Therefore, before blood transfusion, a donor's and the recipient's blood must be tested for compatibility else there will be agglutination or clotting of blood in the recipient vessels which can lead to death.

Blood Group	Antigens on RBC	Antibodies (serum)	Can donate blood to	Can receive blood from
A	A	Anti-B	A and AB	A and O
B	B	Anti-A	B and AB	B and O
AB	A and B	None	AB	All groups
O	None	Anti-A and Anti-B	All groups	O

From the table it is obvious that blood group AB is the universal acceptor and blood group O the universal donor.

rabbits with the blood of a monkey - *Macaca rhesus*.

RHESUS GROUPINGS

The Rh (Rhesus) blood group system (including the Rh factor) is one of thirty current human blood group systems. Clinically, it is the most important blood group system after ABO. Persons who have this antigen are termed as Rh positive (Rh+) and the persons in whom the Rh antigen is absent are designated as Rh negative (Rh-) people. This is called Rh Grouping. During blood transfusion, Rh factor/ Rh antigen is also taken into account. It was discovered by Landsteiner and Wiener in 1940 after immunizing the

Revision Questions

2.1. VARIATION IN POPULATION

1. UME/98/39

After one week of life, the weights of five chicks of same sex hatched simultaneously from the eggs of same hen and fed on the same diet were 45g, 40g, 35g, and 30g. This is an example of

- A. growth rate
- C. Variation

- B. natural selection
- D. mutation

2. UME/2006/22
One advantage of variation in a species population is that individuals
A. with favoured traits become dominant
B. easily teach their reproductive age
C. are easily recognized by mates
D. are better adapted to changes

22.2. TYPES OF VARIATION

3. PCE/2004/43/TYPE:9
Which of the following groups are physiological variations?

- A. height, tongue rolling and blood group
- B. fingerprint, blood group and colour blindness
- C. blood group, tongue rolling and PTC tasting
- D. PTC tasting, skin colour and height

4. UME/93/46
Which of the following is an example of discontinuous variation observed in man?

- A. skin colours
- B. tongue rolling
- C. body weight
- D. height

5. UME/86/39
Which of the following is an example of discontinuous variation?

- A. skin coloration
- B. left-handedness
- C. body weight
- D. height

6. UME/2002/46/TYPE: Y
In a population of living things, the parameters of size, height, weight and colour are examples of

- A. discontinuous variations
- B. continuous variations
- C. non-heritable variations
- D. physiological variations

7. PCE/2003/7/TYPE: N
Arches, loops, whorls and compounds are types of variation in

- A. finger prints
- B. hair colour
- C. blood group
- D. eye colour

8. PCE/2000/41/TYPE: A
The ability in plants and animals to resist certain diseases is
A. a physiological variation
B. a morphological variation
C. a continuous variation
D. an acquired variation

9. UME/2010/41/TYPE: D
A dilute solution of phenylthiocarbamide tastes bitter to some people and is tasteless to others. This is an example of
A. discontinuous variation
B. morphological variation
C. continuous variation
D. taste bud variation

10. UME/2009/40
A phenotypic character with that intermediate forms that can be graded from one extreme to the other is referred to as
A. discontinuous variation
B. continuous variation
C. a mutant
D. a genome

22.3. CAUSES OF VARIATION

11. UME/81/27
Normally any character shown by an organism is due to the effects of
A. hormones and chromosomes
B. chromosomes
C. mutations
D. hormones and genes
E. genes and environment

12. UME/95/39
Human height is an example of a feature which depends on both
A. genotype and phenotype
B. mother's genotype and environmental factors
C. genotype and environmental factors
D. phenotypic and environmental factors.

13. UME/2006/18
Hassan and Hussain are identical twins but Hassan grows taller and fatter than Hussain. This is probably because
A. Hassan inherits genes for tallness and fatness from the father
B. they have dissimilar genotypes
C. Hassan is endowed with genes for shortness and thinness
D. they are raised in different environment

14. UME/2008/40
Which of the following is a major cause of variation among organisms?
A. inbreeding
B. backcrossing
C. sexual reproduction
D. gene dominance

22.4. APPLICATION OF VARIATION

22.4.1. CRIME DETECTION

15. PCE/95/45
One of the best applications of variation in the detection of crime is found in the use of
A. size of the criminal
B. fingerprints of the criminal
C. colour of the eyes of the criminal
D. skin colour of the criminal

22.4.2. BLOOD TRANSFUSION

16. UME/81/29
A person whose blood can be donated to all other people must have the blood group
A. O
B. AB
C. B
D. A
E. none of the above

17. UME/98/36
In blood transfusion, agglutination occurs when
A. white blood cells from two individuals meet
B. two different antibodies meet
C. two different antigens meet
D. contrasting antigens and antibodies meet

18. UME/2008/41
The Rhesus factor of blood was first identified in a category of
A. monkeys

- B. human females
- C. human males
- D. chimpanzees

19. PCE/01/91/TYPE: 4

The possible blood groups of offspring that can be produced from a man in blood group O and a woman in blood group AB are

- A. A and O
- B. A and AB
- C. O and AB
- D. A and B

20. PCE/06/18/TYPE: A

What is the number of recognizable groups in the ABO blood system?

- A. 3
- B. 4
- C. 1
- D. 2

22.4.3. DETERMINATION OF PATERNITY

21. PCE/2001/42/TYPE: 4

If a baby with blood group B is claimed to be born to parents with blood group O, it can be implied that

- A. the paternity of the body is questionable
- B. the paternity of the baby is certain
- C. blood group B is dominant over O
- D. B and O exhibit co-dominance

22. UME/2004/23/TYPE: 4

Paternity disputes can most accurately be resolved through the use of

- A. DNA analysis
- B. finger printing
- C. tongue rolling
- D. blood group typing

23. UME/2009/36

An accurate identification of a rapist can be carried out by conducting a

- A. RNA analysis
- B. DNA analysis
- C. blood group test
- D. behavioural traits test

MISCELLANEOUS QUESTIONS (22)

24. Characteristics that could have intermediate forms such that they could be graded from one extreme to the other is referred to as

- A. intermediate variation
- B. discrete variation
- C. discontinuous variation
- D. continuous variation (PCE/92/44)

25. The difference in height, weight, skin colour and texture of hair between two individuals is an example of

- A. genetic variation
- B. physiological variation
- C. morphological variation
- D. anatomical variation (PCE/97/40)

26. Discontinuous variation is evidenced in

- A. variation in the body weight
- B. variation in height
- C. the ability to roll the tongue
- D. the ability to determine colours (PCE/98/41/TYPE: C)

27. Which of the following indicates physiological variation?

- A. appearance of different colours in flowers of the same species
- B. differences in the eye colours of students in a class
- C. differences in the ability to taste phenylthiocarbamide
- D. differences in the uniqueness of finger prints (PCE/99/43/TYPE: E)

28. A person with type O blood can donate to a patient with type A because the donor's blood

- A. lacks antigens
- B. anti-A antibodies
- C. lacks anti-B antibodies
- D. has both anti-A and anti-B antibodies (UME/95/40)

29. If a baby with blood group B is claimed to be born to parents with blood group O, it can be implied that

- A. the paternity of the baby is questionable
- B. the paternity of the baby is certain
- C. blood group B is dominant over O
- D. B and O exhibit co-dominance (PCE/2001/42/TYPE: 4)

30. The finger prints of man can be classified into

- A. arches, simple, complex and compound
- B. arches, loops, whorls and compound
- C. arches, loops, simple and compound
- D. arches, simple, whorls and compound (PCE/97/41)

CHAPTER 22

1. C	2. A	3. C	4. B	5. B
6. B	7. A	8. A	9. A	10. B
11. E	12. C	13. D	14. C	15. B
16. A	17. D	18. A	19. D	20. B
21. A	22. A	23. B	24. D	25. C
26. C	27. C	28. A	29. A	30. B

CHAPTER 23

HEREDITY (GENETICS)

Understand the following after reading this chapter:

Inheritance of characters in organisms
Chromosome- the basis of heredity
Probability in genetics
Applications of the principle of heredity

DEFINITION OF TERMS IN GENETICS

- A -

albinism the genetically inherited condition in which there is a marked deficiency of pigmentation in skin, hair, and eyes. An individual with these traits is an "albino." Since the gene for albinism is recessive, it only shows up in the phenotype of homozygous recessive people. This is a pleiotropic trait.

alleles alternate forms or varieties of a gene. The alleles for a trait occupy the same locus or position on homologous chromosomes and thus govern the same trait. However, because they are different, their action may result in different expressions of that trait.

amino acids small molecules that are the components of proteins. There are 20 different kinds of amino acids in living things. Proteins are composed of different combinations of amino acids assembled in chain-like molecules. Amino acids are primarily composed of carbon, oxygen, hydrogen, and nitrogen.

autosome a chromosome not involved in sex determination. The diploid human genome consists of 46 chromosomes, 22 pairs of autosomes, and 1 pair of sex chromosomes (the X and Y chromosomes).

- B -

back cross this is the crossing of an organism with the homozygous recessive organisms from the original parent generation. Test cross and back cross is used to determine the genotype of organisms showing dominant phenotype.

- C -

carrier an individual who is heterozygous for a trait that only shows up in the phenotype of those who are homozygous recessive. Carriers often do not show any signs of the trait but can pass it on to their offspring. This is the case with haemophilia.

chromosomes thread-like, gene-carrying bodies in the nucleus of a cell. Chromosomes are composed primarily of DNA and protein. They are visible only under magnification during certain stages of cell division. Humans have 46 chromosomes in each somatic cell and 23 in each sex cell.

codominance the situation in which two different alleles for a trait are expressed unblended in the phenotype of heterozygous individuals. Neither allele is dominant or recessive, so that both appear in the phenotype or influence it. Type AB blood is an example. Such traits are said to be codominant.

- D -

diploid a full set of genetic material, consisting of paired chromosomes one chromosome from each parental set. Most animal cells except the gametes have a diploid set of chromosomes. The diploid human genome has 46 chromosomes.

dominant allele an allele that masks the presence of a recessive allele in the phenotype. Dominant alleles for a trait are usually expressed if an individual is homozygous dominant or heterozygous.

DNA (deoxyribonucleic acid) a large organic molecule that stores the genetic code for the synthesis of proteins. DNA is composed of sugars, phosphates and bases arranged in a double helix shaped molecular structure. Segments of DNA in chromosomes correspond to specific genes.

- E -

evolution genetic change in a population of organisms that occurs over time. The term is also frequently used to refer to the appearance of a new species.

- F -

F₁ generation the first offspring (or filial) generation. The next and subsequent generations are referred to as F₂, F₃, etc.

- G -

gene flow the transference of genes from one population to another, usually as a result of migration. The loss or addition of individuals can easily change the gene pool frequencies of both the recipient and donor populations—that is, they can evolve.

gene pool all of the genes in all of the individuals in a breeding population. More precisely, it is the collective genotype of a population.

genes units of inheritance usually occurring at specific locations, or loci, on a chromosome. Physically, a gene is a sequence of DNA bases that specify the order of amino acids in an entire protein or, in some cases, a portion of a protein. A gene may be made up of hundreds of thousands of DNA bases. Genes are responsible for the hereditary traits in plants and animals.

genetic drift evolution, or change in gene pool

resulting from random chance. Genetic drift rapidly in small populations. In large populations, random deviations in allele frequencies in one generation are more likely to be cancelled out by random deviations in the opposite direction.

-L- Locus the location of a gene or transcription unit on a chromosome.

-M- meiosis cell division in specialized tissues of ovaries and testes which results in the production of sperm or ova. Meiosis involves two divisions and results in four daughter cells, each containing only half the original number of chromosomes—23 in the case of humans.

Mendelian genetics inheritance patterns which can be explained by simple rules of dominance and recessiveness of genes.

modifying gene a gene that can alter the expression of another gene in the phenotype of an individual.

monozygotic twins (identical twins) twins that come from the same zygote are essentially the same genetically. Differences between monozygotic twins later in life are virtually always the result of environmental influences rather than genetic inheritance. Fraternal twins may look similar but are not genetically identical.

multiple-allele series a situation in which a gene has more than two alleles. The ABO blood type system is an example. Multiple-allele series only partly follow simple Mendelian genetics.

mutation an alteration of genetic material such that a new variation is produced. For instance, a trait that has only one allele (A) can mutate to a new form (a). This is the only mechanism of evolution that can produce new alleles of a gene.

-O- ovum (plural ova) a female sex cell or gamete.

-P- phenotype the observable or detectable characteristics of an individual organism—the detectable expression of a genotype.

pleiotropy the situation in which a single gene is responsible for a variety of traits. The collective group of symptoms known as sickle-cell trait is an example.

polygenic trait a trait that is determined by genes at two or more loci. Simple Mendelian rules of dominance do not apply to the complex interaction of these genes. As a result, phenotypes may appear as apparent blends or intermediate expressions. Human skin and hair color are polygenic traits. Many polygenic traits are also influenced by environmental factors.

probability the likelihood that a specific event will occur. Probability is usually expressed as the ratio of the number of actual occurrences to the number of possible occurrences.

proteins any of a large number of complex organic molecules that are composed of one or more chains of amino acids. Proteins can serve a wide variety of functions through their ability to bind to other molecules. Structural proteins may be enzymes, hormones, antibodies, structural

proteins, and so on.

-K- karyotype photomicrograph of an individual's chromosomes arranged in a standard format showing the number, size, and shape of each chromosome type, used in low-resolution physical mapping to correlate gross chromosomal abnormalities with the characteristics of

chromosomes.

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-K- karyotype photomicrograph of an individual's chromosomes arranged in a standard format showing the number, size, and shape of each chromosome type, used in low-resolution physical mapping to correlate gross chromosomal abnormalities with the characteristics of

chromosomes.

to different alleles, the phenotype or trait is dominant or recessive. Traits are said to be

existing of paired alleles, each parental set of chromosomes has 46 chromosomes.

presence of a dominant allele for a trait is dominant.

genetic molecule, synthesis of proteins, and various cellular structures, and to specific

organisms that are used to refer

iteration. The first to as F₂, F₃,

population. The loss of a gene pool or populations—

individuals in a collective

at specific

a gene is a

portion of a

thousands

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hereditary

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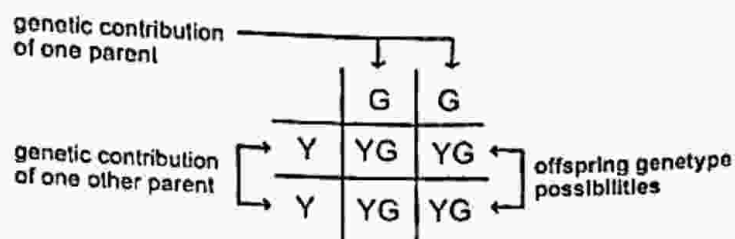
hereditary

hereditary

hereditary

hereditary

components, or gas-transporting molecules. proteome the full complement of proteins produced by an individual (or a species). It is estimated that each human produces approximately 90,000 types of proteins. Punnett square a simple graphical method of showing all of the potential combinations of offspring genotypes that can occur and their probability given the parent genotypes. See example below. Punnett squares are commonly used by genetics counselors to predict the odds of a couple passing on particular inherited traits.



purebred offspring that are the result of mating between genetically similar kinds of parents--the opposite of hybrid. Purebred is the same as true breeding.

- R -

recessive allele an allele that is masked in the phenotype by the presence of a dominant allele. Recessive alleles are expressed in the phenotype when the genotype is homozygous recessive (aa).

- S -

sex cell gamete, either a sperm or an ovum. Sex cells are produced by the meiosis process.

sex-controlled gene a gene that can be inherited by both genders but is usually expressed differently in males and females.

sex-limited gene a gene that can be inherited by both genders but is usually expressed in only males or females.

sickle-cell trait a genetically inherited recessive condition in which red blood cells are distorted resulting in severe anemia and related symptoms that are often fatal in childhood. Sickle-cell trait is the result of a pleiotropic gene. Sickle-cell trait is also known as sickle-cell anemia.

somatic cell any cell in the body except those directly involved with reproduction. Most cells in multicellular plants and animals are somatic cells. They reproduce by mitosis.

sperm a male sex cell or gamete.

- T -

test cross this is the crossing of an organism with the homozygous recessive organisms e.g. crossing genotype Tt against tt.

- X -

X-linked referring to a gene that is carried by an X sex chromosome.

- Z -

zygote a "fertilized" ovum. More precisely, this is a cell that is formed when a sperm and an ovum combine their chromosomes at conception. A zygote contains the full complement of chromosomes (in humans 46) and has the

potential of developing into an entire organism.

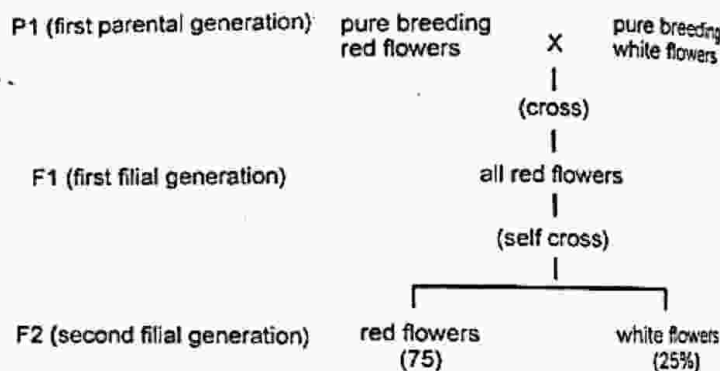
INHERITANCE OF CHARACTERS IN ORGANISMS (CLASSICAL GENETICS)
Classical genetics is the study of inheritance of characteristics at the whole organism level. It is also known as transmission genetics or Mendelian genetics, since it was pioneered by Gregor Mendel.

Gregor Mendel

Mendel (1822-1884) investigated inheritance in pea plants and published his results in 1866. They were ignored at the time, but were rediscovered in 1900, and Mendel is now recognised as the "Father of Genetics". His experiments succeeded where other had failed because:

- Mendel investigated simple well-defined characteristics (or traits), such as flower colour or seed shape, and he varied one trait at a time. Previous investigators had tried to study many complex traits, such as human height or intelligence.
- Mendel use an organism whose sexual reproduction he could easily control by carefully pollinating stigmas with pollen using a brush. Peas can also be self-pollinated, allowing self crosses to be performed. This is not possible with animals.
- Mendel repeated his crosses hundreds of times and applied statistical tests to his results.
- Mendel studied two generations of peas at a time.

A typical experiment looked like this:

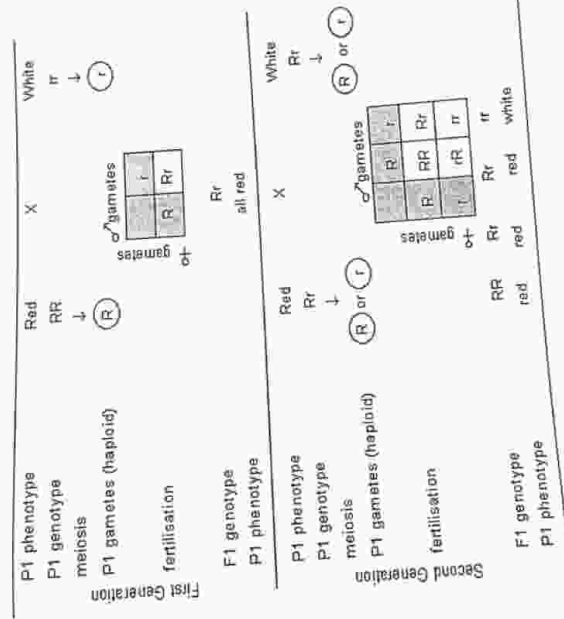


Mendel made several conclusions from these experiments:

1. There are no mixed colours (e.g. pink), so this disproved the widely-held blending theories of inheritance that characteristics gradually mixed over time.
2. A characteristic can disappear for a generation, but then reappear the following generation, looking exactly the same. So a characteristic can be present but hidden.
3. The outward appearance (the phenotype) is not necessarily the same as the inherited factors (the genotype) For example the P1 red plants are not the same as the F1 red plants.
4. One form of a characteristic can mask the other. The two forms are called dominant and recessive respectively.
5. The F2 ratio is always close to 3:1. Mendel was able to explain this by supposing that each individual has two versions of each inherited factor, one received from each parent. We'll look at his logic in a minute.

genes are now called alleles. So in the example of pea plants, there is a gene for flower colour. The two alleles are "red" and "white". One allele is dominant, and the other is recessive. In each parent, and the two alleles are found on the homologous chromosomes. With two alleles there are three possible genotypes: RR (homozygous dominant), Rr (heterozygous), and rr (homozygous recessive). (or phenotypes):

Name	Phenotype
homozygous dominant	Red
homozygous recessive	White
heterozygous	Red



- If the offspring all show the dominant trait then the parent must be homozygous dominant.
- If the offspring are a mixture of phenotypes in a 1:1 ratio, then the parent must be heterozygous.

The Dihybrid Cross
Mendel also studied the inheritance of two different characteristics at a time in pea plants, so we'll look at one of his dihybrid crosses. The two traits are seed shape and seed colour. Round seeds (R) are dominant to wrinkled seeds (r), and yellow seeds (Y) are dominant to green seeds (y). With these two genes there are 4 possible phenotypes:

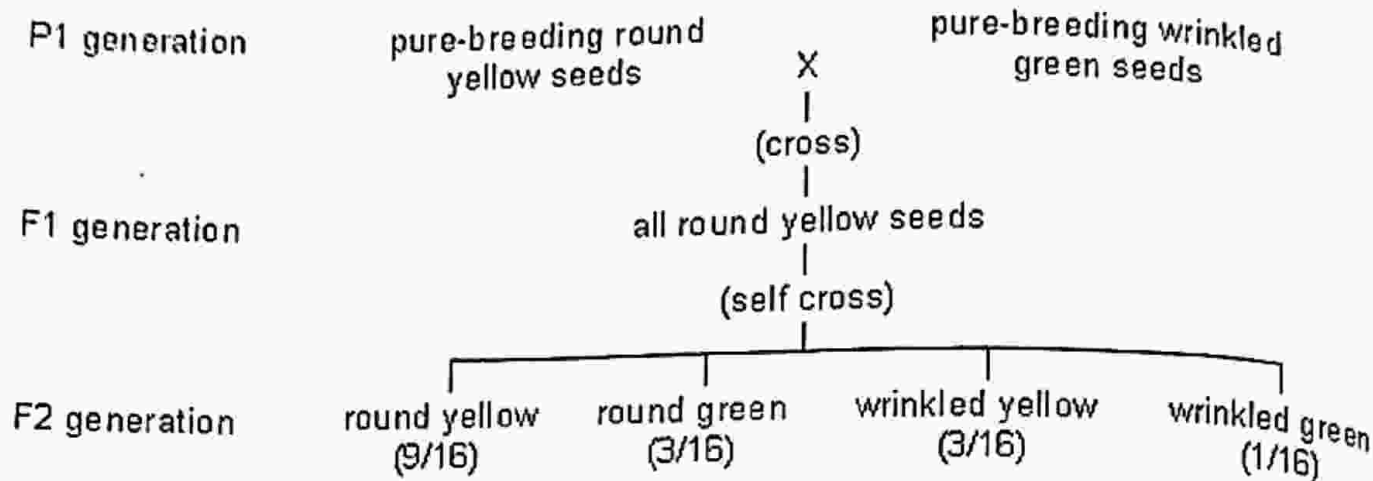
Genotypes	Phenotype
RRYY, RRYy, RrYY, RrYy	round yellow
RRyy, Rryy, rrYY, rrYy	wrinkled yellow
rryy	wrinkled green

At fertilisation any male gamete can fertilise any female gamete at random. The possible results of a fertilisation can most easily be worked out using a Punnett Square as shown in the diagram. Each of the possible outcomes has an equal chance of happening, so this explains the 3:1 ratio (phenotypes) observed by Mendel.

This is summarised in Mendel's First Law, which states that individuals carry two discrete hereditary factors (alleles) controlling each characteristic. The two alleles segregate (or separate) during meiosis, so each gamete carries only one of the two alleles.

The Test Cross
You can see an individual's phenotype, but you can't see its genotype. If an individual shows the recessive trait (white flowers in the above example) then they must be homozygous recessive. If they show the dominant trait then they could be homozygous dominant or heterozygous. You can find out which by performing a test cross with a pure-breeding homozygous recessive. This gives two possible results:

Mendel's dihybrid cross looked like this:



All 4 possible phenotypes are produced, but always in the ratio 9:3:3:1. Mendel was able to explain this ratio if the factors (genes) that control the two characteristics are inherited independently; in other words one gene does not affect the other. This is summarised in Mendel's second law (or the law of independent assortment), which states that alleles of different genes are inherited independently. We can now explain the dihybrid cross in detail.

First Generation	P1 phenotype	round yellow	X	wrinkled green
	P1 genotype	RRYY		rryy
	P1 gametes	(RY)		(ry)
	fertilisation	♂ gametes RY ry ♀ gametes RY RrYy		
		F1 genotype	RrYy	
F1 phenotype	all round yellow			

Second Generation	F1 phenotype	round yellow	X	round yellow		
	F1 genotype	RrYy		RrYy		
	F1 gametes	(RY) (Ry) (rY) (ry)		(RY) (Ry) (rY) (ry)		
	fertilisation	♂ gametes RY Ry rY ry ♀ gametes RY RRYY RRYy RrYY RrYy Ry RRYy RRyy RrYy Rryy rY RrYY RrYy rrYY rrYy ry RrYy Rryy rrYy rryy				
		F2 phenotype	round yellow	round green	wrinkled yellow	wrinkled green
		F2 genotype	RRYY RRYy RrYY RrYy	RRyy Rryy	rrYY rrYy	rryy
		F2 ratio	9	3	3	1

The gametes have one allele of each gene, and that allele can end up with either allele of the other gene. This gives 4 different gametes for the second generation, and 16 possible genotype outcomes.

Dihybrid Test Cross

There are 4 genotypes that all give the same round yellow phenotype. Just like we saw with the monohybrid cross, these four genotypes can be distinguished by crossing with a double recessive phenotype. This gives 4 different results:

Original genotype	Result of test cross
RRYY	all round yellow
RrYY	1 round yellow : 1 round green
RRYy	1 round yellow : 1 wrinkled yellow
RrYy	1 round yellow : 1 round green: 1 wrinkled yellow: 1 wrinkled green

CHROMOSOME- THE BASIS OF HEREDITY

(MOLECULAR GENETICS)

Molecular Genetics (or Molecular Biology), which is the study of heredity at the molecular level, and so is mainly concerned with the molecule DNA. It also includes genetic engineering and cloning, and is very trendy.

Structure of DNA

The three-dimensional structure of DNA was discovered in the 1950's by Watson and Crick. The main features of the

structure are:

- DNA is double-stranded, so there are two polynucleotide strands alongside each other. The strands are antiparallel, i.e. they run in opposite directions (5' 3' and 3' 5')
- The two strands are wound round each other to form a double helix.
- The two strands are joined together by hydrogen bonds between the bases. The bases therefore form base pairs, which are like rungs of a ladder.
- The base pairs are specific. A only binds to T (and T with A), and C only binds to G (and G with C). These are called complementary base pairs. This means that whatever the sequence of bases along one strand, the sequence of bases on the other strand must be complementary to it. (Incidentally, complementary, which means matching, is different from complimentary, which means being nice.)

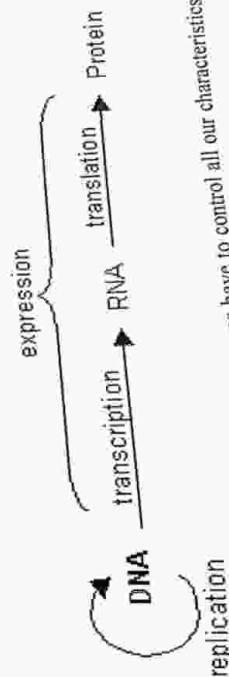
Function of DNA

DNA is the genetic material, and genes are made of DNA. DNA therefore has two essential functions: replication and expression.

- Replication means that the DNA, with all its genes, must be copied every time a cell divides.
- Expression means that the genes on DNA must control characteristics. A gene was traditionally defined as a factor that controls a particular characteristic (such as flower colour), but a much more precise definition is that a gene is a section of DNA that codes for a particular protein. Characteristics are controlled by genes through the proteins they code for, like this:



Expression can be split into two parts: transcription (making RNA) and translation (making proteins). These two functions are summarised in this diagram (called the central dogma of genetics).



One knows exactly how many genes we humans have to control all our characteristics, the latest estimates are 60,000. The sum total of all the genes in an organism is called the genome. This shows the estimated number of genes in different organisms:

RNA

RNA is a nucleic acid like DNA, but with 4 differences:

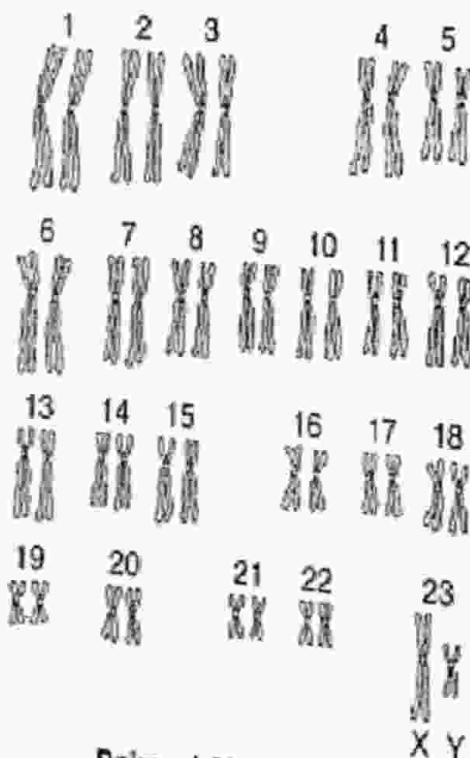
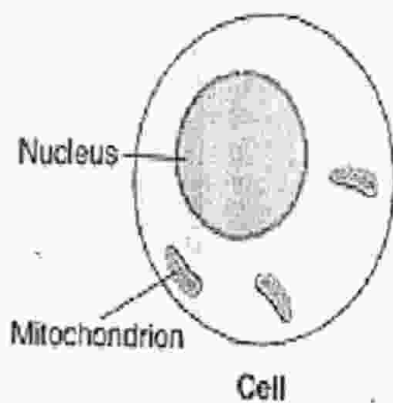
- RNA has the sugar ribose instead of deoxyribose
- RNA has the base uracil instead of thymine
- RNA is usually single stranded
- RNA is usually shorter than DNA

Messenger RNA (mRNA)

mRNA carries the "message" that codes for a particular protein from the nucleus (where the DNA master copy is) to the cytoplasm (where proteins are synthesised). It is single stranded and just long enough to contain one gene only. It has a short lifetime and is degraded soon after it is used.

Ribosomal RNA (rRNA)

rRNA, together with proteins, form ribosomes, which are the site of mRNA translation and protein synthesis. Ribosomes have two subunits, small and large, and are assembled in the nucleolus of the nucleus and exported into the cytoplasm.

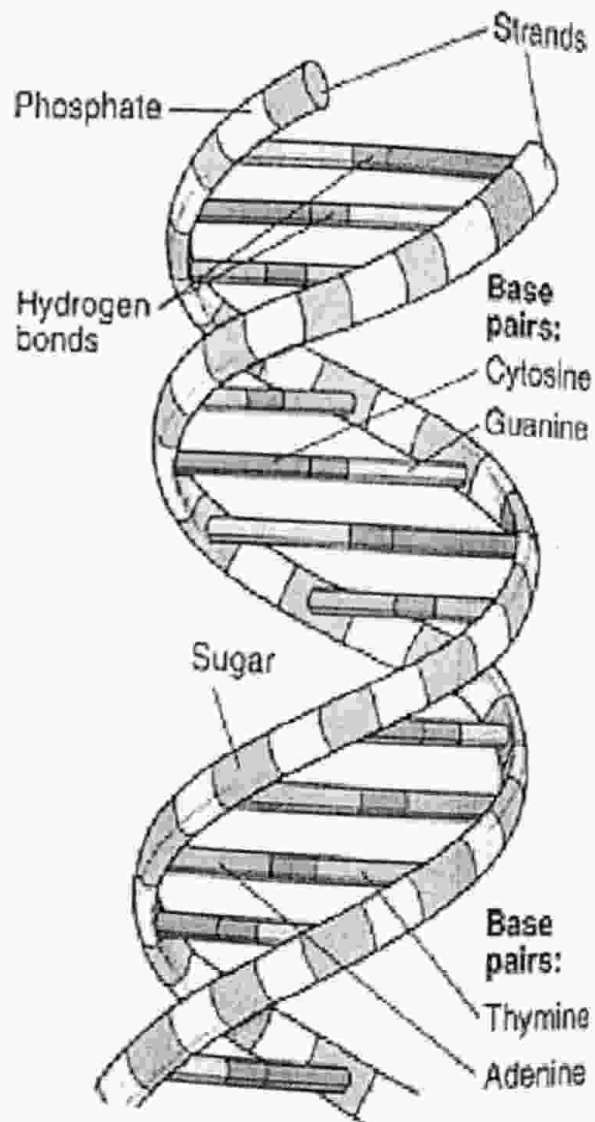
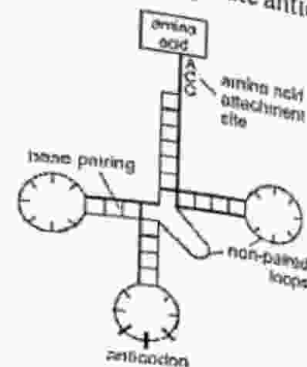


Pairs of Chromosomes
in a Human Cell

Chromosome and DNA

Transfer RNA (tRNA)

tRNA is an "adapter" that matches amino acids to their codon. tRNA is only about 80 nucleotides long, and it folds up by complementary base pairing to form a looped cloverleaf structure. At one end of the molecule there is always the base sequence ACC, where the amino acid binds. In the middle loop there is a triplet nucleotide sequence called the anticodon. There are 64 different tRNA molecules, each with a different anticodon sequence complementary to the 64 different codons. The amino acids are attached to their tRNA molecule by specific enzymes. These are highly specific, so that each amino acid is attached to a tRNA adapter with the appropriate anticodon.



DNA Double Helix

- A. Thomas Morgan B. Gregor Mendel
C. Charles Darwin D. Robert Hooke
7. PCE/99/44/TYPE: E
The segregation of genes during gamete formation and the recombination of genes at fertilization are explained in Mendelian
A. first and second laws respectively
B. first law only
C. second law only
D. second and first law respectively.
8. UME/2006/23
Mendel's second law of inheritance states that
A. alleles combine randomly
B. alleles segregate independently
C. alleles separate predictably
D. chromosomes segregate independently
9. UME/2003/39/TYPE: A
An organism that has been extensively used to test the chromosome theory of heredity is
A. *Homo sapiens* B. *Drosophila melanogaster*
C. *Zea mays* D. *Musca domestica*
10. UME/79/33
If you cross-breed a tall variety (Tt) of maize with a short variety (tt), the ratio of tall to short plants in the offspring will be
A. 2:1 B. 1:2 C. 1:1 D. 1:3 E. 3:1
11. UME/81/5
What is the genetic ratio of a cross between a homozygous tall plant and a homozygous dwarf plant?
A. 0 tall : 4 short B. 3 tall : 1 short
C. 2 short: 2 tall D. 1 tall : 1 short
E. 4 tall : 0 short
12. UME/83/49
What is the genetic ratio of the F₂ generation if members of F₁ generation are allowed to self pollinate?
A. 1 tall : 3 short B. 3 tall : 1 short
C. 1 tall : 1 short D. 4 short : 0 tall
E. 4 tall : 0 short
13. UME/84/20
A man with a normal haemoglobin (AA) marries a woman who has sickle-cell haemoglobin (SS). They have a child who has a sickle-cell trait. Which of the following genotypes could be associated with the child's haemoglobin?
A. AA B. OO
C. AO D. AS
E. SS
14. UME/87/42
If a plant, homozygous for round and yellow (RRYY), is crossed with a wrinkled green type (rryy) all of the resulting seeds will be

- D. round and greenish yellow
15. UME/92/42
If R and r denote the genes for a character, the offspring of the cross between RR and Rr
A. RR, 2Rr, rr B. 2RR, 2rr
C. 2RR, 2Rr D. 4Rr
16. UME/94/44
If a woman's genotype is Tt Qq Rr what will be the content of her eggs?
A. TQr, tqr B. TQR, tqr
C. TqR, tQR D. tQr, TQR
17. UME/99/44/TYPE: D
If the offspring of a cross between a brown mouse (bb) and a black mouse (BB) are allowed to interbreed, how many different genotypes would result?
A. 2 B. 3 C. 4 D. 5
18. UME/88/37
A red-coloured flower when crossed with a white-coloured one produced pink flowers. This is an example of
A. complete dominance B. blending inheritance
C. interaction of genes D. back crossing
19. UME/2004/25/TYPE: 4
In a Mendelian cross of red and white varieties of the four o'clock plant, the F₂ generation expresses incomplete dominance by having flowers which are
A. multicolored B. pink
C. red D. white
20. UME/2009/42
In Mendelian inheritance, discontinuous characters are controlled by the
A. centromeres B. alleles
C. chromosomes D. chromatids
21. UME/2001/38/TYPE: T
Both recessive and dominant characters are found
A. on different chromosomes in the cell
B. at the same locus of a homologous chromosome
C. at different loci of a homologous chromosome
D. on the same chromatid in a chromosome
22. PCE/96/41
An example of incomplete dominance in the inheritance of blood group in human population is
A. OO B. OA C. OB D. AB

	Gametes			
	T	t	II	IV
T	TT	Tt	II	IV
t	Tt	tt	II	IV

PROBABILITY AND GENETICS

Probability is the study of the likelihood of the occurrence of a particular event or offspring. The chance or probability that an event will take place can be expressed as a fraction ($1/4$), ratio ($1:4$) or % (25%).
 Probability = $\frac{\# \text{ of chances for an event}}{\# \text{ of possible combinations}}$

THE RULE OF INDEPENDENT EVENTS

events have no impact on future events. The chance of having a girl is $1/2$. If you already have one girl the chance that your next baby will be a girl is still $1/2$. Each event is regarded as an individual event.

THE PRODUCT RULE: the chance that independent events will occur together is the product of their individual probabilities. Thus the chance of having 3 girls in a row is: $1/2 \times 1/2 \times 1/2 = 1/8$ or 12.5% .

These principles only predict theoretical possibilities and there is no certainty that the event will occur.

EXAMPLE:

$Rr \times Rr$ (heterozygous monohybrid cross)

Probability of RR is $1/2$ from mom

$1/2$ from dad

thus $1/2 \times 1/2 = 1/4$

Probability of rr is $1/2$ from mom

$1/2$ from dad

thus $1/2 \times 1/2 = 1/4$

Probability of Rr is $R: 1/2$ from mom and $1/2$ from dad $1/2 \times 1/2 = 1/4$

$r: 1/2$ from mom and $1/2$ from dad $1/2 \times 1/2 = 1/4$

Thus $1/4 + 1/4 = 2/4$ or $1/2$

Our phenotypic ratio of $3:1$ is met, 3 dominant to 1 recessive.

As is our genotypic ratio of $1:2:1$, $1 RR$ to $2 Rr$ to $1 rr$, as seen in a punnet.

Dihybrid crosses are based on the numbers of the heterozygous monohybrid cross, for example:

$DdQq \times DdQq$ (heterozygous dihybrid cross)

Probabilities are based on the monohybrid cross:

1. Production of quality fruits and seeds
2. Production of strains resistant to diseases and climatic factors.
3. Production of high yielding varieties
4. Development of early maturing varieties of crops and animals.
5. Production of improved qualities of animals for sports and recreation.

APPLICATION OF GENETICS TO MEDICINES

1. Diagnosis and Management of certain diseases which tend to run in families.
2. Management of inheritable diseases and genetic counselling.

Revision Questions

23.1. HERITABLE AND NON-HERITABLE CHARACTERS

1. UME/82/11

If a 26 year old blind man married a young one-eyed woman and they had four children how many of them would be blind like their father?

- A. All
 B. 3
 C. 2
 D. 1
 E. none

2. UME/87/41

An example of monohybrid inheritance in man is

- A. astigmatism
 B. cretinism
 C. hyperthyroidism
 D. albinism

3. UME/94/43

In the gene locus for eye colour in humans, the allele for brown-eyes is dominant over the allele for blue eyes. If a homozygous brown-eyed girl has a brother with blue eye, what are the likely phenotypes of their parents eye colour? (*Eye colour is not a sex-linked trait*)

- A. both parents have blue eyes
 B. their father has blue eyes and their mother has brown eyes
 C. both parents have brown eyes

	D	d
--	---	---

F III

D IV

24. UME/2008/43

The genotype ratio of the offspring of the cross is

- A. 1:2
B. 1:2:1
C. 1:1:1:1
D. 3:1

23.3. TERMS USED IN GENETICS

25. UME/83/24

An organism having one pair of identical genes is

- A. heterozygote
B. n hybrid
C. an allelomorph
D. a homozygote
E. a diploid

26. UME/84/8

The character-producing factors in living organisms are

- A. chromomeres
B. alleles
C. chromatids
D. chromosomes
E. genes

27. UME/85/34

The characters by which an organism is recognized are termed its

- A. phenotype
B. genotype
C. morphology
D. anatomy
E. physiology

28. UME/87/40

A gene which expresses itself only in the homozygous condition is

- A. a mutant
B. dominant
C. recessive
D. lethal

29. UME/89/37

When the two alleles present in an organism are of the same type, the genotype is described as

- A. heterozygous
B. heterogamous
C. homozygous
D. homologous

30. UME/2010/45/TYPE: D

A pair of genes that control a trait is referred to as

- A. recessive
B. dominant
C. hybrid
D. an allele

CHROMOSOMES-THE BASIS OF HEREDITY

31. UME/98/41.

Correct increasing order for the cell components visible for heredity is

- A. gene, chromosome, DNA, nucleus, gene
B. gene, chromosome, nucleus, DNA, gene
C. gene, nucleus, chromosome, DNA, gene
D. gene, nucleus, chromosome

32. E/85/53

Hereditary material in cell is known as

- A. B. CNS
C. D. ATP

33. U/89/36

At cell division, two strands of chromosomes are at a point called

- A. B. chromatide

C. centromere

D. aster

34. UME/90/36

In an organism where the 2n number of chromosomes is 16, the number of chromosomes in each gamete will be

- A. 32
B. 16
C. 8
D. 4

35. UME/91/38

The exchange of genes between homologous chromosomes is called

- A. test cross
B. back cross
C. crossing over
D. mutation

36. UME/95/41

The DNA molecule is a chain of repeating

- A. nucleosides
B. nitrogenous bases
C. sugar phosphates
D. nucleotides

37. UME/95/42

The specific number of chromosomes in each somatic cell is represented by

- A. 2^N
B. 23
C. 2N
D. n

38. UME/2003/43/TYPE: A

If a DNA strand has a base sequence TCA, its complementary strand must be

- A. ATG
B. GAT
C. AGT
D. TAG

39. UME/2006/19

The two normal types of sex chromosomes are

- A. xxy and xyy
B. xx and xxy
C. xy and xxy
D. xx and xy

40. UME/2010/46/TYPE: D

The chromosome number of a cell before and after the process of meiosis is conventionally represented as

- A. n n
B. n 2n
C. 2n n
D. 2n 2n

41. The nitrogenous base found in the RNA but not in the DNA is

- A. Thymine
B. Adenine
C. Guanine
D. Uracil
E. Cytosine

23.5. PROBABILITY IN GENETICS

42. UME/97/44

The homozygous condition HbS HbS results in sickle cell anaemia whereas HbA HbS has the sickening trait. What is the probability that a couple with the sickening trait will give birth to one normal child?

- A. 1/2
B. 1/4
C. 1/8
D. 0

43. UME/2002/50/TYPE: Y

What proportion of the offspring of a cross between two heterozygous parents will exhibit the recessive condition phenotypically?

- C. vegetative propagation is the best form of reproduction for all crops
D. seed were not physiologically mature at harvest
52. UME/2000/42/TYPE: M
The surest way to combine the best qualities of both parents in the offspring by
A. cross-breeding B. inbreeding
C. selective breeding D. pure breeding
53. UME/2002/45/TYPE: Y
Which a farmer has observed in his herd, the farmer need to
A. feed the animals in the herd with more food
B. cross breed his animals with a different herd
C. inbreed the animal in his herd
D. prevent diseases in his herd.
54. UME/2008/44
Genetically modified food products have not become universally accepted because
A. they are not tasty as others produced by conventional means
B. their effects on human consumers is not yet fully understood
C. they are usually costlier than others produced by conventional means
D. the technology can be applied only in developed countries
- 23.6.2. MEDICINE
55. UME/99/43/TYPE: D
The biological factor that is unique to individual is the
A. DNA B. eye colour
C. blood group D. RNA
56. UME/2000/43/TYPE: M
Blood grouping is human being is derived from its combinations of
A. two different alleles B. four different alleles
C. three different alleles D. two different genes
57. UME/2002/49/TYPE: Y
Genetic counseling is important when a marriage is planned between
A. Rh-woman and Rh-man
B. Rh-woman and Rh-man
C. Rh+ woman and Rh-man
D. Rh+ woman and Rh-man
58. UME/99/42/TYPE: E
Where a child with blood group O has parents with blood group A and B respectively the
A. child does not belong to the father biologically
B. parent are homozygous A and B respectively
C. blood group O is more favoured
D. parents are heterozygous A and B respectively
59. UME/2007/49
Which of the following is true of cloning?
A. it is welcome as an ethically and morally sound science
B. only one cell of the original organism is needed to
45. UME/95/44
The F1 of a cross between a tall and a dwarf plant was tall. The F1 was advanced to F2 how many of 120 F2 plant will be dwarf?
A. 30 B. 60 C. 90 D. 120
46. UME/86/38
If a woman who is a carrier of sickle cell trait (AS) married a man who is a sickler (SS) and they had four children how many of them would be normal
A. three B. two C. one D. none
47. UME/89/38
If parents with blood group AB and OO produce six children
A. three of them will have group B
B. two of them will have group A
C. all the offsprings will have O
D. none of them will have group A
48. UME/94/46
A man who has the trait for colour blindness marries a normal woman. What percentage of their children would be sufferers, carriers and normal respectively?
A. 25%, 25% and 50% B. 25%, 50% and 25%
C. 50%, 25% and 25% D. 25%, 37.5% and 37.5%
49. UME/2003/41/TYPE: A
A man and his wife are both heterozygous for the sickle cell trait. The likely percentage of their offspring that will be either carriers or sicklers is
A. 50% B. 25%
C. 75% D. 100%
- 23.6. APPLICATIONS OF THE PRINCIPLES OF HEREDITY
23.6.1. AGRICULTURE
50. PCE/2003/4/TYPE: N
The knowledge of genetic is applied in agriculture to improve
A. character variations and disease resistance
B. yield, quality and disease resistance
C. variation, yield and disease resistance
D. yield, character and quality
51. UME/92/44
Pawpaw seed collected from a tree with many desirable agronomic qualities did not give rise to plants of desirable characters as the parent because
A. seeds are not reliable for propagating plant
B. uncontrolled out crossing can introduce unwanted variability
- the process of cloning is similar to the process of asexual reproduction. Cloning involves the use of a single cell or a small piece of tissue to create a new organism that is genetically identical to the parent. This can be done in several ways, including somatic cell nuclear transfer (SCNT) and embryo splitting. Cloning has many applications, including in agriculture, medicine, and conservation. However, it also raises ethical concerns, particularly regarding the potential for human cloning. The process of cloning is a complex one, involving many steps and a high degree of precision. It is a field of research that is constantly evolving, and there is much to be learned about the possibilities and risks of cloning.
61. Which of the following is a true statement about the father of the child?
A. The father is the biological father.
B. The father is the legal father.
C. The father is the social father.
D. The father is the genetic father.
62. A man who is a carrier of the sickle cell trait (AS) and a woman who is a carrier of the sickle cell trait (AS) have four children. What is the probability that one of the children will be a carrier of the sickle cell trait?
A. 25%
B. 50%
C. 75%
D. 100%
63. A heterozygous blue-eyed man (Bb) and a heterozygous blue-eyed woman (Bb) have four children. What is the probability that one of the children will be blue-eyed?
A. 25%
B. 50%
C. 75%
D. 100%
64. A nucleic acid is a polymer of nucleotides. Which of the following is not a nucleic acid?
A. DNA
B. RNA
C. Protein
D. Carbohydrate
65. Homocystinuria is a genetic disorder. Which of the following is not a symptom of homocystinuria?
A. Intellectual disability
B. Osteoporosis
C. Marfan syndrome
D. Diabetes
66. One of the following is not a function of the cytoskeleton.
A. Cell division
B. Cell movement
C. Cell signaling
D. Cell death
67. In the following, which is not a function of the cytoskeleton?
A. Cell division
B. Cell movement
C. Cell signaling
D. Cell death

initiate the process

C. the clone is similar to but not exactly like the original organism

D. it involves the asexual multiplication of the tissues of the original organism

60. UME/2009/44

A health condition that is known to have resulted from gene mutation

A. haemophilia

C. sickle cell anaemia

B. colour blindness

D. anaemia

MISCELLANEOUS QUESTIONS (23)

61. Which of the following is the best explanation for a child which is phenotypically short and born of two tall parents?

A. The father possesses a gene for shortness

B. The mother possesses a gene for shortness

C. Nature makes the child short

D. Both parents possess genes for shortness
(UME/2005/44/TYPES: D)

62. A man who has the trait for colour blindness marries a normal woman. What percentage of their children would be sufferers, carriers and normal respectively?

A. 25%, 25% and 50%

B. 25%, 50% and 25%

C. 50%, 25 and 25%

D. 25%, 37.5 % and 37.5% (UME/94/46)

63. A heterozygous brown eyed woman (Bb) marries a blue-eyed man (bb) and produces four children. The proportion of the eye colour of their children is

A. 25% brown, 75% blue

B. 20% brown, 80% blue

C. 75% brown, 25% blue

D. 50% brown, 50% blue (PCE/2005/25/TYPES: P)

64. A nucleotide of DNA consist of

A. 5-carbon sugar, inorganic base and phosphoric acid

B. 5-carbon sugar, nitrogenous base and phosphoric acid

C. 5-carbon sugar, ditrogenous base and phosphoric acid.

D. 4-carbon sugar, inorganic base and phosphoric acid
(PCE/2005/25/TYPES: P)

65. Homozygous and heterozygous traits that show dominance can be distinguished by

A. test cross

B. a progeny cross

C. inbreeding

D. interbreeding
(PCE/2005/21/TYPES: P)

66. One advantage of out breeding is that it increase

A. hybrid vigor

B. the rate of mutation

C. the rate of fertility

D. hybrid size
(PCE/2005/22/TYPES: P)

67. In DNA structure, the nitrogenous base Adenine complementary pairs with

A. Thymine

B. Uracil

C. Cytosine

D. Guanine
(PCE/2002/50/TYPES: E)

68. A pure stock of red and white flowered pea plants is crossed. If red colour is dominant over white, what will be the phenotype of flowers of the first filial generation?

A. 75% red and 25%, white

B. 59%, white and 50%. red

C. 100% red

D. 100% white (PCE/2008/44/TYPES: N)

69. Gene flow indicates the

A. interbreeding is possible between members of different species

B. interbreeding is possible between of the same species

C. some genes may spread more than others.

D. random changes alter genetic information
(PCE/2001/50/TYPES: 4)

70. A gene which can only express its characters in a homozygous condition is said to be

A. recessive

B. dominant

C. co-dominant

D. segregated (PCE/2000/42/TYPES: A)

71. What is the probability that two consecutive pregnancies in a newly wedding couple will result in male issues?

A. 1

B. $\frac{1}{4}$

C. $\frac{1}{2}$

D. $\frac{1}{4}$ (PCE/2000/43/TYPES: A)

72. The genetic constitution of the children whose parents are both sickle cell trait carriers

A. AA, AS, AS, SS

B. AA, AS, AA, AS

C. AS, SS, SS, AS

D. AS, AA, SS, SS
(PCE/98/42/TYPES: C)

73. The number of chromosomes in the nucleus of a fertilized human ovum is

A. 23

B. 26

C. 46

D. 92 (PCE/97/42)

74. The abnormality that is caused when the body cell has more than the normal number of chromosomes is

A. mongolism

B. mutation

C. cretinism

D. haemophilia (PCE/97/4)

75. A gene which expresses itself only in the homozygous conditions is

A. recessive

B. an alleles

C. dominant

D. a mutant (PCE/06/19/TYPE: N)

76. Genetic traits that are concerned primarily with forms and structures are said to be

A. morphological B. ecological

C. phenotypical D. physiological
(PCE/06/19/TYPE: A)

77. A pure-breed black rabbit is crossed with a pure-breed white rabbit and all the F₁ offspring have black hair. The F₂ offspring are

A. hybrids

B. zygote

C. autosomes

D. homozygous
(PCE/06/22/TYPE: A)

CHAPTER 23

1. E	2. D	3. C	4. B	5. D
6. B	7. A	8. A	9. B	10. C
11. E	12. B	13. D	14. B	15. C
16. B	17. B	18. B	19. B	20. B
21. B	22. D	23. D	24. B	25. D
26. E	27. A	28. C	29. C	30. D
31. B	32. E	33. C	34. C	35. C
36. D	37. C	38. C	39. D	40. C
41. D	42. B	43. A	44. D	45. A
46. D	47. A	48. A	49. C	50. B
51. B	52. B	53. C	54. B	55. A
56. C	57. A	58. D	59. C	60. C
61. D	62. A	63. D	64. B	65. A
66. A	67. A	68. C	69. B	70. A
71. C	72. A	73. C	74. A	75. A
76. C	77. A			

SEX DETERMINATION AND SEX-LINKED CHARACTERS

SEX DETERMINATION

SEX is determined by the sex chromosomes (X and Y). They are called heterosomes or sex chromosomes, while the other 22 pairs are called autosomes. In humans the sex chromosomes are homologous in females (XX) and non-homologous in males (XY). Using monohybrid-cross to show that there will always be a 1:1 ratio of males to females. Sex is determined solely by the sperm. There are techniques for separating X and Y sperm, and this is used for planned sex determination in farm animals using In-Vitro Fertilization (IVF).

SEX-LINKED CHARACTERS

Characters controlled by genes located on the X chromosomes are usually described as sex-linked characters. The male has only one gene for any sex-linked character. Whatever genes, dominant or recessive, that are on the X chromosome in the male are therefore expressed in the phenotype.

Examples of sex-linked characters in man are:

1. **Baldness:** lack of hair on the head.
2. **Haemophilia:** a disease in which the blood takes an abnormally long time to clot thereby resulting in excessive bleeding even in a small cut;
3. **Colour blindness:** inability to recognize some colours usually red and green.
4. **Muscular dystrophy:** gradual weakening of the muscles.

The genes for these characters are recessive.

A female has two genes for each sex-linked trait, one on each of the sex chromosomes. If she is homozygous, the trait will be expressed. If she is heterozygous, she will be normal but she carries the abnormal trait involved. A female who is heterozygous for a recessive sex-linked trait is known as a carrier.

CHROMOSOME ABNORMALITY

A chromosome anomaly, abnormality or aberration reflects an atypical number of chromosomes or a structural abnormality in one or more chromosomes. A Karyotype refers to a full set of chromosomes from an individual which can be compared to a "normal" karyotype for the species via genetic testing. A chromosome anomaly may be detected or confirmed in this manner. Chromosome anomalies usually occur when there is an error in cell division following meiosis or mitosis. There are many types of chromosome anomalies. They can be organized into two basic groups, **numerical** (alteration in chromosome number) and **structural** (alteration in chromosome structure) anomalies.

ALTERATIONS IN CHROMOSOME NUMBER:

Nondisjunction occurs when either *homologues fail to separate* during anaphase I of meiosis, or *sister chromatids fail to separate* during anaphase II. The result is that one

gamete has 2 copies of one chromosome and the other has no copy of that chromosome. (The other chromosomes are distributed normally).

If either of these gametes unites with another during fertilization, the result is **aneuploidy** (abnormal chromosome number)

A trisomic cell has one extra chromosome ($2n + 1$) = example: trisomy 21. (**Polyploidy** refers to the condition of having three homologous chromosomes rather than two)

A **monosomic** cell has one missing chromosome ($2n - 1$) = usually lethal except for one known in humans: Turner's syndrome (monosomy XO).

The frequency of nondisjunction is quite high in humans, but the results are usually so devastating to the growing zygote that miscarriage occurs very early in the pregnancy. If the individual survives, he or she usually has a set of symptoms - a **syndrome** - caused by the abnormal dose of each gene product from that chromosome.

1. **Human disorders due to chromosome alterations in autosomes (Chromosomes 1-22).** There only 3 trisomies that result in a baby that can survive for a time after birth; the others are too devastating and the baby usually dies in utero.

- A. Down syndrome (trisomy 21)
- B. Patau syndrome (trisomy 13)
- C. Edward's syndrome (trisomy 18)

2. **Nondisjunction of the sex chromosomes (X or Y chromosome):** Can be fatal, but many people have these karyotypes and are just fine!

- a. Klinefelter syndrome: 47, XXY males.
- b. 47, XYY males.
- c. Trisomy X: 47, XXX females.
- d. Monosomy OX (Turner's syndrome)

ALTERATIONS IN CHROMOSOME STRUCTURE: Sometimes, chromosomes break, leading to 4 types of changes in chromosome structure:

1. Deletion
2. Duplication
3. Translocation
4. Inversions.

Revision Question

1. UME/80/42

If a baby is a female, her mother's ovum must have been fertilized by a sperm carrying the chromosome

- A. X
- B. XY
- C. YY
- D. Y

2. UME/82/10
A married couple have 20 children and they are all girls, which of the following is the CORRECT explanation?
A. the woman is incapable of producing male children
B. the man's sperms are very weak
C. the man is not athletic enough
D. the Y component of the man's sex chromosomes was always involved
E. the X component of the man's chromosomes was always involved.

3. UME/87/44
In which of the following crosses will all the female offspring be colour blind?
A. colour blind mother X colour blind father
B. colour blind mother X normal father
C. carrier mother X colour blind father
D. carrier mother X normal father

4. UME/91/40
Which of the following characteristics is NOT sex-linked?
A. river blindness
B. baldness
C. haemophilia
D. colour blindness

5. UME/92/40
Which of the following is true of the children of a haemophilic man who marries a woman that is not haemophilic and does not carry the trait?
A. all their sons will be haemophilic
B. all their daughters will be haemophilic
C. all their daughters will be carriers
D. all their sons will be carriers.

6. UME/92/45
Women do not suffer from colour blindness because
A. the trait is sex-linked
B. only men are colour blind
C. the genes are recessive and sex-linked
D. the genes occur on both the X and Y chromosomes

7. UME/95/45
All the sons of a colour-blind woman will be colour blind regardless of the state of the father because
A. the egg determines the phenotype of the son
B. sons inherit the sex chromosomes of the mother
C. the father's sex chromosome is weaker in son
D. sex-linked traits express dominance in females.

8. UME/2004/24/TYPE:4
Sex-linked genes are located on
A. X- and Y- chromosomes
B. homologous chromosomes
C. X-chromosome
D. Y-chromosome

9. PCE/99/40/TYPE:E
A male child cannot inherit any of his father's sex-linked traits because
A. Y chromosomes is usually genetically empty
B. the male child cannot receive Y-chromosome from his

father
C. X-chromosome is dominant over Y-chromosome
D. Y-chromosome is dominant over X-chromosomes from his father.

10. PCE/2003/5/TYPE: N
All the sons of a colour-blind woman will be colour-blind regardless of the their father because
A. sex-linked traits express dominance in females
B. the father's sex chromosome is weaker than that of the mother
C. the egg determine the phenotype of the sons
D. the sons inherit the gene for colour-blindness from their mother.

11. PCE/97/45
The pair of colours that cannot be distinguished by a person with traits of colour blindness is
A. white and blue
B. yellow and green
C. red and blue
D. red and green

12. UME/2002/47/TYPE: Y
If X^n is the dominant alleles for normal vision and X^n the recessive allele for colour-blindness, a boy with the genotype X^nY will
A. have normal vision
B. colour blind
C. be totally blind
D. be a carrier of colour-blindness

13. UME/2003/38/TYPE: A
If the pair of alleles for baldness is given as Bb, a female carrier will be denoted by
A. $X^B X^b$
B. $X^B X^B$
C. $X^b Y$
D. $X^B Y$

14. UME/2001/39/TYPE T
The probability of a baby being a boy or a girl depends on the contribution of the
A. father's sex cell
B. father's somatic chromosomes
C. mother's sex cell
D. mother's X-chromosomes

15. PCE/99/41/TYPE:E
In man, sex determination is based on the XY-chromosome system. An individual that carries only one X-chromosome with no allele (XO) will be a
A. sterile female
B. normal male
C. sterile male
D. normal female

16. UME/2007/50
An example of a sex-linked trait is the
A. ability to grow long hair in females
B. colour of the skin in humans
C. ability to roll the tongue

D. possession of facial hair in adult human

17. UME/2010/48/TYPE:D

At what stage in the life history of a mammal is the sex of an individual set?

- A. at puberty
- B. at birth
- C. at conception
- D. at adolescence

18. PCE/04/46/TYPE: 9

In humans, baldness is peculiar to the male because the trait is

- A. carried by the x-chromosomes
- B. carried by the y-chromosomes
- C. Heterogametic
- D. homogametic

19. PCE/07/47/TYPE: A

The result of mating between a colour-blind male (X^cY) and a normal female (X^cX^c) is

- | | | |
|----------------------|---|-----------------|
| A. 3 carrier females | : | 1 normal male |
| B. 1 normal female | : | 3 carrier males |
| C. 2 normal female | : | 2 normal males |
| D. 2 carrier females | : | 2 normal males |

20. PCE/07/43

Which of the following is responsible for contributing the chromosome that determines the sex of an unborn offspring?

- A. The father
- B. The mother
- C. The grandfather
- D. The grandmother

MISCELLANEOUS QUESTIONS (24)

21. Which of the following characteristics is NOT sex-linked?

- A. albinism
- B. baldness
- C. haemophilia
- D. colour blindness (UME/97/45)

22. Examples of water-borne and sex-linked diseases are

- A. taeniasis and malaria
- B. cholera and gonorrhoea
- C. typhoid and syphilis
- D. dracunculiasis and haemophilia (UME/2000/46/TYPE:M)

23. Colour blindness is a defect which occurs

- A. more often in women than men
- B. more often in men than women
- C. only in aged people
- D. more in young adolescents (PCE/96/45)

24. The pair of colours that cannot be distinguished by a person with traits of colour blindness is

- A. white and blue
- B. yellow and green
- C. red and blue
- D. red and green (PCE/97/45)

25. Characters expressed by a gene located on only one chromosome are said to be

- A. recessive
- B. dominant
- C. sex-linked
- D. alleles (PCE/98/45/TYPE:C)

26. In man, sex determination is based on the sex chromosome system. An individual that carries only one X-chromosome with no allele (XO) will be a

- A. sterile female
- B. normal female
- C. sterile male
- D. normal female (PCE/97/41/TYPE:E)

CHAPTER 24

1.A	2.E	3.A	4.A	5.C
6.C	7.B	8.C	9.A	10.D
11.D	12.A	13.A	14.A	15.A
16.A	17.C	18.A	19.D	20.A
21.A	22.D	23.B	24.D	25.C
26.A				

ADAPTATION FOR SURVIVAL

Understand the following after reading this chapter:

- Structural adaptation
- Adaptive colouration and its function
- Behavioural adaptation in social animals

COMPETITION

Competition is the interaction between organisms in a community in order to obtain mutually required resources. There are two types:

Intra-Specific Competition: This occurs between members of the same population.

Inter-Specific Competition: This occurs between members of different populations.

ADAPTATIONS FOR SURVIVAL

There are three major types of adaptation viz; structural adaptation, adaptive colouration and behavioural adaptation.

Structural adaptation for protection and defense:

1. **Stinging:** This is used to inflict pains and ward off enemies by some animals e.g. bees, ants, scorpions.
2. **Shells:** These serve as hard protective cover for some animals e.g. snails and tortoise.
3. **Claws:** These are the pointed nails of some animals used for scratching and defence e.g. birds.

Structural adaptations for obtaining food:

1. Modification of mouth parts e.g. beaks for feeding in birds.
2. Trapdoor used by bladderwort to obtain food.

Structural adaptation for securing mates

1. Red head of male lizards to attract the female
2. Beautiful tail feathers of some birds e.g. turkey, peacock
3. Beautiful colouration of some insects e.g. butterfly.

Structural adaptation for water conservation in:

Plants

1. Possession of small leaves or spines e.g. Acacia
2. Possession of thick cuticle over the leaves e.g. citrus
3. Absence of leaves e.g. cactus
4. Possession of sunken stomata e.g. Oleanders
5. Presence of water storage tissues e.g. cactus.

Adaptations to dry habitats in:

Plants

Plants in different habitats are adapted to cope with different problems of water availability.

Mesophytes

plants adapted to a habitat with adequate water

Xerophytes

plants adapted to a dry habitat

Halophytes

plants adapted to a salty habitat

Hydrophytes

plants adapted to a freshwater habitat

Animals

1. Possession of exoskeleton e.g. chitin in arthropods
2. Possession of scales e.g. reptiles
3. Possession of shell e.g. snails
4. Possession of hairs e.g. Mammals

2. ADAPTIVE COLOURATION

1. **Bright colours** in insect pollinated flowers.

2. **Cryptic (concealing/camouflage)**

Protective colouration in some animals to deceive enemies or predators by blending their background and be less visible. The different types are:

- a. **Colour change** to blend with the surrounding e.g. chameleon
 - b. **Disruptive colouration:** Possession of spots and stripes to break up the familiar shapes of their bodies, enabling them to go undetected e.g. leopard, tiger, giraffe.
 - c. **Counter shading:** Possession of lighter colour on the ventral surface and a darker colour on the dorsal surface e.g. fish
3. **Warning or aposematic colouration:** It consists of bright and conspicuous colours possessed by some animals advertising themselves as poisonous and distasteful.

3. BEHAVIOURAL ADAPTATIONS

Behavioural adaptations of predators

1. Hide and wait position
2. Chasing and overtaking the prey with greater speed
3. Setting up traps to catch the prey e.g. spider

Behavioural Adaptations for protection from Predators

1. Swaying in the air
2. Feigning death
3. Secretion of offensive and repulsive odour
4. Retraction into a burrow when threatened by enemies
5. Escaping by running or flying away from enemies

Behavioural Adaptations against unfavourable Climatic Condition

1. **Dormancy:** Inactive state of an organism to withstand unfavourable climatic condition.
2. **Hibernation:** Inactive state or deep sleep in some animals during cold season (winter) e.g. bat, fox.
3. **Aestivation:** Inactive state in some animals during hot weather (drought) e.g. snail, frog
4. **Migration:** Outright relocation to more suitable environment

SOCIAL ANIMALS BEES

The bee caste system include the:

1. **Queen:** This is the fertile female which lays eggs
2. **Drones:** These are the fertile males which mate with queen
3. **Workers:** These are the sterile females which perform several functions such as feeding older larvae with honey and pollen, and foraging. However, their functions varies with age and needs of the colony

	Climbing	Feet like these help birds, like woodpeckers, climb trees. Notice the sharp nails for digging into the wood, and the back toes so that the bird doesn't topple backward.
	Swimming	Webbed feet help birds, like ducks, paddle through the water more efficiently.
	Running	For running quickly, birds like emus, often have three toes, all of which face forward.
	Perching	Feet with four toes, one of which is in the back, are useful for perching on tree branches. Birds, like blue jays, wrap their toes around the branch to help balance.
	Grasping	Predatory birds, like hawks, have clawlike feet called talons for grabbing their prey.
	Scratching	Chickens, and other birds that scratch in the dirt for insects, usually have feet with four toes, all of which have strong nails for digging into the ground.

TERMITES

The termite caste is composed of the:

1. **Reproductive:** Which consist of the king (male) queen (female) and winged reproductive whose function is mainly to reproduce
 2. **Workers:** These perform among other functions searching for food, feeding other members of the colony, constructing and repairing the nest.
 3. **Soldiers:** These consist of the nasute and mandibulate soldiers whose function is to defend and protect the colony
- Economic importance of termites**
They destroy wooden materials and products.
They attack some crop plants e.g. rubber trees, oil palm.
They help the process of decay and recycling of nutrients.
They aerate the soil by burrowing thereby increasing soil fertility. They can serve as food. Their termitaria (ant-hill) are used for surfacing tennis courts.

Revision Questions

1. UME/2001/47/TYPE: T

The ability of an organism to live successfully in an environment is known as

- A. resistance
- B. competition
- C. succession
- D. adaptation

2. UME/2001/50/TYPE: T.

For heterotrophic organisms, competition is least caused by the inadequacy of

- A. mates
- B. space
- C. light
- D. nutrients

3. PCE/92/45

Which of these processes will occur when a thousand maize plants are planted on a 1m² plot?

- A. association
- B. inter-specific competition
- C. succession
- D. intra-specific competition

4. PCE/98/46/TYPE: C

The competition for an available resource by different species in an area leads to

- A. primary succession
- B. intra-specific competition
- C. secondary succession
- D. inter-specific competition

5. UME/2004/36/TYPE: 4

Which of the following is an example of intra-specific competition?

- A. Bean and potato shoots growing out through the same soil
- B. Lizard and an ant-eater chasing an insect
- C. Worker termite and a soldier in a limited space
- D. Hawk and an eagle targeting the same chicken

6. UME/2007/13

Which of the following factors can bring about competition in a population?

- A. dispersion
- B. emigration
- C. drought
- D. mortality

7. UME/2008/45

A major adaptive feature of endoparasites is the

- A. loss of the organ of movement
- B. presence of claws
- C. loss of the central nervous system
- D. presence of piercing mouthparts

8. PCE/07/44

Competition may occur in a rice field if

- A. the environmental temperature increase
- B. the field become weedy
- C. nematodes are absent
- D. nematodes are present

9. PCE/07/45

Communities at early successional stages are often characterized by

- A. high competitive ability
- B. high biological diversity
- C. low productivity
- D. high productivity

25.1. STRUCTURAL ADAPTATION FOR:

25.1.1 OBTAINING FOOD

10. UME/91/46

Which of these is NOT an adaptive features for arboreal life?

- A. possession of a long tail
- B. possession of claws
- C. possession of teeth
- D. counter shading in coat colour

11. UME/95/46

The slender, long and slightly curved beak of the sun-bird is an adaptation for feeding on

- A. nectar
- B. small seeds
- C. big seeds
- D. insects

12. UME/98/44

The group of insects that have mouth parts adapted for both piercing and sucking is

- A. cockroaches, aphids and mosquitoes
- B. aphids, houseflies and moths
- C. mosquitoes, tse-tse flies and aphids
- D. aphids, beetles and grasshoppers.

- B. Thick leaves and the feathers
- C. Thick leaves and the feathers
- D. Lethery cuticle and the feathers

31. UME/98/48
In the whistling pine, leaves are reduced to brown scales and young stems are green. This is an adaptation for
A. obtaining food
B. conserving materials
C. storing water
D. reducing transpiration

32. UME/2000/50 TYPE: M
Which of the following structural features are adapted for uses other than water conservation?
A. Succulent stems
B. Scales in animals
C. Spines in plants
D. Feathers in birds

33. UME/2001/48 TYPE: T
The most important adaptation for xerophytes is the ability of the protoplasm to
A. Resist being damaged by loss of water
B. Store sugar and minerals in the vacuoles
C. Absorb water and swell
D. Shrink from the cell wall.

34. UME/2003/45 TYPE: A
The presence of sunken stomata and the folding of leaves are adaptations to
A. Prevent entry of pathogens
B. Prevent guttation
C. Remove excess water
D. Reduce water loss

35. PCE/96/46
Which of the following set of structures is good at temperature regulation in vertebrates?
A. skins, scales and feathers
B. feathers, carapace and skin
C. hairs, skin and feathers.
D. scales, hairs and feathers.

36. PCE/99/34 TYPE: E
One of the adaptations shown by hydrophytes is the
A. Lack of special absorbing and conducting tissues
B. Development of thick mechanical tissues
C. Possession of stomata on the under surface of the leaf
D. Development of thick cuticles

37. UME/2007/16
An example of a fish that aestivates is
A. shark
B. croaker
C. lung fish
D. cat fish

38. UME/2008/33
Stomata of some plant are sunken and protected by hairs. These are features of
A. mesophytes
B. epiphytes
C. hydrophytes
D. xerophytes

39. UME/2009/47
Which of the following animals is most adapted for water conservation?
A. earthworms
B. mammals

C. flatworms
D. insects

40. PCE/97/48
Thermal gapping is a common phenomenon in
A. lions
B. rabbits
C. birds
D. alligators

25.2. ADAPTATIVE COLOURATION AND ITS FUNCTIONS

41. UME/2003/49 TYPE: T
An example of cryptic colouration is the
A. Bright marks on a poisonous tropical frog on variegated leaves
B. Bright colour of an insect pollinated flower
C. Mottled colours on moths that rest on lichens
D. Green colour of a plant.

42. UME/2001/49 TYPE: T
A green snake in green grass is able to escape notice from predators because of its.
A. disruptive colouration
B. counter shading
C. warning colouration
D. cryptic colouration

43. UME/2004/37 TYPE: 4
The spots and stripes of the leopard and tiger are examples of
A. warning colouration
B. counter shading
C. cryptic colouration
D. disruptive colouration

44. UME/95/48
The colour of the ventral surface of a fish is lighter than that of the dorsal. This is mainly
A. an adaptation for movement
B. an adaptation for camouflage
C. for attracting mates
D. for regulating body temperature

45. UME/99/48 TYPE: D
The very bright colours in some types of Mushroom
A. are a warning that they may be poisonous
B. indicates that they are very tasty
C. attract potential transporters of their spores
D. perform the same function as bright colours in flower

46. PCE/2001/49 TYPE: T
The conspicuous colouration of flowers in plants is to aid
A. Fertilization
B. Seed formation
C. Pollination
D. Seed dispersal

47. UME/2009/48
The specialized pigment cells that are involved in colouration and colour change in animals are the
A. xanthophyll
B. chematophores
C. chlorophyll
D. melanin

25.3. BEHAVIOURAL ADAPTATION IN SOCIAL ANIMALS

48. PCE/2002/23 TYPE: 5
The chemical produced by ants for communication are the

A. Hormones
C. Vitamins

B. Enzymes
D. Pheromones

49. UME/98/47

In the honey bee colony, the drones are

- A. sterile males with reduced mouth parts
- B. fertile males with reduced mouth parts
- C. sterile males with well-developed mouth parts
- D. fertile males with well-developed mouth parts

50. UME/2001/46/TYPE: T

The role of male honey bee is to

- A. clean the hive
- B. ventilate the hive
- C. mate with the queen
- D. care for the young

51. UME/2002/24/TYPE: Y

A feature of the caste systems of bees and termites is that

- A. the workers are sterile
- B. the kings are bigger than the queens
- C. only the workers perform duties
- D. nuptial flight is performed by all members

52. PCE/92/50

Which of the following should be killed in order to destroy a termite colony?

- A. The king
- B. The queen
- C. The workers
- D. The soldiers

53. UME/2004/48/TYPE: 4

The association between termites and the cellulose-digesting protozoan in their guts is an example of

- A. Mutualism
- B. Saprophytism
- C. Commensalism
- D. Parasitism

54. PCE/2000/48/TYPE: A

The caste of the termites that have protozoans living as symbionts in their guts are the

- A. Soldiers
- B. Queens
- C. Reproductive
- D. Workers

55. PCE/97/49

Workers in a bee hive perform a waggle dance to indicate that food is

- A. less than 100m away
- B. more than 100m away
- C. Not available
- D. Abundant

56. UME/97/47/TYPE: D

The loud cry made by a brooding hen when a predator is around is meant to

- A. alert the poultry attendants
- B. attract cocks to come and fight
- C. advertise the boundaries of its territory to intruders
- D. warn chicks and other chickens of impending danger

57. UME/2000/49/TYPE: M

Complex social behaviour and organization are found mostly in

- A. Insects
- B. Birds
- C. Reptiles
- D. Mammals

58. UME/2003/47/TYPE: A

The inactive state exhibited by an animal during hot dry

seasons is termed

- A. Aestivation
- C. Resting

- B. Dormancy
- D. Hibernation

59. UME/2009/49

During the dry season in the tropics, the body metabolism of some animals slows to a minimal level in a process referred to as

- A. hibernation
- C. dormancy

- B. aestivation
- D. senescence

60. UME/2010/34/TYPE: D

I. Adoption of appropriate nocturnal habits

II. Burrowing

III. Adjusting their internal body temperature

IV. Possession of many sweat pores

Which of the above are ways in which desert animals adapt to extreme heat of the environment?

- A. II and III only
- C. I, II and III only

- B. I and II only
- D. I and IV only

MISCELLANEOUS QUESTIONS (25)

61. Any feature which improves the chances of an organism surviving in its surrounding is called?

- A. Speciation
- B. Adaptation
- C. Ecology
- D. Predation

(PCE/94/48)

62. Among the vertebrates, mammals are generally the most successful because they

- A. are capable of internal fertilization
- B. have a well-developed brain
- C. are warm-blooded
- D. have a well-developed heart

(PCE/2005/49/TYPE: D)

63. When an animal has a dark-coloured dorsal surface and a light-coloured ventral surface this is an adaptation called?

- A. Concealment colouration
- B. Countershading
- C. Colour blending
- D. Disruptive colouration

(UME/2005/48/TYPE: D)

64. Long, sharp curved claws for catching and gripping prey are found in

- A. Ducks
- B. Domestic fowls
- C. Hawks
- D. Weaver-birds

(PCE/95/48)

65. An organism which uses 'shamming death' as a method of self preservation against its predator is the

- A. Ladybird beetle
- B. Praying mantis
- C. Chameleon
- D. Stick insect

(PCE/96/48)

66. The purpose of aestivation among animals is to?

- A. Conserve water and food
- B. Reproduce in large numbers
- C. Survive unfavourable conditions
- D. Hide from enemies

(PCE/98/48/TYPE: C)

67. One of the adaptations shown by hydrophytes is the

- A. Lack of special absorbing and conducting tissues
- B. Development of thick mechanical tissues
- C. Possession of stomata on the under surface of the leaf
- D. Development of thick cuticles

(PCE/99/34/TYPE: E)

68. Disruptive colouration is demonstrated in
 A. Spots and stripes of a giraffe
 B. Mixed colours on goats and sheep
 C. Changing colours of chameleon
 D. Counter-shading in fishes
 (PCE/2000/74/TYPE: A)

69. The caste of termites that have protozoans living as symbionts in their guts are the
 A. Soldiers
 B. Queens
 C. Reproductives
 D. Workers
 (PCE/2000/48/TYPE: A)

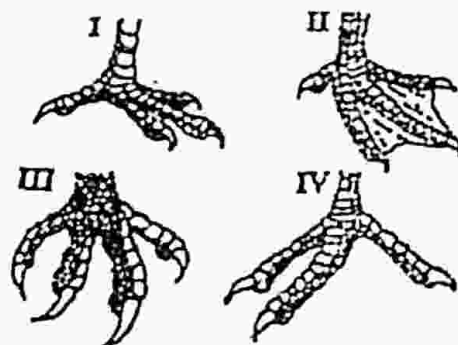
70. Intraspecific competition is more severe than interspecific because of
 A. Identical needs
 B. Predation
 C. Unidentical needs
 D. Co-existence
 (PCE/2002/24/TYPE: A)

71. Which of the following best illustrates behavioural adaptation?
 A. Flocking
 B. Camouflage
 C. Defence
 D. Countershading
 (PCE/2004/49/TYPE: 9)

72. A dance by a worker bee indicating that food is close to the hive is called?
 A. waggle dance
 B. figure of eight dance
 C. round dance
 D. the straight run
 (PCE/2001/48/TYPE: 4)

73. The long limbs of antelopes are an adaptation for
 A. obtaining food
 B. escape and defence
 C. scaring other animals
 D. structural adaptation
 (PCE/03/2/TYPE: N)

Use the diagram below to answer questions 74 and 75



74. An animal with the type of foot represented II is
 A. Pigeon
 B. Hen
 C. Duck
 D. Vulture (PCE 2001/46/Type 4)

75. An adaptation of the foot for catching prey is shown in
 A. IV
 B. III
 C. II
 D. I (PCE 2001/47/Type 4)

CHAPTER 25

1. D	2. C	3. D	4. D	5. C
6. C	7. A	8. B	9. C	10. C
11. A	12. C	13. D	14. B	15. B
16. D	17. B	18. A	19. A	20. C
21. A	22. B	23. B	24. C	25. C
26. C	27. C	28. A	29. A	30. D
31. D	32. D	33. A	34. D	35. A

CHAPTER 26

EVOLUTION

Understand the following after reading this chapter:

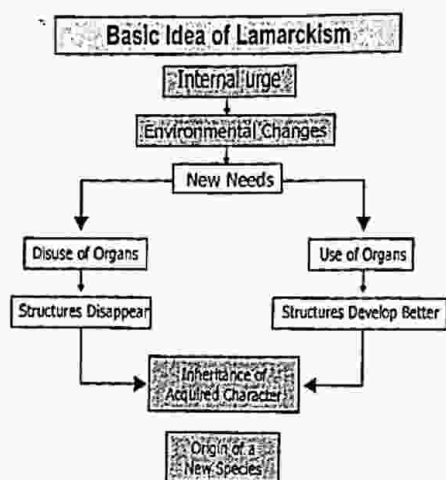
Theories of evolution
Evidences of evolution
Patterns of evolution
Forms of evolution

Evolution (or more specifically biological or organic evolution) is the change over time in one or more inherited traits found in populations of individuals.

THEORIES OF EVOLUTION

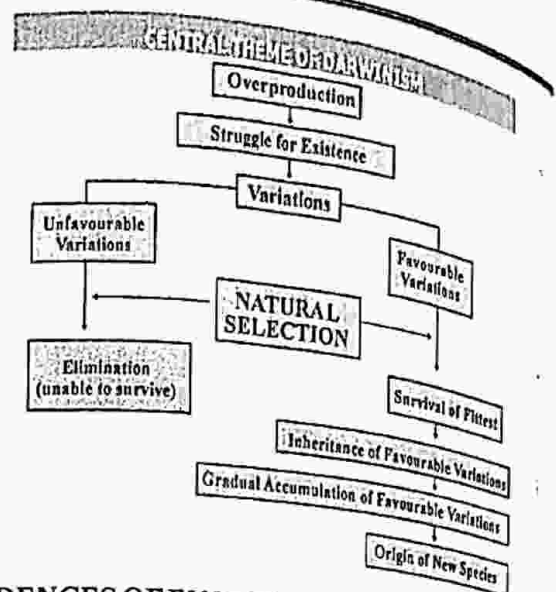
Lamarck's Theory (Lamarckism)

Lamarck proposed that individuals lose characteristics that they do not use, and develop further on characteristics that they use a lot (use and disuse). And that these change would be inherited (inheritance of acquired traits). For example, an athlete would have stronger kids or that giraffes stretching for branches generation after generation would lead to them having longer necks.



Darwin's Theory (Darwinism) or Theory of Natural Selection

In Darwinian evolution, genetically programmed traits which increase organisms' chances of surviving, do survive more, and spread through the population. Natural selection states that: 1. More organisms are produced than can survive because of limited resources 2. Struggle for existence 3. Variation in traits, some of which are heritable 4. Some traits are better adapted than others 5. Survival of the fittest.



EVIDENCES OF EVOLUTION

Fossil records: Fossils are remains of organisms or their impressions preserved in earth's sedimentary rocks. Their ages can be determined by radioactive dating. The study of fossil is called palaeontology.

Biochemical similarities: the chemical make-up and biochemical process of all organisms are similar e.g. all organisms use energy in the form of ATP.

Comparative anatomy: body parts of organisms show structural similarity, progressive complexity and modifications e.g. all organisms have a cellular structure, all vertebrates show a basic body plan, there is a progressive complexity among vertebrates in the number of heart chambers. Vestigial parts like the appendix in man which functions as caecum in herbivores also show evolution.

Embryology: during early development, the embryo of an organism resembles that of its ancestors.

DIFFERENT PATTERNS OF EVOLUTION

Groups of species undergo various kinds of natural selection and, over time, may engage in several patterns of evolution: convergent evolution, divergent evolution, parallel evolution, and coevolution.

Convergent evolution

Convergent evolution is the process in which species that are not closely related to each other independently evolve similar kinds of traits. For example, dragonflies, hawks, and bats all have wings. None of these organisms owes its wings to genes inherited from any of the others. Each kind of wing evolved independently, suggesting that the trait of flight is a useful one for the purpose of survival and reproduction. These independently evolved wings are called *analogous structures*.

Divergent evolution

Divergent evolution is the process in which a trait held by a common ancestor evolves into different variations over

time. A common example of divergent evolution is the vertebrate limb. Whale flippers, frog forelimbs, and your own arms most likely evolved from the front flippers of an ancient jawless fish. Because they share a common evolutionary origin, these are examples of *homologous structures*.

An important consequence of divergent evolution is *speciation*, the divergence of one species into two or more descendant species.

Parallel evolution

Parallel evolution is sometimes difficult to distinguish from convergent evolution. Parallel evolution occurs when different species start with similar ancestral origins, and then evolve similar traits over time. This kind of thing happens because the two different species, though they do not necessarily share a common ancestor, experience similar kinds of environmental pressures and survive only by undergoing similar adaptations. A classic example of parallel evolution is found among plants, in which several similar but distinct forms of leaf evolved in parallel and are evident today.

Coevolution

Coevolution occurs when closely interacting species exert selective pressures on each other, so that they evolve together in a kind of conversation of adaptations. Examples of coevolution are common among predator-prey and host-parasite pairs. More picturesque examples of coevolution occur among hummingbirds and the flowers from which they seek nectar and unwittingly pollinate.

FORMS OF EVOLUTION

There are five main forms of evolution: natural selection, mutation, genetic drift, gene flow, and non-random mating.

Natural selection is the process by which genetic mutations that enhance reproduction become and remain more common in successive generations of a population.

Mutations are changes in the DNA sequence of a cell's genome, which can be caused by radiation, viruses, transposons and, mutagenic chemicals as well as errors that occur during meiosis or DNA replication. Mutation theory of evolution was proposed by Hugo de Vries in 1901.

Genetic drift refers to evolution occurring through random changes in allele frequency over time. It depends on chance alone.

Gene flow refers to the recombination and spread of genes in a population as they interbreed. Gene flow occurs in a non-random manner.

Nonrandom mating occurs when the probability that two individuals in a population will mate is not the same for all possible pairs of individuals. When the probability is the same, then individuals are just as likely to mate with distant relatives as with close relatives — this is random mating. Nonrandom mating can take two forms:

Inbreeding - individuals are more likely to mate with close relatives (e.g. their neighbors) than with distant relatives. This is common.

Out breeding - individuals are more likely to mate with

distant relatives than with close relatives. This is less common.

Revision Questions

6.1. THEORIES OF EVOLUTION

1. UME/94/41

The differences and similarities among living things account for:

- | | |
|----------------|--------------|
| A. Diversity | B. Stability |
| C. Competition | D. Evolution |

2. UME/92/49

The theory of natural selection was developed by

- | | |
|-----------------------|-----------------------|
| A. Lamarck and Darwin | B. Wallace and Mendel |
| C. Darwin and Wallace | D. Mendel and Lamarck |

3. UME/2002/25/TYPE: Y

In his theory of evolution, Darwin implied that

- A. The struggle for existence among living organisms is sporadic
 B. The most successful organisms are those that best adapt to their environment
 C. Organs of the body which are not regularly used by an organism will disappear
 D. Any traits acquired by an organism during its lifetime can be passed on to its offspring.

4. UME/98/49

The best explanation for the theories of natural selection is that

- A. All organisms have equal capacity for survival in their habitats.
 B. Organisms have varying capacities for survival in their habitats.
 C. Organisms compete for resources and better competitors survive and thrive.
 D. Habitats allow only organisms that will not have to compete for survival.

5. UME/99/49/TYPE: D

Which of the following is one of Lamarck's theories?

- A. Some variations are more favourable to existence in a given environment than others.
 B. All living organisms are constantly involved in a struggle for existence
 C. The size of a given population remains fairly constant.
 D. New species originate through the inheritance of acquired traits.

6. UME/2003/50/TYPE: A

An argument against Lamarck's theory of evolution is that

- A. Acquired traits cannot be passed onto the offspring.
 B. Disuse of body part cannot weaken the part.
 C. Disused part is dropped off in the offspring
 D. Traits cannot be acquired through constant use of body parts.

7. UME/97/50

Long neck in giraffe is used to illustrate the theory of

- | | |
|--------------------|----------------------|
| A. Use and disuses | B. Origin of species |
| C. Origin of life | D. Natural selection |

8. PCE/97/50

The cutting of the tails of mice in successive generation by

Wallace was used to disprove a theory postulated by

- A. Darwin
- B. De Vries
- C. Mendel
- D. Lamarck

9. UME/2000/47/TYP:M

The mutation theory of origin of evolution was propounded by

- A. Gregory Mendel
- B. Hugo de Vries
- C. Jean Lamarck
- D. Charles Darwin.

10. UME/98/50

The basic point of impact by changes which produces mutation is the

- A. Gametes
- B. Chromosomes
- C. Phenotype
- D. Zygote.

11. UME/2008/50

The natural process that produces adaptive evolutionary changes is

- A. mutation
- B. gene flow
- C. genetic draft
- D. natural selection

12. UME/2009/50

According to Darwin, the driving force behind evolutionary change is

- A. natural selection
- B. genetic drift
- C. mutation
- D. gene flow

13. PCE/06/38/TYP:A

According to Charles Darwin, the main cause of natural selection is

- A. gene flow
- B. genetic drift
- C. physiological pressure
- D. environmental pressure

26.2. EVIDENCE OF EVOLUTION

14. UME/99/50/TYP:D

From an evolutionary standpoint, the older a fossil-bearing rock is the more likely to contain

- A. Aves as opposed to amphibians
- B. Invertebrates as opposed to vertebrates
- C. Angiosperms as opposed to algae.
- D. Vertebrates as opposed to invertebrates

15. UME/94/50

The anatomical evidence usually used in support of the evolutionary among whales, humans, birds, and dogs is the possession of

- A. thick skin
- B. pentadactyl limb
- C. tail
- D. epidermal structure

16. UME/93/50

Which of the following organisms has lost the pentadactyl limb structure?

- A. Bat
- B. Fish
- C. Frog
- D. Pigeon.

17. UME/95/49

The least evidence of the theory evolution is provided by the study of

- A. anatomy
- B. ecology
- C. geology
- D. embryology

18. UME/95/50

From which groups of animals are mammals generally believed to have most recently evolved?

- A. Reptiles
- B. Fishes
- C. Amphibians
- D. Birds.

19. UME/98/17

The heart of the adult frog consists of

- A. two auricles and two ventricles
- B. one auricle and one ventricle
- C. two ventricles and one auricle
- D. one ventricle and two auricles.

20. UME/2002/25/TYP:Y

The structure that is common in the embryos of mammals, amphibians, birds, fishes and reptiles, and which is an evidence of their common ancestry is the

- A. eye
- B. chorion
- C. allantois
- D. gill slits

21. UME/2003/44/TYP:A

Which of the following requires the use of carbon-dating to prove that evolution has occurred?

- A. Biochemical similarities
- B. Molecular records
- C. Fossil records
- D. Comparative anatomy

22. UME/2004/39/TYP:4

An evidence of the relationship between living organisms and their extinct relatives can best be obtained from

- A. paleontology
- B. embryology
- C. comparative anatomy
- D. comparative physiology

23. PCE/93/50

Which of the following is a vestigial structure?

- A. Appendix in man
- B. Wings of birds
- C. Forelimbs of toads
- D. Wings of cockroaches

24. PCE/95/50

From which of the following is the most direct evidence of evolution?

- A. Comparative anatomy
- B. Embryology
- C. Paleontology
- D. Geographical

25. PCE/98/50/TYP:C

The evidence of evolution that is based on the structural affinities of organisms is

- A. embryology
- B. biochemical similarities
- C. comparative anatomy
- D. fossil records

26. PCE/2000/50/TYP:A

Evolutionary relationship are reflected in natural classification based on structure that are

- A. homologous
- B. heterozygous
- C. homozygous
- D. analogous.

27. PCE/2002/25/TYP:5

The similarity that occurs between the wing of a bat and the human arm as a result of common evolutionary origin is called

- A. embryology
- B. paleontology
- C. morphology
- D. homology.

28. UME/2008/36

Which of the following is the most advanced evolutionary development in plants?

- A. possession of unicellular structures
- B. development of flowers
- C. dispersal of spores
- D. development of secondary thickening

29. PCE/06/36/TYPE: A

The description of how species are evolutionarily related to one another by modern systematic is based on

- A. fossil records and comparative anatomy
- B. fossil records and physiology
- C. fossil records only
- D. comparative anatomy only

MISCELLANEOUS QUESTIONS (26)

30. In evolution, Larmark's theory is known as theory of

- A. mutation
- B. inheritance of acquired characters
- C. natural selection
- D. survival of the fittest through competition

(PCE/94/50)

31. Which of the following belongs to the group that first attempted to colonize the terrestrial habitat?

- A. Hawk
- B. Lizard
- C. Gorilla
- D. Toad

(PCE/95/46)

32. Darwin's explanation of the way evolution occurs implies that

- A. God determines which species should evolve
- B. Organisms have evolved simultaneously
- C. Those traits used most often, persist longer
- D. Progressive adaptations enable perpetuation of species

(PCE/98/49/TYPE: C)

33. Which of the following represents the most ancestral in the evolutionary development among plants?

- A. Development of fruits
- B. Dispersal of spores
- C. Unicellular plant body
- D. Development of vascular system

(PCE/99/46/TYPE: E)

34. An example of a vestigial structure in the human body is the

- A. Appendix
- B. Heart
- C. Diaphragm
- D. Small Intestine

(PCE/2005/3/TYPE: P)

35. The most convincing evidence of evolution from embryology is the presence of

- A. Head
- B. Blood
- C. Notochord
- D. Tail

(PCE/2005/5/TYPE: P)

36. The pentadactyl limb arrangement of vertebrates is an evolutionary fact derived from

- A. Morphological differences
- B. Serology

C. Embryology

D. Comparative anatomy

(PCE/2004/50/TYPE: 9)

37. An evidence of a common ancestry for fishes, amphibians, reptiles, birds and mammals is the

- A. Possession of wings by birds and bats
- B. Cold-bloodedness of fishes, amphibians and reptiles
- C. Possession of gill clefts in vertebrate's embryos
- D. Possession of scales by fishes and reptiles

(UME/2005/50/TYPE: D)

38. An organism with a two-chambered heart is

- A. Fish
- B. Man
- C. Toad
- D. Bird

(PCE/2005/50/TYPE: P)

39. The evidence that supports the advancement of ferns over mosses is derived from

- A. comparative anatomy
- B. molecular records
- C. biochemical similarities
- D. physiological records

(UME/2006/34)

40. Which of the following vertebrates has the most simple structured heart?

- A. reptile
- B. fish
- C. mammal
- D. amphibian

(UME/2006/35)

41. The theory which supports the view that the large muscles developed by an athlete will be passed on to the offspring was proposed by

- A. Lamarck
- B. Darwin
- C. Mendel
- D. Pasteur

(UME/2007/14)

42. The flippers of a whale and the fins of a fish are examples of

- A. coevolution
- B. continuous variation
- C. convergent evolution
- D. divergent evolution

(UME/2010/50/TYPE: D)

43. Gene flow indicates that

- A. inbreeding is possible between members of different species
- B. interbreeding is possible between members of the same species
- C. some genes may spread more than others
- D. random changes alter genetic information.

(PCE 2001/50/TYPE: 4)

44. Which of the following is not included in the theory of natural selection

- A. Struggle for existence
- B. Offspring show variation
- C. Inheritance of adaptive structures
- D. Disappearance of functionless organs.

45. The order of evolutionary trend in plants is

- A. Bryophyta, Thallophyta, Pteridophyta and Spermatophyta

CHAPTER 26

1. D	2. C	3. B	4. C	5. D
6. A	7. A	8. D	9. B	10. B
11. D	12. A	13. D	14. B	15. B
16. B	17. B	18. D	19. D	20. D
21. C	22. A	23. A	24. C	25. C
26. A	27. D	28. D	29. A	30. B
31. D	32. D	33. C	34. A	35. C
36. D	37. C	38. A	39. A	40. B
41. A	42. D	43. B	44. D	45. B
46. C	47. B			

- B. Thallophyta, Bryophyta, Pteridophyta and Spermatophyta
 C. Spermatophyta, Pteridophyta, Bryophyta and Thallophyta
 D. Bryophyta, Pteridophyta, Spermatophyta and Thallophyta.

46. A vestigial organ in human is
 A. earlobe
 B. toe bone
 C. tail bone
 D. spleen.

47. Natural selection is a consequence of
 A. distribution of organism
 B. adverse condition
 C. variation in organism
 D. inbreeding.

MODEL QUESTIONS SET I

1. A similarity between plant and animal cells is the possession of a
 A. cell wall
 B. nucleus
 C. definite shape
 D. chloroplast

2. Which of the following is true of the epithelial tissue in animals?
 A. it binds other tissue together
 B. it forms a protective covering over the body surface
 C. it fills spaces in the body and provides support
 D. it transmit impulses to various parts of the body.

3. Amoeba gets rid of excess water that enters the cells through the
 A. pseudopodia
 B. endoplasm
 C. contractile vacuole
 D. oral groove

4. The criteria used in classifying living organisms into the five kingdoms include complexity of the
 A. organism's behaviour and its body size
 B. organism's body and its habitat
 C. organism's body and its mode of nutrition
 D. cell structure and its body size.

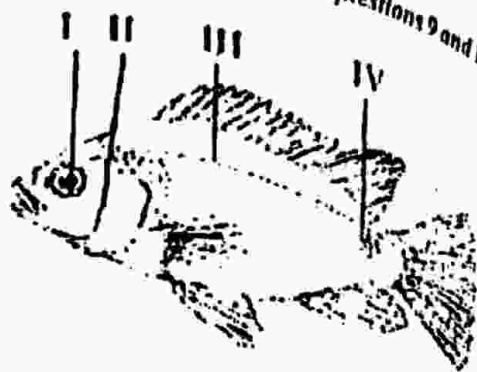
5. In which of the following aquatic habitats is spirogyra found?
 A. The lagoon
 B. A pond
 C. The ocean
 D. A river

6. In cockroach, the organ which is sensitive to smell is the
 A. pedipalp
 B. mandible
 C. maxilla
 D. antenna

7. A typical life cycle of plants involves alternation of the
 A. diploid sporophytes with the haploid gametophytes
 B. haploid sporophytes with the diploid gametophytes
 C. diploid sporophytes
 D. haploid gametophytes

8. The main feature that distinguishes a bacterial cell from an eukaryotic cell is the
 A. cell wall
 B. cell membrane
 C. nuclear material
 D. cytoplasmic material

Use the diagram below to answer questions 9 and 10.



9. the structure used for detection of vibration in water is labeled
 A. I
 B. II
 C. III
 D. IV

10. The structure labeled II is the
 A. neck
 B. head
 C. operculum
 D. gill.

11. The body cavity of mammals is separated into two regions by the
 A. thoracic cavity
 B. abdominal cavity
 C. placenta
 D. diaphragm.

12. The widest uninterrupted zone in the transverse section of a herbaceous dicot stem is the
 A. cortex
 B. xylem
 C. pith
 D. phloem

13. Plants that have special devices for trapping and digesting insects are referred to as
 A. parasite
 B. saprophytic
 C. carnivorous
 D. symbiotic

14. The most suitable combination of food for a strict vegetarian includes
 A. carrot, cheeses and cucumber
 B. eggs, beans and yam
 C. beans, rice and mushrooms
 D. tomatoes, rice and fish.

15. Which of the following processes takes place in the ileum of humans?
 A. Absorption of water
 B. Absorption of digested food
 C. Secretion of hormone
 D. Secretion of saliva.

16. The hardest material in the body of a mammal is the
 A. dentine
 B. pulp cavity
 C. enamel
 D. cement

17. A steroid that is found in the mammalian blood is
 A. globulin
 B. albumin
 C. cholesterol
 D. fibrinogen

18. The uptake of mineral ions into the root cells in plants is by
 A. active transport
 B. simple diffusion
 C. osmosis
 D. translocation

19. In the lungs, the air sacs are surrounded by a cluster of
 A. bronchioles B. capillaries
 C. trachea D. ribs
20. In mammals, the lungs serve as the major excretory organ for
 A. urea B. salts
 C. carbon (IV) oxide D. water
21. Muscles are attached to bones by the
 A. ligament B. cartilage
 C. tendon D. connective tissue.
22. If a germinating seed is attached horizontally on a revolving klinostat, the effect on the seedling after three days is that the
 A. plumule will curve upwards while the radicle will curve downwards
 B. plumule and radicle will not show any visible curvature
 C. radicle will curve downwards while the plumule will not show any curvature
 D. plumule will curve upwards while the radicle will not show any curvature.
23. In entomophilous flower, the function of the large petals is to
 A. secrete large quantity of nectar
 B. protect the stigma
 C. attract insects
 D. attach the flower to the shoot.
24. The hormone in the mammalian females that induces ovulation is?
 A. testosterone B. oestrogen
 C. progesterone D. leutenizing hormone
- Use the diagram below to answer questions 25 and 26.
-
25. The optimum temperature for the absorption of ions by X and Y is between
 A. 10-80°C B. 40-50°C
 C. 50-60°C D. 50-80°C
26. What is the title for the graph?
 A. relationship between temperature and growth in X and Y.
 B. Effect of temperature on the rate of ion uptake in X and Y
 C. rate of growth of X and Y in a medium
 D. Interaction of temperature and ions absorbed in a medium.
27. The cells that transmit messages to the effectors are the
 A. sensory receptors B. relay neurons
 C. motor neurons D. sensory neurons.
28. The ecological condition that favours breeding of blackfly is
 A. slow flowing streams
 B. water in ponds and swamps
 C. water in small containers
 D. fast flowing streams.
29. The pair of abiotic factors specific to the distribution of organisms in aquatic habitats is
 A. salinity and humidity
 B. salinity and turbidity
 C. turbidity and temperature
 D. turbidity and rainfall.
30. The protozoans living symbiotically in the stomach of termites is responsible for the
 A. digestion of cellulose taken in by the termites
 B. digestion of protein taken in by the termites.
 C. supply of mineral salts to the termites
 D. supply of vitamins to the termites.
31. Carbon in the form of carbon (IV) oxide is removed from the atmosphere during
 A. evaporation B. transpiration
 C. photosynthesis D. respiration
32. Breathing roots is an adaptive feature for plants that grow in the
 A. mangrove swamp B. Sahel savanna
 C. rain forest D. Guinea savanna.
33. Which of the following is the most common arboreal dweller?
 A. Moths B. Mosquitoes
 C. Butterflies D. Blackflies
34. The Obudu cattle ranch in Nigeria is located in the
 A. tropical rainforest
 B. Guinea savanna
 C. highland of montane forest
 D. Sudan savanna
35. Which of the following is most important in the adoption of biological control for pests?
 A. The knowledge of agricultural practices by the farmer.
 B. The relative abundance of plants and animals in the farm.
 C. The knowledge of the breeding habits of the predator.
 D. The predator-prey relationship in the ecological community.
36. Excess growth of algae in a pond will result in the depletion of
 A. nutrients B. water
 C. nitrogen D. oxygen

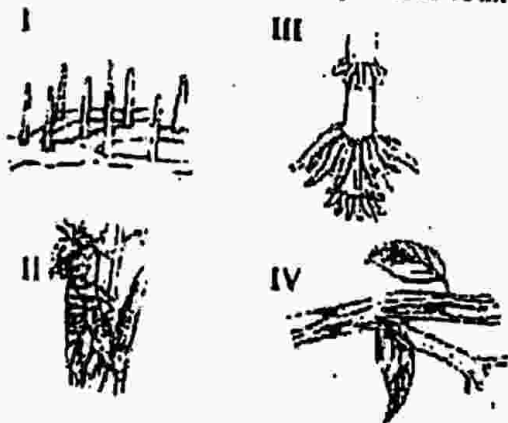
37. The soil that has the lowest water retention capacity is
 A. clay B. sand
 C. loam D. laterite.
38. The loss of soil fertility through leaching is caused by
 A. deforestation through agriculture
 B. overgrazing by animals
 C. draining of the nutrient contents
 D. rill and gully erosions.
39. Non-renewable resources can be conserved by
 A. afforestation B. deforestation
 C. recycling D. sanitation.
40. Which of the following parasites is transmitted by houseflies?
 A. *Plasmodium* sp. B. *Vibrio* sp.
 C. *Trypanosoma* sp. D. *Schistosoma* sp.
41. Which of the following can lead to variation in living organisms?
 A. Budding B. Fertilization
 C. Fragmentation D. Sporulation
42. The knowledge of blood group genotype is very important in counseling before marriage so as to prevent.
 A. genetic diseases B. communicable diseases
 C. agglutination reactions D. infertility in couples

43. One of the major criticisms of the Mendelian laws arises from the fact that
 A. one trait is often controlled by many pairs of genes.
 B. a single factor inheritance is never a reality
 C. complete dominance is always possible
 D. complete dominance is never a reality.

44. In a cross involving a heterozygous black goat (Bb) and a white goat (bb), what is the probability that the offspring will be Bb?
 A. $\frac{1}{4}$ B. $\frac{1}{2}$ C. $\frac{3}{4}$ D. 1

45. The exchange of genes between homologous chromosomes is referred to as
 A. mutation B. crossing-over
 C. back-cross D. test-cross.

Use the diagram below to answer questions 46 and 47.



46. An adaptation for obtaining air in an aquatic habitat is shown in
 A. I B. II C. III D. IV

47. Which of the following is represented in II?
 A. Stilt roots of mangrove
 B. Breathing roots of mangrove
 C. Prop roots of maize
 D. Aerial roots of an epiphyte

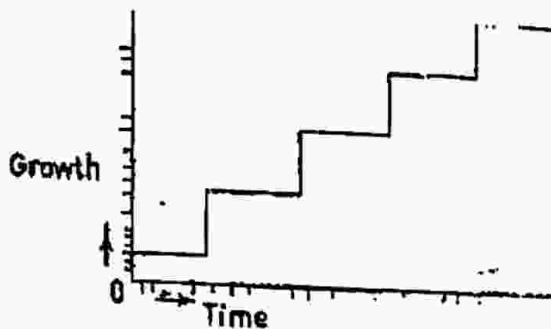
48. Counter-shading in fishes, toads and snakes is an important adaptation for
 A. regulating body temperature
 B. warning and attacking predators
 C. securing mates for reproduction
 D. concealing the animals from predators

49. Which of the following is an adaptation for conserving water in plants?
 A. broad leaves with hairs.
 B. leaves with sunken stomata and thin cuticle.
 C. leaves with sunken and thick cuticle.
 D. reduced leaves with numerous stomata.

50. Which of the following castes is absent in a termite colony?
 A. Queen B. Drone.
 C. Worker. D. King.

MODEL QUESTIONS SET II

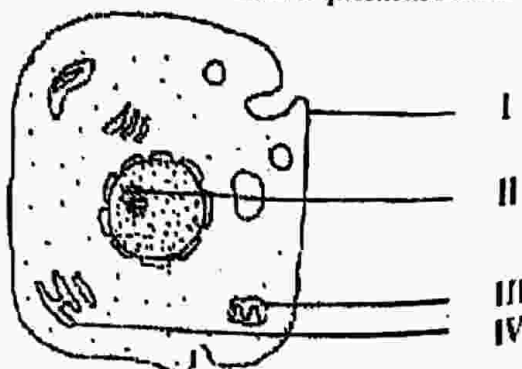
Use the diagram below to answer questions 1 and 2.



1. The growth pattern illustrated above is an
 A. isometric growth B. Automatic growth
 C. Intermittent growth D. Hyperbolic growth

2. Which of the following groups of animals shows this pattern of growth?
 A. cestoda B. reptilia
 C. amphibia D. insecta

Use the diagram below to answer questions 3 and 4.



3. The organelle responsible for heredity is labeled
A. I B. II C. III D. IV

4. The part labeled IV is the
A. mitochondrion B. cell wall
C. endoplasmic reticulum D. nucleus.

5. Which of the following is the lowest category of classification?
A. class B. species
C. family D. genus.

6. Plants that show secondary growth are usually found among the
A. thallophytes B. pteridophytes
C. monocotyledons D. dicotyledons

7. The fungi are a distinct group of eukaryotes mainly because they have
A. spores B. no chlorophyll
C. many fruiting bodies
D. sexual and asexual reproduction.

8. An arthropod that is destructive at the early stage of its life cycle is
A. butterfly B. mosquito
C. bee D. millipede.

9. An animal body that can be cut along its axis in any plane to give two identical parts is said to be
A. radically symmetrical B. bilaterally symmetrical
C. asymmetrical D. symmetrical

10. Which of the following possesses mammary gland?
A. Dogfish B. Whale
C. Shark D. Catfish

11. The feature that links birds to reptiles in evolution is the possession of
A. feathers B. beak
C. skeleton D. scales.

12. Countershading is an adaptive feature that enables animals to
A. fight enemies B. remain undetected
C. warn enemies D. attract mates.

13. Which of the following plant structures lacks a waterproof cuticle?
A. leaf B. stem
C. root D. shoot

14. In the mammalian male reproductive system, the part that serves as a passage for both urine and semen is the
A. urethra B. ureter
C. bladder D. seminal vesicle.

15. In plants, which of the following is required in minute quantities for growth?
A. copper B. potassium
C. phosphorus D. sodium.

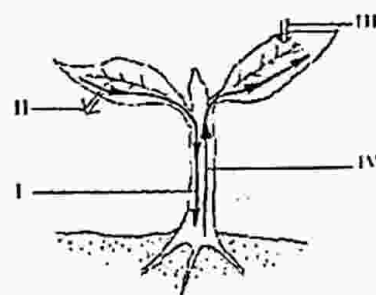
16. Which of the following organisms is both parasitic and autotrophic?

- A. Sundew B. *Loranthus*
C. *Rhizopus* D. Tapeworm

17. A function of the hydrochloric acid produced in the human stomach during digestion is to
A. neutralise the effect of bile
B. coagulate milk protein and emulsify fats
C. stop the action of ptyalin
D. break up food into smaller particles.

18. Which of the following is a polysaccharide?
A. Glucose B. Sucrose.
C. Maltose. D. Cellulose.

Use the diagram below to answer questions 19 and 20.



19. Transportation in the xylem is represented by
A. I B. II C. III. D. IV

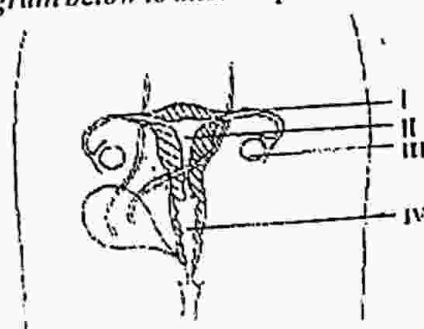
20. The arrow labeled II represents the
A. release of oxygen
B. intake of carbon (IV) oxide
C. movement of photosynthates
D. movement of nutrients

21. In the kidney of mammals, the site of ultrafiltration is the
A. uriniferous tubule B. Bowman's capsule
C. loop of Henle D. renal tubule.

22. Which of the following is involved in secondary thickening in plants?
A. Collenchyma and xylem cells.
B. Vascular cambium and collenchyma cells.
C. Vascular cambium and cork cambium.
D. Cork cambium and sclerenchyma.

23. An example of a fruit that develops from a single carpel is
A. okro B. tomato
C. bean D. orange

Use the diagram below to answer questions 24 and 25.



24. The developing embryo is usually contained in the part labeled

- A. IV B. III C. II D. I

25. The function of the part labeled III is to

- A. produce egg cells
B. protect sperms during fertilization
C. secrete hormones during coitus
D. protect the developing embryo.

26. Plant growth can be artificially stimulated by the addition of

- A. gibberellin B. kinin
C. abscisic acid D. ethylene

27. The autonomic nervous system consists of neurons that control the

- A. voluntary muscles B. heart beat
C. tongue D. hands

28. Plants of temperate origin can be grown in tropical areas in the vegetation zones of the

- A. rain forest B. Guinea savanna
C. Sudan savanna D. montane forest.

29. The water cycle is maintained mainly by

- A. evaporation of water in the environment
B. evaporation and condensation of water in the environment
C. condensation of water in the environment
D. transpiration and respiration in plants.

30. Organisms living in an estuarine habitat are adapted to

- A. withstand wide fluctuation in temperature
B. survive only in water with low salinity
C. withstand wide fluctuations in salinity
D. feed only on phytoplankton and dead organic matter.

31. The presence of stilt roots, pneumatophores, sunken stomata and salt glands are adaptive features of plants found in the

- A. tropical rainforest B. mangrove swamps
C. grassland D. montane forest.

32. Which of the following animals can exist solely on the water they get from food and metabolic reaction?

- A. Forest arboreal dwellers B. Desert dweller.
C. Forest ground dwellers D. rainforest dwellers

33. The most likely first colonizers of bare rock are

- A. mosses B. ferns
C. lichen D. fungi

34. The carrying capacity of habitat is reached when the population growth begins to

- A. increase slowly B. increase exponentially
C. slow down D. remain steady.

35. The abiotic factors that control human population include

- A. disease and famine B. space and rainfall
C. flooding and earthquake D. temperature and disease

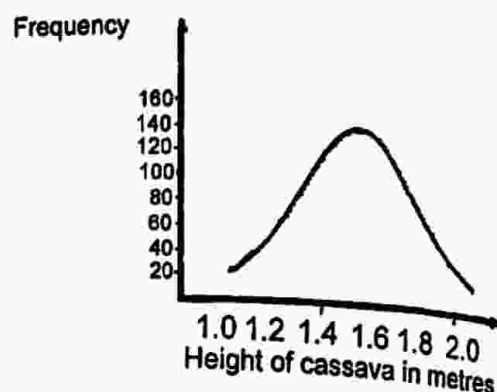
36. An indigenous method of renewing and maintaining soil fertility is by

- A. clearing farms by burning
B. planting one crop type
C. adding inorganic fertilizers yearly
D. crop rotation and shifting cultivation.

37. The diseases caused by water-borne pathogens include

- A. gonorrhea and poliomyelitis
B. typhoid and syphilis
C. tuberculosis and cholera
D. typhoid and cholera.

Use the diagram below to answer questions 38 and 39.



38. The graph illustrates

- A. the highest frequency for height of 2 metres
B. a discontinuously varying character
C. a continuously varying character
D. total yield in a cassava farm.

39. the largest number of cassava plants has an approximate height of

- A. 1.4m B. 1.6m C. 1.8m
D. 2.0m

40. Which of the following is true in blood transfusion?

- A. A person of blood group AB can donate blood only to another person of blood group AB.
B. Persons of blood groups A and B can donate or receive blood from each other.
C. A person of blood group AB can receive blood only from person of blood group A or B.
D. A person of blood group O can donate only to a person of blood group O.

41. A yellow maize is planted and all the fruits obtained are of yellow seeds. When they are cross-bred, yellow seeds and white seeds are obtained in a ratio 3:1. The yellow seed is said to be

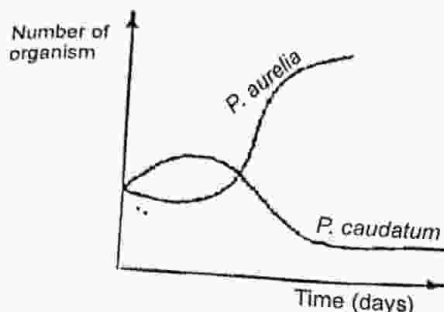
- A. non-heritable B. sex-linked
C. a recessive trait D. a dominant trait.

42. When a colour-blind man marries a carrier woman. What is the probability of their offspring being colour blind?

- A. 25% B. 50% C. 75% D. 100%

43. The correct base pairing for DNA is
 A. adenine ↔ thymine and guanine ↔ cytosine
 B. adenine ↔ guanine and thymine ↔ cytosine
 C. adenine ↔ cytosine and guanine ↔ thymine
 D. adenine ↔ adenine and cytosine ↔ cytosine

Use the diagram below to answer questions 3 and 4.



44. The type of interaction shown is referred to as
 A. interspecific competition
 B. intraspecific competition
 C. mutualism
 D. cooperation.
45. Which of the following statements is true of the interaction?
 A. *P. aurelia* is better adapted for obtaining food than *P. caudatum*
 B. *P. caudatum* is better adapted for obtaining food than *P. aurelia*
 C. both organisms cannot coexist.
 D. both organisms cannot reproduce.

46. The short thick beak in birds is an adaptation for
 A. crushing seeds
 B. sucking nectar
 C. tearing flesh
 D. straining mud.

47. The basking of *Agama* lizards in the sun is to
 A. change the colour of their body
 B. raise their body temperature to become active
 C. fight to defend their territories
 D. attract the female for courtship.

48. The significance of a very large number of termites involved in nuptial swarming is to
 A. provide birds with plenty of food
 B. ensure their perpetuation despite predatory pressure.
 C. search for a favourable place to breed
 D. ensure that every individual gets a mate.

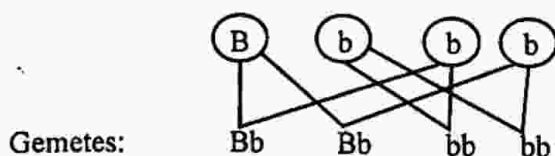
49. The use and disuse of body parts and the inheritance of acquired traits were used to explain
 A. Darwin's theory
 B. Lamarck's theory
 C. genetic drift
 D. gene flow.

50. From his study of Galapagos finches, Darwin derived his theory of evolution from
 A. comparative anatomy
 B. comparative physiology
 C. fossil remains
 D. comparative embryology.

ANSWERS TO MODEL QUESTIONS SET 1

1. B = Plant and animal cell have the nucleus in common.
2. B = Epithelial tissue in animals form a protective covering over the body surfaces.
3. C = Amoeba removes excess water (osmoregulation) via the contractile vacuole
4. D = Complexity of cell structure (prokaryotic or eukaryotic) and size are the criteria used in classifying living organisms into the five kingdoms namely: Monera, Protista, Fungi, Plantae, Animalia.
5. B = Spirogyra floats in masses in ponds, lakes, and slow moving streams. They are commonly called pond scum or water silk.
6. D = Cockroach antenna, also known as feelers, are responsible for their sense of smell.
7. A = The sporophyte is diploid (2n) while the gametophyte is haploid (n).
8. C = The bacterial cell is a prokaryotic cell, their nuclear material is scattered in the cytoplasm.
9. C = I - Eye, II - operculum (gill cover), III - lateral line IV - scales. The lateral line is used to detect vibration in water.
10. C
11. D = The diaphragm separate the thoracic cavity from the abdominal cavity.
12. A = Xylem is interrupted from phloem by the cambium. The cortex is uninterrupted. The cortex is wider than the pith.
13. C = Carnivorous plants (or insectivorous plants)
14. C = A vegetarian takes diets obtained strictly from plants. The body needs carbohydrate, proteins and lipids basically but also needs vitamins. Beans (proteins), rice (carbohydrate) mushroom (carbohydrate, fat, protein and vitamins)
15. B = Absorption of digested food takes place in the ileum.
16. C = The tooth enamel is the hardest and most highly mineralized substance in the human body. Enamel's primary mineral is hydroxyapatite, which is a crystalline calcium phosphate. The large amount of enamel accounts not only for its strength but also for its brittleness. Dentine is less mineralized and less brittle.
17. C = Cholesterol is a steroid in mammalian blood plasma. Globulin, albumin and fibrinogen are proteins in the blood plasma.
18. A = Active transport is the movement of a substance against its concentration gradient (from low to high concentration). It requires energy in the form of ATP. Example include the uptake of glucose in the human intestines and uptake of mineral ions in the root hair cells in plant.
19. C = The air sacs are the alveoli, which are surrounded by the blood capillaries for effective diffusion of gases. Capillaries have thin walls.
20. C = The percentages of gases in inhaled and exhaled air respectively is Nitrogen (78% & 78%), Oxygen (21% & 17%), Carbondioxide (0.04% & 4%) and water vapour (0.86% & 3%)

21. C = Tendon - muscle to bone. Ligament - bone to bone.
 22. A = The plumule will curve upward (shoots of plants are negatively geotropic) while the radicle will curve downward (roots of plants are positively geotropic).
 23. C = Entomophilous flowers (flowers that need pollinators) commonly have larger showy petals to attract insects or birds or large white petals to attract bats. Anemophilous plants do not need pollinators, they are pollinated by wind.
 24. D = Leutenizing hormone (LH) induces ovulation, that is why there is LH surge at ovulation.
 25. C = There was optimum absorption of by X and Y between 50-60°C, 80mg of ions was absorbed by X (optimum) and at 60°C, 140mg of ions was absorbed by Y (optimum).
 26. B = The best title for the graph is the effect of temperature on the rate of ion uptake (absorption) in X and Y.
 27. C = Motor neurons - CNS to effectors. Sensory neurons - receptors to CNS.
 28. D = Black fly, *Simulium damnosum* breeds in fast flowing streams.
 29. B = Salinity and turbidity are unique to aquatic habit.
 30. A = The protozoan digest the cellulose taken in by the termite while the termite provide food and shelter for the protozoan.
 31. C = Photosynthesis removes CO₂.
 32. A = Pneumatophores (breathing roots) of the white mangrove allow intake of atmospheric air.
 33. C
 34. A = Obudu cattle ranch is situated deep in the tropical and mountainous rainforest of Northern Cross River State.
 35. D = The predator - prey relationship in the ecological community e.g. cat to prey on rat
 36. D = Oxygen is depleted by alga overgrowth in water bodies
 37. B = Sand has the lowest water retention capacity. Clay has the highest.
 38. C = During leaching, flood water (from heavy rainfall) permeates the soil and carries along with it, dissolved nutrients into the depth of the soil where roots cannot reach.
 39. C = Non-renewable resources can be conserved by recycling.
 40. B = Houseflies are vectors of *Vibrio* sp., the bacteria that causes cholera.
 41. B = Sexual reproduction (fertilization) will result in variation because of gene combination.
 42. A = Genetic diseases such as sickle cell anaemia.
 43. A = A trait is often controlled by many pairs of genes
 44. B = parents: Bb vs bb



Probability of Bb is 2/4 i.e. 1/2

45. B = Chromosomal crossover (or crossing over) is an exchange of genetic material between homologous chromosomes.

46. A = I - Breathing root of mangrove,
 II - Aerial roots of an epiphyte,
 III - Prop roots of maize,
 IV - Dicot leaves.
 47. D
 48. D = Countershading helps to conceal animals from predators making them remain unnoticed.
 49. C = Leaves with sunken stomata and thick cuticle with reduce loss of water by transpiration, helping to conserve water.
 50. B = The drone is found in the bee colony.

ANSWERS TO MODEL QUESTIONS SET II

1. C = Growth patterns include:
 (i) Isometric growth: when parts of an organism grow at the same rate as the whole organism e.g. plant leaves.
 (ii) Allometric growth: When parts or organs of an organism grow at different rates from the whole organisms e.g. the head and brain of man during the early stage of life.
 (iii) Intermittent growth: This is a step-like growth; it occurs in arthropods due to moulting.
 2. D.
 3. B = The gene is the unit of inheritance, it is the constituent of the chromosome, located in the nucleus.
 4. C = I - Cell membrane,
 II - Nucleolus,
 III - Mitochondria,
 VI - Endoplasmic reticulum
 5. B = The highest category of classification is kingdom and the lowest is species.
 6. D = Secondary growths is found in the dicotyledons because of the presence of cambium
 7. C = Fungi are distinct among other eukaryotes because of many fruiting bodies - sporocarps, a multicellular structure of which spore - producing structures, such as basidia or asci, are borne. The presence of spores, sexual and asexual reproduction; and of lack of chlorophyll is not unique to the fungi.
 8. A = Butterfly (adult form) helps in pollination of flower while its larva form - caterpillar, is destructive.
 9. A = Radial symmetry will give a mirror image when cut along its axis in any plane. Bilateral symmetry will only give a mirror image, when cut along its axis in one plane.
 10. B = Whale is a mammal. Mammals are distinct for their pinna (external ear), mammary gland, hair and sebaceous gland).
 11. D = Birds and reptiles have scales.
 12. B = Countershading helps to conceal animals from predators making them remain unnoticed.
 13. C = The epidermis of roots lacks the waxy cuticle found in the parts of the plant. This is so to allow effective absorption of water and nutrients by the hair cells.
 14. A = The urethra is a conduit for both urine (excretory) and semen (reproductive).
 15. A = Copper is a trace element needed in small quantity. Major elements are: nitrogen, potassium, phosphorus

magnesium and sulphur.

Question: Nigerian Police Pick Criminals Most

Parasitism is a partial parasite (obligate), it has a symbiotic relationship with host such as mango. It also has chloroplast in its leaves for photosynthesis - autotrophic nutrition.

Hydrochloric acid (HCl) in the stomach is to halt action of ptyalin from the mouth, which can only function in an alkaline medium. It also kills bacteria in food.

Monosaccharides - glucose, fructose, ribose.

Saccharides - sucrose, maltose, lactose.

Polysaccharides - starch, cellulose.

I and II - Translocation of nutrients,

III - Transpiration of water vapor,

IV - Transportation of nutrients in the xylem.

C = Movement of photosynthate (translocation).

B = Ultra filtration takes place in the Bowman's capsule.

B = Secondary thickening is as a result of the activity to two lateral meristems, the cork cambium and vascular cambium. The cause the lateral growth of the stem and root.

C = Bean is a legume. A legume is formed from one carpel and split along two sides.

C = I - Oviduct (fallopian tube) - passage of eggs, site of fertilization.

II - Uterus - where fertilized egg develop.

III - ovary - produces eggs

IV - vagina - where sperm is deposited.

25. A

26. A = Gibberellins promotes growth.

27. B = The autonomic nervous system consists of neurons that control involuntary (reflex) actions e.g. heart beat.

28. A = plants of temperate origin can be grown in tropical areas in the vegetation zones of the rainforest. There are two types of rain forest, temperate and tropical.

29. B = Water leaves the earth crust via evaporation and transpiration (in plants), respiration (in plants and animals) and returns as rain via condensation and precipitation.

30. C = An estuary is a ecological zone where river and sea water meet, thus to establish brackish conditions. Brackish water has a salinity which fluctuates with the tides and the wet and dry seasons.

31. B = Plants of mangrove have stilt root (red mangrove) pneumatophores (white mangrove), sunken stomata and salt glands.

32. B = Desert dwellers strictly conserve water.

33. C = The first colonizers of a bare rock surface are the lichen. They breakdown the rocks to provide soil for subsequent plants (mosses).

34. D = Carrying capacity is the number of individuals an environment can support without significant negative impacts to the given organism and its environment. Below carrying capacity, populations typically increase while above, they typically decrease. Population remain fairly steady at carrying capacity.

35. C = Flooding and earthquake are purely abiotic factors. Disease is a biotic factor because it involve living

organisms e.g. vectors

36. D = Indigenous methods refers to natural methods and shifting cultivation

37. D = Typhoid is caused by *Salmonella* sp. And cholera is caused by *Vibrio* sp. both of which are carried in water. Gonorrhoea and syphilis are sexually transmitted. Tuberculosis is air-borne while poliomyelitis can also be water-borne.

38. C = The graph illustrates continuous variation in height.

39. B = 1.6m

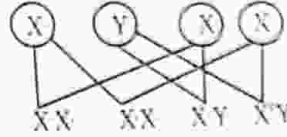
40. A = Blood group AB is a universal recipient but can only donate to AB blood group.

41. D = Yellow is a dominant trait.

42. B = Colour blindness is a sex-linked traits.

Parents: Colour blind man X^cY vs carrier women X^cX

Gametes:



Offspring:

Carries female, colour blind female, normal male, colour blind male respectively.

43. A = AT; GC i.e. adenine thymine & guanine cytosine

44. A = *P. aurelia* and *P. caudatum* are two different species of *Paramecium*. Hence, it is interspecific competition.

45. A = From the graph, it is obvious that *P. aurelia* is better adapted for obtaining food than *P. caudatum*.

46. A = A short thick beak in birds is an adaptation for crushing nuts (seeds).

47. B = *Agama* lizards basks in the sun to raise their body temperature to become active. They are poikilothermic.

48. B = Large numbers of termite participate in the nuptial swarming in order to ensure perpetuation despite predatory pressures.

49. B = Lamarck propose the theory of use and disuse; and the inheritance of acquired traits

50. A = Charles Darwin made a comparative anatomical studies of the Galapagos finches to illustrate evolution.

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1. When two genes for the same character are contained in the same individual, the character that shows is known as
 A. important character B. dominant character
 C. superior character D. controlling character

2. In Nigeria, a tropical rainforest can be found in
 A. Sokoto B. Kaduna
 C. Abuja D. Abia

3. The sum total of the biotic and abiotic factors that affect living things is referred to as
 A. environment B. lithosphere
 C. hydrosphere D. atmosphere

4. Hygrometer is an ecological instrument that measures
 A. rainfall B. humidity
 C. temperature D. light

5. *Candida vaginalis* is a
 A. bacterium B. fungus
 C. virus D. Protozoan

6. Which one of the following is air borne?
 A. Malaria B. Yellow fever
 C. Cholera D. Tuberculosis

7. All of these are vertebrates except
 A. lizard B. rat
 C. star fish D. tilapia

8. A box was left in the lawn for two days when the box was removed, the grass under had turned yellow due to lack of
 A. carbon dioxide B. light
 C. oxygen D. water

9. Which of the following deaminates excess amino acids?
 A. Duodenum B. Ileum
 C. Liver D. Kidney

10. In which of the following floral parts does meiosis occur?
 A. Anther B. Petal
 C. Receptacle D. Style

11. When an organism moves its whole body towards a stimulus, the organism is said to exhibit
 A. tropic movement B. trophic movement
 C. tactic movement D. nastic movement

12. A stable self sustaining environment produced by an interaction between the biotic and abiotic components its best described as
 A. a niche B. a community
 C. a habitat D. an ecosystem

13. Which of the following is not present in the nucleus of a cell
 A. Chromosomes B. Nucleolus
 C. Genes D. Mitochondria

14. The release of useful substances from the cells of an organism is called
 A. excretion B. evacuation
 C. metabolism D. secretion

15. The head of the femur articulates with the pelvis and the
 A. glenoid cavity B. olecranon process
 C. acetabulum D. caroid process

16. The mammalian cervical vertebrae invariably numbers
 A. 4 B. 7
 C. 12 D. 15

17. What mechanism is responsible for the rise of water to the top of tall trees?
 A. Transpiration pulls B. Root pressure
 C. Osmosis D. Capillarity

18. Antibodies in mammalian blood are formed by
 A. platelets B. white blood
 C. red blood cells D. livers

19. Which of the following liquids supplies cells in the tissues of a mammal with oxygen and nutrients?
 A. Blood B. Plasma
 C. Serum D. Lymph

20. The cockroach and grasshoppers have mouth parts adapted for
 A. sucking B. piercing and sucking
 C. biting and chewing D. biting and lapping

21. The Sponging and lapping mouthparts occurs in
 A. housefly B. butterfly
 C. cockroach D. mosquito

22. Which of the following is not an example of sex-linked character?
 A. Baldness B. Colour blindness
 C. Haemophilia D. Height

23. Which of these is a vestigial structure?
 A. Appendix B. Caecum
 C. Pancreas D. Rotundus

24. Which of the following air pollutants causes acid rain?
 A. sulphur dioxide B. lead oxide
 C. hydrogen dioxide D. carbon dioxide

25. Which of the following does not have a well-developed tissue?
 A. Moss B. Fern
 C. Whispering tree D. Maize

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26. Which is the largest single cell in the body?
 A. a neuron B. the ovum
 C. muscle D. Cell

27. Blue-green algae belongs to the phylum
A. cyanophyta B. schizophyta
C. chlorophyta D. chrysothyma
28. The concept of antibiotics started with the concept of
A. Gregor Mendel B. Mary Slessor
C. Louis Pasteur D. Alexander Fleming
29. The botanical name for yellow yam is
A. *Discorea cayensis* B. *Discorea ratundata*
C. *Discorea alata* D. *Discorea dometorum*
30. Which of the following tissues is made up of dead cells?
A. Xylem vessels B. Cambium
C. Mesophyll D. Palisade
31. Which is not a function of the liver?
A. storage of iron
B. formation of bile
C. breakdown of excess amino acids
D. excretion of urea from the blood
32. Alkaline pyrogallol was used in an experiment. That experiment must have been connected with
A. respiration B. photosynthesis
C. transpiration D. excretion
33. The enzyme that breakdown sugar cane is
A. lipase B. Ptyalin
C. invertase D. peptidase
34. The largest single bone is the
A. scapula B. humerus
C. femur D. skull
35. Auxins are produced at the
A. root and stem apices
B. young leaves and nodes
C. flower buds D. leaves apices
36. The hormone which tones up the muscle of a person in times of danger is from the
A. thyroid gland B. pancreas
C. adrenal gland D. sebaceous gland
37. A diet with a high concentration of iodine will probably be needed by a patient suffering from a malfunction of the
A. thyroid gland B. adrenal gland
C. nervous system D. Circulatory
38. Which of the following group of factors is completely abiotic?
A. soil, water, bacteria, salinity
B. salinity, tide, plankton, turbidity
C. wind, altitude, humidity, light
D. conifers, wind, pH, rainfall
39. Nitrogen fixing bacteria and cowpea demonstrate an ecological association known as
A. predation B. parasitism
C. mutualism D. Commensalisms.
40. Plants that live in salty water areas are called
A. hydrophytes B. xerophytes
C. halophytes D. salinophytes
41. A sample of soil was put into a measuring cylinders and water was added to it. After the mixture was shaken, the cylinder was left undisturbed for one hour. This experiment was probably performed to
A. compare the capillarity of different samples
B. find out the relative densities of different soil samples
C. find out the water retaining capacity of the soil
D. demonstrate the presence of air in the samples
42. A farmer X working in a swamp did not eat any food not drink any water. Which of these diseases can he not contract
A. bilharzia B. malaria
C. cholera D. sleeping sickness
43. Which of the following elements may be added to drinking water to lessen dental decay?
A. calcium B. phosphorus
C. fluorine D. chlorine
44. A man with normal haemoglobin marries a woman who has sickle cell haemoglobin. They have a child who had sickle cell trait. Which of the following genotype could be associated with the child's haemoglobin?
A. SS B. AS
C. AO D. AA

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45. Hydra is able to perform all the following except
A. feeding B. photosynthesis
C. moving D. egestion
46. Spherical bacterial cells arranged in chains are called
A. staphylococci B. streptococci
C. cocci D. bacilli
47. One of the following factors that must be considered for safe blood transfusion is
A. social class of the donor
B. rhesus factors of the donor and the recipient
C. age of the recipient
D. nationality of the donor
48. The theory of use and disuse of organs was pro pounded by
A. Charles Darwin B. Jean Lamark
C. Louis Pasteur D. Robert Hooke
49. Which of these is not a function of the kidney?
A. production of urine B. production of bile
C. osmoregulation D. removal of urea
50. Auxins are produced in the
A. petioles of leaves
B. epidermis of roots and shoots
C. parenchyma of roots and shoots
D. apical regions of roots and shoots

51. Which of the following mammalian features acts as a shock absorber to developing embryo?

- A. Umbilical cord
- B. Amniotic fluid
- C. Placenta
- D. Amnion

52. The maintenance of a constant internal environment is called

- A. environmental regulation
- B. balance in nature
- C. peristalsis
- D. homeostasis

53. People in the blood group which can receive blood from another blood group fall into

- A. AB blood group
- B. O blood group
- C. A blood group
- D. B blood group

54. Which of the following crop disease is caused by a fungus?

- A. Groundnut Rosette
- B. Black pod disease of cocoa
- C. Yam Mosaic
- D. Cassava Blight

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55. One of the function of xylem

- A. Strengthening the stem
- B. Manufacturing food
- C. Conducting manufactured food
- D. None of above

56. People suffering from myopia

- A. can see near objects clearly
- B. can see far away objects clearly
- C. cannot see any object clearly
- D. are colour-blind

57. The cilia in paramecium are used for

- A. Respiration
- B. Locomotion
- C. Protection
- D. Excretion

58. Which of these types of skeleton is most appropriate to the cockroach

- A. Hydrostatic skeleton
- B. Exoskeleton
- C. Endoskeleton
- D. Cartilaginous skeleton

59. When proteins are broken down they produce

- A. Oxygen
- B. Carbohydrate
- C. Energy
- D. Amino acids

60. The function of lenticel is

- A. To receive excess water in the plant
- B. To absorb water from the atmosphere
- C. For gaseous exchange
- D. To absorb light

61. Which of the following is characteristic of the animal cell

- A. presence of large vacuoles
- B. possession of a cellulose cell wall
- C. absence of large vacuoles
- D. presence of large vacuoles

62. In the life history of Schistosoma (Bilharzia) one of the

following is the intermediate host

- A. Man
- B. Snail
- C. Mosquito larva
- D. Fish

63. The hormone which tones up the muscles of a person in the time of danger is from the

- A. thyroid gland
- B. pancreas
- C. adrenal gland
- D. spleen

64. The study of the organisms and the environment of abandoned farmland is the Ecology of

- A. A community
- B. A population
- C. A species
- D. An ecosystem

65. At fertilization

- A. One chromosome from the male joins another from the female
- B. One gene from the male combines with the another from the female
- C. The male nucleus fuses with the female nucleus
- D. One set of chromosome combines with another set from the female

66. The neck region of the tapeworm (Taenia spp) is responsible for the

- A. The production of eggs
- B. The storage of eggs
- C. The formation of new segments
- D. The development of the suckers

67. The movement of molecules from a region of higher concentration to one of lower concentration is

- A. Diffusion
- B. Transpiration
- C. Osmosis
- D. Plasmolysis

68. The region of cell division in a root is

- A. Root cap
- B. Endodermis
- C. Xylem
- D. Meristem

69. Which of the following is not an excretory organ?

- A. Lungs
- B. Kidney
- C. Leaf
- D. Large intestine

70. The part of the mammalian brain responsible for maintaining balance is

- A. Medulla oblongata
- B. Cerebellum
- C. Optic lobe
- D. Cerebrum

71. A sugar solution was boiled with Fehling's solutions A and B and the colour remain blue. The sugar tested was

- A. Glucose
- B. Maltose
- C. Fructose
- D. Sucrose

72. The blood vessel which carries digested food from the small intestine to the liver is the

- A. Renal vein
- B. Renal artery
- C. Hepatic artery
- D. Hepatic portal vein

73. The maize grain is regarded as a fruit and not a seed because

- A. It is covered by a sheath of leaves
- B. The testa and fruit wall fuse after fertilization

- C. It has both endosperm and cotyledon
D. The pericarp and seed coat are separate

74. Identical twins are produced under one of the following conditions

- A. Two ova fertilized at the same time by the sperm
B. One ovum fertilized, divides to give two embryos
C. Two ova fertilized by one sperm
D. One ovum fertilized by sperms

75. Which of the following hormones is produced during fright or when agitated?

- A. Insulin
C. Thyroxine
B. Adrenalin
D. Pituitary

76. Grasses → Grasshopper → Lizards → Snakes → Hawks. In the food chain, the organisms which are the least in number are

- A. Grasses
C. Lizards
B. Hawks
D. Snakes

77. One significant difference between roots and stem is that

- A. Branch root originate in the pericycle while branch stems do not
B. Stems are always below the ground while roots are always under the ground
C. Stems are positively geotropic while roots are negatively geotropic
D. Stems are sometimes used for storage while roots are never so use

78. The arrangements below are steps in protein digestion. Which is the correct sequences

a-polypeptides b-protein c- amino acids
d-peptones

- A. a → b → c → d
C. b → d → a → d
B. c → d → c → b
D. b → d → a → c

79. Partially digested food ready to leave the stomach is called

- A. chyme
C. glycogen
B. curd
D. paste

80. The particles found in the blood which play an important role in blood clotting are called

- A. platelets
C. monocytes
B. red blood cells
D. granulocytes

81. In a cross between a normal male and a female carrier for haemophilic disease, the percentage of their sons expected to suffer the disease as hemophiliac is

- A. 25%
C. 70%
B. 50%
D. 45%



82. The structure labeled 4 is for

- A. feeding
C. excretion
B. attachment
D. respiration

83. As the tadpole develops the structure labeled 5

- A. grows longer
C. becomes the hind legs
B. becomes shorter
D. remains unchanged

84. The gill rakers of fishers take part in

- A. feeding
C. swimming
B. respiration
D. diffusion

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85. Where is energy produced in a cell

- A. Nucleus
C. Mitochondria
B. Lysosomes
D. Nucleolus

86. Which of the following organisms does not exist as a single free-living cells

- A. Amoeba
C. Chlamydomonas
B. Euglena
D. Volvox

87. Euglena is an autotrophic organism because it

- A. has flagella
C. can manufacture its food
B. has plant and animal features
D. moves fast

88. In which of the following organisms does a single cell perform all function of active movement, nutrition, growth excretion and photosynthesis?

- A. Paramecium
C. Euglena
B. Amoeba
D. Hydra

89. What is the function of contractile vacuole in Paramecium?

- A. produces enzymes
C. stores and digests food
B. gets rid of excreta
D. gets rid of excess water

90. The ability of organisms to maintain a constant internal environment is known as

- A. diuresis
C. plasmolysis
B. endosmosis
D. homeostasis

91. Which of the following is the medium of transportation of nutrients within unicellular organism?

- A. Lymph
C. Protoplasm
B. Plasma
D. Serum

92. In aerobic respiration, oxidative phosphorylation takes place in

- A. cytoplasm
C. mitochondrion
B. lysosome
D. ribosomes

93. Bryophytes are different from flowering plants because they

- A. are simple small plant
B. carry out alternation of generation
C. possess small
D. possess no vascular tissue

94. In lower plants like mosses, the structure which performs the functions of roots of higher plants is called
 A. roots hairs B. rhizoids
 C. hyphae D. roots

95. Which of the following components of an ecosystem has the greatest biomass?
 A. primary producers B. primary consumers
 C. secondary consumers D. tertiary consumers

96. The name of a bacterium which derives its energy from oxidizing nitrites into nitrates is
 A. nitrosomonas B. azotobacter
 C. nitrobacter D. Escherichia coli

97. Potometer is used to measure
 A. rate of osmosis B. rate of diffusion
 C. rate of transpiration D. rate of photosynthesis

98. Meiotic cell division ensures that
 A. many similar cells are produced
 B. chromosome number of cells is halved
 C. cells produced are doubled
 D. cells produced possess the same chromosome number

99. The stem of young herbaceous plants keep upright mainly by
 A. Osmotic pressure B. Turgidity
 C. Transpiration pull D. Root pressure

100. Which of the following tissues is not found in the stem and root of monocotyledons?
 A. xylem B. cambium
 C. pith D. pericycle

101. Fruit enlargement can be induced by spraying young ovary with
 A. gibberellins, ethylene and abscisic acid
 B. auxin, cytokinin and gibberellin
 C. auxin, cytokinin and abscisic acid
 D. auxin, ethylene and gibberellin

102. A dry, indehiscent, winged fruit formed from one carpel is known as
 A. schizocarp B. caryopsis
 C. samara D. nut

103. A fruit which develops without fertilization is described as
 A. simple B. aggregate
 C. multiple D. parthenocarpic

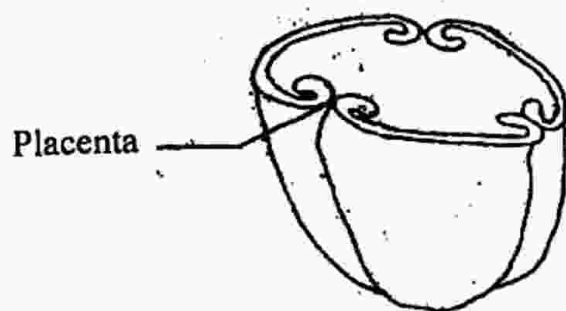
104. A dwarf plant can be stimulated to grow to normal height by the application of
 A. thyroxine B. gibberellin
 C. insulin D. Kinin

105. The condition known as cretinism is caused by the deficiency of
 A. Vitamin A B. Insulin
 C. Thyroxine D. Vitamin C

106. The difference between viviparous and oviparous animals is

A. possession of yolked eggs
 B. laying and brooding of eggs
 C. possession of yolk less egg
 D. laying of unfertilized egg

107. The following are features of the tropical rainforest except
 A. loose and moist soil
 B. short trees growing beneath tall trees
 C. scanty trees with small leaves
 D. presence of many animals



Transverse section of an ovary
 108. The type of placentation shown in the figure above is
 A. parietal B. marginal
 C. axile D. free-central

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109. An amoeba moving towards a crumb of cake in a pond most likely exhibits
 A. Phototropism B. Chemotaxis
 C. Thermotaxis D. Nastic movement

110. Which of the following cells would probably contain the greatest number of Golgi bodies
 A. muscle cell B. secretory cell
 C. nerve cell D. white blood cell

111. A group of similar cells performing the same function is
 A. an enzyme B. an organ
 C. a tissue D. an organelle

Structures found in cells are listed below:
 i. Cell wall ii. Cell membrane
 iii. Chloroplast iv. Cytoplasm
 v. Nucleus vi. Sap vacuole

112. Which of these structures are found in both animal cells and plant cells?
 A. i, ii and v B. i, ii and v
 C. ii, iii and v D. ii, iv and v

113. Which of the following is not present in the nucleus of a cell
 A. Chromosomes B. Nucleolus
 C. Mitochondrion D. Genes

114. A plant which grows on another plant without apparent harm to the host plant is called
 A. A parasite B. An epiphyte

113. The petals of a flower are collectively called

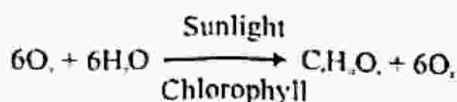
A. Sepal
B. Capsule
C. Corolla
D. A predominate

114. The petals of a flower are collectively called

A. Sepal
B. Capsule
C. Corolla
D. A predominate

115. Osmosis can be defined as diffusion of

- A. water molecules from an area of high concentration to an area of low concentration
B. water molecules from dilute solution to a concentrated solution across a permeable membrane
C. Water molecules from a concentrated solution to a dilute solution through a semi-permeable membranes
D. Water molecules from a dilute solution to a concentrated solution through a semi-permeable membrane



117. The oxygen given off during the process in the above equation is derived from

- A. Sunlight
B. Water
C. Carbondioxide
D. Atmosphere

118. When testing a leaf for starch why is it first placed in boiling water?

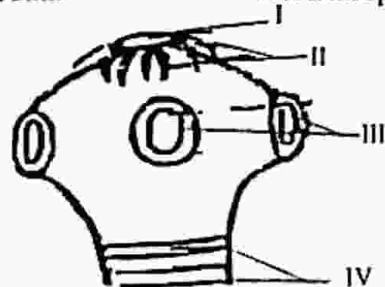
- A. To extract the chlorophyll
B. To remove colour from the leaf
C. To dissolve the starch
D. To stop chemical reaction

119. Each of the following is an arthropod except the

- A. crab
B. scorpion
C. spider
D. snail

120. The largest phylum in animal kingdom is

- A. Cnidaria
B. Mollusca
C. Chordata
D. Arthropoda



121. Which reference to the figure above, which of these are correct

- A. I and II are proglottids and hooks
B. I and III are rostellum and suckers
C. III and IV are hooks and proglottids
D. II and IV are hooks and rostellum

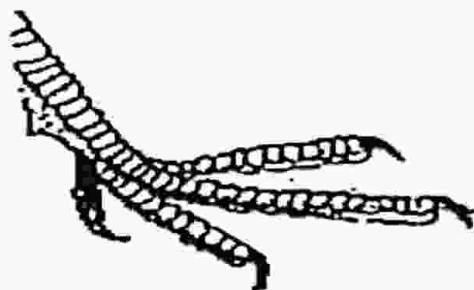
122. Which of the following statement is not correct about the function of each group of mammalian vertebrae?

- A. Caudal vertebrae support the tail and provide attachment for tail muscles
B. Thoracic vertebrae articulate with the ribs
C. Lumbar vertebrae provide attachment for abdominal muscles

D. Sacral vertebrae support the skull and allow nodding and rotating movement

123. In the adult toad, gaseous exchange takes place through

- A. Buccal cavity, skin and spiracle
B. Buccal cavity, bladder and lungs
C. Buccal cavity, skin and lungs
D. Gills, skin and buccal cavity



124. The foot of the bird shows above is strong and has strong claws on its digits. This implies that the bird

- A. is a scavenger
B. is a bird of prey
C. uses the foot to supplement wing action
D. uses the foot to scratch the soil

125. Which of the following is not a means of conservation

- A. Replacing harvested nature timber trees with their seedlings
B. Prevention of poaching
C. Controlling excessive deforestation
D. Burning of vegetation before cropping

126. One of the following statements is not true of viruses

- A. They are micro-organisms
B. They are smaller than bacteria
C. They can be seen with an ordinary light microscope
D. They cause tobacco disease, polio and smallpox

127. The brain and the spinal cord constitute the

- A. autonomic nervous system
B. sympathetic nervous system
C. somatic nervous system
D. central nervous system

128. Which of the following parts of the mammalian brain is involved in taking the decision to run rather than walk

- A. Cerebellum
B. Medulla oblongata
C. Midbrain
D. Cerebrum

129. Which part of the ear is responsible for the maintenance of balance

- A. Cochlea
B. Sympathetic nervous system
C. Eustachian tube
D. Semi-circular canals

130. Which of the following factors is a decomposer?

- A. Spirogyra
B. Mushroom
C. Chlamydomonas
D. Paramecium

131. Which of the following is not a type of supporting tissue in plant
 A. Cambium
 C. Paramedian
 B. Collenchyma
 D. Xylem

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132. The medium of transpiration in higher plant is
 A. cell sap
 C. phloem
 B. water
 D. cytoplasmic fluid

133. The association between protozoa and termites is an example of
 A. Symbiosis
 C. Saprophytism
 B. Parasitism
 D. Predation

134. The organ that secrete both digestive enzymes and hormones is
 A. Salivary gland
 C. Spleen
 B. Pancreas
 D. Liver

135. Deficiency of calcium in a child diet causes
 A. Rickets
 C. Goitre
 B. Cretinism
 D. Scurvy

136. The part of the nephridium through which waste products are discharged in the
 A. Muscular tube
 C. Nephridiopore
 B. Narrow tube
 D. Nephrostome

137. Property of the enzymes excludes
 A. Specificity
 C. Inexhaustibility
 B. Clarity
 D. Sensitivity

138. Chlamydomonas is
 A. a blue-green algae
 C. a clammy algae
 B. a green algae
 D. a red algae

139. What substance make up the genetic material
 A. deoxyribonucleic acid
 C. Adenosine
 B. Ribonucleic acid
 D. Proteins

140. The table shows three processes that contribute to transport across cell surface membranes which processes are active and which are passive?

	Diffusion	Endocytosis	Osmosis
A	a	a	a
B	p	a	a
C	p	a	p
D	p	p	p

Key: a = active process p = passive process

141. Which statement is the best explanation of why sucrose rather than glucose is transported by phloem?
 A. Sucrose can pass through plant cell surface membranes more easily
 B. Sucrose is a disaccharide and a more easily converted to starch
 C. Sucrose is a non-reducing sugar, so is less reactive
 D. Sucrose synthesis requires less energy

142. What will break an ionic bond between amino acids?
 A. Condensation
 C. Hydrolysis
 B. High temperature
 D. pH change

143. From which organelle are nucleic acid absent?
 A. Chloroplast
 C. Mitochondria
 B. Golgi body
 D. Ribosome

144. The first in milk produced by the mother for a new-born baby contains antibodies. What do these antibodies provide?
 A. Artificial active immunity
 B. Artificial passive immunity
 C. Natural active immunity
 D. Natural passive immunity

145. Which protein has a fibrous structure?
 A. Amylase
 C. Hemoglobin
 B. Collagen
 D. Insulin

146. A girl has blood group A and her brother has blood group B. Which combination of genotypes cannot belong to their parents.

	Mother	Father
A.	$I^A I^A$	$I^B I^O$
B.	$I^A I^B$	$I^A I^B$
C.	$I^O I^O$	$I^A I^B$
D.	$I^B I^O$	$I^A I^O$

147. In which form is carbon dioxide mainly transported in blood?

- A. as carbamino-hemoglobin
 C. as hydrogen carbonate
 B. as carbonic acid
 D. in solution

148. Which cell component is absent from all prokaryotic cells?

- A. cell surface membrane
 C. endoplasmic reticulum
 B. cell wall
 D. flagellum

149. Which feature prevents xylem vessels from collapsing?

- A. continuous lumen
 C. narrow lumen
 B. lignified wall
 D. pitted wall

150. What causes an increase in the heart rate during exercise, excitement or stress?

- A. accumulation of lactate in the muscle
 B. decreased concentration of oxygen in the blood stream
 C. hormonal action on receptors in the heart
 D. increased concentration of glucose in the blood stream

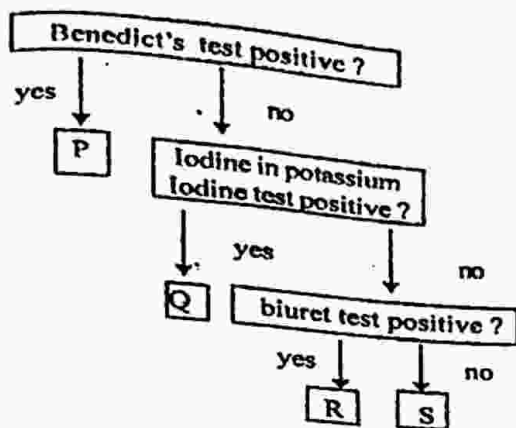
151. What limits the number of trophic levels in a food chain?

- A. biomass of the autotrophs
 B. efficiency of energy conversion between levels
 C. net productivity of the ecosystem
 D. species diversity in the ecosystem

152. For what process is the large surface area of the cristae in the mitochondria important?

- A. Energy radiation
 C. Gaseous exchange
 B. Enzyme reaction
 D. Protein synthesis

153. Various steps are identified using the following procedure



What could the four substances be ?

	P	Q	R	S
A	Glucose	Starch	Protein	Lipid
B	Glucose	Sucrose	Starch	Protein
C	Sucrose	Protein	Lipid	Starch
D	Sucrose	Starch	Lipid	protein

154. Which part of the lipid molecule contributes most to the thickness of a surface membrane

- A. glycerol B. hydrocarbon chain
C. hydrophilic head D. phosphate group

155. Food tests are carried out on four solutions. Which solution contains only sucrose and protein

solution	Benedict's test	Acid Hydrolysis Benedict's test	Iodine in KI Solution
A	-ve	-ve	+ve
B	-ve	-ve	-ve
C	+ve	+ve	-ve
D	-ve	+ve	+ve

156. Which of the following defines an ecological niche

- A. the habitat in which an organism finds its food supply
B. the habitat in which an organism finds the most suitable climate
C. the relationships between an organism and other species
D. the way in which the environment is exploited by an organism

157. Which region of the kidney nephron is the main site of glucose reabsorption

- A. Bowman's capsule
B. Proximal convoluted tubule
C. Distal convoluted tubule
D. Glomerulus

Below are names of groups used when classifying organisms.

- I. phylum II. kingdom III. class
IV. family V. species VI. genus
VII. order

Use it to answer question 158

158. The correct order when grouping organism is

- A. I, II, III, IV, V, VI and VII
B. I, II, VI, V and VII
C. II, I, III, VII, IV, VI and V
D. II, IV, I, VI, VII, II and V

159. Elephantiasis is a disease caused by

- A. filarial worms B. hookworms
C. flukes D. tapeworms

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160. One of these is not found in urine

- A. Water B. Sodium chloride
C. Nitrogenous compounds D. Calcium chloride
E. Nitrogenous salts

161. An organism with a pair of indistinguishable genes is a

- A. heterozygote B. hybrid
C. allelomorph D. homozygote
E. diploid

162. The fruit formed from a single flower having several free carpels is called

- A. multiple fruit B. simple fruit
C. aggregate fruit D. dehiscent fruit
E. indehiscent fruit

163. The function of ossicles (maleus, incus and stapes) in the mammalian ear to

- A. transmit vibrations
B. regulate pressures
C. support of the inner ear
D. maintain balance during motion
E. secrete oil

164. "Jointed skeleton" is absent in the

- A. Cockroach B. Spider
C. Millipede D. Snail
E. House fly

165. Mucor and Spirogyra can be put in a group because they

- A. are unicellular
B. have spores that are dispersed by wind
C. can live independent lives
D. reproduce sexually
E. have bodies made up of thallus and filaments alternatively

166. The organ through which nourishment and oxygen diffuse into an embryo called

- A. Amnion B. Chorion

C. Umbilical cord
E. Placenta

D. Oviduct

167. A tapeworm fastens itself to the intestine of its host with

- A. Neck & sucker
- B. Hooks & suckers
- C. Rostellum & suckers
- D. Proglottids & neck
- E. Rostellum, hooks & suckers

168. Which of these is false about the piliferous layer of a root? It

- A. has a thin cuticle
- B. is the outermost layer of the cortex
- C. may bear root hairs
- D. breaks down with age
- E. is replaced by cork in old roots

169. Anaerobic respiration in yeast produces

- A. CO₂ & ethanol
- B. CO₂ & H₂O
- C. CO₂ & O₂
- D. CO₂ & glucose
- E. Ethanol & H₂O

170. Which of these set of factors is completely abiotic?

- A. turbidity, tide, salinity, plankton
- B. pressure, pH, soil, insect
- C. water, soil, bacteria, salinity
- D. pH, bamboos, wind rainfall,
- E. light, altitude, wind, humidity

171. In which of these are flagella and cilia found?

- A. flatworms
- B. protozoa
- C. coelenterates
- D. annelids
- E. nematodes

172. The three important organs that are situated close to the stomach are

- A. liver, kidney & gall bladder
- B. pancreas, liver & kidney
- C. gall bladder, pancreas & spleen
- D. liver, kidney & spleen
- E. kidney, gall bladder, liver

173. The plantain reproduces asexually by

- A. spores
- B. buds
- C. fragments
- D. sucker
- E. flowers

174. The major function of swim-bladder in fish is

- A. Breathing
- B. swimming
- C. diving
- D. repelling enemy
- E. buoyancy

175. The part of the central nervous system concerned with answering an examination question is the

- A. spinal cord
- B. cerebellum
- C. oryx
- D. cerebrum
- E. medulla oblongata

176. Wind pollinated flowers usually have

- A. long styles

- B. sticky stigmas
- C. small and short stigmas
- D. rough pollen grains
- E. short styles and pollen

177. The radicle of a bean seedling grown most rapidly in the region

- A. of the root tip
- B. below the top soil
- C. just around the root tip
- D. just below the root tip
- E. just above the root tip

178. A key similarity between nervous and hormonal system is that both

- A. involve chemical transmission
- B. have widespread effect
- C. shed chemicals into the blood stream
- D. evoke rapid response
- E. eliminate response

179. The bone of the neck on which the skull rest is

- A. odontoid
- B. axis
- C. occipital
- D. atlas
- E. patella

180. A child with blood group genotype different from those of both parents and with a mother of genotype OO, can only have a father of genotype

- A. A
- B. B
- C. OO
- D. AB
- E. AA

181. A true climax community

- A. changes from year to year
- B. persists until the environment changes
- C. is the first stage in plant succession
- D. consists of latest trees and small animals
- E. is in a state of perurbation

182. In a predator food chain involving secondary and tertiary consumers, the organisms become progressively

- A. smaller
- B. equal in number
- C. larger and fewer along the food chain
- D. parasitized along the food chain as consumers get bigger
- E. sparse in distribution

183. Which of these is an adaptation to a xerophytic environment?

- A. fleshy tissue with reduced leaves
- B. extensive surface roots & broad leaves
- C. thick barks and broad leaves
- D. rough leaves and shallow root system
- E. stunted growth and surface roots

184. When it is cold, the blood vessels of the skin

- A. dilate to increase blood flow to the skin
- B. constrict to reduce the amount of blood flowing to the skin

- C. dilate to reduce the amount of blood flowing to the skin
D. constrict to increase the amount of blood flowing to the skin

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185. The table shows three processes that contribute to transport across cell surface membranes which processes are active and which are passive?

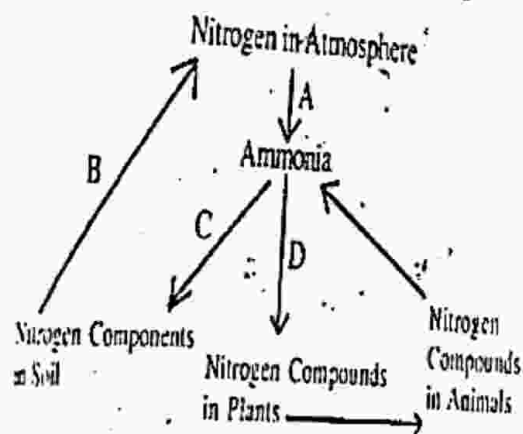
	Diffusion	Endocytosis	Osmosis
A	a	a	a
B	p	a	a
C	p	a	p
D	p	p	p

Key: a = active process p = passive process

186. Which statement is the best explanation of why sucrose rather than glucose is transported by phloem?

- A. Sucrose can pass through plant cell surface membranes more easily
B. Sucrose is a disaccharide and is more easily converted to starch
C. Sucrose is a non-reducing sugar, so is less reactive
D. Sucrose synthesis requires less energy

187. The diagram represent part of the nitrogen cycle. Which process is carried out by nitrifying bacteria?



188. From which cell organelle are nucleic acid absent?

- A. Chloroplast B. Golgi body
C. Mitochondrion D. Ribosome

189. What will break an ionic bond between amino acids?

- A. Concussions B. High temperature
C. Hydrolysis D. pH change

190. The drawings show stages of the mitotic cell cycle



In which order do the stages occur?

	First				Last
A	2	1	3	5	4
B	2	4	1	5	3
C	4	2	1	3	3
D	4	2	3	1	5

191. The first in milk produced by the mother for a new-born baby contains antibodies. What do these antibodies provide?

- A. Artificial active immunity
B. Artificial passive immunity
C. Natural active immunity
D. Natural passive immunity

192. Which protein has a fibrous structure?

- A. Amylase B. Collagen
C. Haemoglobin D. Insulin

193. In which form is carbon dioxide mainly transported in blood?

- A. As carbamino-haemoglobin B. As carbonic acid
C. As hydrogen carbonate D. In solution

Below are names of groups used when classifying organisms.

- I. phylum II. kingdom III. class IV. family
V. species VI. genus VII. order

Use it to answer question 194

194. The correct order when grouping organism is

- A. I, II, III, IV, V, VI and VII
B. I, II, VI, V and VII
C. II, I, III, VII, IV, VI and V
D. II, IV, I, VI, VII, II and V

195. During sexual reproduction, the macronucleus in paramecium

- A. divide into 2 parts B. divide into 4 parts
C. breaks into 4 parts D. fuses with the micronucleus

196. Which of the following tissues represents supporting tissue?

- A. columnar epithelium B. cartilage
C. striated muscle D. nervous tissue

197. The creamy semi-fluid of food in the stomach after some hours known as

A. chyle
B. chyme

D. Choline

108. Which of the following structures is not in the male reproductive parts?

A. Uterine tube
C. Epididymis

B. Prostate gland
D. Seminal vesicle

109. The organelle in Amoeba which performs a similar function with that of the kidney in vertebrates is the

A. nucleus
C. mitochondrion

B. contractile vacuole
D. pseudopodium

200. Steady spread of the unpleasant odour from dead rat is a physical process called

A. cyclosis
C. osmosis

B. diffusion
D. plasmolysis

201. De-oxygenated blood from the head region first enters the heart through the

A. right auricle
C. left auricle

B. right ventricle
D. left ventricle

202. Which of the following is not a characteristic of enzymes?

A. they are protein which are activated by co-enzymes
B. they are specific and can act only on specific substrates
C. They are organic
D. Their reactions are irreversible

203. The best way of protecting natural resources is by

A. creating game and forest reserves
B. enforcing conservation laws
C. embarking on tree planting
D. prohibiting lumbering forest reserves

204. One basic characteristic feature of viruses is that they grow

A. as colonies
C. inside living cells

B. as jelly-like materials
D. as plant

205. Which feature is a characteristic of prokaryotic organism?

A. A cell
C. Mitochondria

B. Circular DNA
D. Rough endoplasmic reticulum

206. What is meant by resolution in light microscopy?

A. the product of the magnifications of the eyepiece and the objective lenses
B. the shorter distance between objects that can be seen as separate
C. the size of the smaller object that can be seen
D. twice the wavelength of the light used to illuminate the specimen

207. From which organelle are nucleic acids absent?

A. Chloroplast
C. Mitochondria

B. Golgi body
D. Ribosome

208. For what process is the large surface area of the cristae in the mitochondria important?

A. Energy radiation

B. Enzyme reaction
C. Gaseous exchange
D. Protein synthesis

209. Which level of protein structure maintains the globular shapes of enzymes?

A. primary
C. tertiary

B. secondary
D. quaternary

210. What bonds hold substrate to the active site of an enzyme?

A. disulphide
C. hydrogen

B. glycoside
D. peptide

211. Which of the following set of plants has a taproot system?

A. maize, bamboo, waterleaf
B. waterleaf, balsam, wheat
C. waterleaf, cowpea, balsam
D. wheat, rice and cowpea

212. Which adaptation would increase the efficiency of active transport of carbohydrates from a plant cell?

A. area where the cell wall is thin
B. increase permeability of the cell wall
C. large surface area of the cell surface membrane
D. large volume of the cell vacuole

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213. The nutritive layer of the eye in mammals is

A. choroid
C. cornea

B. conjunctiva
D. sclera

214. Ultrafiltration in the kidney takes place in the

A. Bowman's capsule
C. Loop of Henle

B. Pelvis
D. Proximal convoluted tubule

215. Which of the following bones is not a component of the forelimb?

A. olecranon
C. tibia

B. ulna
D. humerus

216. The condition in which the anther matures before the stigma is called

A. protandry
C. hypogyny

B. epigyny
D. protogyny

217. In most true ferns, sporangia are grouped into

A. indusium
C. fronds

B. sori
D. prothalli

218. The ratio of carriers to sicklers in the F₂ generation derived from a parental cross of two carriers of haemoglobin S gene is

A. 3:1
C. 2:1

B. 1:3
D. 1:2

219. In which part of a leguminous plant can bacteria like Azotobacter be found?

- A. spongy mesophyll
- B. root nodules
- C. stem internodes
- D. stem nodes

220. In a dicotyledonous stem, companion cells are found close to the

- A. endodermal cell
- B. sieve tubes
- C. xylem vessels
- D. pericyclic fibres

221. The position occupied by an organism in a food chain is referred to as

- A. trophic level
- B. niche level
- C. energy level
- D. feed level

222. The ventricles of the mammalian heart are more muscular than the auricles because the

- A. auricles have smaller capacity
- B. ventricles are larger in size
- C. ventricles pump blood to distant organs
- D. ventricles receive more blood

223. The young shoot of a plant is referred to as

- A. Radicle
- B. Plumule
- C. Bud
- D. branch

35. A	37. C	38. D	39. D	40. C
41. C	42. D	43. D	44. B	45. A
46. C	47. B	48. D	49. D	50. C
51. A	52. B	53. A	54. D	55. B
56. D	57. D	58. A	59. B	60. B
61. B	62. B	63. B	64. C	65. A
66. C	67. A	68. A	69. D	70. A
71. A	72. C	73. B	74. C	75. B

CHAPTER 26

1. D	2. C	3. B	4. C	5. D
6. A	7. A	8. D	9. B	10. B
11. D	12. A	13. D	14. B	15. B
16. B	17. B	18. D	19. D	20. D
21. C	22. A	23. A	24. C	25. C
26. A	27. D	28. D	29. A	30. B
31. D	32. D	33. C	34. A	35. C
36. D	37. C	38. A	39. A	40. B
41. A	42. D	43. B	44. D	45. B
46. C	47. B			

ANSWERS TO POST UME QUESTIONS

1. B	2. D	3. A	4. B	5. B
6. D	7. C	8. B	9. C	10. D
11. C	12. D	13. D	14. D	15. C
16. B	17. A	18. B	19. B	20. C
21. A	22. D	23. A	24. A	25. A
26. B	27. B	28. D	29. A	30. A
31. D	32. A	33. C	34. C	35. A
36. C	37. A	38. C	39. C	40. C
41. C	42. C	43. C	44. B	45. B
46. B	47. B	48. B	49. B	50. D
51. B	52. D	53. A	54. B	55. A
56. A	57. B	58. B	59. D	60. C
61. C	62. B	63. C	64. A	65. A
66. C	67. A	68. D	69. C	70. B
71. D	72. D	73. B	74. B	75. B
76. B	77. A	78. D	79. A	80. A
81. A	82. B	83. B	84. A	85. C
86. D	87. C	88. C	89. D	90. D
91. C	92. C	93. D	94. B	95. A
96. C	97. C	98. B	99. B	100. B
101. B	102. C	103. D	104. B	105. C
106. B	107. C	108. A	109. B	110. B
111. C	112. D	113. C	114. B	115. D
116. D	117. B	118. C	119. D	120. D
121. B	122. D	123. C	124. D	125. D
126. C	127. D	128. D	129. D	130. D
131. C	132. A	133. A	134. B	135. A
136. C	137. C	138. B	139. A	140. D
141. C	142. C	143. B	144. D	145. B

146. A	147. C	148. C	149. B	150. C
151. B	152. B	153. A	154. D	155. D
156. A	157. B	158. C	159. A	160. E
161. D	162. C	163. A	164. D	165. D
166. E	167. E	168. A	169. A	170. E
171. B	172. C	173. D	174. E	175. D
176. A	177. A	178. A	179. D	180. D
181. B	182. C	183. A	184. B	185. D
186. C	187. C	188. B	189. C	190. D
191. D	192. B	193. C	194. C	195. A
196. B	197. B	198. A	199. B	200. B
201. A	202. A	203. B	204. C	205. B
206. B	207. B	208. B	209. D	210. C
211. C	212. C	213. A	214. A	215. C
216. A	217. B	218. C	219. B	220. B
221. A	222. C	223. B		

CHAPTER ONE

177. C
182. A
187. C
192. B
197. B
202. A
207. B
212. C
217. B
222. C
178. A
183. B
188. A
193. B
198. C
203. A
208. B
213. A
218. C
223. B
179. B
184. D
189. E
194. B
199. C
204. B
209. D
214. A
219. B
224. B

9. All living organisms require an environment in which to live. They obtain energy from the environment and use it to grow and reproduce. They also use energy to perform other functions such as movement, photosynthesis, and respiration. Only plants and animals are able to perform these functions.
10. A non-living object like a crystal can also grow in size. However, in non-living things, growth is external because it takes place due to the deposition (addition) of more particles on the outer surface of the crystal. Growth in living organisms is internal as a result of the addition of new protoplasm.
11. Onion bulb (plant cell) possesses cellulose cell wall which is absent in cheek cell (animal cell).
12. Golgi complex (or Golgi apparatus) modifies proteins and also transports organic materials in and out of the cell.
13. The nucleus contains thread-like structures called chromosomes. They carry the genes. The genes are made up of chromatin threads which join to form chromatids. A chromosome consists of two strands of chromatids.
14. A matured plant cell has a cellulose cell wall, hence has a rigid shape.
15. The reservoir of energy for the cell is carried by the mitochondria, not the nucleus.
16. Centriole is peculiar to animal cells while middle lamella, starch grains and cellulose cell wall are peculiar to plant cells.
17. B = Endoplasmic reticulum transports substances within the cytoplasm and the nucleus.
18. B = 1-cell membrane; 2 = rough endoplasmic reticulum; 3 = nucleolus; 4 = mitochondria; 5 = Golgi complex; 6 = nuclear membrane.
19. C = The nucleolus is a dark area in the nucleus where ribosomal RNA is being made.
20. C = The mitochondria is also known as the powerhouse of the cell because they generate most of the cell's supply of ATP use as a source of chemical energy.
21. C = The chromosome contains genes, which are the units of inheritance.
22. B = Cell vacuole in plant cell is surrounded by the tonoplast.
23. C = Cell wall and chloroplast are peculiar to plant cells.
24. B = Ribosome is responsible for protein synthesis.
25. A = see 16
26. A = Cytoplasm in unicellular organisms e.g. Amoeba is differentiated into ectoplasm and endoplasm just as the protoplasm is differentiated into nucleus and cytoplasm.
27. A = The role of lysosomes enables it to detoxify the cell of harmful substances. They are also called the

CHAPTER TWO

1. A = Prokaryotic cells lack a defined nucleus; B = membrane - bounded organelles but they possess complex cytoplasm.
2. A = The pioneer organism in a plant succession followed by foliose lichen and then mosses.
3. D = Kingdom Phylum Class Order Genus
4. As we go down the levels, the variety of life decreases and similarity of organisms increases.
5. B = Kingdom - monera - have - prokaryotic bacterial and blue green algae. Their nuclei e.g. chromosomes are found scattered cytoplasm as they lack a nuclear membrane.
6. B = Blue - green algae are prokaryotic
24. E = The cell membrane is made up of proteins and lipids in the ratio of 1:59. It is also called the plasma membrane or plasmalemma.
25. C = Cell Tissue Organ System Multicellular organism.
26. C = Aggregates of cells form tissue; aggregates of tissues form organ; aggregates of organs form system. Well coordinated systems and organs form an organism.
27. C = An amoeba is a single cell. An unbird chicken egg is also a single cell.
28. B = Rods and cones are cells present in the retina - tissue in the eye.
29. B = Onion bulb is an organ - aggregates of epithelial tissues.
30. B = Tissue are aggregates of cells performing one or more functions.
31. C = Kingdom protista contains protists (Protozoa e. Amoeba and protophyta e.g. Chlamydomonas) a eukaryotic unicellular organisms. They have specialized tissues and organs.
32. B = A virus possesses strand of nucleic acid which may be DNA or RNA.
33. A = Presence of DNA or RNA in viruses is responsible for the transmission of characters.
34. B = Mitochondria and chloroplast are responsible for cellular respiration and photosynthesis respectively.
35. A = The nucleus is a central organelle because it contains genetic materials used in controlling or activities in organism.
36. D = Water is a medium of all metabolic reactions in the body of living organisms.
37. A = Mitochondria with mitochondria in cells that require more energy for its activities. Sperm cells contain many mitochondria to provide energy for their propulsive movement.
38. D = Ribosomes are attached to the walls of the endoplasmic reticulum.
39. B = All cells have plasmalemma (cell membrane).
40. A = A bacterial cell (a prokaryotic cell) has no nuclear membrane - bounded organelle hence its cytoplasm is continuous with the nucleus.
41. B = Non-living things have relatively low content compared to living things.

monera, protophytes are plant-like protists e.g. Chlamydomonas- kingdom protista. All protists are eukaryotes and not prokaryotes.

7. D = Bacteria are prokaryotes. All prokaryotes are non-nucleated.
8. B = Bacteria will not survive in salt water.
9. C = Eukaryotic cells walls are usually made of cellulose or chitin (fungi). In contrast, most bacterial cell walls contain peptidoglycan a network of modified sugar polymers.
10. E = All of the above
11. D = Malaria is caused by a plasmodium, a protozoan. Pneumonia can be caused by different types of microorganisms, including bacteria, viruses and fungi.
12. B = Putrefying bacteria are usually anaerobic. They decompose animal proteins resulting in amines such as putrescine and cadaverine which have a putrid odour. They cause food poisoning.
13. D = Bacteria toxin are the most powerful human poisons known and retain high activity at very high dilutions.
14. D = Immunity is the state of having sufficient biological defenses to avoid infection, disease or other unwanted biological invasion. The case in the question gives an acquired immunity.
15. C = Pellicle is an animal-like feature in euglena. Although the possession of flagellum is also animal-like, it is present in some plants like Chlamydomonas.
16. D = The contractile vacuole is for osmo-regulation.
17. D = Plant characteristics of Euglena include: (i) chloroplast, therefore, nutrition is holophytic (autotrophic); can be holozoic in the absence of light (ii) store food as starch-like carbohydrate, paramylon (iii) Possesses pyrenoid, an organelle found in many algae, which functions in the synthesis of paramylon.
18. C = Spirogyra is non-motile, Euglena moves by means by cilia and chlamydomonas moves by means of flagella.
19. C = The mushroom bears a microscopic "fruiting body" which consist of a foot (hyphae) which attaches it to the substrate, a stalk or stipe which bears an umbrella shaped portion at its end known as pileus or cap.
20. B = Lichen is formed by the combination of algae and fungi; the algal part is found in between two fungi parts; while the alga produces food by photosynthesis, the fungus protects the alga and also provides water for photosynthesis, hence, the two are in a symbiotic relationship.
21. C = Spore production and dispersion occur in all the classes of organisms mentioned.
22. E = Fungi lack chlorophyll; hence, they cannot photosynthesize. Conjugation occurs in algae and fungi only while mosses, ferns and some algae show alternation of generation. All of them reproduce by forming spores and not seeds.
23. B = Fungi are multinucleated and multicellular with exception of yeasts, which is unicellular.
24. B
25. C
26. D = Evolutionary trend is as shown: Schizophyta

(Bacteria) → Thallophyta (Algae) → Bryophyta (Mosses and Liverworts) → Pteridophyta (Ferns) → Spermatophyta (seed plants). Spermatophyta is further divided into Gymnosperms (conifers) and Angiosperms (Flowering plants).
Memory Tip: Samuel The Biologist Planted Sugarcane.

27. D = Bryophytes and Pteridophytes inhabit moist terrestrial habitat (require moisture for fertilization). Spermatophytes (Gymnosperms and Angiosperms) live exclusively on dry land.
28. C
29. E
30. D = Thallophytes (Algae) and Bryophyte (Mosses and Liverworts) are non-vascular.
31. B = The dominant phase in the life cycle of a bryophyte is the sporophyte.
32. C
33. C = Only Pteridophytes and Spermatophytes have their vegetative bodies differentiated into true roots, stems and leaves.
34. C
35. A
36. B = Sporophyte is the dominant phase in the life cycle of mosses and ferns, that show alternation of generation
37. B = Whistling pine is a gymnosperm and have cones, which are naked seeds.
38. D
39. B
40. B
41. C
42. A = Double fertilization is peculiar to flowering plants leading to the formation of fruit and seed.
43. D
44. D
45. D = This is because insects can live in almost all habitats and can survive unfavourable conditions and their mode of reproduction, undergoing metamorphosis makes them very difficult to eradicate.
46. B = An animal is said to be radially symmetrical when it can be divided into two equal halves along any plane e.g. hydra.
47. B = The bodies of coelenterates are composed of two clearly marked layers of cells, an outer ectoderm and a inner endoderm i.e. they are diploblastic animals.
48. B = Coelenterates have a cavity within their body, the enterogastrocavity with a single opening to the outside (mouth) for ingestion and egestion.
49. B
50. D = Coelenterates reproduce asexually by budding
51. B
52. B = The cnidoblast is an ectodermal cell in which there is a large sac, the nematocysts, that are used for offense and defense.
53. D = Coelenterates are pseudocoelomates; they lack true coelom or body cavity called enteron.
54. B
55. E
56. A = The hypostome is the most apical structure, the domed portion of head where the mouth opens.
57. E = The testes are usually located below the tentacles

58. B = above buds.
59. A = The mouth is usually located below the bud i.e. a little distance above the basal foot on foot.
60. A = ectodermal cells derived from the mesodermal epithelial cells.
61. C = Interstitial cells form the reproductive organs and buds.
62. A.
63. C = Excess water enters the enteron (body cavity) which is then removed via the mouth.
64. A = An example of symbiotic association.
65. C = There are 8 forms of movement in hydra namely: gliding, floating, swimming, walking, climbing, dragging, looping (forming 180 degree angle plane, most common type) and somersaulting.
66. E = Hydra is an hermaphrodite but prevent self-fertilization by protandry i.e. testis and sperms mature first before the ovary.
67. D.
68. C = Coelenterates and platyhelminthes.
69. C = Annelids are metamerically segmented while nematodes are not.
70. C = As a result of their metameric segmentation.
71. D = Cocoon in earthworm is secreted by the clitellum. The clitellum is made up of about five or six glandular segments. Eggs are laid in the cocoon.
72. D = The soil is ground up in the gizzard.
73. D.
74. C = Phylum Arthropoda is divided into four subphyla namely: Crustacea (e.g. crab), Insecta (e.g. cockroach), Arachnida (e.g. spider) and Myriapoda (e.g. millipedes and centipedes) **Memory tip: CIAM.**
75. B.
76. B = Snails have tentacles, earthworms have cheateae and millipedes have antennae.
77. D = Arthropods have a segmented body, exoskeleton made of chitin.
78. C = The thoracic section of the cephalothorax in crayfish has eight pairs of jointed appendages. The first three pairs are maxillipedes or "jaw-feet". They help to hold food material during feeding.
79. A = On the head of insects are a pair of antennae (not two pairs) and a pair of compound eyes.
80. C = The characteristics described are all true except in C. Each thoracic segment bears a pair of joints legs. The three NOT two.
81. D.
82. A.
83. C = Pedipalp, also known as feeler, is used for feeling.
- NOTE:** Spiders have spinnerettes on the abdomen from which the silk used in web-making comes. Not all spiders make webs, some hunt their prey.
84. C = Arachnids have no antennae.
85. D = A centipede has poison claws which it use to hunt its prey while millipedes feed on dead and decaying organic matter. A centipede possesses a pair of legs at each abdominal segment while millipede possesses two pairs of legs at each abdominal segment.
86. C = Insects have two compound eyes while millipedes have two simple eyes (ocelli).
87. B = Snail body is covered with hard calcareous (not

phosphorus) shell secreted by the mantle into which the head and foot retracts.

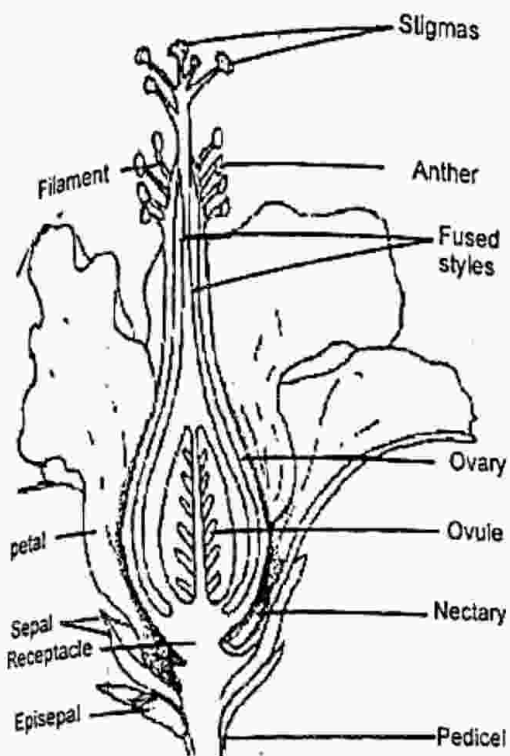
88. C = Mollusca exist as free living organisms. Although some serve as intermediate host of parasitic worms.
89. D.
90. B = respiration in mollusca is through gills or a lung which is within the mantle cavity. The upper surface of the mantle cavity is vascularized in land snail, which used muscular contractions to pump air in and out of the small respiratory pore at the anterior edge of the mantle cavity.
91. B.
92. B.
93. A = Viviparity is when fertilized eggs are produced, developed and nourished inside the body of the mother through placenta. Examples are all placenta mammals.
- Oviparity is the laying of eggs in which the embryos hatch outside the body of the parents. Examples are amphibians, most fish (e.g. tilapia) and birds (e.g. hen, pigen duck).
- Ovoviviparity is when fertilized eggs are produced, developed and nourished inside the body of the mother without placenta and hatch before or soon after eggs are laid. Examples are certain fishes and reptiles.
94. B = The first terrestrial vertebrates are reptiles; they evolved from amphibians (inhabit land and water).
95. D.
96. D.
97. A.
98. B.
99. B = Volvox exists as a colony.
100. D = Hyphal wall of fungi are made of the chitin.
101. B.
102. D.
103. B.
104. B.
105. C.
106. B = Aves (birds) are completely oviparous.
107. C.
108. C.
109. A.
110. C.
111. C.
112. B = Obelia is a coelenterate.
113. C = Metameric segment.
114. B.
115. B.
116. B.
117. C.
118. A.

CHAPTER THREE

1. D = Spirogyra is a filamentous algae, the cells are joined end to end. Each cell can carry out nutrition, respiration, growth and reproduction independently.
2. C = Spirogyra have unbranched filaments (straight filament)
3. E = Pyrenoids are associated with the formation of starch.
4. B = Gametes are dissimilar usually "male" and "female" gametes.
5. B = Some species of spirogyra have rhizoids which they use to attach to surfaces.
6. D = Spirogyra cells is coated with a mucilage layer and gives a slippery feeling.
7. A
8. E = Autotrophic (holophytic) because of the presence chlorophyll.
9. A
10. D
11. C
12. C
13. B = The vegetative body of fungi is called a mycelium.
14. C = The rhizoid is the root-like structure of the mucus and rhizopus. It is formed from the hyphae.
15. C = The cell wall of fungi is not made of the normal cellulose found in other plants but chitin and a special type of cellulose called fungal cellulose.
16. B = Fusion takes place between two gametangia, which are multinucleated cells formed from branched hyphae of each.
17. B = Structurally, there is no difference. Sporangophore is a vertically growing reproductive hypha, which grows away from the substrate.
18. A = Fungi e.g. *Mucor*, *Rhizopus* and yeast are non-green. They lack chlorophyll.
19. B = In some fungi, the filaments are divided into cells by cross walls (septate) e.g. *Penicillium*, but in others e.g. *Rhizopus* the cross walls are absent (non-septate) or coenocytic.
20. B
21. C
22. C = Carbohydrate is absorbed as glucose as in other heterotrophic organisms.
23. C = In fungi, carbohydrate food reserve is stored in the form of glycogen and not starch.
24. D = Rhizoid perform similar functions as roots of plants absorption.
25. A = In bryophytes, the male and female sex organs, the antheridia and archegonia respectively are produced in the gametophyte.
26. C
27. A = Asexual stage is the sporophyte generation while the sexual stage is the gametophyte generation.
28. A = Male gametangium (antheridium); female gametangium (archegonium). This is also true for fern.
29. D = Gametophyte stage: prothallus antheridium (spermatozoid) and archegonium (egg cell). Sporophyte stage: Fern plant *Sporangium* spore
30. D = An inflorescence is a reproductive structure found in flowering plants (angiosperms) consisting of many flowers while strobilus is the reproductive structure of gymnosperms consisting of a number of sporophylls.
31. D = The sporophyte on ferns is the visible fern plant; it is diploid and independent of the gametophyte.
32. B = I - Archegonium (female sex organ); II - Antheridium (male sex organ) III - Rhizoids.
33. B = The structure contains both male and female gamete. It is therefore equivalent to the gametophyte generation of moss.
34. C = The sporophyte is the dominant plant. Other descriptions are true of the prothallus.
35. D = Sporophyte = diploid (asexual reproduction). Gametophyte = haploid (sexual reproduction).
36. D = Antherozoid + archegonium = Zygote (inside haploid archegonium).
37. B = Each spore germinates into a bisexual prothallus - gametophyte generation.
38. B = Dryopteris is a fern.
39. B = Many sporangium are organized into a sorus with or without a cover or indusium.
40. B = Spore dispersal is effected by means of the annulus and stomium under the control of changes in humidity. The hygroscopic annulus absorbs water from the air when the humidity is high.
41. C
42. B = Each sporangium is made up of a head and a stalk. The head is ovoid with a wall made up of thick-walled cells called annulus. A few cells of the wall however remain thin-walled forming the stomium.
43. B
44. E
45. C
46. D
47. A = Duration of sunlight determines the flowering period of a plant.
48. D = Axil is the angle between the leaf III and the stem in which the axillary bud II borne. Node is where leaves are attached. Internodes is the part between two successive nodes.
49. D = Photosynthesis occurs in the mesophyll cells of the leaf chlorophyllous parts in the shoot system. Photosynthesis is least likely to occur in the root (IV).
50. C = The characteristics combined in C are true for a monocot. They also have fibrous root system.
51. D = Root hairs develop in the region of differentiation (region of maturation), hence the region or zone of differentiation is also called the root hair region.
52. B
53. D = Cassava has tuberous root where food is stored.
54. D = White mangrove have breathing roots (pneumatophores) which allow the intake of atmospheric air.
55. C = Valamen is a spongy, multiple epidermis that covers the roots of some epiphytes. It is many cell layers thick and capable of absorbing atmospheric moisture and nutrients.
56. C = No secondary thickening in monocot because of the absence of cambium.
57. C = Medullary rays lie between two adjacent vascular bundles
58. D = Collenchyma cell walls are thickened at the corner while parenchyma cells consists of loosely packed, with air spaces between

Sexual Reproduction in Flowering Plants

Flower structure



Structure of an hibiscus flower

Sexual reproduction in flowering plants centres around the flower. Within a flower, there are usually structures that produce both male gametes and female gametes.

Sepal: protect the flower bud.

Petal: brightly coloured and scented and may have nectaries which are all used to attract insects, petals in wind pollinated flowers are tiny, and used for pushing the bracts (leaf-like structures) apart to expose stamens and stigma.

Anther: has pollen sacs with pollen grains which contain the male nucleus (male gamete).

Stigma: platform on which pollen grains land.

Ovary: hollow chamber, ovules grow from the walls.

Pollen tube: pollen grain lands on stigma and creates a tunnel down the style, through the micropyle, to the ovules.

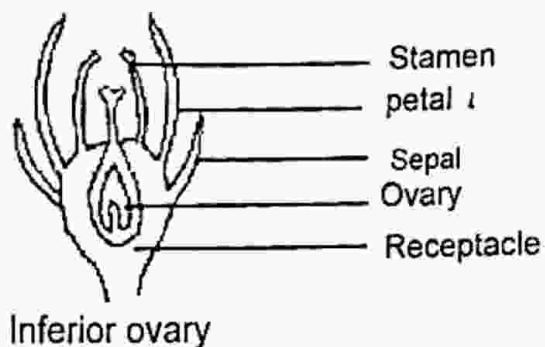
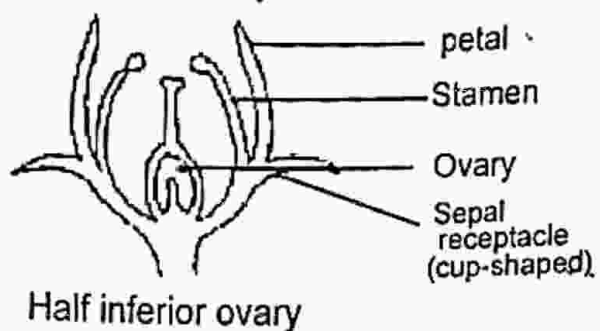
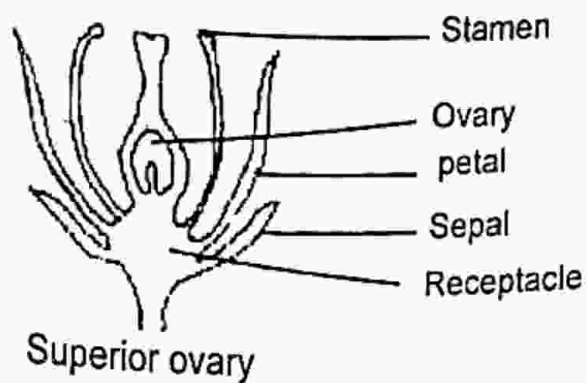
Structure of non-endospermic seed:

Development of the ovule and female gamete

Inside the ovary there may develop one or more ovules. Each ovule begins life as a small projection into the cavity of the ovary. As it grows and develops it begins to bend but remains attached to the ovary wall by a placenta.

At the start, the ovule is a group of similar cells called the nucellus. As it develops, the mass of cells differentiates to form an inner and an outer **integument**, surrounding and protecting the nucellus within, but leaving a small opening called the **micropyle**.

At the centre of the ovule is an **embryo sac** containing the haploid egg cell (the female gamete).

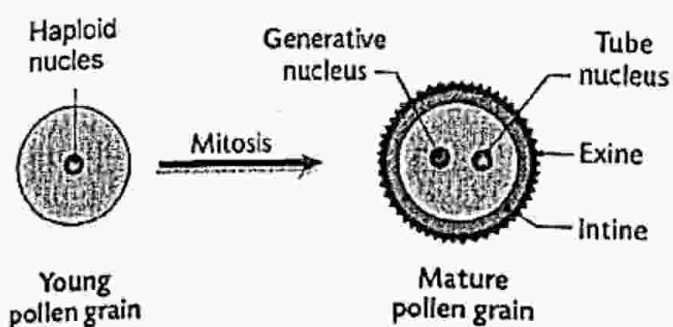


Types of ovary

Development of the male gamete

Each anther contains 4 pollen sacs. Many pollen grains develop inside each pollen sac. It begins with a mass of large pollen mother cells in each pollen sac. All are diploid.

In each pollen grain the wall thickens and forms an inner layer (the **intine**) and an often highly sculptured outer layer (the **exine**). The surface pattern is different on pollen grains from different species. When the pollen grains are mature, the anther dries out and splits open (a process called **dehiscence**) and the pollen is released.



ANSWERS TO POST TIME QUESTIONS

1. B	2. D	3. A	4. B	5. B
6. D	7. C	8. B	9. C	10. D
11. C	12. D	13. D	14. D	15. C
16. B	17. A	18. B	19. B	20. C
21. A	22. D	23. A	24. A	25. A
26. B	27. B	28. D	29. A	30. A
31. D	32. A	33. C	34. C	35. A
36. C	37. A	38. C	39. C	40. C
41. C	42. C	43. C	44. B	45. B
46. B	47. B	48. B	49. B	50. D
51. B	52. D	53. A	54. B	55. A
56. A	57. B	58. B	59. D	60. C
61. C	62. B	63. C	64. A	65. A
66. C	67. A	68. D	69. C	70. B
71. D	72. D	73. B	74. B	75. B
76. B	77. A	78. D	79. A	80. A
81. A	82. B	83. B	84. A	85. C
86. D	87. C	88. C	89. D	90. D
91. C	92. C	93. D	94. B	95. A
96. C	97. C	98. B	99. B	100. B
101. B	102. C	103. D	104. B	105. C
106. B	107. C	108. A	109. B	110. B
111. C	112. D	113. C	114. B	115. D
116. D	117. B	118. C	119. D	120. D
121. B	122. D	123. C	124. D	125. D
126. C	127. D	128. D	129. D	130. D
131. C	132. A	133. A	134. B	135. A
136. C	137. C	138. B	139. A	140. D
141. C	142. C	143. B	144. D	145. B

10. C
15. A
10. C
5. B
0. B
5. A
0. A
1. B

146. A	147. C	148. C	149. B	150. C
151. B	152. B	153. A	154. D	155. D
156. A	157. B	158. C	159. A	160. E
161. D	162. C	163. A	164. D	165. D
166. E	167. E	168. A	169. A	170. E
171. B	172. C	173. D	174. E	175. D
176. A	177. A	178. A	179. D	180. D
181. B	182. C	183. A	184. B	185. D
186. C	187. C	188. B	189. C	190. D
191. D	192. B	193. C	194. C	195. A
196. B	197. B	198. A	199. B	200. B
201. A	202. A	203. B	204. C	205. B
206. B	207. B	208. B	209. D	210. C
211. C	212. C	213. A	214. A	215. C
216. A	217. B	218. C	219. B	220. B
221. A	222. C	223. B		

them.

59. B
60. B = Periderm cells are cork cells in the stems of dicots. The periderm is produced by the phellogen (cork cambium) toward the outside of the stem.
61. A = Meristematic tissue continues to rapidly divide producing undifferentiated cells.
62. C = See page 36.
63. E = The leaves of onion are modified into a storage organ known as bulb.
64. B = Bulbs like garlic and onion store food in the form of sugar rather than as starch.
65. E = *Bryophyllum pinnatum* can be easily propagated by leaves because their leaves can develop buds and adventitious roots at the nodes along the margin. The buds develop into new plants.
66. C
67. B = Trichocyst in paramecium is for defence.
68. C = The contractile vacuole is present in both Amoeba and Paramecium.
69. D = Contractile vacuole perform a similar function to kidney, osmoregulation.
70. E = The cell membrane is also known as plasmalemma and provides a layer of protection to cell. It forms the outermost part of membrane in amoeba. Pellicle is the membrane covering that protects the paramecium like skin. It is responsible for the definite shape in paramecium.
71. A = Amoeba excrete ammonia through cell membrane (body surface) in dissolved form.
72. A = Excretion of ammonia- body surface; Osmoregulation- contractile vacuole.
73. A = Amoeba feeds predominantly on bacteria and diatom.
74. C = Contractile vacuoles are osmoregulatory in function as they are involved in the removal of excess water from the body.
75. C
76. B = Cysts formation is a means of survival under adverse conditions.
77. B
78. E = Amoebic dysentery (amoebiasis) is an infection of the intestine (gut) caused by an amoeba called *Entamoeba histolytica*, that among other things, can cause severe diarrhoea with blood.
79. C
80. C.
81. B = The surface of the paramecium is covered with cilia and contain bottle-shaped organelles, the trichocysts.
82. C
83. E = Cilia in the gullet create a swirling current of water which drives food mostly bacteria into the mouth-pore (cytostome). Food is egested through the anal-pore (cytopyge).
84. E = As in Amoeba, Paramecium removes metabolic waste by diffusion through the body surface.
85. C = The contractile vacuole primarily expels excess water absorbed by osmosis. However, little nitrogenous wastes (ammonia) accompany the water.
86. D = Man is usually the primary host of tapeworm. Pig

and cow are the secondary host.

87. B = Two species of tapeworm that commonly in man are *Taenia solium* in Pig (with hook) and *Taenia saginata* in Cow (without hook).
88. A = The excretory ducts and the nerve cords p down the entire length of the body.
89. D = I - hook, II - sucker, III - head or scolex, IV neck, V - young proglottids.
90. C = Proglottids, which are flat segments budded from the neck (IV) behind the scolex or head.
91. C = Tapeworm feeds by absorbing partly digest food in the host's intestine.
92. B
93. C = Have microtriches on their body surface for food absorption.
94. D
95. E
96. C = Eggs in the mature proglottid of the tapeworm are stored in the uterus.
97. D
98. B = It bores into the blood stream through the walls of the alimentary canal.
99. A
100. D
101. A
102. D
103. A
104. C
105. B
106. C = Samuel The Biologist Planted Sugarcane.

CHAPTER FOUR

1. B = The structure labeled I is the proboscis.
2. D = Proboscis is modified for piercing and sucking.
3. C = The larval stage of a mosquito is called wriggle because it moves by rapid wriggling.
4. C = Oil prevent the larval and pupae from breathing, hence they will die of suffocation.
5. A = The female *Culex* mosquito transmits *Wuchereria bancrofti*, a roundworm, causing filariasis (elephantiasis).
6. C = Saliva injected during mosquito bite is principally to prevent the host's blood from clotting.
7. D = Female *Anopheles* - malaria; female *Aedes* - yellow fever and dengue fever; female *Culex* - elephantiasis (filariasis).
8. A = *Anopheles* mosquito have air-floats on eggs, which are laid singly. *Culex* mosquito lay eggs in batches which float easily.
9. B and C = *Anopheles* larva breath through the Spiracle, and anal gills on the 8th and 9th abdominal segment, *Aedes* larval breath through the siphon (breathing tube) connected to the tracheal system located on the 8th abdominal segment.
10. B = Movement is by the tail paddle (fins).
11. D = Female *Anopheles* mosquito transmits plasmodium, a protozoan parasite causing malaria.
12. C = mosquito undergoes complete metamorphosis. Cockroach undergoes incomplete metamorphosis.
13. D
14. B = Elytra is the fore-wing borne on the mesothorax

- (2nd segment of the thorax).
15. B = The correct sequence that describes the insect segmented leg is: Coxa (i), Tronchanter (iii), Femur (ii), Tibia (iv), Tarsus and Claws (v).
 16. D = The claw (v) of insects is used for gripping or holding.
 17. C = The mandibles or jaws are highly sclerotised, used for chewing.
 18. C = The style is what differentiates the male cockroach from the female. Present in male and absent in female.
 19. E = Metamorphosis is the stages which certain animals like insects, toads and frogs undergo before they attain adulthood.
 20. D = After mating, female cockroach lays its egg in a honey egg - case called ootheca formed in the body.
 21. C = The new emergent is called nymph.
 22. C = The nymph are wingless, small in size and sexually immature compared to the adult.
 23. C = The cerci has mechanoreceptive hairs called setae that are sensitive enough to detect the change in air pressure cause by fast approaching object, like a predator, hence an organ of defence.
 24. D = The cockroach antenna, also known as feelers, are responsible for the sense of smell, without which the cockroach will not be able to locate its food.
 25. B = I - compound eye, II - antenna, III - proboscis or sucking tube.
 26. C
 27. A = The mouthpart of housefly is modified for lapping and sponging. The proboscis is formed from a modified labium; it is a sponge - like organ composed of two halves called labella.
 28. D = The head of a maggot (housefly larva) has a hook with which the larva tears up food and drags itself along.
 29. B = The hind wing of housefly is modified as halteres also called balancers.
 30. E
 31. D = Malaria is transmitted by female Anopheles mosquito. Housefly could spread diseases such as typhoid fever, diarrhea, dysentery, cholera and poliomyelitis.
 32. A = Termite can digest cellulose with the help of protozoan that live in worker termite's gut. This is symbiotic relationship.
 33. B = In termites, new king and queen are produced from the adult reproductives from the same colony.
 34. B = The lack of pigment in the workers suggest the common, though misleading name of "white ant" for these insects.
 35. C
 36. B = The termite colony will no longer exist without a queen.
 37. A
 38. D = Termites live in termitarium built with clay.
 39. A = The queen is distinguished from other caste by the prominent and extended abdomen.
 40. B
 41. A = Adult butterfly helps in pollination whereas the larva (caterpillar) feeds on crops thereby destroying them and reducing their yield.
 42. D = II - silk thread III - Spiracle IV - Silk pad.
 43. A
 44. C = I - antenna, II - compound eyes, III - mouth p. IV - coiled proboscis.
 45. A = Mouthpart of butterfly.
 46. A = Butterflies have thread-like antenna with a club like knob at the end.
 47. A
 48. C
 49. A = Mosquito larva are filter feeders.
 50. A
 51. D
 52. C = Cockroaches and grasshoppers undergo incomplete metamorphosis.
 53. C
 54. A = Grasshopper larva (nymph) destroy plant leaves; Grasshopper adult also destroy plant leaves.
 55. B
 56. D = Moths are insects closely related to the butterfly. They are destructive in their larva (caterpillar) stage.
 57. C = Ecdysis or moulting is the shedding of the exoskeleton
 58. A
 59. D = Except in the head, each segment of insect is covered by a dorsal tergum and a ventral sternum.
 60. A
 61. C = Spiders have no antennae. Antennae in cockroach and butterfly are long.
 62. C = Nocturnal insects are active at night e.g. cockroach.
 63. B.
 64. B = The exoskeleton (cuticle) is secreted by the epidermis and is composed partly of chitin and partly of protein.
 65. B = The tracheal openings in insect is the spiracle found on the abdominal wall.
 66. B = Elytra, the first forewing is borne on the mesothorax (2nd segment of the thorax).
 67. D = The abdomen in insect is devoid of appendages except for a pair of jointed anal cerci which project laterally, one on either side, at the posterior end.
 68. A
 69. E = trumpet (respiratory trumpet) is found in the pupa of mosquito not all insect.
 70. E = No stated differences in the options.
 71. E = Adult insect reproduce to perpetuate the generation of the species.
 72. A = Both have biting mouth parts.
 73. C = Moulting in insects is the periodic shedding or replacement of outer cuticle (older cuticle).
 74. D
 75. C = Insect activities gradually increases with temperature.

CHAPTER FIVE

1. C = Fishes are cold-blooded (poikilothermic) animals whose body temperature varies according to changes in their external surroundings, others include amphibians and reptiles.
2. None of the options is correct = cartilaginous fishes have 5 to 7 pairs of gills without operculum (gill cover). Bony fishes have gills covered by an opercula flap. II and V = cartilaginous fishes IV = bonefishes.
3. A = Stream-lined shape with a neck minimizes friction during passage in water.
4. C = The swim bladder contains gas whose volume is adjustable with changing pressure; it keeps the fish buoyant. Only found in bony fishes.
5. B = The dorsal, ventral and anal fins help to stabilize the fish so that it does not roll in water.
6. E = The lateral line contains sensory cells that allow the fish to detect vibrations in water and to distinguish between differences in water.
7. C = Paired fins in bony fishes are equivalent to the limbs of higher vertebrates. They are the pelvic and pectoral fins.
8. B = Gill rakers prevent food and big particles from passing through the gills slits.
9. B = The pectoral and pelvic fins are used for steering, balancing and slowing movement.
10. B = Lung fish breath with the gills in water and also has a swim bladder that stores air for respiration
11. B = Pectoral fins lie just behind the gill covers.
12. C = The urea raises the body fluid osmotic pressure in order to prevent dehydrating.
13. B = Fertilization in cartilaginous fish is internal and only a few large yolky eggs are produced, each protected in horny egg cases. Development is direct. It has cartilaginous vertebrae.
14. B = Operculum is only found in bony fishes.
15. D
16. C = A fish moves forward by a combination of undulatory contractions of the body muscles and beating of the tail from side to side.
17. E = The swim bladder do not store excretory material. In some fishes, the swim bladder lies against the ear and act as an amplifier to enhance sound detection.
18. C = Olfactory lobe is more developed in the cartilaginous fishes because they depend more on their sense of smell. However, it is the cerebellum and optic lobes in the bony fishes.
19. B = Cerebellum controls muscular activities.
20. B = A, C, and D are true
21. B. Development of fore limb is the last thing in the metamorphosis of toads.
22. E = The eggs laid in strings of jelly are without shell.
23. A = The long and coiled intestine is present in a young tadpole to enable it feed on weeds (herbivorous diet).
24. A = Fertilization is external in toad. During reproduction the male mounts on the female and as she extrudes her eggs into the water, the male discharges sperm over them.
25. C = The jelly spaces of the eggs is for aeration and also protects them from injury, predators and water uptake.
26. B = Poison glands are located on the dorsal surface of the skin to ward off predators.
27. B = Nictitating membrane covers the eyes and protect them from water-borne dirt.
28. B = Amphibians do not have scales, feathers or hairs.
29. B = The permeable skin allowed loss of water via osmosis.
30. E = see 27.
31. C = There are at least two types of skin secretion in toads, namely (1) granular, also called parotid gland, which secrete toxic, milk-white fluid and (2) mucus gland, which secrete non toxic, clear liquid. The granular gland facilitate defence mechanism while the mucus gland facilitate thermoregulation.
32. B = The web of the feet is part of the skin. They breath through: skin, mouth and lungs.
33. E = The forelimbs in male toad possess swollen copulatory (nuptial) pad on the first inner finger. In breeding season the copulatory and becomes dark vary thick and sticky. It is not called breeding pad.
34. E = It does not provide the eggs with mineral salt.
35. C = Toads do not have ciliated skin at any point of development.
36. D = Egg to adult.
37. C = The animal with the mentioned features is a lizard (reptile).
38. A = pre-anal pad, (anal pores) are present in male rainbow lizard (*Agama agama*) but absent in the female.
39. C = Lizards have dry scaly skin, which protect their body from desiccation.
40. A = Lizards eggs, unlike birds eggs, have a thin, pliable leathery shell.
41. B = The gular fold in lizard is lowered during courtship to attract mate or when the animal is frightened.
42. B = Homodont dentition, i.e. teeth all of same type.
43. E = Internal fertilization by means of copulatory organs.
44. B = Amphibians have oval stream-lined bodies with no neck.
45. B = Feathers which cover the body of birds also maintain their body temperature. They are homeothermic, having uniform body temperature.
46. E = Webbed feet is an adaptation for swimming.
47. D = I - Double shell membrane, II - Germinal disc, III - Chalaza IV - Germinal disc. Chalaza suspend the yolk in the abdomen.
48. B
49. A = Down feather gives insulation.
50. A = Wing bears feathers.
51. A = Duck beaks is adapted for scooping (from mud, pond and water) and sieving food.
52. C = Birds show parental care e.g. building of nest for eggs, feeding the newly hatched, protecting the young from predators.
53. E = All true
54. A = Skin of bird is waterproof, non-gland (except preen glands) at the base of the tails
55. C = The spaces on the bird's body without feather tracts are referred to as apteria.
56. C
57. E. Feathers perform all the stated roles.

58. E = The uropygial gland (oil gland, preen gland or rump gland). It performs so many functions in birds including water proofing and keeping the skin, feathers and bill supple; the oils is said to have an antibacterial function.
59. A = The serrated edges helps to scoop and sieve food.
60. B
61. D = Mammals have their bodies covered with hairs.
62. A = The skin of mammals have the following glands: sebaceous, sweat and mammary glands.
63. C = Mouth of rodents have long curved incisors (Chisel-shaped) for gnawing food.
64. E = The organ is placenta, nourishment and oxygen diffuse to the developing embryo through the umbilical cord.
65. D = Evolutionary trend in vertebrate is
PARAM: Pisces (fish), Amphibians, Reptiles, Aves (birds), Mammals.
66. D = Ovoviviparity (viviparity), is a mode of reproduction in animals in which the embryos develop inside eggs that are retained within the mother's body until they are ready to hatch. Examples are found in some species of fish (cartilaginous fishes and reptiles).
67. A = The dark hairy covering of the rat body aids camouflage (disruptive colouration) and escape from predators.
68. A = I – Whisker, II – Pinna of ear, III – Tail, IV – Hair. I (whisker) are photosensitive for feeling the way in the dark burrow.
69. A = Viviparous reproduction occurs only in mammals. It means development of the embryo inside the body of the mother, eventually leading to live birth, as opposed to laying eggs.
70. B
71. A = caudal fin is the tail fin.
72. C = The fore limbs of toad are adapted for absorbing shock on land while the hind limbs are adapted for hopping from the ground.
73. D
74. B = Fishes, Amphibians, and Reptiles are cold blooded (poikilothermic); Birds and mammals are warm blooded (homeothermic).
75. C.
76. A = birds and reptiles have bony structure of head.
77. B = Pectoral muscles assists the feathers for flight.
78. B = Bones of birds are light because they are thin and hollow for ease of flight.
79. D
80. B = pigeon are monogamous, one sexual partner.
5. D = Cambium is present between the phloem and xylem. Cambium cells are constantly dividing and develop into cells of different types (secondary thickening).
6. B
7. C = Sieve tubes lie along with the companion cell. They, along with phloem parenchyma constitute the phloem tissue.
8. B = Collenchyma is living at maturity but thick walled. Xylem vessels are dead elongated tubes. Sieve tubes are un lignified, elongated living cells. Sclerenchyma is dead at maturity but thick walled.
9. D = The guard cells control the opening and closing of the stomata through which air and water move in and out of the mesophyll layer of leaf.
10. B = Mesophyll parenchyma also called photosynthetic parenchyma is the principal site of photosynthesis. It contains two cells namely palisade cell (main site of photosynthesis) and spongy cells.
11. D = Secondary thickening is made possible by the cambium.
12. A = The vascular bundles of the leaf, which are continuous with those of the stem, form the veins.
13. B = I – Pericycle, II – Root hairs, III – Xylem IV – Phloem. Root hairs are used for absorption of water and salt.
14. D = Pericycle I, which is inside the endodermis. The endodermis is also referred to as starch sheath. It stores starch hence when stained with iodine solution, it turns blue-black.
15. C = See 2.
16. A and B = Monocot stems have no cambium, hence cannot undergo secondary thickening. Monocot stems also lack pith.
17. B = The pericycle lies between the endodermis and the vascular bundle.
18. A = In dicot stem, between the epidermis and pericycle is the cortex.
19. A = Xylem and phloem are conducting tissues in plants similar to artery and vein in animals.
20. A = Xylem comes before phloem from inside to outside.
21. E = The organ is placenta, nourishment and oxygen diffuse to the developing embryo through the umbilical cord.
22. C = Lungs and heart are in the thoracic cavity also called the anterior half.
23. A = The heart is enclosed in the pericardial cavity.
24. B = Left auricle receives oxygenated blood from the lung via the pulmonary vein. Right auricle receives deoxygenated blood from the body through the venae cavae.
25. D = Hepatic portal vein starts and ends in a capillary network. It links the villi of the intestine to the liver.
26. D = The thoracic and abdominal cavities are separated by the muscle, diaphragm.
27. C = Mammals are unique for the glands in their skin namely sweat, sebaceous and mammary glands.
28. C = Heterodont, i.e. of different types.
29. D = Perineum separates the anus from the scrotum in the male and the anus from the vulva (the labial opening to the vagina) in the female.

CHAPTER SIX

1. A = Unlike the epidermis of the stem, the piliferous layer of the root has no cuticle.
2. B = A dicot stem and monocot root have the central pith in common.
3. A = In the root, the xylem and phloem strands are arranged on alternate radii while in the stem, they are arranged on the same radii.
4. C = Memory Tip: The yag Xyle p oze C n Produce Emotional Pleasure Called Ecsiasy.

32. B = Bile duct helps the digestion of fat by emulsification of fat.
33. C = The pancreatic duct opens into the duodenum where it releases its secretions.
34. B = The caecum has a blind ending called appendix.
35. D = The pancreas is both a ductless gland and an accessory organ of digestion.
36. C = The liver is the biggest organ. The skin is the most extensive organ.
37. D = Urea formation takes place in the liver.
38. A = The heart pumps blood round the body.
39. D = HCl is secreted by the parietal cells concentrated in the middle part of the stomach. Cells in the pyloric region secrete gastrin and mucus.
40. C = Thyroid gland is located on the trachea in the neck.
41. C
42. A = The liver stores iron.
43. C = Absorption of water takes place in the large intestine colon, as ingested food passes through it.
44. C = Urine enters the urinary bladder through the ureter and leaves through the urethra.
45. B = I – Upper epidermis, II – Palisade mesophyll, III – Bundle sheath, IV – Spongy mesophyll. Palisade cells are positioned towards the upper surface of the leaf and contain the largest number of chloroplast per cell in plants. This makes them the primary site of photosynthesis in plant leaves.
46. B = Palisade cells have a very large surface area in order to absorb more light during photosynthesis.
47. B = I – Endodermis, II – Phloem tissue, III – Xylem tissue, IV – Epidermis.
48. D = Xylem tissue, III, transport water and mineral salts.
49. A = The stele consists of vascular bundle, it is bounded on the outer side by one-or-two cell thick layer of thin walled cells called pericycle.
50. C = Stomata is found between two guard cells to the lower epidermis.
51. A
52. A
53. A = Sclerenchyma cell are dead at maturity, collenchyma, parenchyma and endodermis are alive at maturity.
54. C
55. B

CHAPTER SEVEN

- B = Insects are attracted to insectivorous plants, not to pollinate it, but to feed on the secretions.
- D = Saprophytes digest their substrate outside their body before absorption. This is extracellular digestion.
- D = Fungi lack chlorophyll hence they are heterotrophic, they are either saprophytes or parasites.
- B = Nitrobacter is a nitrifying bacteria – chemosynthetic, *Ascaris* is an obligate parasite in man – parasitic, *Homo sapien* (Humans) ingest food and digest within the body – Holozoic.
- D = Hydra exhibit both intra- and extra-cellular

- Housefly is saprozoic, digest food extracellularly by secreting saliva on food.
6. B = Pitcher plant is a carnivorous (insectivorous) plant which feed on insects. This is a heterotrophic mode of nutrition. Other examples are Sundew, Bladderwort.
 7. C = Tapeworm is an obligate parasite in the intestine of man.
 8. A = A haustoria or sucker is a developed structure with which plant parasites like *Cassytha* absorbs nutrient from its host's conducting tissues.
 9. C = see 1.
 10. B
 11. D = Insectivorous plants such as sundew and bladderworts are primarily autotrophs because they have chloroplasts, hence they are both autotrophic and heterotrophic.
 12. D = In chemosynthesis, energy from the oxidation of inorganic molecules is used to generate organic molecules as food.
 13. C
 14. E = The oxygen given off during photosynthesis is derived from water.
 15. A = The light stage of photosynthesis involve four steps namely: (1) Activation of chlorophyll (2) Photolysis of water (3) Transfer of NADP (Nicotinamide Adenine Dinucleotide Phosphate) (4) Formation of ATP (Adenosine triphosphate).
 16. D = Formation of glucose using energy from NADPH (Nicotinamide Adenine Dinucleotide Phosphate Hydrogen) is the dark stage of photosynthesis.
 17. C = NaHCO_3 enriches the water with carbondioxide.
 18. A = Glucose – direct product, starch – first visible product.
 19. A = In bright light because photosynthesis occurs faster than respiration hence, in dim light there may be equal rates.
 20. C = Light energy and chlorophyll are the principal requirement for photosynthesis.
 21. C = Photosynthesis occurs in the part of plants (shoot system only) that contain chlorophyll.
 22. A = In the dark stage, carbondioxide, absorbed by leaves, combines with a 5-carbon compound, ribulose phosphate, to form an unstable 6-carbon sugar.
 23. B = The dark stage of photosynthesis takes place in the stroma of the chlorophyll.
 24. A = Photosynthetic pigments include: Chlorophylls (a and b) – in green plants, carotenoids (carotenes and xanthophylls) – in higher plants and also present in the green algae and brown algae, phycobilins -- only in red algae and cyanobacteria.
 25. C = The pigments of chloroplasts absorb blue and red light most effectively, and transmit or reflect green light, which is why leaves appear green.
 26. C = Chlorophyll contains metallic magnesium.
 27. A = Carotenoids are accessory pigment of photosynthesis, there are two classes: carotene (yellow) and xanthophylls (orange).
 28. B = CO_2 in gaseous form as CO_2 molecule.
 29. C = Nitrogen is required for formation of protein, enzymes and nucleic acid.
 30. D = Magnesium and iron are required for the

- formation of chlorophyll.
31. B = Yellowing of leaves chlorosis, could be due to the deficiency of nitrogen, magnesium, iron, sulphur.
 32. A: In addition to carbon, Hydrogen and oxygen. Major elements required by plants are: Nitrogen, Phosphorus, Potassium, Calcium, Magnesium, Sulphur. **Memory tip: Nigerian Police, Pick Criminals, Most Saturdays.**
 33. D
 34. B = Lack of phosphorus will cause the symptoms mentioned.
 35. A = Trace or micro-elements are needed in small amount for the formation of pigments and enzymes.
 36. B = Poor growth and yellowing of leaves are the deficiency symptoms of iron. Little growth and yellowing of leaves are the deficiency symptoms of magnesium.
 37. B = see 34
 38. C = Nitrogen is needed for protein synthesis.
 39. C = Food is the material requirement of organisms.
 40. B = Gastrin is a hormone secreted in the stomach.
 41. E = Some vitamins are catalysts or enzymes in important metabolic processes. They yield no energy.
 42. C = Deficiency of iodine causes goitre.
 43. D = Large deficiency of protein in growing children results in kwashiorkor, a disease conditioned characterized by distended abdomen and lean body.
 44. B = Carbohydrates are made of sugar units. They contain carbon (C), hydrogen (H) and oxygen (O) with H and O in the ratio 2:1, formula is $C_x(H_2O)_y$.
 45. B = Calcium is needed for strong bones, teeth and also contraction of muscles. Deficiency results in retarded growth.
 46. C = Fats, carbohydrates and proteins are the foundational organic substances in human body.
 47. A = Vitamin A is required for normal vision and keep skin healthy.
 48. B = Vitamin K is required for the synthesis of blood clotting protein
 49. B = Carbohydrates (main energy) source yields 17 kJ/g. Proteins (energy during starvation) yields 17 kJ/g. Lipids (high energy) yield 39 kJ/g.
 50. A
 51. D
 52. D
 53. B = Deficiency of vitamin E causes sterility.
 54. C = Osmotic pressure of the body fluids is regulated by mineral salts. Also needed for proper functioning of the body.
 55. E = Lactose is a disaccharide sugar that is found most notably in milk and is formed from galactose and glucose.
 56. D = Blood sugar.
 57. D = Sugar solution boiled with Fehling solution A and B, colour turns from blue to green and finally to orange (red) as a precipitate. Except for sucrose, unless HCl is added to it (hydrolyses) before the experiment.
 58. E = Sample reducing sugar (e.g. glucose) can be tested by adding Fehling's reagent in neutral or alkaline medium and boil. Red precipitate is formed.
 59. E = Lipids (fats and oils) shows a red colour with sudden (III) solution
 60. C = Biuret's test for protein.
 61. A = Biuret's test for protein.
 62. B = Test for starch.
 63. D = Benedict's test for reducing sugar e.g. glucose
 64. C = Dental formula for man is C, rabbit is A, sheep and cow is B, dog is D.
 65. B = Incisors – cutting, canines – tearing; premolars and molars – grinding and crushing.
 66. E = The total number of molar teeth in man is 12, check the dental formula, multiply the upper and lower teeth by 2 and add up.
 67. A = In herbivores, canines are inconspicuous or absent the toothless gap is called a diastema. Canines are used for tearing food.
 68. B = Pulp cavity contains connective tissues, nerves and blood vessels.
 69. A = 1, 2, 3, and 4 are molar, premolar, canine and incisor respectively.
 70. B
 71. D = Enamel (outermost) dentine pulp cavity (innermost).
 72. A = Last upper premolar and first lower molar form the carnassial teeth in advanced carnivores.
 73. A = Absorption of food takes place in the ileum. Food is not absorbed in the stomach except alcohol.
 74. E = Cardiac sphincter, upper part of the stomach. Food leaves through the pyloric sphincter.
 75. C = Rectum, which terminates as anus.
 76. E = Excretion of urea is by the kidneys.
 77. A = Oral cavity, Oesophagus, Stomach, Duodenum, Jejunum, Ileum, Large intestine (Colon), Caecum, Rectum.
 78. A
 79. A = Bile from gallbladder emulsify lipids to increase the surface area for digestion.
 80. D = Kidneys are not close to the stomach.
 81. D
 82. C = Liver and pancreas.
 83. B = The entrance and exit of the stomach is guarded by cardiac and pyloric sphincters respectively.
 84. D = The stomach secretes pepsin, which converts proteins to peptide.
 85. B = The pancreas secretes hormones (insulin and glucagon) and enzymes (amylase, trypsin and lipase).
 86. B = Colon is the large intestine.
 87. D = The pH of the entire alimentary canal is alkaline, except the stomach, which is acidic.
 88. D = Epiglottis prevent the passage of food into the larynx of the trachea.
 89. C = Renin coagulates milk protein (casein) before it is acted upon by pepsin.
 90. D = Protein → Peptones (peptides) → Polypeptides → Amino acids.
 91. A = The food particles acted upon by enzymes is referred to as the substrate.
 92. E = Enzymes are inactivated by extreme temperature and pH.
 93. A = Chyme and chyle are partially digested food in the stomach and intestine respectively.
 94. B = Carbohydrate digest fastest.
 95. C = The pancreatic juice contains amylase

(amylase), trypsinogen and lipase (steapsin) for the digestion of carbohydrate, protein and lipids respectively.

96. C
97. C = Deamination leads to the formation of urea.
98. A
99. A
100. A
101. C = Absorption of digested food materials takes place in the ileum via the villi while absorption of water takes place in the colon (large intestine).
102. D
103. A = Carbon, Hydrogen and Oxygen are common to carbohydrates, proteins and lipids.
104. D
105. C = Hepatic portal vein carries blood from the alimentary canal to the liver. It starts and ends in a capillary network.
106. D
107. A
108. A = Osmic acid test is used to test for lipids.
109. B = Photosynthesis and chemosynthesis are autotrophic mode of nutrition.
110. D = With millon's reagent on heating, proteins gives a deep red colour or precipitate.
111. B
112. D = I - Enamel, II - Dentine, III - Gum, IV - Pulp cavity. Dentine is extremely sensitive to heat and cold.
113. A
114. C
115. B = Lipase is an enzyme that digest lipids.
116. B
117. A
118. D
119. C = Osteomalacia is the softening of the bones caused by the defective bone mineralisation. In children it is known as rickets. It could be caused by deficiency of calcium, vitamin D or phosphate.
120. C = Polished rice lacks vitamin B1 (thiamin). The deficiency of vitamin B1 causes beri-beri.
121. C

CHAPTER EIGHT

- B = Small (unicellular) organisms have large surface area (S.A) to volume ratio (V.R) and do not need a transport system because diffusion is efficient, large (multicellular) organisms have low surface area to volume ratio, so require a transport system.
- B = Organism IV has the lowest S.A/V.R (1:5=0.2). Therefore, it will have the most complex transport system.
- A = Pulmonary artery carries deoxygenated blood.
- D = The sequence of blood flow is: Heart → arteries → capillaries → venules → veins → Heart.
- A
- B = The lymph is carried in lymphatic vessels to the subclavian veins in the neck, where it is passed back into the blood.
- E
- D = Blood in the right ventricle will leave the heart to

the lungs for oxygenation via the pulmonary artery. Oxygenated blood will return to the left auricle via the pulmonary vein and will finally leave the heart via the aorta, to the toe without passing through the liver.

9. B = Hepatic portal vein begins and ends with capillaries.
10. C
11. C = Superior (anterior) vena cava returns blood from structures above the heart back to the heart.
12. C = The wall of the left ventricle is further thickened to enable it to pump blood with sufficient pressure round the body.
13. D
14. A
15. C
16. A = Systemic circulation starts from the left ventricle to reach all parts of the body except the lungs and back to the right auricle. Pulmonary circulation starts from the right ventricle to the lungs through the pulmonary artery and back to the left auricle through the pulmonary vein.
17. B
18. B = The sieve tubes of the phloem tissue translocate manufactured food and mineral ions from leaves or storage organs to other parts of the plant.
19. C
20. D = The xylem tracheids and vessels transport water and mineral salts from the soil to the leaves by osmosis.
21. C = The xylem vessels will appear red because the ink travels only in the xylem.
22. E = Red blood cells (erythrocytes) are non-nucleated at maturity.
23. B = Blood clotting begins with the exposure of blood to air from a cut and ends with the trapping of red blood cells and forming of clot.
24. B = Lymph nodes produce certain white blood cells, lymphocytes, and filter out bacteria and other particles.
25. B = The lymph is much like the blood plasma but it lacks plasma proteins and red blood cells. It contains white blood cells.
26. A = The main function of blood is to transport excretory materials from tissues and supply of oxygen to tissues.
27. A = The exchange of nutrients and metabolic products occurs between the lymph and body cells.
28. B = Oxygen is carried by the erythrocytes.
29. B = The cell sap is the medium of transpiration in higher plants.
30. E = A potometer is an instrument for measuring the rate of transpiration of a plant shoot at a very short interval and under different environmental conditions. The limitation of a potometer is that it only measures the water absorption and not the transpiration rate directly.
31. D = Without the phloem tissue, translocation is not possible.
32. C = Increase in temperature, light intensity and air movement increase the rate of transpiration. Increase humidity lowers transpiration rate.
33. D = A transparent polythene is not semi permeable

- hence it will not enhance osmosis.
34. C = Water from conducting tissues in the veins moves into the mesophyll cells, from there to the substomatal cavity and not to the atmosphere through the stomata pores.
 35. C = Mineral salts are absorbed into the plant roots against their concentration gradient by active transport, a process that requires energy. As they become concentrated in the root cell, they tend to diffuse to other parts of the plant.
 36. D = Movement of mineral salts is hardly affected because mineral salts are not transported by the phloem. The plant dies after several weeks.
 37. D = Guttation is the loss of water in a liquid form through the margins of leaves of plants growing in a warm and moist soil in a humid atmosphere.
 38. D = The correct route of water movement in plants is the sequence shown in D.
 39. A = Translocation is the transport of manufactured food substances from the leaves to all parts of the plants.
 40. C = X will shrink as a result of plasmolysis. Y will become turgid as a result of osmosis.
 41. A
 42. D
 43. C = Blood serum is the fluid expressed from clotted plasma; it can be defined as plasma deprived of fibrin but which contains other plasma contents.
 44. A = The function is to reduce friction.
 45. B = Water vapour – transpiration, water droplets – guttation.
 46. C = The Pigment, haemoglobin, imparts dark red colour to blood.
 47. B = Double circulation takes place in mammals. It involves pulmonary and systemic circulation.
 48. C
 49. D = Osmosis regulates the opening and closing of stomata. The stomata open when the guard cells are turgid and close when the guard cells are flaccid.
 50. B = Inferior (posterior) vena cava returns blood from structure below the heart back to the heart.
 51. D = Leucocytes (white blood cells) defend the body against diseases.
 52. A = High osmotic pressure of root cells is able to exert an influence on the absorption of water.
 53. C = Heart to liver – hepatic artery.
 54. A = Semi-lunar valves (pulmonic valves) one located at the opening of the aorta and the other at the opening of the pulmonary artery, they prevent blood from flowing back into the ventricles.
 55. A = Plasma proteins are proteins found in blood plasma, including albumin, fibrinogen and globulin. Albumin, alongside others, aid in regulating the water balance between the blood and tissues.
 56. C
 57. A = The insect circulatory system is an open system, where blood flows freely inside the body cavity without the aid of blood vessels (arteries and veins)
 58. D = Turgor pressure pushes the plasma membrane against the cell wall of plant bacteria and fungi cells as well as those protist cells which have cell walls.
 59. D = Vapour form
 60. D
 61. E = The rate of water absorption was affected.
 62. A = Phloem tissue becomes impaired.
 63. A
 64. B = Red blood cells do not fight bacteria.
 65. C = Heat is produced more in the liver and the muscles than any other parts of the body. This heat is distributed round the body through the flow of blood.
 66. D
 67. B
 68. B
 69. C = The heart is enclosed in a membranous structure called pericardium.
 70. D = The ventricles are thicker than auricles because they transport blood while the auricle receives blood. The left ventricle is further thicker than the right ventricle.
 71. C
 72. C = Arteries take blood away from the heart. So the colour of the blood and its oxygen content are not determining factors.
 73. B = Capillaries possess very thin walls to allow diffusion of gases.
 74. E = Carotid artery carries blood to the head.
 75. E = The celiac artery supplies oxygenated blood to the liver, stomach, abdominal esophagus, spleen and the superior half of both the duodenum and the pancreas.
 76. A = A portal venous system occurs when a capillary bed pools into another bed through veins, without first going through the heart. Examples of such system include the hepatic portal system and the hypothalamic portal system.
 77. D = The left subclavian artery supplies blood to the left arm while the right subclavian artery supplies blood to the right arm.
 78. C = Lymph nodes manufacture lymphocytes, a type of white blood cell.
 79. E = Oxygen is not transported in the lymph.
 80. E = Fatty acids and glycerol enter the blood stream through the subclavian vein via the lymphatic system.
 81. A = The iliac artery supplies blood to the lower limbs and gonads.

CHAPTER NINE

1. C = Pyruvic acid produced during glycolysis, in the absence of oxygen, is converted to CO_2 and ethanol in plants and lactic acid in animals.
2. D = After about 6-12 hours, the rise in the coloured water level in the glass tube of the animal specimen will be higher than in the glass tube of the plant specimen. Thus, the experiment is a comparison of oxygen uptake (respiratory rates) in plants and animals.
3. C = Manometer, used to measure the pressure of liquids and gases.
4. A = Glycolysis in plants and animals yields pyruvic acid.
5. A = Respiration oxidizes food to release energy.
6. E = Crustaceans (e.g. crab) respire by means of gills.
7. E = Insects (e.g. grasshopper) respire by means of tracheal tubes.

8. D = Epiglottis is a regulatory structure that leads to the larynx (voice box).
9. A = Respiratory surfaces must not be dry and freely permeable.
10. D = In mammals, air moves from mouth and nose → pharynx → trachea → bronchi → bronchioles → alveoli. Each alveolus is surrounded by network of capillaries and gas exchange takes place in it.
11. A = Spiders respire by means of lungbooks.
12. C = Gas exchange in Aves (birds) occurs in the air sacs of the lungs.
13. B = Mammals respire by means of lungs and whales is a mammal.
14. A = In insects, gas exchange occurs in the tracheoles of the trachea (air tubes) which open out as spiracles.
15. C = Oxygenated water is drawn in through the mouth, the dissolved oxygen is then absorbed by the blood capillaries in the gills and deoxygenated water flows out through the operculum.
16. D = Toads and frogs respire by means of gills at tadpole stage; skin (cutaneous), mouth (buccal) and lungs (pulmonary) at adult stage.
17. B
18. E = During inhalation volume (size) of the thoracic cavity increase while the pressure in the lungs decrease.
19. B
20. C = Increase in air pressure of the lungs occur during exhalation, while the diaphragm relaxes.
21. D = Exhaled air is warm, has more CO_2 , water vapour and less oxygen.
22. D = Diaphragm separate the thorax and abdomen.
23. A = Contraction of the intercostal muscles move rib cage upward and forward.
24. C = Oxygen will diffuse from lungs (higher concentration) to the blood (lower concentration).
25. D = The presence of intercellular air spaces in the cells of the stem and leaf allow the circulation of air.
26. C
27. C = Lenticels are found in woody stems, while stomata are found in leaves. They help exchange gases.
28. D = CO_2 releases by respiring plant cells will be used for photosynthesis.
29. C = Increase in solute concentration in the guard cell makes the guard cell become turgid and thus the stomatal pores open. Decrease in solute concentration in the guard cell makes the guard cell become flaccid and thus the stomatal pores close.
30. B = During Krebs's cycle, hydrogen is released. Hydrogen reduces oxygen into water vapor.
31. A = During Krebs's cycle, the oxidative part of respiration takes place in the mitochondria.
32. A = CO_2 the gas produced during tissue respiration can be identified by calcium hydroxide (lime water), which it turns milky.
33. C = The significance of tissue respiration is the oxidation of food and the release of useable energy.
34. B = Artificial respiration is an emergency procedure where breathing is maintained artificially.
35. C = The bubbles of CO_2 released is what make the

- dough to rise.
37. E = This is due to anaerobic respiration of an organism such as yeast.
38. A
39. D = 2ATP molecules of energy is produced during anaerobic respiration.
40. D
41. D = Cramp and muscle fatigue are caused by the accumulation of lactic acid and the resulting increase in acidity on the muscle due to insufficient oxygen during strenuous exercise.
42. C = 2ATP molecules of energy and alcohol.
43. D = Glycolytic pathway – cytoplasm. Krebs's cycle – mitochondria.
44. D
45. A = Each alveolus is surrounded by a dense network of capillaries for effective gas exchange.
46. D
47. A = Yeast carry out anaerobic respiration.
48. C
49. D
50. C = The trachea in insects open outside as spiracles on the cuticle.
51. D = Gas exchange in earthworm is via the skin by diffusion.
52. D = Respiratory substrates are monosaccharide (simple sugar) primarily glucose.
53. C = Glycolysis – anaerobic. Krebs's cycle – aerobic.
54. B
55. D
56. D
57. E = In the dark, respiring plant release large quantity of CO_2 because photosynthesis is not taking place. CO_2 is used for photosynthesis.
58. D = Oxygen is soluble in water. Plant root hairs absorb the oxygen in solution with water.
59. C = Air enters the insect's body through the spiracle.
60. B = Tracheoles are supplied with capillaries, non-cuticularized.
61. D = The gill filaments are richly supplied with blood vessels i.e. vascularised, to enhance the utilization of oxygen.
62. D = Gill cover or operculum.
63. E = When water enters the mouth of the fish (by opening of mouth and lowering the floor of the mouth and closing the operculum), the oral valves are closed, the floor of the mouth is raised and the opercula are open, water is forced through the pharynx over the vascularized gills where gaseous exchange occurs.
64. C
65. B = Air sac function as an oxygen reservoir.
66. B = The palate complex.
67. B
68. B = Mammals do not have air sacs.
69. D = The larynx is the 'voice box' having the vocal cords.
70. E = The ribs, intercostal muscles and diaphragm.
71. B = Diaphragm flattens during inspiration and become dome shape during expiration.
72. E = In the lungs, there is an exchange of gases by diffusion via the alveoli.
73. A

CHAPTER TEN

1. C = The leaf is not an excretory organ.
2. B
3. D = Flame cells are only for excretion in flatworms.
4. B = All insects carry out excretion with the malpighian tubule.
5. B = Earthworm is an annelid.
6. A = The Bowman's capsule and glomerulus of the nephron form the malpighian capsule.
7. A = The proximal convoluted tubule, loop of Henle and distal convoluted tubule form the uriniferous tubules.
8. C = Concentration of sodium chloride.
9. C = Malpighian corpuscle – cortex; loop of Henle and collecting duct – Medulla.
10. C = I – Glomerulus, II – Bowman's capsule, III – Distal convoluted tubule, IV – Proximal convoluted tubule.
11. A = In the proximal convoluted tubule, Na^+ is reabsorbed primarily with HCO_3^- and organic solutes such as glucose and amino acid, 67% of Na^+ is reabsorbed. In the distal convoluted tubule, Na^+ is reabsorbed primarily with Cl^- but without organic solutes, 5% of Na^+ is reabsorbed.
12. D = The segment of the loop of Henle are responsible for counter current multiplication, which is essential for the concentration and dilution of urine, 25% of Na^+ is reabsorbed; 3% is reabsorbed from the collecting duct.
13. B = Glomerulus Bowman's capsule Proximal convoluted tubule Henle's loop Distal convoluted tubule Collecting tubule.
Memory tip: Great Britain Prosecuted Hitler's Deputy Commander.
14. C
15. D = The Glomerulus and Bowman's capsule form the malpighian body, which is located in the cortex.
16. B = Ureter links the kidney to the bladder, the bladder opens out through the urethra.
17. A = The sweat gland helps the skin to remove Na^+ as sweat.
18. B = I – Sebaceous gland, II – Nerve fibre, III – Blood vessel V – Sweat gland.
19. A
20. D = The malpighian layer is an undulatory row of meristematic cells. Cells contain melanin pigment
21. A = The sebaceous gland is a sac of glandular cells which opens into the hair follicle, secretes oil which keeps hair supple and water proofs the skin.
22. D = Sap in a medium of transportation in higher plants. Oxygen is a waste product from photosynthesis.
23. D = The main waste products formed in plant are water in form of vapour from transpiration and respiration, carbon dioxide from respiration and oxygen from photosynthesis.
24. C = Anthocyanin imparts colours to plant flowers on accumulation.
25. A = Gaseous excretory products are released through the stomata and lenticels e.g. CO_2 from respiration, oxygen from photosynthesis and water vapour from

- transpiration.
26. C = CO_2 released as waste product of respiration used for photosynthesis.
27. C = Tannins are used for tanning leather. They are obtained from mangrove bark.
28. C = Faeces is an undigested remains of food. It expelled via the anus.
29. D = Insects excrete uric acid as a means of conserving water.
30. B
31. C
32. D
33. B = Uric acid, sweat and urea are not excretory products in plants.
34. D
35. B = Allantois helps the embryo exchange gases and handle liquid wastes.
36. E = Excretion is the removal of waste product of metabolism. Not all waste product are poisonous although may become poisonous on accumulation.
37. C
38. D = The kidneys absorb less water on cold days.
39. D =
40. C = The processes of ultrafiltration, reabsorption and secretion in a nephron summarizes the excretion of urine.
41. E
42. A
43. E
44. A = The pyramids consists of loops of Henle and the collecting tubules of the nephron. The pyramids are found in the medulla.
45. C = Outer zone – cortex. Inner zone – medulla.
46. E = All are parts of the kidney.
47. B
48. A = The lungs eliminates CO_2 and water vapour.
49. E = On a wet and cold day water reabsorption by the kidney is less.
50. C
51. D = The liver produce urea from deamination of amino acids.
52. E
53. D
54. E
55. B = Some insect excrete nitrogenous waste within their fat bodies as uric acid to conserve nitrogen.
56. D = Nitrogenous waste is not lost during metamorphosis.
57. A = Contractile vacuole remove excess water.
58. B = The gills remove CO_2 and water vapour during respiration
59. E

CHAPTER ELEVEN

1. D = Xylem conduct water and ions; and also support the plant.
2. A
3. D = Parenchyma cells support the plant by turgor.
4. A = Phloem translocate food from the leaves to other parts of the plants. The phloem parenchyma is a storage and packing tissue.
5. B = Sclerenchyma in plants is the functional

- equivalent of bone in animals, that is, sclerenchyma and bone provide elastic support (flexible and inflexible).
6. C = Lignin is largely responsible for the strength and rigidity of plants. Sclerenchyma is lignified to provide strength and rigidity.
 7. C
 8. A = Haptotropism or thigmotropism is a response to touch e.g. on touching a support, tendrils of a climbing plant twine round it.
 9. D = Roots of plants are positively hydrotropic and geotropic.
 10. D = There is a greater concentration of auxin on the shaded side of the shoot, which in turn causes increased cell elongation and thus bending of the shoot.
 11. A - Taxism involves the whole organism.
 12. B
 13. D
 14. A
 15. A
 16. D = Nastic movements are to general (diffuse) stimuli such as touch.
 17. E
 18. B = Nastic movement is non-directional.
 19. D = The breathing roots of mangrove trees grow against the direction of gravity (negatively geotropic).
 20. D = Plants that grow horizontally are neutrally geotropic.
 21. A = The klinostat is used to show that plant roots respond positively to gravity.
 22. C = Illuminating a young seedling from the top will give a diffuse source of light (not unilateral). So, the plant will remain straight.
 23. D = The differential changes in turgor in the lower and upper side of the pulvinus of the *Mimosa pudica* causes the dropping of the leaves. This is elicited by touch.
 24. A = Negative geotropism helps the plant shoot to grow upright against the direction of gravity.
 25. C = There is a greater elongation of cells on the shaded side as a result of accumulation of auxin on the shaded side.
 26. C = Light is responsible for the formation of carbohydrate in the guard cells, which in turn increases the concentration of the guard cells and cause the turgidity of these cells by osmosis. The stomata open when the guard cells are turgid.
 27. E = All the options are true.
 28. C = Scales, carapace, and hoofs are external skeleton. But carpals are bones of the wrist.
 29. A = Bones and cartilage constitute the internal skeleton of animals.
 30. C = Earthworms have a fluid-based skeleton called hydrostatic skeleton.
 31. A = Bone to bone - ligament. Muscle to bone - tendon.
 32. C = Exoskeleton is an outer skeleton made up of chitin; it is present in arthropods. Maggot and caterpillar are larvae of housefly and butterfly respectively. Earthworm has hydrostatic skeleton.
 33. C = Axial skeleton is made up of skull, vertebral column, ribs and sternum.
 34. E = Appendicular skeleton is made up of girdles and limbs.
 35. C = Atlas before Axis.
 36. B = Odontoid process of the axis fits into the neural canal of the atlas. The odontoid process makes the axis bone peculiar.
 37. C = The pectoral and pelvic girdle are part of the appendicular skeleton. The pectoral girdle has scapular. Ischium is present in the pelvic girdle.
 38. D = The pre-and-post zygapophysis articulate successive vertebrae to allow bending movement and prevent rotational movement in the vertebral column.
 39. C = The spinal cord is enclosed in the neural canal.
 40. D = Ball and socket joint allow movement in about 360° direction.
 41. A
 42. C = Transportation is not a function of skeleton.
 43. A
 44. B
 45. D = The axis has two post-zygapophyses but not pre-zygapophyses. The atlas has no zygapophyses. The atlas shape allow the head to nod 'yes' and the axis shape allows the head to shake 'no'.
 46. B = The neural canal encloses the spinal cord.
 47. E = The chief peculiarity of the atlas is that it has no body but axis has a body. They both have vertebral canals for passage of arteries, neural spine, and transverse processes with varying degree of modifications.
 48. A = Thoracic vertebra has long neural spine backwardly directed.
 49. B = In between consecutive vertebrae are compressible cartilage pads known as intervertebral discs which allow the vertebrae to move slightly.
 50. B = The lumbar vertebrae provide attachment to abdominal muscles (not back muscles) and bears considerable weight of the body.
 51. C = Sacral vertebrae usually fused to form a mass of bone called the sacrum.
 52. B = Ulna - longest bone of the lower arm. Humerus - longest bone of the upper arm.
 53. C = The longest bone of the thigh and the body is the femur.
 54. A
 55. D = The synovial fluid secreted by the synovial membrane lubricates the joints.
 56. B = Swimming to a warm region from a cold region by bacteria is a positive response to temperature. Positive thermotaxis.
 57. C = Hydrostatic skeleton is found in earthworm (class: Oligochaeta, phylum: Annelida).
 58. C = Thigmotropism is a movement on which an organism (plant) moves or grows in response to touch or contact stimuli.
 59. B
 60. C
 61. B = I - Cervical vertebrae, II - Lumbar vertebrae
 62. A = Chitin is embedded in a protein matrix and strengthened by mineralization, usually in the form of calcium carbonate.
 63. B = Vertebrates have endoskeleton.
 64. B = The biceps contract and triceps relax when the

fore arm is raised.

- 65. A
- 66. C = *Mimosa pudica*, Venus fly trap and Pitcher plant show nastic movement. It is non-directional.
- 67. B = Cork cells are bark cells. The cells are dead at maturity. Cork is also called phellem
- 68. D
- 69. D

CHAPTER TWELVE

1. B = Spirogyra reproduces asexually (vegetatively) by fragmentation. Fragment of a matured spirogyra breaks off to live an independent life.
2. A = Exponential increase in population is common in microorganisms that reproduce by binary fission.
3. A = The nucleus, during binary fission, divides by mitosis into two.
4. B = Budding is a form of asexual reproduction common to both hydra and yeast. A bud is an outgrowth of the parent organism which develops into a new organism.
5. NIL = *Mucor/Rhizopus* – asexual reproduction by spore formation. *Amoeba/Paramecium* – asexual reproduction by binary fission. *Spirogyra* – asexual reproduction by fragmentation. *Penicillium* – asexual spore (conidium). *Ascaris* – sexual reproduction. No option contain group that exclusively reproduce sexually, which is the implication of the question.
6. A = Banana, yam, pineapple and cassava are normally propagated asexually. Banana (suckers) pineapple (off sets); yam (stem tuber); cassava (stem cutting).
7. A
8. D = In asexual reproduction, there is no exchange of genetic material, hence no variation among the offspring.
9. B = Rhizomes are elongated horizontal underground stems swollen with food reserves e.g. cannalily, ginger.
10. E = Stem tubers are swollen terminal ends of axillary branches which goes down into the soil. They are enlarged and swollen with food e.g. Irish potato.
11. A = A corn is a short globular underground stem swollen with food and have lateral buds along the nodes. E.g. cocoyam.
12. D
13. B = Bulbs have condensed conical stems from which leaves arise at its dorsal side. The leaf bases are swollen with food e.g. onion, garlic, tulip.
14. D = Runners are horizontal stems which creep on the ground producing roots and aerial shoots wherever they touch the ground. When the internodes die out, the shoots become independent plants e.g. sweet potato.
15. E = Suckers are shoots produced from below the ground level by underground stems or roots running near the surface e.g. plaintain, banana.
16. D = Cassava is propagated by stem cutting.
17. E = When the leaf of Bryophyllum falls on moist soil, buds and adventitious roots develop at the notches along its margin. The buds develop into new plants.
18. D = The offspring are better adapted to the environment of origin but may not survive in changing environment conditions due to lack of variation.
19. A = A scion is a detached living portion of a plant (as a bud or shoot) joined to a stock in grafting. It needs the stock to survive.
20. E = Stock – old established plant; scion -- a desired plant.
21. D
22. D = Each micronucleus divides mitotically into two and later into four.
23. B = Spindle fibres are formed during metaphase. Each chromosome is seen attached to spindle fibres at the centromere.
24. D = Parthenogenesis is a form of asexual reproduction where growth and development of embryos occur without fertilization.
25. B
26. C = The protruding protoplast is active and may be called the male cell, while the other is passive (its contents stay in the same cell) and may be called the female cells.
27. A = The chiasmata is thought to be the point where two homologous non-sister chromatids exchange genetic material during chromosomal crossover during meiosis. It is visible in the diplotene stage of prophase I.
28. A = During anaphase II, centromeric division occurs and the sister chromatids are separated. They start to migrate towards the opposite poles.
29. D = Cleavage is a period of cell proliferation, which converts the unicellular zygote into a multicellular embryo. It is a form of mitotic division.
30. D = Pair of homologous chromosomes segregates during anaphase I of meiosis so that diploid (2n) chromosome can be halved to haploid (n) in order to restore the genetic content of zygote to that of the parents.
31. E = At metaphase, the chromosomes are individually arranged on the metaphase plane (metaphase plate) or the middle of the cell.
32. C = The pistil (gynoecium) is made up of an ovary, one or more styles and one or more stigmas.
33. A = Petals are highly coloured and so attract pollinating agents.
34. D = Protection and photosynthesis.
35. A = Monocarpous flowers are composed of one carpel (a simple pistil). The terms apocarpous or syncarpous refers to compound pistils (one or more carpel). Apocarpous flowers contain two or more distinct carpels. In syncarpous flowers, two or more carpels are fused together.
36. B
37. B = An inflorescence may be defined as cluster flowers, all flowering arising from the main stem axis or peduncle.
38. D = Actinomorphic (regular) are flowers that are radially symmetrical in that they are able to be bisected into similar halves in more than one vertical plane, forming mirror images. Zygomorphic (irregular) are flowers that are bilaterally symmetrical so that they are able to be bisected into similar halves only one plane, forming mirror images.

39. C = Dichogamy is having pistils and stamens that matures at different times, thus promoting cross-pollination rather than self-pollination. Homogamy is the maturation of pistil and stamen of a flower at the same time, ensuring self-pollination.
40. D
41. A
42. A = Dichogamous flowers are either protandrous or protogynous. protandry is when the stamens mature before the pistil. Protogyny is when the pistil matures before the stamen.
43. E
44. D = Wind pollinated flowers have long styles so that the stigma can project from the flower catching pollen grains flowing in the air.
45. C = They do not have sticky grains or stamen.
46. A = Cross pollination allows variable characteristics.
47. A = Insect visit in other to feed but pollinate in the process of feeding.
48. B = Crotalaria is protandrous; when a bee alights on the wing petals, its body weight pushes down the wing petals and the keel; the pollen in the keel is pushed out of it into the undersurface of the bee by the rounded anthers.
49. B = After fertilization, a flower dries, its petals drop, the top of the pistil falls off and the ovary swells as the seeds form inside it. After its ovules are fertilized, the ovary becomes the plant's fruit.
50. A = Flowers with inferior ovaries are termed epigynous, the flowers are above. Flowers with superior ovaries are termed hypogynous, the flower are below.
51. B = Double fertilization ensures the formation of fertile embryo (diploid) and endosperm (triploid). Diploid (male gamete + egg cell). Triploid (male gamete + definitive nucleus).
52. B = See 24.
53. D = During fertilization, there are three nuclei in the pollen tube – a tube nucleus and two male nuclei from generative nucleus.
54. A = I – Integuments, II – Polar nuclei, III – Egg nucleus, IV – Two synergids, V – Micropyle. Micropyle V allows the passage of pollen tube and pollen nucleus.
55. C = Female gamete is the egg nucleus. One male nucleus fuses with the egg nucleus to form zygote (2n), the other migrates to the centre of the embryo sac and fuses with the two polar nuclei to form a triploid (3n) primary endosperm cell.
56. A = Simple fruits are produced from a single flower in which the pistil is made up of one carpel or several fused carpels.
57. D = Indehiscent fruits – non-splitting fruits. Dehiscent fruits – splitting fruits.
58. C = Caryopsis – pericarp and seed coat have become fused together e.g. grains of cereals (maize, guinea corn, millet) and grasses.
59. B = Follicle – formed from one carpel, split along one side only e.g. cnestus, magnolia. Legume – formed from one carpel, split along two sides (dorsal and ventral) e.g. crotalaria. Capsule – formed from two or more fused carpels and split longitudinally.
60. C = Drupe – Pericarp consists of 3 layers: epicarp – thin outer skin mesocarp – middle fleshy or fibrous layer; and endocarp – hard inner layer enclosing the seed.
61. D = Multiple or compound fruits are produced from many flowers, the ovaries are fused together, e.g. pineapple and fig.
62. B
63. D – In parietal placentation, ovules are arranged along walls of a syncarpous ovary. In marginal placentation, ovules are arranged along the joined edges of a single carpel. In axile placentation, ovules are arranged at the centre of the ovary of a syncarpous ovary.
64. B
65. D
66. B = A maize grain is a fruit and not a seed because the testa and the fruit wall fuse after fertilization.
67. E = An aggregate fruits is formed from a single flower with several free carpels.
68. D = A schizocarp is a dry fruit formed from a syncarpous ovary (fused) in which the carpels break into several parts each containing one seed, i.e. mericarps.
69. B = Groundnut is a legume, the pericarp is not hard and tough. Nut have tough, hard and stony pericarp e.g. cashew nut.
70. C
71. B = A tree fruits is formed from fertilized ovary. Seeds are found from ovules.
72. C
73. C = Complete – possess all the floral parts; regular (actinomorphic) can be cut into two halves from any plane (radially symmetrical); hermaphroditic – possesses both stamen (androecium) and carpel (gynoecium); inferior ovary (perigynous) – outer floral part appears to be above the ovary.
74. A = Samara is an achene with pericarp extended to form wings e.g. combretum.
75. A
76. A = Castor oil seed is a dicotyledonous seed while maize is a monocotyledonous seed.
77. B = An aggregate achene or aggregate fruit.
78. C
79. D
80. C = 1 carpel = follicle and legume; 2 or more fused carpels = capsule.
81. C = Possession of floss (fine silky thread) around seeds e.g cotton and silk cotton enhance their dispersal by wind.
82. A = Matured pararubber explode, dispersing seed up to 15 metres. Mango and oil palm and dispersed by man. Tecoma is dispersed by wind. Triumfetta fruits stick to the fur of animals that disperse them.
83. B = I – Ureter (from the kidney), II – Prostate gland, III – Vas deferens IV – Urethra.
84. C = Vasectomy is achieved by severing the vas deferens (sperm duct).
85. B = Oviduct and uterus are part of female reproductive system clasper is found in some fishes.

86. B = The epididymis store spermatozoa which are released on ejaculation.
87. E = Seminal vesicle (vesicula seminalis) secrete a significant proportion of fluid that ultimately become semen.
88. B = *Vas deferens* transport sperm. Sperm is transferred from the vas deferens into the urethra, collecting secretions from the male accessory sex glands such as the seminal vesicles, prostate gland and the bulbourethral glands which form bulk of the semen.
89. C = Urine and spermatozoa pass through the urethra
90. E = The bladder store urine only.
91. D = see 89
92. D
93. B = I – Ovary, II – Uterus, III – Cervix, IV Vagina V – Vulva.
94. B = Fertilized egg is implanted and developed in the uterus, II.
95. C = The ovum matures in the graafian follicle, where it is released at ovulation into the uterus. It is transported to the uterus and unless fertilization occurs, and out through the vagina.
96. D = The placenta functions as a selective maternal – fetal barrier. The placenta is an organ that connects the developing foetus to the uterine wall to allow nutrient uptake, waste elimination and gas exchange via the mother's blood supply.
97. D = Fertilization typically takes place on the upper part of the oviduct (fallopian tube).
98. C = Sex determination occur at fertilization.
99. A = Identical twins – One ovum fertilized by one sperm. Non identical twins – two ova fertilized by two sperms.
100. D = An ectopic pregnancy is an abnormal pregnancy that occur outside the womb (uterus). The baby (foetus) cannot survive, and often does not develop at all in this type of pregnancy.
101. C = Parturition is the process of delivering the baby and placenta from the uterus to the vagina to the outside world. Also call labour and delivery.
102. C = Once superficial implantation occurs, the embryo begins receiving nourishment directly from the cells lining of the mother's uterus. Later when the yolk sac is formed it provides nutrients from the mother to the embryo before the placenta is ready to function.
103. D = The outer cells of the blastocyst form a covering of two membranes, an outer chorion and inner amnion containing amniotic fluid which protects the embryo from shock and desiccation.
104. A
105. B = Birth control pills used by females contain oestrogen, which depress follicle stimulating hormone (FSH); and progesterone, which thickens the cervical muscles so that entry of sperms into the uterus is impeded.
106. A = Cervix is a muscular ring closing the lower end of the uterus.
107. B = Amniotic fluid is found in pregnancy.
108. C =
109. A = The placenta is responsible for gas exchange (lungs), removal of waste products (kidneys) and nutrition (digestive system).
110. C = Some placenta hormones are human chorionic somatomatotropin (hCS), human chorionic gonadotropin (hCG), progesterone and oestrogen.
111. B = Chorion – outermost membrane; amnion – inner membrane.
112. D = Groundnut is a fruit, fruits are product of fertilized ovary (sexual reproduction).
113. D
114. D
115. A
116. C
117. A = Pineapple crown of leaves are called offsets. They can be used to propagate pineapples.
118. D = Stigma before anther – protogyny; anther before stigma – protandry.
119. E = Yam is an underground. Underground stems are parts of plants which helps it to reproduce. It stores food for the plant and MUST be underground.
120. C = The petiole secures the leaf to the stem.
121. E = The leaves of the onion are modified into a storage organ known as bulb.
122. B = Food is stored in bulbs (e.g. onion and ginger) as sugar not as starch.
123. E = *Bryophyllum pinnatum* can be easily propagated because the leaf is capable of reproducing vegetatively by producing buds.
124. B
125. E = Binary fission is a method of reproduction in unicellular organisms by splitting into two.
126. D = Gonads are reproductive organs. Testes in male, ovary in female. They produce gametes – reproductive cells.
127. C = Lizards do not show parental care.
128. D = Airspaces are characteristics of seeds or fruits dispersed by water so that they can float easily without sinking.

CHAPTER THIRTEEN

1. C = Growth is an irreversible increase in the size and weight of an organism by the addition of new protoplasm.
2. A = Without the addition of new protoplasm, growth cannot be said to have occurred.
3. D = Changes in dry weight gives a reliable estimate of growth.
4. B = Growth occurs when anabolism (build up of complex molecules) exceeds catabolism (breakdown of complex molecules).
5. D = Optimum growth is attained when the organism can reproduce itself.
6. B
7. B = The radicle is a miniature root; a little above the root tip is the actively dividing meristematic cells, the outermost layer of which forms the root cap and the piliferous layer.
8. A = Etiolation occurs when plant grows in the dark. It results in lengthening of internodes, reduced and pale yellow leaves due to absence of chlorophyll.
9. C = In the curve, III represent the highest growth rate, I represent birth.
10. C = Environment resistant is greatest at the point with

- little growth IV. V represent negative growth or senescence (old age). The shape of the curve is sigmoid.
11. B = Unilateral illumination of shoot causes auxin to move away from the light source. This gives a greater concentration of auxin on the shaded side of the shoot, which in turn causes increased cell elongation and thus bending of the shoot.
 12. B = Growth in animals is equal in all parts of the body (intercalary) and limited to certain periods in life. Growth in plants is restricted to growing regions (apical meristem) and continuous throughout life.
 13. D
 14. B = During germination, the embryo develops into a seedling.
 15. D = 1 – Fused pericarp and testa (seed coat), 2 – Endosperm, 3 – Embryo, 4 – Scutellum, 5 – Point of attachment to cob.
 16. C = The endosperm stores food which nourish the embryo and the growing parts.
 17. B = A dormant seed is a viable but dry seed that can remain for a long period without germinating. Dormancy ensures germination during a favourable season. Seed dormancy may be reduced by growth inhibitors, impermeability of the testa, delayed development and concentration of seed.
 18. B = In epigeal germination, the cotyledons are carried above the soil due to the elongation of the hypocotyl. The hypocotyl is the part of the embryo just below the cotyledon.
 19. C
 20. D = In hypogeal germination, the cotyledons remain in the soil due to the elongation of the epicotyls. The epicotyls is the part of the embryo just above the cotyledon.
 21. B
 2. B = The micropyle is a tiny hole through which water is absorbed during germination.
 3. B = Water helps to activate enzymes; it provides a medium for metabolic reactions and it transports nutrients to the growing regions of the embryo.
 - C = Intrinsic factors such as growth hormones – Auxin (indole acetic acid), gibberellins; and enzymes are most critical for plant growth.
 - A = Condition necessary for germination are: sufficiency of water, adequate supply of oxygen, suitable temperature, food or nutrient and light.
 - C = Oxygen is needed for tissue respiration, leading to the production of energy.
 - 4
 - 3 = Alkaline pyrogallol is used in the experiment to show that oxygen is necessary for germination, its function is to absorb oxygen.

= Endospermous seed are swollen with stored food e.g. maize seed.

= Auxin (indole acetic acid) a powerful growth stimulant is effective in extremely low concentrations.

35. B
36. C = Heat is produced as seeds respire.
37. E = The sequence of events in seed germination is as follows: (i) Seed absorbs water and swells (ii) Testa splits (iii) Radicle appears and grows downwards (iv) Plumule appears and grows upwards.
38. C
39. A
40. E = Transition zone
41. E. In epigeal germination e.g. bean and castor oil, the cotyledons, when exposed to light, become green and carry out photosynthetic function.

CHAPTER FOURTEEN

1. E
2. C
3. A = Receptor to CNS – sensory neuron, CNS to effector – motor neuron.
4. D
5. D
6. A
7. C
8. B
9. C
10. A = A neuron is made up of cell body, dendrites and axon.
11. B
12. C
13. C = Another classification of various receptors divides them into (i) teleceptors “distance receivers”, which are concerned with events at a distance; (2) exteroceptors, which are concerned with external environment near at hand; (3) interoceptors, which are concerned with the internal environment; and (4) proprioceptors, which provide information about the position of the body in space at any given instance.
14. D = Blinking of the eye is an involuntary (reflex) action, controlled by the medulla oblongata.
15. E = The primary function of the peripheral nerve is to transmit signals from the spinal cord to the rest of the body or to transmit sensory information from the rest of the body to the spinal cord.
16. B = The olfactory lobes of the brain eventually interpret the sensations of smell.
17. E = Hypothalamus is responsible for temperature regulation.
18. C = The spinal cord is an extension of the medulla oblongata of the brainstem.
19. None of the above. The left cerebellar hemisphere. Coordinates the left arm and leg.
20. B = The brain and spinal cord is covered by meninges, which have 3 layers, an outer pia matter, middle dura matter and inner arachnoid matter.
21. B = The name ventral fissure was given because it is wide.
22. E = The dorsal root ganglion (or spinal ganglion) is a nodule on a dorsal root that contains cell bodies of neurons in afferent spinal nerves.
23. A = The grey matter contains neural cell bodies, (nerve cells) while the white matter contains myelinated nerve fibres. It is white because of the myelin.

24. D
25. A
26. D = Motor neuron only exist in the CNS (brain and spinal cord).
27. C = 1 - node of Ranvier
28. D = Na^+ influx and K^+ efflux causes depolarization which leads to generation of action potential.
29. A = Transmission of impulse across a synapse is a chemical process. Acetylcholine is a neurochemical secreted by the synaptic knob which facilitate neuromuscular transmission (nerve to muscle).
30. B
31. D
32. C = Ganglion contains cell bodies of neurons.
33. C = 1 - sensory neuron, 2 - relay neuron, 3 - grey matter, 4 - motor neuron.
All the cell bodies of the neurons in the spinal cord are found in the grey matter (3) while the nerve fibres of the cell bodies are present in the white matter
34. A = In reflex action impulse flows as shown: Receptor in skin sensory neuron (1) relay neuron (2) motor neuron (4) effector in muscle.
35. E = 3-4-1-5-2
36. A = I - Choroid, II - Fovea (yellow spot), III - Vitreous humour, IV - Optic nerve. The choroid nourishes the eye.
37. C
38. D = Outer sclera, middle choroid and inner retina.
39. B = In the retina are the cone-shaped and rod-shaped cells for bright and dim vision respectively.
40. A = During image formation, light rays are bent most at the air/cornea surface and the lens. NOTE: The aqueous and vitreous humour also bend light rays.
41. A = When the gaze is directed at a near object, the ciliary muscle contracts. This decreases the distance between the edges of the ciliary body and relaxes the lens ligaments, so that the lens springs into a more convex shape (bulges).
42. D
43. C = Ciliary muscles and the lens tend to lose their elasticity at old age.
44. B = In myopia (short-sightedness), image is formed in front of the retina.
45. B = Nocturnal animals (e.g. bat) have more rods than cones. Diurnal animals (e.g. man) have more cones than rods.
46. A = In correcting myopia, concave lens is used to diverge incoming rays and bring them to focus on the retina. In correcting hypermetropia, convex lens is used to converge incoming rays to bring to focus on the retina.
47. A = That is long sightedness. Corrected with convex lens.
48. D = I - Auditory meatus, II - Ear ossicles, III - Oval window, IV - Cochlea, V - Eustachian tube. Hearing is brought together by I, II, IV.
49. D
50. C = Sound entering the hear follow this path: Auditory meatus, eardrum (tympanum), ear ossicles, oval window.
51. B = Outer ear - pinna and auditory canal; middle ear - tympanum, ear ossicles, and eustachian tube; inner ear

- semicircular anal, ampulla, utricle, saccule, cochlea, and auditory nerve.
52. A = Vibrations are amplified and carried across oval window by the ear ossicles.
53. D = Tip - sweet, sides of centre - salty; side back - bitter.
54. D = The taste buds are sensitive to the four primary tastes.
55. D = Chemicals in the surrounding air dissolve in mucus film and stimulate the chemoreceptors which send off sensory impulses through the olfactory nerves to the brain.
56. B = I - Sebaceous gland, II - Nerve fibre, III - Blood vessel, IV - Sweat gland.
57. A
58. D = Endocrine system secretes hormones. Hormones are responsible for chemical coordination.
59. A = Endocrine system releases hormones (chemical); nervous system, for example, neuromuscular transmission, releases acetylcholine. Therefore, they both involve chemical transmission.
60. A = Growth hormones are secreted in the anterior pituitary gland, which is in the brain.
61. D = Antidiuretic hormone (ADH) and oxytocin are released from the posterior pituitary gland.
62. B = Deficiency of thyroxine (from thyroid gland) causes cretinism - poor physical and mental development.
63. B = Thyroid gland secrete thyroxine, which regulates metabolic rate, growth; mental and physical development.
64. B = Adrenalin is released during emergency to increase heart beat, and elicit some sympathetic actions which increase efficiency in danger.
65. C = The islet of Langerhans of the pancreas secrete insulin and glucagon. They decrease and increase blood glucose respectively.
66. C
67. D = I - Pituitary gland, II - Thyroid and parathyroid gland, III - Adrenal gland; and IV - Gonad.
68. B = Parathyroid gland secrete parathormone which raise the level of calcium ions in the blood.
69. B = Growth hormone is also called somatotrophic hormone.
70. B = Insulin is secreted by the beta cells of the islet of Langerhans in the pancreas.
71. A
72. B
73. B = Phototropism is caused by the unequal distribution of auxin, auxin accumulate on the shaded side of the shoot.
74. D = Abscissic acid is a growth inhibitor. Options A, B, and C contain abscissic acid.
75. C = Auxins are produced in the shoot and root apices, hence control apical dominance.
76. A = Gibberellins do not induce the formation of adventitious root, unlike auxin. This is why auxin can be used in stem cutting.
77. B = Auxin have no effect on the growth of dwarf plants but if gibberellins was applied to them, they will grow to normal height.
78. C = Gibberellins have their effect on plant stem in the

same way as auxins, by stimulating cell elongation in particular. Rapid cell division is by cytokinin.

79. C = Ethylene induces ripening of fruits. Unripe fruits are exposed low level of ethylene to ripen them for the supermarket shelves. The effect of ethylene on the plant is to raise the rate of respiration and this cause the ripening of fruits.
80. C = Auxin can affect the metabolism of the cells, making them to respire excessively. These effects cause the death of the plant. As a result, synthetic auxins can be used as very effective weed killers. They are absorbed much more effectively by broad-leaved (dicotyledonous) plants than by monocotyledonous plants and are therefore particularly useful for removing broad-leaved weeds from monocotyledonous cultures as in lawns and fields.
81. D = Both gibberellins and cytokins break seed dormancy.
82. D = Both auxins and gibberellin are produced in the meristems of apical buds and embryo; and are both responsible for cell elongation and apical dominance. NOTE: Cytokinin and auxin are both responsible for cell division.
83. B and C = Homeothermic animals are able to maintain a relatively constant body temperatures. When the environment is cold, the body needs to gain heat. Conduction, convection and radiation can cause both heat loss and heat gain to the body, evaporation is a mechanism of heat loss only, in which a liquid is converted to gas. Sweat is also a mechanism of heat loss only.
84. A = Osmoregulation is the process by which a constant balance of osmotic pressure is maintained in the body of an animal by control of salts, ions and water content.
85. A = Poikilothermic animals are also called cold blooded e.g. fish, amphibian and reptiles while homeothermic animals are also called warm blooded e.g. birds and mammals.
86. D
87. C = Homeostasis is a feedback reaction that ensures a fairly constant internal environment. It is a reflex action.
88. A = When the body is too hot, it decreases heat production and increase heat loss by peripheral dilation, the dilation of blood vessels in the skin. When these vessels dilate, large quantities of warmed blood from the core of the body are carried to the skin, where heat loss may occur via radiation, convection and conduction. Sweating also causes cooling.
89. B = Increase osmotic pressure of blood (increase Na^+) will cause the increase in reabsorption of water and decrease reabsorption of salts by the kidney nephron.
90. C = Decrease osmotic pressure of blood (increase water) will cause the increase in reabsorption of salts and decrease in the reabsorption of water by the kidney nephron.

complex body movements and positions of the head in man to maintain balance.

95. C
96. A
97. C = The malpighian layer is a row of meristematic cells, they produce new epidermal cells by mitosis.
98. B = The cells of the malpighian layer also contain melanin pigment, which gives the skin its colour.
99. A = The removal of the pancreas would affect the metabolism of glucose due to insulin deficiency and digestion of carbohydrate, protein and lipids due to the absence of amylase, trypsin and lipase.
100. C
101. C
102. C = The neurons are the functional unit of the nervous system.
103. C = Synapses are the small gaps occurring between the nerve fibres.
104. A
105. B = The neurolemma (neurilemma or sheath of Schwann) is a nucleated cytoplasmic layer of Schwann cells that surrounds the myelin sheath of axon in the PNS.
106. C = The Nodes of Ranvier are small gaps between the myelin sheath along the length of a nerve fibre. There is no neurilemma in the CNS.
107. E = None of the above. Reflex actions are involuntary.
108. B
109. E = Cranial nerves are nerves from the brain while spinal nerves are nerves from the spinal cord.
110. C = The taste buds and olfactory organs respond to chemical stimuli.
111. D
112. D = The vitreous humour is more viscous (thicker) than the aqueous humour.
113. D
114. C
115. E = The rays of light that enters the eyes are refracted by the cornea, lens, aqueous humour and vitreous humours.
116. A
117. B = The anatomy of the inner ear is dominated by large fluid-filled spaces. The bone tube (sometimes called the bony labyrinth) is filled with fluid called perilymph.
118. C
119. B = The ossicles are made up of malleous, incus and stapes.
120. E = The eustachian tube connect the ear to the oropharynx. Vibration in the ear is lost down the Eustachian tube.
121. C
122. A = The function of the cochlea is to convert vibration by using cilia and fluid filled chambers to send nerve impulses to the brain in order to convert these vibrations to sound (hearing).
123. D
124. D
125. C

CHAPTER FIFTEEN

1. E = Ecosystem is a self supporting unit consisting of the biotic (living) and the abiotic (non-living) components that are in a continuous interaction.
2. B = Population is defined as a collection of similar organisms that are found in the same habitat while a species is defined as a collection of similar organisms that are capable of interbreeding freely to produce a viable offspring.
3. C
4. B = A community consists of different population of organisms living together in a particular habitat.
5. C = Physical factors are abiotic factors.
6. C = Guinea grass = 15; *Ipomea* spp = 5
Sida spp = 7; *Imperata* spp = 23 Total = 50
% of *Imperata* spp = $23/50 \times 100 = 46\%$
7. B = A caterpillar and an aphid living in different parts of the same plant are occupying different ecological niches – the smallest area within an habitat.
8. A
9. C = A biosphere is the part of the earth and its atmosphere in which living organisms exist or that is capable of supporting life.
10. B
11. B
12. A
13. C = Abiotic factors are non-living factors, plankton, insect, conifers and bacteria are living things.
14. A = Biotic factors are living factors. Temperature, water and soil are non-living things.
15. A = Terrestrial and aquatic habitats will have these in common; rainfall, temperature, pressure, light, wind, and pH.
16. D = Soil factors are referred to as edaphic factors.
17. A = pH affects the distribution of animals in marine and fresh water habitats.
18. A = An environment is everything surrounding an organism that affects it. Physical environments are abiotic, this, with the biotic environment, affects an organism.
19. C = Intertidal zone is exposed to wave action and the organism are subjected to desiccation.
20. B = Density dependent factors affect population growth by either increasing or decreasing as population density changes e.g. predation, food, competition, diseases. See page 123.
21. B = Competition is a biotic factor (living) which affects any habitat with limited resources.
22. C = A turbid pond will impair the penetration of water thereby limiting the intensity of light needed for photosynthesis.
23. A = The main problem of intertidal organisms is exposure and the resulting drying out (desiccation) when the tide ebbs; to overcome this, organisms in sandy and muddy shores burrow into the soft substratum; those with shells e.g. molluscs, withdraw into their protective shells.
24. D = Marshes and swamps are characterized by high salinity, tidal movement, water-logged soil and anaerobic soil resulting in high acidity (low pH).
25. B
26. C

27. B = Transect method is one of the methods used for estimating the number and types of plants in a particular area.
28. B
29. A = Microscope is used to magnify and view organisms or part of organisms.
30. B
31. C
32. A
33. C = Microscope, potentiometer, chemical balance, telescope are not used to measure ecological factors.
34. B
35. D
36. C = Capture and recapture methods are used to monitor the census of bird, fish, and insect population. A quadrant method is used to determine the frequency or population of inactive animals like snail, millipedes or aphids on a leaf.
37. C = Barnacles and limpets are found in the marine habitat. They compete for space in the intertidal zone.
38. A = Deserts have low, irregular rainfall, low temperature at night and high temperature by day, low relative humidity, strong-wind and high intensity of sunlight, very sparse vegetation. Temperature is the most important of these abiotic factors.

CHAPTER SIXTEEN

1. D
2. C = X is a saprophyte – feeds on dead decaying organic matter. Y is a substrate.
3. D = Symbionts are unrelated.
4. B = Epiphytes derives only support from host without causing any damage.
5. A = Saprophytism is between a saprophyte (living organism) and a substrate (non-living organism).
6. B = Lichen is a symbiotic association between algae and fungi.
7. A = Nitrogen fixing bacteria are found in the root nodules of leguminous plants.
8. A = *Mycorrhiza* is a symbiotic association between fungi and roots of higher plants. The fungi aid the roots in transport of inorganic nutrients from the soil into the plant while the plant supplies the fungi with organic nutrients.
9. C = Mistletoe is a partial plant parasite. It is parasitic due to its possession of a modified root called haustoria to obtain inorganic nutrient when it pierces the host's stem. It is autotrophic because it can carry out photosynthesis.
10. A = *Hydra viridis* have *zoochlorellae* algae cells in their bodies. These algae photosynthesize, producing sugars and other things which are both used by the algae and the Hydra. Hydra also supplies the algae with its mineral requirements.
11. D = Bacteria in the caecum of ruminants help to produce cellulase for the digestion of cellulose diets of the ruminants and when digested, they produce vitamin B and K. The bacteria in turn, gain shelter, protection and ideal environment.
12. D = See 8.

B = All food chains and webs begins with a producer (an autotroph).
 E = The number of individual organisms decreases progressively down the trophic level in the food chain i.e. from producers to consumers.
 A = Only a fraction of a solar energy that reaches the earth's surface is absorbed by the plants not all the solar energy.
 B = Decomposers help to recycle nutrients e.g. fungi and bacteria.
 C = Mouse is an omnivore (consume both plant and animal materials).
 A = In a pyramid of number, hawk will have the least population compared to the others.
 C = Every trophic level begins with producers (autotrophs) and then the consumers (heterotrophs).
 B = The concentration of non-biodegradable substance will increase as it goes higher up the food chain. E.g. DDT in aquatic body, fish (2nd trophic level) 0.001 ppm, seagull (3rd trophic level) 0.01 ppm.
 B = As we move forward in a food chain, there is a decrease in the number of individuals.
 C = The highest energy is at the level of producers, there is always a great loss of energy as food is transferred from one trophic level to the next.
 C = Pyramid of numbers represents the number of individuals at each trophic level of food chain at a particular time.
 B = Pyramid of energy represents the rate of flow of food energy through each trophic level of a food chain.
 D = The most efficient is from the sun to the producers while the least efficient is from consumers to decomposers.
 C = Organisms become larger and fewer along the food chain.
 D =
 B = Algae (producer), tadpole (primary consumer), Tilapia (secondary consumer), shark (tertiary consumer).
 D = Second law of thermodynamics.
 D = Energy transfer decreases along the trophic level. This leads to a reduction in the population size of organisms at the higher trophic level. Pyramid of biomass represents the total wet or dry mass of an organism in each trophic level. It takes into account the size and number of individual organism.
 B = Carbondioxide is released into the atmosphere during respiration, combustion, decay and volcanic eruption.
 C = Carbondioxide will be fairly maintained by the balance between photosynthesis (depletes CO_2) and combustion (replenishes CO_2).
 A = I - Evaporation, II - Transpiration, III - Precipitation, IV - Surface run-off.
 B =
 A = Evaporation releases water to the atmosphere where it condense in the cloud. Precipitation returns water to the earth as rain.
 C = Capillary water is held in capillary pores of the

- soil. Plant roots are able to absorb it, hence it is the available water for plants.
39. A = During thunderstorm, nitrogen in the atmosphere combines with oxygen to form nitrogen (II) oxide; with the addition of more oxygen, nitrogen (IV) oxide is formed. This dissolves in water to form trioxonitrate (V) acid HNO_3 , which reacts with bases in the soil to form nitrates.
 40. C
 41. D = Nitrification is the conversion of ammonia or ammonium salts into nitrites by *Nitrosomonas* and nitrite into nitrates by *Nitrobacter*.
 42. C = Leguminous plants have symbiotic nitrogen fixing bacteria in their root nodules; the bacteria fix nitrogen thereby enriching the soil with organic nitrogen.
 43. A = Denitrification is the process by which soil nitrates are reduced to atmospheric nitrogen through the action of denitrifying bacteria. All other processes would help to add nitrate to the soil.
 44. D = See 41.
 45. D = Dead plants and animals are acted upon by saprophytes to form ammonia. The ammonia is converted into nitrites by *Nitrosomonas* and nitrite is converted into nitrates by *Nitrobacter*.
 46. A
 47. A = Most denitrifying bacteria respire anaerobically, e.g. *Thiobacillus denitrificans*, obtain its energy by anaerobically oxidizing sulphur compounds; nitrates is used as the oxidizing agent and in the process it is reduced to gaseous nitrogen.
 48. B = Denitrification is the reduction of nitrates in the soil into nitrogen gas by denitrifying bacteria e.g. *Escherichia coli* and *Thiobacillus denitrificans*.
 49. B
 50. A = Soil nitrite *Nitrobacter* Soil nitrate.
 51. E = An example of symbiont living in the root nodules of leguminous plant is *Rhizobium*, not given in the option.
 52. C = The *Rhizobium* obtain photosynthetically derived carbohydrates and an ideal environment to live.
 53. E = Nitrates are made into proteins in plants.
 54. C = Putrefaction is one of the seven stages in the decomposition of the body of a dead animal. Some of the end products of putrefaction are: ammonia and ammonium compounds, hydrogen sulphide and hydrogen phosphide.
 55. D = organisms that are not dependent are capable of surviving on their own i.e. autotrophs. In parasitism, commensalism and symbiosis, organisms either depend on each other or one depending on the other while the other is either unaffected or harmed.
- | | | |
|-------|-------|-------|
| 56. D | 57. D | 58. D |
| 59. D | 60. A | |
61. C = Zooplankton form autotrophs in the aquatic habitat.
 62. A = Mineralization refers to the process where an organic substance is converted to an inorganic substance.
 63. D = *Pseudomonas denitrificans* is a denitrifying bacteria.
 64. A

CHAPTER SEVENTEEN

1. B = In plants with floating leaves, stomata may be found only on the upper epidermis; submerged leaves may lack stomata entirely.
2. B = Scorpion and ant-lion are not present in fresh water.
3. D = Brackish water is formed when freshwater and seawater mixes at estuaries. Brackish water has salinity which fluctuates with tides, wet and dry seasons. In the lagoon, for example, the salinity of the water increases during the dry seasons and decreases during the rainy season.
4. C = Halophytes – plants adapted to salty environment. Xerophytes – plants adapted to dry habitats. Hydrophytes – plants adapted to aquatic habitats. Mesophytes – plants that live in an environment with adequate water supply. Epiphytes – plants which depends on another plants for support only.
5. A = Aerenchymatous tissues are tissues of thin-walled cells with large, air-filled intercellular spaces found in the roots and stems of some aquatic and marsh plants for buoyancy.
6. A = Air cavities (or air-filled intercellular spaces) which enhances buoyancy to float on water.
7. D = Stilt roots of the red mangrove enable it to stand firm in strong ocean winds and keep the branches above high tide levels. Pneumatophores (breathing roots) of the white mangrove allow intake of atmospheric air.
8. A = In mangrove plants, seed germinate on the parent plant, i.e. are viviparous, and develop long radicle.
9. B = Pneumatophores (breathing roots) of the white mangrove.
10. D = There are more species (specie diversity) in the rainforest than other habitats.
11. C = Littoral zone is along the edge of the stream.
12. B = Tilapia is found in freshwater. Tortoise – land and fresh water. Achatina is the land snail.
13. B = Abiotic factors are physical or non-living factors.
14. C = Date-palm thrive best in almost rainless districts like in grassland and desert.
15. C = Water conservation is an important adaptation required by terrestrial organisms particularly in the Sahel savanna and desert. Food, predator and space affects all organisms regardless of the habitat.
16. A
17. A = Rainfall is the most important ecological factor that affects biotic community in a terresrial habitat.
18. B
19. C = Reduction in the number of stomata on the leaves reduces the loss of water vapour via transpiration.
20. C
21. D
22. A

CHAPTER EIGHTEEN

1. C = Guinea savanna can be found in Kaduna, Enugu, Benue, and Kwara.
2. A = Rivers state (Port-Harcourt) is an estuarine and Yobe state (Damaturu) is a Sahel. Moving from: estuarine → rain forest → Guinea savanna → Sahel.
3. B = Up north is Sahel, down south is estuarine.
4. C = Non-epiphytic ferns and fern allies live in the

- shade created by the canopy formed by the tall broad-leaved trees.
5. C = Rain forest is rich in epiphytes and climbers. They use tall trees as support to reach for the light.
6. D = Savanna trees possess thick, fire-resistant barks. In the Sahel, plant bark posses thorns.
7. C = Southern Guinea savanna has more rainfall and tall grasees compared to the Northern Guinea savanna.
8. C = Savanna grasses have underground stems (*Rhizomes*) helping them to root the soil properly.
9. B = Guinea savanna is between the rain forest and Sudan savanna.
10. D
11. A = Cactus is a plant found in the desert and mammals also undergo aestivation for long periods in the desert.
12. A = Perennating organs are organs possessed by plants that enable them to survive from one growing season to the next e.g. bulb, runner, sucker, corm e.t.c.
13. A = The greatest species diversity occur in the tropical rainforest. Transect method is difficult to use in estuarine. It is also difficult to use in Guinea and Sahel savanna because of few plant species.
14. C = Northern Guinea savanna and Sahel savanna can be found in Sokoto state.
15. Trees are planted as shelter belts in the Sahel to reduce the harmattan wind.
16. A = Desert plants have devices to reduce transpiration e.g. reduced leaves and stomata
17. C = Desert has low rainfall and high temperature.
18. A Relative humidity is the amount of moisture in the atmosphere; it is high when there is high rainfall and low temperature, and low when there is low rainfall and high temperature.
19. B = Katsina state falls in the Sahel savanna, which is very susceptible to desert encroachment.
20. D = Countershading is more important in some aquatic organisms like fishes.
21. C = Between the littoral and benthic zone, light can only penetrate the littoral zone while benthic zone is dark and cold.
22. B = Grasses are predominant in Guinea savanna, which grow tall during the rainy season with sparse trees.
23. C
24. A
25. D = Cactus plant is structurally adapted to survive by possession of swollen stems that store food and water; and the leaves are reduced into thorns.
26. C = Mambilla plateau is a high grassland plateau with an average elevation of about 1,524 metres found in Taraba state of Nigeria. It is the highest plateau in Nigeria.
27. D = Mangrove swamp in Nigeria is restricted to the tropical rainforest.
28. D = Montana vegetation is typical of Adamawa and Jos plateau.
29. C = Tropical deserts has a temperature ranges from 27°C to 33°C in summer; to 15°C in winter. The absence of cloud in desert skies result in hot days and very cold nights, rainfall is low and fast blooming plants during rainy season.

CHAPTER NINETEEN

1. B = Species are group of similar organisms that can interbreed freely. Population deals with the total numbers of organisms of the same species in a habitat.
2. C = Density dependent factors affect population growth by either increasing or decreasing as population density changes. Those that can limit a population include competition, disease, predation, parasitism, food shortage.
3. D = Population will increase when there is high birth rate, low mortality rate, high immigration rate and low emigration rate.
4. D = Seed dispersal provides means of preventing overcrowding; seeds are dispersed away from the parent plant.
5. B
6. C = Capture and recapture methods are used to monitor the census of bird, fish, and insect population. Quadrant method is used to determine the frequency of population of plants and inactive animals like snail, millipedes or aphids on leaf.
7. B = Overcrowding can be caused by immigration (movement into), natality (birth), high survival rate. Overcrowding can be reduced by emigration (movement out of), mortality, and low survival rate.
8. D = Turbidity is peculiar to aquatic habitat. It is the depth of visibility in water, which is a measure of penetration of light.
9. B = Exponential (logarithmic) growth doubles the size of population at a constant interval. It occurs in bacteria.
10. D = Population density is the average number of individuals of a species per unit area of habitat. $\text{Density} = \text{Population size} / \text{Area of habitat}$. Population frequency is how often a species occur at different sites in its habitat.
1. B = Population equilibrium is attained when there is a balance in nature (Predator-prey) as a result of the attainment of its carrying capacity.
2. B = Food scarcity increases mortality rate by increasing competition.
- B = Population dynamics is the changes (increase or decrease) in population size over time.
- C = The pioneering organism in a plant succession on bare rock surfaces are some species of crustose lichen, followed by foliose lichen and then mosses after the lichen have changed the rock to soil.
- C = Primary succession starts on sites which have not previously borne vegetation.
- = Competition will lead to succession by the emergent organisms.
- = In the given trend, Spirogyra is the simplest plant and water lily is the most advanced.
- = Climax forms the final stage of succession. It is characterized by stable composition of plant and animal species.
- = Primary succession involves the formation of new land.
- Secondary succession is faster than primary succession because soil is already present.

20. A = As succession approaches climax, there is an increasing stability.
21. C
22. D = Growing of vegetation on an abandoned farmland is an example of secondary succession.
23. B = Intra-specific competition arises when organisms of the same species compete for a limited resources.
24. B = In a normal succession, species diversity increases until it becomes fairly constant due to the attainment of a climax community.
25. D
26. A
27. B = Space by organism in an habitat
= $\frac{\text{Area of habitat}}{\text{Number of organisms}}$
28. B = Territorial behaviour helps organisms to guard against intruders thereby preventing overcrowding.

CHAPTER TWENTY

1. B = Clay contains the finest particles. Silt is finer than sand but still feels gritty.
2. B = Granite is a form of igneous rock and not a soil type it contains readily recognized mineral crystals such as muscovite, quartz, biotite and hornblende.
3. D = Mineral particles in the soil are derived from the chemical breakdown of rock particles or from decomposing organic matter in a less degree.
4. D = By sedimentation, heavier particles settle at the bottom leaving lighter ones to suspend.
5. D = Clay soil is the most difficult to plough in a wet season because of its sticky nature.
6. E = The experiment is to compare the capillarity of various soil samples; capillarity is high in clay, low in sand and average in loam.
7. C = The experiment is to determine the percentage of humus in soil sample. The presence of humus release the smoke.
8. A = After sedimentation, particles settles in this order (light to heavy): humus, clay, silt, sand, gravel (stones).
9. D = Clay has high water retaining capacity. Although humus can also retain water, it has air spaces for soil aeration. Sand, however, is very porous; hence a soil devoid of sand and low in humus with high clay content would retain the highest water.
10. D = Laterite is a red soil; it is common in the tropics. It is produced as a result of leaching; the soil tends to be acidic because lime potash from the topsoil are leached away by rain action.
11. B = Water retention in soil is greatest in clay soil and poorest in sandy soil. It indicates the capillarity of the soil.
12. B = Nitrates are particularly prone to loss through leaching due to their high solubility in water.
13. A = Irrigated lands after some years become unfertile.
14. A = Continuous cropping reduces soil fertility.
15. B = Leaching is the removal of soluble material from a substance, such as soil or rock, through the percolation of water.
16. C = Gravel (>2mm); feel coarse, sand (2- 0.05 mm); feels gritty, silt (2-0.05mm); feels like flour but still gritty, clay (<0.002 mm); feels sticky when wet.

17. D = Weight of crucible alone = 5 gm
 Weight of crucible + fresh soil before being heated = 10 gm
 Weight of fresh soil being heated is (10-5)gm = 5 gm
 Weight of crucible + fresh soil after being heated = 8 gm
 Weight of water in the soil is (10-8) gm = 2 gm
 Percentage % of water = $\frac{2}{5} \times \frac{100}{1} = 40\%$

18. C =
 Volume of air = vol. of water + vol. of soil added - vol. of water + soil after stirring
 = $(500 + 350) \text{ cm}^3 - 800 \text{ cm}^3$
 = $850 \text{ cm}^3 - 800 \text{ cm}^3$
 = 50 cm^3
 % of air in soil sample = $\frac{\text{Volume of air}}{\text{Volume of soil + air}} \times 100\%$
 = $\frac{50 \text{ cm}^3}{350 \text{ cm}^3} \times 100\%$
 = 14.28%

19. D (Closest option). From the given parameter
 Mass of soil = (29-10)g = 19g
 Mass of soil + water = (29-18)g = 11g
 % of water loss = $\frac{\text{Mass of water}}{\text{Mass of soil + water}} \times 100$
 = $\frac{11}{19} \times 100$
 = 57.89%

20. A = All soil have humus in common in varying degree.
 21. D = The spaces between clayey soil particles are very tiny because of its fine nature.
 22. D = Sands have a specific heat capacity.
 23. E = Fewer nutrients because of its poor aeration.
 24. D
 25. A = Erosion can be reduced by ridging across slope this slows down the flow of water and less soil is removed.
 26. D = Mixed grazing leads to overgrazing which exposes the soil to erosion.
 27. B = The process by which lime is added to clay soil is called flocculation. Lime is any material containing high amount of calcium or calcium and magnesium. Flocculation reduces soil acidity, encourages good granular soil structure for better root development.
 28. C
 29. C = Soil microorganisms help in the recycling of nutrients like carbon and nitrogen.
 30. D = Liming makes the soil more porous by flocculation of the soil particles thereby improving capillary action.
 31. A = Microbes conserve and renew soil fertility by the cycling of nutrients. They also decay dead organic matter.
 32. B = Excessive application of fertilizer alters soil pH.
 33. C
 34. A (Closest option) = soil erosion can be controlled by:
 i. Vegetation protection;
 ii. Contouring of sloping ground;
 iii. Terracing of slopes;
 iv. Mulching;

v. Strip cropping.

35. B = Bacteria and fungi are decomposers. They are largely responsible for the decay of dead organic matter and its subsequent recycling.
 36. A = Crop rotation can maintain soil fertility, it is a practice whereby farmland is planted with different crops in successive seasons.
 37. B
 38. D = To a soil scientist humus is the organic, non-cellular, long lasting component of the soil. It may also have some trace elements.
 39. C = Soil water can be classified into 3: Gravitational water – occupies the larger pores of the soil. It is of no use to plants because is easily pushed down by gravity. Capillary water – is held in capillary pore of the soil. Plant roots are able to absorb it; hence it is the available water for plant. Hygroscopic water – The water held tightly at the surface of soil colloid particle, it can be utilized by some microorganism not by plants.
 40. A = Soil contains SWOLAM = small rock particles, water, organic matter (humus), living organisms, air (oxygen and CO_2), and mineral salt.
 41. A = Cover crops such as grasses check leaching.
 42. C
 43. C = Soil enhance plant growth, which are the primary producers that maintain life.
 44. B = Leguminous cover crops enrich the soil, the residues are also rich in organic nitrogen from decomposing nodules.
 45. B = Percentage of water in soil = $\frac{\text{Mass of water}}{\text{Mass of soil}} \times \frac{(x-y) \times 100}{(x)}$

46. C
 47. D = Soil pH tends to decrease (increase acidity) with the repeated use of nitrogen fertilizers.
 48. B = Amount of soil water is given as: Mass of crucible and soil, X_2 g – Mass of crucible and soil after heating, X_1 g.

CHAPTER TWENTY ONE

1. A = Cholera can only be contacted by eating or drinking contaminated food or water.
 2. E = Syphilis and gonorrhea are both caused by bacteria.
 3. D = Cholera is caused by bacteria. Malaria is transmitted through the bite of female *Anopheles* mosquito. Syphilis is caused by bacteria. Sleeping sickness is caused by a protozoan (*Trypanosome*). Small pox is a viral disease, it causes skin blister and is transmitted through close contact with an infected person.
 4. D = The life cycle of Bilharzia involves two hosts, man and a water snail. Man is the primary host and snail is the secondary host.
 5. D = Control of Onchocerciasis is mainly by the destruction of the blackfly (vector) and by treating infected person with drugs.
 6. A = Swollen shoot disease of cocoa is a viral disease.
 7. D = Blackfly, vector of Onchocerciasis breeds in fast flowing water.

2019-08-14

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D = Malaria - female *Anopheles* mosquito; cholera -
housefly; river blindness (Onchocerciasis) - blackfly.
B = Bilharzias can be controlled by destroying the
water snails that belong to the class of mollusca and by
ensuring that water bodies are not contaminated by
urine or faeces of infected persons.

D = Guinea worm is caused by drinking infected
water. It can be controlled by provision of portable
water.

A = Elephantiasis is transmitted through the bite
female *Culex* mosquito while Taeniasis gets into man
through pig or cow meat.

D = Vaccination or inoculation is the introduction of
the mild form of the disease - causing organisms into
the body. This stimulates the production of antibodies
which help to build immunity against the disease in
the body.

D = The construction of dam may lead to an increase
in the prevalence of the diseases that are associated
with water such as malaria (mosquitoes breed in
water), bilharzias (has water snail as the intermediate
host) and onchocerciasis (blackflies breed in fast
flowing streams).

A = Typhoid fever is caused by *Salmonella typhi*.

B = Bird flu is caused by virus.

A = The infective cercariae of (Schistosome can
penetrate the bladder (*Schistosoma haematobium*)
causing bloody urine.

B = River blindness (Onchocerciasis) is transmitted
blackfly, *Simulium damnosum*.

A = Poliomyelitis is caused by a virus.

B = Immunization is the process of rendering an
animal resistant to infection artificially, by conferring
either passive immunity or by vaccination.

D = *Schistosoma haematobium* penetrate the bladder.
The larval stage is called Miracidium which swims
with the aid of cilia.

D = Infective plasmodia of the malaria is the
sporozoite.

D = An obligate parasite is completely dependent on
its host. *Taenia* (tape worm) cannot survive without
host man.

A = Tuberculosis is a bacteria infection transmitted by
droplet infection in coughs, sneezes and conversation.

D = Crowded habitation favours the spread of
tuberculosis.

C = Only meningitis (inflammation of the meninges)
is not associated with water among the options.

D = Female *Aedes* mosquito transmits a virus causing
yellow fever

D = *Plasmodium falciparum* causes malaria, which is
transmitted by female *Anopheles* mosquitoes.

C = United Nations Children Fund (UNICEF) is an
international organization set up to improve health
and welfare of children and nursing mothers.

D = Refuse, when not burnt in incinerators, gives out
smoke at a level that is detrimental to the health of
living organisms.

D = Applying chemicals to water bodies would cause
water pollution.

B = Loud disco music constitute noise pollution.

C = Carbon monoxide poisons tissues by combining

with haemoglobin to form carboxy haemoglobin,
which reduces the affinity of haemoglobin for oxygen,
a condition that may lead to suffocation.

33. B = Some petrol often contain tetraethyl lead, thereby
putting people who suck it at risk of increasing their
blood concentration of lead.

34. C = Sulphur (IV) oxide is an acidic poisonous gas; it
can pollute air, water and soil because it dissolves in
moist air to form acid, when inhaled, it irritates and
damages the sensitive living part of the eyes, air
passages and lungs. It dissolves in water to form acid
rain which dissolves the aluminum salts in soil causing
them to build up to toxic levels in underground water
supplies; this affects sources of drinking water.

35. C = Oil is an imported pollutant of the marine
environment in Nigeria because of the perpetual oil
spillage.

36. D = Chlorofluorocarbon (CFC) react with the Ozone
(O₃) in the upper atmosphere converting it to oxygen,
thereby depleting the ozone layer.

37. C = Agro-chemicals are used in the farms e.g.
fertilizers when in excess, they are washed by rain into
fresh water bodies thereby causing water pollution.

38. A

39. D = Radioactive materials can be carcinogenic
(substances that cause cancer).

40. A = Carbon monoxide is issued from car exhaust,
which has strong affinity for haemoglobin.

41. B = Greenhouse effect is a gradual increase in
temperature of the earth due to excessive amount of
heat absorbing gases such as CO₂. Trees absorb CO₂
for photosynthesis. Deforestation is cutting of trees.

42. A = Excess CO₂ in the atmosphere could lead to global
warming or greenhouse effect.

43. A = The degree of water pollution can be tested using
the biochemical oxygen demand (BOD). This test
measures the amount of oxygen bacteria needed to
break down the organic matter in a water sample in a
fixed period of time.

44. A = Activities that releases excess CO₂ into the
atmosphere can result in green house effect (global
warming) said as burning of fossil fuel e.g. coal.

45. A

46. D = Non-biodegradable pollutants cannot be broken
down into simpler harmless substances e.g. plastics,
glass, tin, and metal scraps. They are better controlled
by recycling.

47. D

48. C = FEPA is empowered to prevent indiscriminate
destruction of the environment that may cause
pollution.

49. B = Water, soil, wildlife, forest are all renewable
resources. Wildlife are very important in conservation
because they are under perpetual threat by man's
activities.

50. A = Mineral resources are non-renewable hence the
need to conserve them.

51. B = Poaching is the illegal catching or shooting of
animals.

52. B = Bush burning leads to the destruction of diverse
species of organisms.

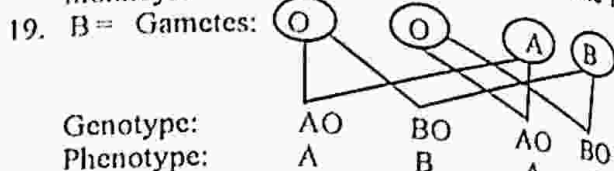
53. B

54. C
 55. A = Malaria is caused by -- *Plasmodium*, which is a protozoan.
 56. C
 57. A = Recycling is used for non-biodegradable wastes such as glass, metal scraps, plastics. Organic matter is biodegradable.
 58. C = River dams, acid rain, sewage are not sources of air pollutants.
 59. A
 60. C = Nuclear energy is powered by radioactive materials such as uranium obtained from the earth crust - nonrenewable.
 61. B = Oxides of sulphur (SO_2) and nitrogen (NO_2)
 62. D
 63. D = *Schistosoma*, commonly known as blood - flukes and biharzia. *Schistosoma haematobium* is commonly referred to as bladder fluke because it enters the bladder.
 64. A
 65. C
 66. C
 67. B
 68. B
 69. D

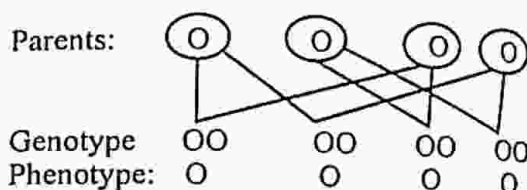
CHAPTER TWENTY TWO

- C = Variation accounts for the essential differences seen among organisms of the same species.
- A = Individuals with unfavourable traits naturally go into extinction leaving individuals with favourable traits to dominate.
- C = Physiological variations are variations in the functioning and activities of living organisms. These include: blood group, tongues rolling and PTC tasting. They are also discontinuous variation.
- B = Discontinuous variation does not offer an intermediate form e.g. tongues rolling, you are either a tongue roller or not.
- B = Left-handedness is a discontinuous variation. You can either be left handed or not.
- B = Continuous variations show gradual transition between two extreme form as a result, there are intermediate e.g. size, height, weight and colour.
- A = The different fingerprint pattern are arches, loops, whorls and compound or double whorl.
- A = Disease resistance is a physiological variation.
- A = See question 3.
- B
- E = Any character shown by an organism is due to the genes inherited from the parents, the environmental factors such as nutrition, education, climate etc
- C = An organism is a product of its inherited genes and environment.
- D = Identical twins inherit the same gene hence the same genotype but can show distinct variations when raised in different environments.
- C = Sexual reproduction give rise to variation while asexual reproduction do not.
- B = Fingerprint is unique to individuals though individuals may have same pattern.
- A = Blood group O is a universal donor.
- D = Agglutination occurs when contrasting antigen (e.g antigen A) combine with antibodies (e.g antibody B)

18. A = Rhesus factor was first identified in the Rhesus monkeys.



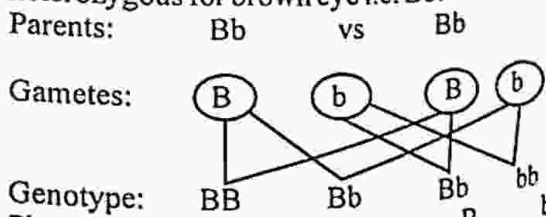
20. B = In the ABO blood system, the recognizable groups are: A, B, AB and O (4 groups)
 21. A = Parents with blood group O cannot produce a child with any blood antigen (A or B)



22. A = DNA analysis is the most accurate to determine paternity.
 23. B
 24. D
 25. C
 26. C
 27. C
 28. A = Blood group O is a universal donor because it lacks blood antigen (A and B). Blood group AB is a universal recipient because it lack blood antibodies (Anti-A and anti-B).
 29. A
 30. B

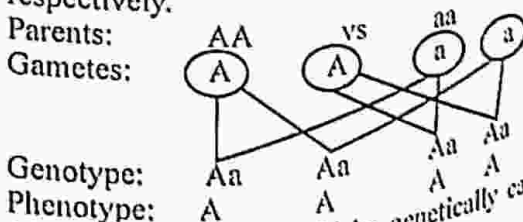
CHAPTER TWENTY THREE

- E = None of the children will be blind because blindness is not an inherited trait.
- D = A monohybrid inheritance is a single characteristic that is controlled by the gene.
- C = Let the gene for brown and blue eye "B" and "b" respectively. Genotype for homozygous brown eyed girl and blue eyed boy is BB and bb respectively to get these two genotypes, the parents must be heterozygous for brown eye i.e. Bb.



Homozygous brown = BB
 Homozygous blue eye = bb

4. B = Albino is a recessive trait. Therefore let the gene for normal male and albino female be AA and aa respectively.



5. D = Heredity is the transmission of inherited character from parents offspring via genes. Good yield is a function of the gene and its transferable from

generation to generation.

B = Gregor Mendel was known to be the father of genetics

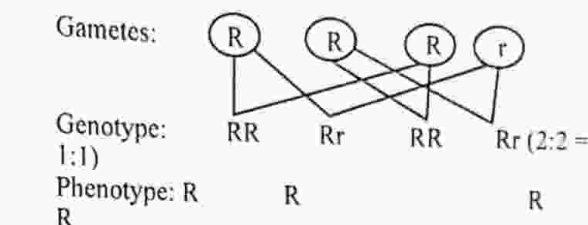
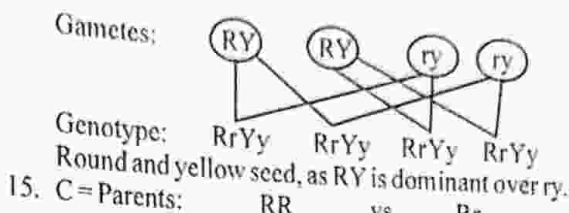
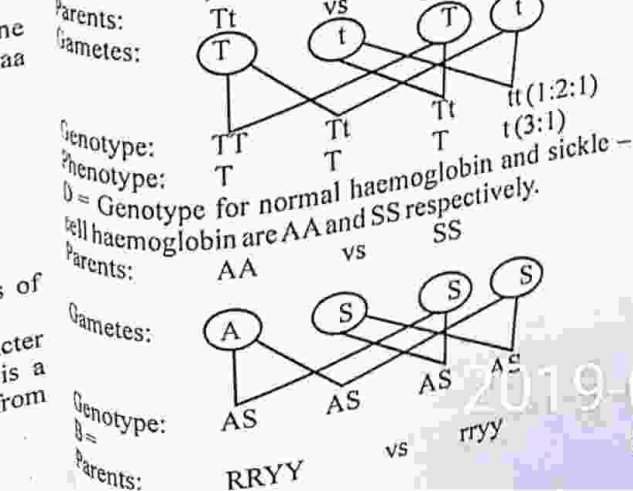
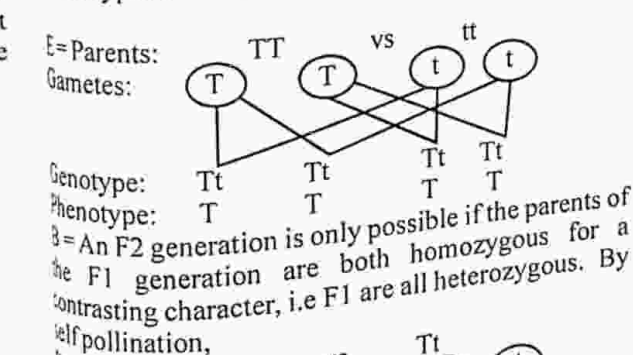
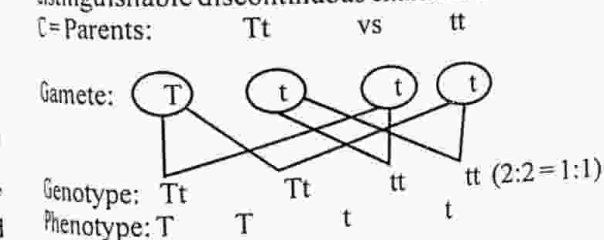
A = The key words in Mendel's first and second law are segregation (separation) and assortment (recombination) respectively.

Mendel's first law (also called the law of segregation) states that during the formation of reproductive cells (gametes), pairs of hereditary factors (genes) for a specific trait separate so that offspring receive one factor from each parent.

Mendel's second law (also called the law of independent assortment) states that each member of a pair of homologous chromosomes separates independently of the members of other pairs so the results are random.

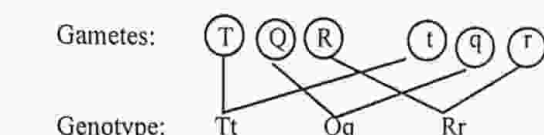
Mendel's third law (also called the law of dominance) states that one of the factors for a pair of inherited traits will be dominant and the other recessive, unless both factors are recessive.

A.
B = For most genetic studies, fruit fly, *Drosophila melanogaster*, was found suitable because it is small and could be kept in the laboratory in large number, it has a short lifecycle of two weeks; it has just four homologous pairs of chromosomes; the male and the female can be easily distinguished and it has many distinguishable discontinuous characteristics.

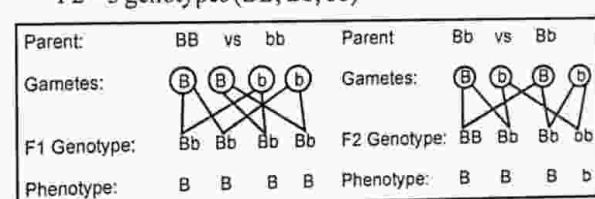


16. B = The gene content of the woman's eggs would be TQR , tqr

Parents: TQR vs tqr



17. B =
F1 = 1 genotype (Bb).
F2 = 3 genotypes (BB , Bb , bb)



18. B = Blending inheritance is an example of incomplete dominance where neither allele is dominant over each other. So the heterozygous genotype has its own phenotype.

19. B
20. B = According to Mendelian inheritance, the alternative form of a gene for a character that expresses the character in a different way is called allele. E.g. blood group in man (a discontinuous variation) is controlled by a gene I, with the alleles I^a I^b.

21. B = Alleles are two alternative genes at the locus in homologous chromosomes.

22. D = Incomplete dominance is a situation in which both member of an allelic pair are expressed in an in a heterozygous situation. In AB blood group neither antigens is dominant over the other.

23. D = I - TT, II - Tt, III - TT, IV - tt.

24. B = 1:2:1

25. D

26. E = Genes are discrete unit of inheritance that determines hereditary traits.

27. A

28. C = Recessive genes are always suppressed in the presence of dominant genes as a result are only expressed in the homozygous form.

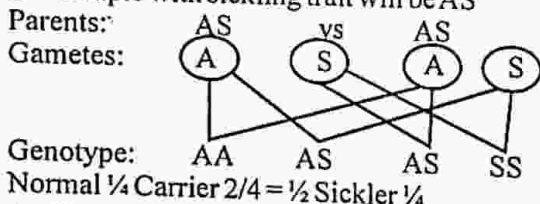
29. C

30. D

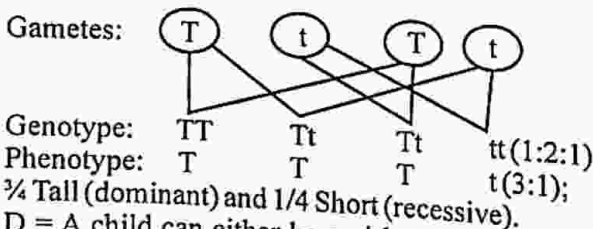
31. B

32. I

33. C
34. $C = 2n$ is a diploid number, whereas in gametes, there are haploid numbers (n) of chromosome; therefore, in an organism where $2n = 16$, $n = 16/2 = 8$.
35. $C =$ Crossing over occurs when chromatids at meiosis exchange portions with their homologous partners. It occurs in the late prophase stage of meiosis.
36. $D =$ DNA- Deoxyribonucleic acid is made up of repeating units called nucleotides. Each nucleotide consists of a five-carbon sugar, a phosphate group and organic nitrogen containing base; Adenine, Guanine, Cytosine and Thymine.
37. $C =$ Somatic cells contain diploid chromosome denoted as $2N$.
38. $C =$ In a DNA strand, guanine will only pair with cytosine (i.e. G to C) and adenine with thymine (A to T). A DNA strand with a base sequence TCA will have a complementary strand of AGT.
39. $D = XX$ - Female; XY - Male.
40. $C =$ After meiosis the number of chromosomes changes from diploid ($2n$) to haploid (n).
41. $D =$ In RNA, uracil is found instead of thymine. In DNA, A and G are purines; T and C are pyrimidines.
42. $B =$ Couple with sickling trait will be AS



43. $A =$ Heterozygous parents e.g. Tt for height
- Parents: Tt vs Tt

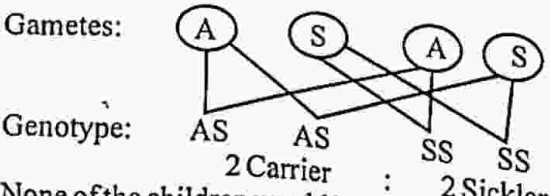


44. $D =$ A child can either be a girl or a boy; hence, the probability of the fifth child being a girl is one out of two.

45. $A =$ F1 All Tt. F2 = TT, Tt, Tt, tt

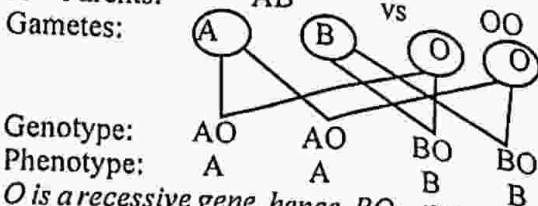
Number of dwarf = $\frac{1}{4} \times 120 =$

46. $D =$ Parents: AS vs AS



None of the children would be normal

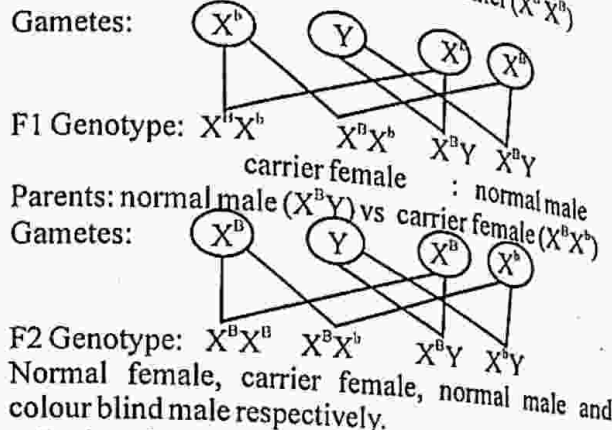
47. $A =$ Parents: AB vs OO



O is a recessive gene, hence, BO will be B and AO will be A. Since they are six, three will fall into group B and the other three into group A

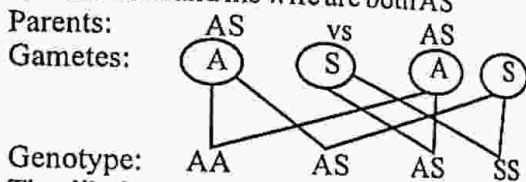
48. $A =$ Parents:

Colour blind father (X^bY) : Normal mother ($X^B X^B$)



$\therefore$ Ratio of sufferers, carriers and normal are 25%, 25%, and 50% respectively.

49. $C =$ The man and his wife are both AS



The likely parentage of their offspring that will be either carriers or sickler is $\frac{3}{4} \times 100 = 75\%$

50. B

52. $B =$ Out breeding (hybridization or crossbreeding) leads to introduction of new traits. It increase the hybrid vigour.

53. $B =$ Inbreeding ensures the preservation of traits within a given population.

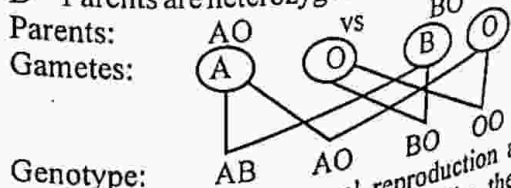
54. B

55. $A =$ The DNA is unique to every individuals.

56. $C =$ Blood grouping in human beings is derived from the combination three different alleles A, B and O. Blood group is a character being controlled by multiple allele.

57. $A =$ People who have Rhesus antigen on the surface of the red blood cell are said to be Rhesus positive RR or Rr and those that lack the antigen are said to be Rhesus negative (rr). Agglutination occurs when a Rhesus negative person receives Rhesus positive blood due to the release of anti- Rhesus antibodies by the immune system of the Rhesus negative person that reacts with the antigen on the red blood cells; agglutination also occurs when a Rhesus negative mother has a Rhesus positive child; hence, genetic counselling is necessary if a Rhesus negative woman marries a Rhesus positive man since there is a probability of the couple having a Rhesus positive child.

58. $D =$ Parents are heterozygous A and B respectively i.e.

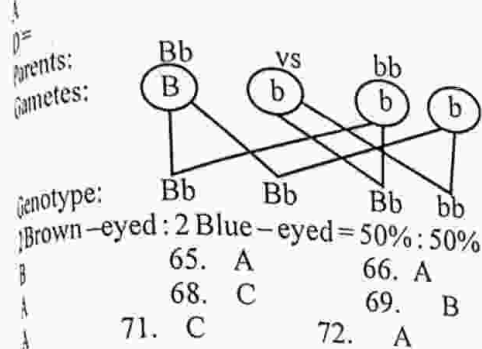


59. $C =$ Offspring of asexual reproduction are called clones. They almost are exactly like the original organism.

60. $C =$ Sickle cell anaemia is caused by a mutation in a gene for haemoglobin that results in an altered haemoglobin molecule.

61. $D =$ Two tall parents can only give birth to a short child

when the two parents possess recessive gene for shortness. The parent must be heterozygous tall (Tt).



C = An ovum has a haploid (n) chromosome but a fertilized ovum has diploid (2n) chromosome (23 pairs or 46).

A = Mongolism is an obsolete name for Down's syndrome which is XXX (Trisomy 21).

C = Phenotype is the observable expression of the characters of an organism resulting from the interaction between its gene and the environment it is the totality of the expressed traits of an organism.

A = A hybrid is the cross between organisms having contrasting traits. Hybrids are produced by the process called hybridization.

CHAPTER TWENTY FOUR

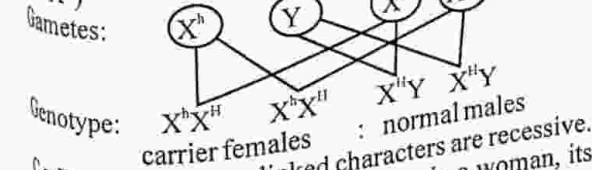
A = XX - female XY - male. The ovum always carry the X chromosome while the sperm cell could carry X or Y chromosome. Therefore to get a female, the ovum must be fertilized by the sperm cell carrying an X chromosome.

A = A cross between a colour blind mother and a colour blind father will give rise to all colour blind offsprings.

A = River blindness is caused by a round worm, *Onchocerca volvulus*, through the bite of the blackfly, *Simulium damnosum*.

C = Haemophilia is a sex-linked trait. It is carried on the X-chromosome. Genotype of haemophilic man can be represented as X^hY

Genotype of normal woman is X^HX^H .
Parents: Haemophilic father (X^hY) : Normal mother (X^HX^H)



C = The genes for sex-linked characters are recessive. When it occurs in the recessive form in a woman, its effect is suppressed by the dominant X chromosome. In man, it is readily expressed in the phenotype because they have one X chromosome, they cannot carry it in the recessive form.

B = Sons inherit the sex chromosomes of the mother and the sex linked trait is on the mother's X-chromosome.

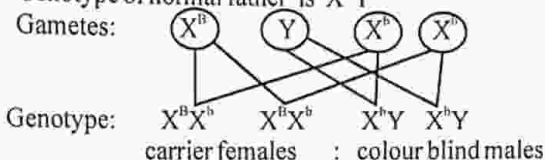
C = Sex linked trait are carried on the X-chromosome.

chromosome.

9. A = Sex-linked traits are ONLY carried on the X-chromosome. Y-chromosome is empty of the traits. Also the male child cannot receive X-chromosome from his father.

10. D = The sons inherited the gene for colour blindness from their mother. As shown below:

Let gene for colour blindness be X^b
Genotype of colour blind mother is X^bX^b
Genotype of normal father is X^BY
Gametes: X^b , Y, X^b , X^b



11. D = Colour blindness is the inability to recognize some colours usually red and green.

12. A A child with X^BY (male) will have normal vision while a child with X^bY (male) will be colour blind. Reason, sex-linked characters are expressed in the recessive form.

13. A = X^BX^b - carrier female, X^BX^B - normal female, X^bX^b - bald female.

14. A = Sex is determined by the contributions of father's sex (X or Y) chromosome.

15. A = Turner's syndrome (monosomy XO) is a sterile female.

16. A = Ability to grow long hair in females is sex-linked because it is carried in the X chromosome. The Y chromosome is deficient in genes. Possession of facial hair is determined by the Y-chromosome (male sexual character).

17. C = Sex is determined at fertilization (conception).

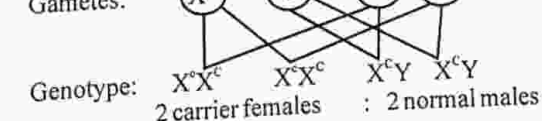
18. A = X chromosome (see question 6).

19. D = Let gene for colour blindness be X^c

Genotype of normal female is X^CX^C

Genotype of colour-blind male is X^cY

Gametes: X^C , Y, X^c , X^c



20. A = See 14.

21. A = albinism is caused as a result of genetic mutation.

22. D = Dracunculiasis caused by *Dracunculus medinensis* is a water-borne disease and haemophilia is sex-linked.

23. B = Sex-linked characters occurs more often in men than women because it can only be expressed in the recessive form.

24. D = See 11.

26. A = See 15.

25. C = Sex-linked.

CHAPTER TWENTY FIVE

1. D

2. C = Autotrophs, especially plants, need light to manufacture food while heterotrophs depend on already manufactured food.

3. D = Intra-specific competitive will set in because they

- are of the same species.
4. D = Inter-specific competition occur among organisms of different species.
 5. C = A worker termite and a soldier termite are of the same species; hence intra-specific competition.
 6. C = Competition sets in when available resources become limited like in drought, fires out-break and other natural disasters.
 7. A = Endoparasites have lost their organs of movement due to disuse. The choice of option D is wrong because not all endoparasites have piecing mouthparts e.g. *Ascaris*.
 8. B = Competition will occur in a rice field when the field become weedy because the resources will become inadequate.
 9. C = Early successional stages are often characterized by low productivity. Older communities are characterized by greater productivity as a result of increasing stability.
 10. C = Arboreal animals are animals living on trees and tree trunks. e.g. flying birds, which lack teeth but they possess beak.
 11. A = Sunbird possesses a long, slender and slightly curved beak for feeding on nectar.
 12. C = Grasshoppers and cockroaches have biting and chewing mouth parts while houseflies possess sponging mouth parts. Only mosquito, tsetse flies and aphids have piecing and sucking moth parts from the lists provided.
 13. D 14. B
 15. B = The keyword here is, large.
 16. D = Mandibles are heavy toothed, jaw-like structures used to cut and crush food materials.
 17. B = Some hydrophytes have air floats in their leaves, stem and petiole for buoyancy e.g. water hyacinth.
 18. A = Camouflage helps an organism to blend with its surroundings so that it is not easily detected. Well developed limbs enhance fast movement and effective vision for good sight.
 19. A = Gills – used for respiration is only efficient in water. Lateral line helps to detect movement in water. Streamlined body with no neck minimises friction during passage in water.
 20. C = Forest species have buttress roots for stability.
 21. A
 22. B = Reptiles have dry, scaly, water proof skin which retards the loss of water.
 23. B = Chameleon changes its colour to defend/ protect itself from predators.
 24. C = Countershading in fish with dark dorsal surface (from top) and silver ventral surface (from bottom) protects them from predators.
 25. C = Red coloured head of matured male lizard which it displays in front of a female is to attract them.
 26. C 27. C
 28. A = Plants growing on dry or arid land are called xerophytes. Some of their adaptive methods are: thick succulent stems for storing water, succulent leaves for water storage, reduction of leaves to spines or scales.
 29. A
 30. D = Plants possess thick waxy cuticle and hairs on leaves to reduce water loss by transpiration. Animals possess feather or modified scales (e.g) birds to prevent the evaporation of water as sweat.
 31. D = Whistling pine (*Casuarina*) bears tiny scale leaves on needle-like branches to reduce water loss by transpiration thereby conserving enough water for use in the long dry period.
 32. D = Feathers in birds are of different types some prevent evaporation of water as sweat some provide thermal insulation e.g. down feather, some are use for flight e.g contour feather.
 33. A = The protoplasm of xerophytes are highly viscous and have high water content in their cells, as result, they can live with little water without being damaged.
 34. D = The presence of sunken stomata or lower number of stomata and folding of leaves is to reduce water loss by transpiration.
 35. A = Scales (in reptiles), feathers (in birds) and skin (in mammals) help in body temperature regulation.
 36. A = Hydrophytes live directly in water; the roots are not long and the xylem tissue, which is for the transportation of water, is feably developed.
 37. C = Lung fish are capable of an amazing form of aestivation. Lungfish are primitive fish that still have lungs, allowing them to breathe air. When lung fish's lake dries up, the fish burrow into the mud, then secretes mucus until its entire body is covered. The mucus dries into a sack that holds moisture in. Even when the mud dries completely, the lung fish stays moist and breaths through a mucus tube.
 38. D
 39. D = The bodies of insects are covered by thick cuticle which prevents water loss by evaporation. Also in a way to conserve water they excrete their nitrogenous waste as uric acid and not as urea.
 40. D = Gaping is a characteristic activity of several species of large reptiles in which the animal lies ashore with its mouth held open for prolonged periods. It is suggested as a thermoregulatory response.
 41. C = Cryptic or concealing colouration helps to keep the prey and predator unnoticed. It is used in camouflage.
 42. D
 43. D = Disruptive colouration are patterns on the bodies of animals that break up the familiar shapes of their bodies, enabling them to go undetected e.g. spots and stripes of the leopard, giraffe, tiger and zebra.
 44. B = Countershading.
 45. A 46. C
 47. B = Chematophores are pigment for colour change. Also called camouflage cells present in chameleon.
 48. D = Pheromones are chemicals produced as messengers that affect the behaviour of other individuals of insets or other animals.
 49. C
 50. C = Drones are the only males in the honey bee colony. They mate with the queen.
 51. A = The worker of the bees and termites are both sterile females. The workers termites are also blind.
 52. B = The practice of removing the queen is a means of destroying the termite colony.
 53. A = The protozoan helps to digest cellulose converting

into sugar which is absorbed by the worker termite. The protozoan in turn feed on the abundant nutrients of the worker termite as well as obtaining shelter in the termite's gut.

D
B = Waggle dance = food more than 100m away.
Round dance = food within 100m.

D
D = Social behaviour is more complex and organized among mammals.

A = Aestivation = dry and hot. Hibernation = wet and cold.

B
B = III and IV are not under the control of the organism. I and II can be adopted to avoid extreme temperatures.

B 62. B 63. B

C
A = The Ladybird beetles are remarkable for shamming death as soon as they are seized.

C 67.A 68.A 69.D

A 71.A 72.C 73.B

C 75.B

CHAPTER TWENTY SIX

D = Evolution is brought about by the differences and similarities among organisms.

C = The theory of natural selection was developed by Charles Darwin and Alfred Russel Wallace in the year 1858 in a paper titled: A Theory of Evolution By Natural Selection.

B = Darwin proposed the theory of natural selection. Animals that are well adapted to the environment are the most successful.

C = Darwin's theory of natural selection suggests the survival of the fittest by better competitors

D = Lamarck proposed the theory of inheritance of acquired characters.

A = The flaw in Lamarck theory is that acquired traits are not heritable as shown by Wallace.

A = Long neck in giraffe was used by Lamarck to explain his theory of use and disuse. He said ancestors of giraffes had short necks; it was by stretching their necks to reach for the leaves on trees that the present-day giraffes became long-necked.

D = See 6.

B = Hugo de Vries propounded the mutation theory in 1901.

B = Mutation is a random change in the DNA (present in the chromosome) that alters genetic information and so introduces new characteristics e.g. albinism, sickle cell-anaemia.

D = Natural selection allows organisms who are better competitors to survive by developing adaptive features.

A = According to Darwin, natural selection is the driving force behind evolution.

D = Environmental pressure e.g. insufficient resources, climatic changes result in natural selection.

B = The idea of evolutionary trend is from simple to complex, unicellular to multicellular. Therefore older fossils have the simplest form of life.

15. B = The pentadactyl limb is modified into wings for flying in birds, flippers for swimming in whales, legs for running and walking in dogs and arms for grasping and holding in human beings and other bipeds.

16. B = The pentadactyl limb structure is absent in pisces (fishes).

17. B = Ecology is the study of organisms in relation to their environment, it has little evolutionary evidence.

18. D = Evolutionary trend in vertebrates is from Pisces to Amphibians to Reptiles to Aves (Birds) to Mammals. Therefore mammals evolved from birds.

19. D = Pisces (1 auricle and 1 ventricle), amphibians (2 auricles and 1 ventricle), reptiles (2 auricles and a partially divided ventricle), aves and mammals (2 auricles and 2 ventricles).

20. D = Evidence from embryology showed that all embryos pass through a gillslit stage, establishing a common ancestry.

21. C = Radioisotope (carbon) dating is used in fossils to estimate ages; arranging the fossils according to their age shows evolutionary trend.

22. A = The relationship between living organisms and their extinct relatives can be obtained by studying the remains of organisms (extinct relatives) preserved mainly in earth's sedimentary rocks. This is called paleontology.

23. A = The appendix in man which functions as caecum in herbivores supports evolution. So also is the tail bone in man.

24. C
25. C = Comparative anatomy gives the structural affinities of organisms.

26. A = Homologous structures are e.g. pentadactyl limb pattern points to common evolutionary origin.

27. D = See 26.

28. D = Angiosperms are the most recent plants. They bear flowers and are divided into monocot and dicot. However, dicots are more advanced than monocots; and are capable of secondary thickening which monocots cannot.

29. A = Fossil records (for extinct organisms) helps to compare with presently living organism. Comparative anatomy compares similarities in forms (for modern organisms).

30. B

31. D = Simple forms before complex ones.

32. D = Natural selection suggests that organisms survive through adaptation and reproduce offsprings that are better adapted.

33. C = Evolutionary trend suggests that life begins from simple forms (unicellular) to complex forms (Multicellular).

34. A 35. C 36. D
37. C 38. A 39. A

40. B

41. A = Inheritance of acquired traits.

42. D = A common example of divergent evolution is the vertebrate limb. Whale flippers, frog forelimbs and human arms, see note on evolution.

43. B = Gene flow occurs among populations of the same species.

44. D = The disappearance of functionless organs was

propounded by Jean Larmarck.

45. B = The evolutionary trend in plants is: Schizophyta, Thallophyta, Bryophyta, Pteridophyta and Spermatophyta. Memory tip: **Samuel The Biologist Planted Sugarcane.**
46. C = The tail bone is a vestigial organ in human, as it has no function.
47. B = Natural selection is a consequence of adverse condition (e.g. food scarcity) due to overpopulation.

GLOSSARY

A
Abcission Separation of an organ from the plant, as in leaf fall, fruit drop and loss of unfertilized flowers.
Accommodation Changing the shape of the lens of the eye to focus a sharp image of the objects on the retina.
Active transport The movement of substances across a biological membrane from a region of their low concentration to one of their high concentration; this movement requires energy.
Adaptation Any change in an organism that makes it better suited to survive a particular environment.
Adventitious root Roots which develop from the stems or leaves of plants.
Aerobic respiration The release of energy from food with the aid of oxygen.
Anaerobic respiration The release of energy from food without the use of oxygen.
Annual A plant that completes its life cycle from germination to flowering within one season.
Anterior The front or head end.
Antibody A protein produced by an animal to counteract the effect of the presence of a foreign protein (antigen) inside the body.
Antiseptic Substance used locally on body tissue that prevents its decay by arresting the growth of microorganism.
Anus The posterior opening of the digestive tract.
Asexual reproduction Reproduction that does not involve fusion of sex cells. It may be by spore formation, vegetative propagation, fission or budding.
Autecology The study of individual organism in relation to their environment.
Autotrophic A method of feeding in which an organism builds up its food from simple inorganic molecules.

B
Balanced diet A course of feeding containing all the essential food nutrients in correct proportions and in adequate amounts for a healthy life.
Binary fission Method of reproduction in unicellular organism by splitting into two.
Bilateral symmetry The property of having two similar sides with definite upper and lower surface and definite anterior and posterior ends.

C
Calorie (C) The amount of heat required to raise the temperature of 1,000 grammes of water by one degree Centigrade.
Capillarity A force causing the rise of a liquid along the surface of a tube or vessel as in the soil.
Capillary Minute blood vessel which carries blood to individual cells.
Carnivore An animal whose diet consists of flesh e.g. cats, leopards, dogs.
Cephalization The concentration of sense organs, nervous tissue (brain) and food catching organs at the anterior end of the body forming a head.
Chitin An insoluble polysaccharide material forming the main component of the cell wall and exoskeleton of

arthropods.
Chromosome Thread-like structure in the nucleus that carries genes; it is made of DNA and protein.
Cloaca A common opening for the digestive, excretory and reproductive organs e.g. in roundworm, toad.
Commensalism An association between two organisms in which one of them, the commensal, derives benefit from the other, the host, which neither benefits nor suffers from the association.
Community Group of plants and animals living in a habitat.
Conduction A process of losing heat through physical contact with another object of body.
Conjugation Sexual reproduction in which the contents of two cells unite as in certain algae and fungi; or exchange of nuclear materials resulting in rejuvenation of the cells as in paramecium and some bacteria.
Convection A process of losing heat through the movement of air or water molecules across the skin.

D
Deamination A process in the liver that remove (NH₂) radical from amino acid and converts the liberated ammonia into urea.
Decomposer An organism that breaks down the organic matter accumulated in the bodies of other organisms.
Diffusion The movement of molecules (liquid or gas) from a place of high concentration to one of low concentration.
Digestion The process of breaking down large insoluble food molecules into small soluble molecules that can be absorbed.
Disinfectant A chemical that frees materials from infection by killing the microbes but not their spores. It is normally used on inanimate objects (c.f. antiseptic).
Dorsal Back or upper side of the body.
Droplet infection Infection that can be spread between persons through airborne particles containing viable microbes such as in coughing and sneezing.

E
Ecosystem The sum total of physical features and living organisms which interact in an environment to produce a stable system.
Emulsification The breaking down of fats into minute droplets.
Endoskeleton Internal skeleton of bone or cartilage typical of vertebrates.
Enzyme A protein catalyst produced by living cells which in small amount promotes chemical reactions e.g. during digestion.
Epicotyl The part of the plumule just above the attachment of seed leaves, (cotyledons).
Epigeal Type of seed germination in which the cotyledons come above soil level.
Epiphyte A plant that grows attached to the surface of another plant to gain a more beneficial position. It is naturally independent of the host. E.g. orchids on tree branches, mosses on tree barks.
Epithelium A layer of cells which covers cavities, tubes and all free surface of the body.
Erosion The loss of topsoil by the action of wind or water.
Etiolation Condition of green plants when grow in

darkness, characterized by thin, elongated stems and small chlorotic (yellow) leaves.

Excretion The removal of waste products of metabolism from the body.

Exoskeleton Hard outer covering which provide support for the body of certain animals especially arthropods.

F

Fertilization The fusion of a male and female nuclei to form a zygote.

Fermentation The process of anaerobic respiration in yeasts and some bacteria. In alcoholic fermentation ethanol and carbon dioxide are produced.

G

Gamete A male or female reproductive cell.

Ganglion An aggregate of nerve cell bodies located outside the central nervous system (CNS).

Gene A part of a chromosome that controls a particular hereditary feature of an individual.

Genotype The genetic characteristics of an organism.

Geotropism The growth of a parts of plant in response to gravity.

Gland A structure (cell, tissue or organ) concerned with the secretion of a specific chemical substance e.g. sebaceous gland in the skin, cells in plants secreting gums, tannins, nectar. Animal glands may secrete their products through ducts (exocrine glands) or directly into the blood as in endocrine glands.

H

Habitat The place where an organism lives; the ecological characteristics in this area are relatively uniform.

Herbivore Animals whose diet consists of plant materials.

Hermaphrodite An organism with both male and female reproductive parts

Heterotroph An organism that can not build up its own food; hence it has other means of deriving its nourishment such as parasitic, saprophytic or holozoic methods.

Homeothermic Having a body temperature that remains fairly constant regardless of changes in the temperature of the environment.

Homologous organs Body organs in different organisms that have a similar origin and structure.

Homologous chromosomes A matching pair of chromosome inherited from male and female parents.

Hormone A chemical substance produced by the ductless gland in the body of organism.

Hybrid An offspring from a cross between parents differing in one or more traits.

Hybridization The crossing of two different varieties to produce a new variety with one or more desirable traits.

Hydrolysis The splitting apart of a molecule with the addition of water.

Hydrotropism The growth movement of a plant in response to water.

Hypocotyls The part of the plumule below the attachment of the cotyledons node. In epigeal germination it grows rapidly and carries the cotyledon above the ground.

Hypogeal The type of seed germination in which the cotyledons remain below soil level and inside the testa.

I

Immunity The power to resist an infective disease through natural or artificial means.

Inoculation Injecting a person or animal with ready made antibodies or weakened virus or germs to stimulate production of antibodies.

K

Kilojoule (KJ) S.I unit of energy (1 calorie = 4184 Kilojoules)

L

Larva An immature form of some animals that undergoes radical change in form to attain the adult form.

Leaching Loss of soluble mineral nutrients from the topsoil to lower levels where plant roots cannot reach as a result of the downward movement of rain or ground water.

Loam A fertile type of soil containing both sand and clay particles in roughly equal proportion.

Lumen A space with the blood vessel through which blood flow.

Meiosis Cell division that results in halving of the chromosome number occurring in the formation of gametes.

Metabolism The physiological and chemical processes in the body.

Metabolite A substance that is produced during metabolism or one that takes part in metabolism.

Metamorphosis Changes in the life cycle of an organism through one or more forms as it matures into the adult form, each form being of different appearance. In *complete metamorphosis* the immature (larval) form is markedly different from the adult form hence there is a radical or complete transformation during the maturation. In *incomplete metamorphosis* the larval form is quite similar to the adult form hence the change is subtle.

Mitosis Cell division that results in two cells identical in chromosome number with the parent cell.

Moulting (Ecdysis) Periodic shedding and replacement of outer covering such as cuticle in arthropods, feathers in birds or skin in reptile. The shedding in arthropods allows growth to occur.

Mutualism An association between two organisms in which both benefits.

Myxoedema A disease condition resulting from low metabolic rate as a result of deficiency or lack of thyroxine.

N

Nastic movement A non-directional movement of a plant in response to a stimulus.

Nymph Young stage of an insect. It resemble the adult except that it is sexually immature and wingless.

Nerve impulse Electrical and chemical changes which pass along a nerve cell fibre (axon)

O

Omnivore An animal whose diet consists of plant or animal materials e.g. man, pig, cockroach.

Organ A collection of different tissues grouped into an identifiable unit and performing one or more function e.g. liver, heart, etc.

ism A living thing.

regulation The process by which a constant balance of osmotic pressure is maintained in the body of an animal.

control of salts, ions and water content.

osis The movement of water by diffusion across a permeable membrane.

ous Animals which lay fertilized eggs; the eggs contain poorly developed embryos with yolk to nourish it.

viviparity is when fertilized eggs are produced, developed and nourished inside the body of the mother without placenta and hatch before or soon after the eggs are laid.

Examples are certain fishes and reptiles

parasite An animal or plant which lives in or on another living organism called the host.

pathogen A disease-causing organism.

perennial A plant that continues to grow from year to year.

peristalsis Waves of muscular contraction of the wall of the alimentary canal that helps to move the food along.

Permeable membrane A membrane that allows substances to pass through it.

Phenotype The visible physical characteristics of an organism.

Photosynthesis The process by which green plants make carbohydrates from carbon dioxide and water using light energy absorbed by the green pigment, chlorophyll.

Phototropism The growth of a plant or plant part in response to light.

Placenta 1. The part of a plant ovary to which ovules are attached. 2. The tissue to which sporangia or spores are attached. 3. The structure formed from embryonic and maternal blood circulation that enables the foetus to depend on maternal systems for obtaining oxygen and nutrients as well as for removal of metabolic waste products.

Plankton The microscopic forms of life in lakes and seas; animals are called zooplankton, plants are phytoplankton.

Plasmolysis Contraction of cell protoplasm from the cell wall when placed in a highly concentrated solution due to osmotic withdrawal of water from the cell sap.

Poikilothermic Having a body temperature that varies with that of the environment. Most animals, except birds and mammals are poikilothermic.

Poison A chemical substance which when absorbed by a living body will kill or gravely harm it.

Predator An animal which hunts, captures and kills other animals for food.

Proboscis A tubular mouth in some insects (e.g. butterfly, housefly) used to suck up food.

R

Radial symmetry The arrangement of the parts of an animal body around a central axis so that any two planes are exactly alike; it is characteristic of coelenterates and echinoderms. Plant or plant organs showing this type of symmetry are termed actinomorphic.

Radiation A form of heat loss through infrared rays.

Roughage Indigestible cellulose material which adds to the size of the food and thereby promotes peristalsis.

S

Saprophyte A plant which feeds by digesting dead

organic material outside the body and absorbing the soluble products.

Secretion Production of chemical substances in the body of a cell or a gland.

Seed The structure that develops from a fertilized ovule of a flowering plant; it contains an embryo protected by one or more coats.

Segmentation The division of the body of an animal into series of similar units, e.g. in tapeworm, earthworm.

Semi-permeable membrane A membrane which allows only water but not dissolved substances to pass through it.

Sex-linked A characteristic determined by a gene on a sex chromosome.

Sexual reproduction Reproduction that involves the fusion of the nuclei of male and female gametes.

Species A group of individuals that are alike in all characteristics and are therefore capable of producing a fertile offspring.

Sperm The male reproductive cell or gamete.

Spiracle A breathing hole found in the surface of an insect's body.

Sporangium A plant part that produces spores e.g. in fungi, mosses and ferns.

Spore An asexual reproductive cell of plants produced in large numbers which may develop directly into a new organism.

Sterilization Process of rendering materials free from any microbial life.

Substrate A material on which an organism grows and secretes enzymes.

Symbiosis A natural association between two different living organisms including parasitism, mutualism and commensalism.

Synapses A gap between two nerve cells across which the nerve impulse passes.

System A number of connected organs and tissues in the body working together for a particular major function.

T

Taxis A locomotory movement of an organism or cell in response to a directional stimulus, such as chemicals or light.

Thallus Plant body without root stem or leaves.

Tissue A group of similar cells, bound together by intercellular materials, performing a particular function.

Toxin A poison produced by certain microorganism.

Translocation The movement of dissolved food materials in the vascular tissue of plants.

Transpiration The loss of water by evaporation from the aerial parts of a plant primarily through the stomata.

Tropism A growth movement in part of a plant whose direction is determined by the direction of the stimulus.

Turgid Description of a plant cells due to presence of water.

V

Vaccination The injecting into the body of weakened pathogen to stimulate the formation of antibodies and confer immunity. The weakened pathogen is called a vaccine.

Variation The essential differences seen among organisms of the same species.

Vascular tissue The tissue concerned with transporting

substances within the body of an organism such as xylem and phloem in plants and blood and lymph in animals.

Vector An agent, such as an insect, capable of transferring a pathogen from one organism to another.

Vegetative propagation Reproduction by any parts of the plant except the seed.

Ventral The belly or lower surface of an organism.

Viviparous Animals which bring out their young alive and well developed; the embryo develops within and derives nourishment from the mother through a placenta. May also be used to describe plants in which the embryo develops on the parents e.g. red mangrove.

W

Watershed A hilly region, usually over a large area, which conducts surface water to streams.

Z

Zygospore A thick-walled resistant spore developing after the fusion of two similar gametes in the reproduction of some algae and fungi.

Zygote The fertilized egg cell in plants and animals.

REFERENCES

Idodo Umeh, G. (1996): College Biology, Benin-city: Idodo Umeh Publishers Limited.

IUPUI, Department of Biology 2003 Class Note.

Odunfa, S.A. (2001): Essentials of Biology, Ibadan: Heinemann.

Regoke O.O. et al (2008): New General Biology for Undergraduates, (Revised Edition), Rajah Dynamic Printers.

Pass At Once Biology for UTME (2009), Wisdomline International (Nig.) Limited.

Rene, F.K. et al (2010): Biology for Dummies, (Second Edition), Dummies Publishers.

Sofola, O.A. (1987): A Revision Text in Medical Physiology, (First Edition), SBO Publishers.

Taylor D.J. et al (1997): Biological Science, (Third Edition), Cambridge University Press.

Usua, E.J (1967): Biology Revision Course, University Press Limited.

www.thestudentroom.co.uk/wiki/Category:A_Level_biology_Revision_Notes